ASNAT User Manual

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Welcome to ASNAT (Air Sensor Network Analysis Tool) -
a free application for retrieval, mapping, summarizing, comparing, plotting and saving subsets of air sensor data read from web services and local files.

Contents

Running ASNAT:

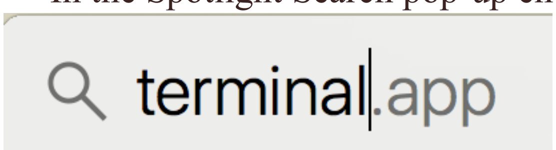
Running `ASNAT` on your local computer requires the free `R` programming environment. If you do not have `R` version 4.2 or later (including `Rscript` and `RTools`) installed on your computer, you will need to install it.

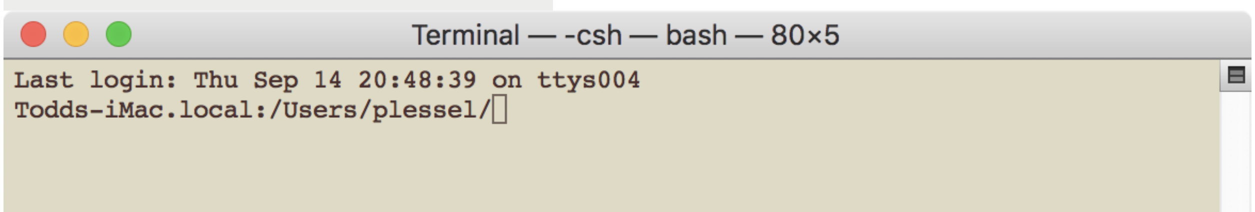
Optional but recommended: Verify R installation:

1. Open a shell window.
   a. On Windows: this is called `Command Prompt (CMD)`.
      Enter **CMD** in the Search area at the bottom of the screen.

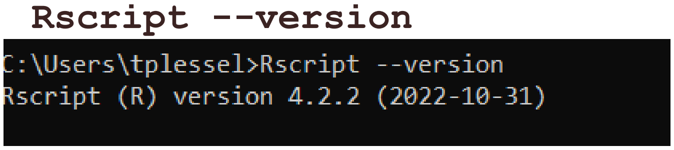
   b. On Mac it is called `Terminal`.
      Click the magnifying glass at the upper-right corner of the screen.

      In the Spotlight Search pop-up enter **`terminal`**





   c. On Linux it is called `xterm`. Click right-mouse button then select Shell.
2. Check that ***R 4.2 or later*** is installed and is in your **path**.
   In the shell enter the following command:
   `Rscript --version`



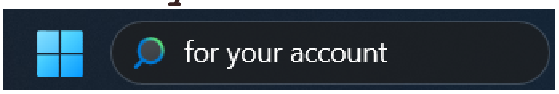
   If the reported version is less than 4.2 then you need to update *`R`* to version 4.2 or later (including *`Rscript`* and *`RTools`*).
   If the '`Rscript`' command is not recognized, but you think you have *`R 4.2`* or later

installed then edit your path to include `Rscript`.

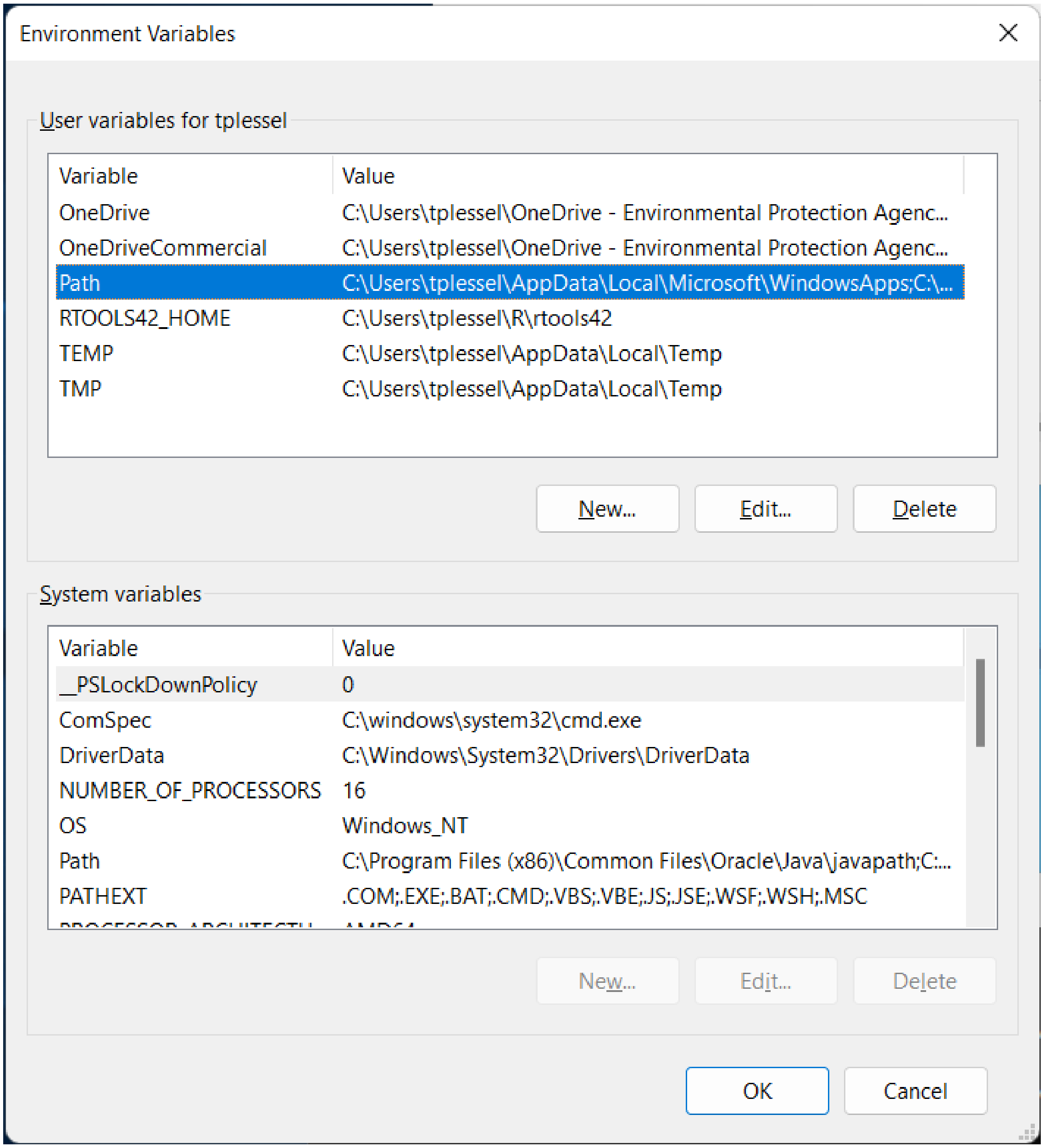
a. On Mac and Linux: edit your **`path`** environment variable in the appropriate file **`~/.cshrc or ~/.profile or ~/.bashrc,`** etc.
Note you do not need root/admin to install *R* in your home directory or edit files in your home directory.

b. On Windows: edit your path as follows:

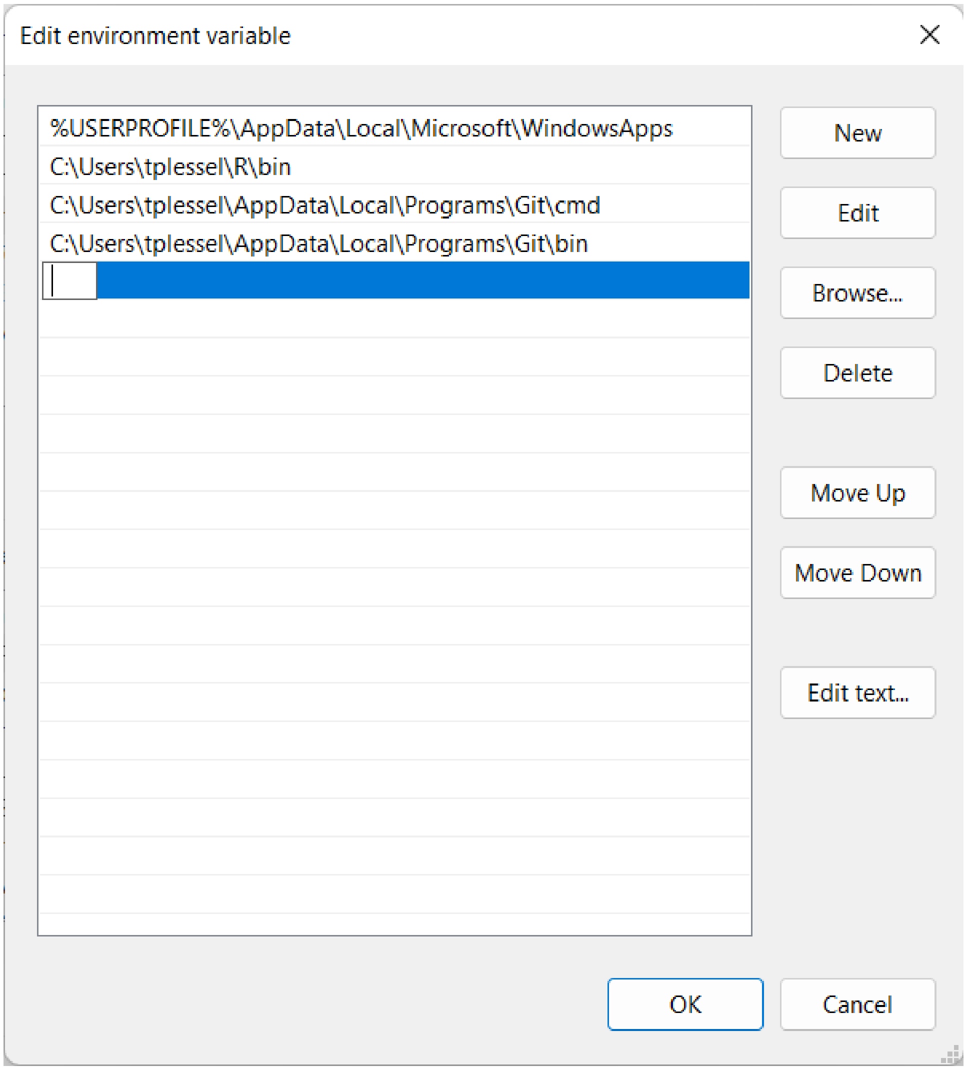
b.1. In the Search area at the bottom of the screen type
`for your account`



b.2. **Click on 'Edit environment variables for your account'**
(Do not attempt to edit system variables.)



b.3. In the popup dialog, **click on Path**.
b.4. Click the **Edit** button.
b.5. Click the **New** button.



b.6. Carefully type the path to the 'R\bin' directory. E.g., C:\Users\logname\R\bin or wherever R is installed on your computer.
b.7. Click the **OK** button.
b.8. Click the **X** in the top-right corner of the dialog window to close it.
b.9. Close the command shell.
b.10. **On Windows you'll need to restart your computer.**

Then open a new shell and check again that

```
Rscript --version
```

now reports 4.2 or greater.

If not ask a system administrator for help installing and setting-up `R 4.2` *or later* including `Rscript` and `RTools`. `Rscript` is a small shell-based *R interpreter*.

(*RTools* is a standard part of *R* that includes a free built-in *C++* compiler (`g++/mingw`) that `R` programs can use to call `C++` functions.

ASNAT is mostly written in *interpreted R* but contains some functions written in *native-compiled C++* which is 100 times faster than the equivalent R functions.)

Download ASNAT:

1. Please use the platform-specific ASNAT zip file. Ensure you have the correct file for your system (e.g., Darwin.x86_64 for Mac with Intel processors, Darwin.arm64 for Mac with M1 processors).

Next, move the `ASNAT.zip` file to your *home directory* if it was downloaded elsewhere such as a `Download` directory/folder.

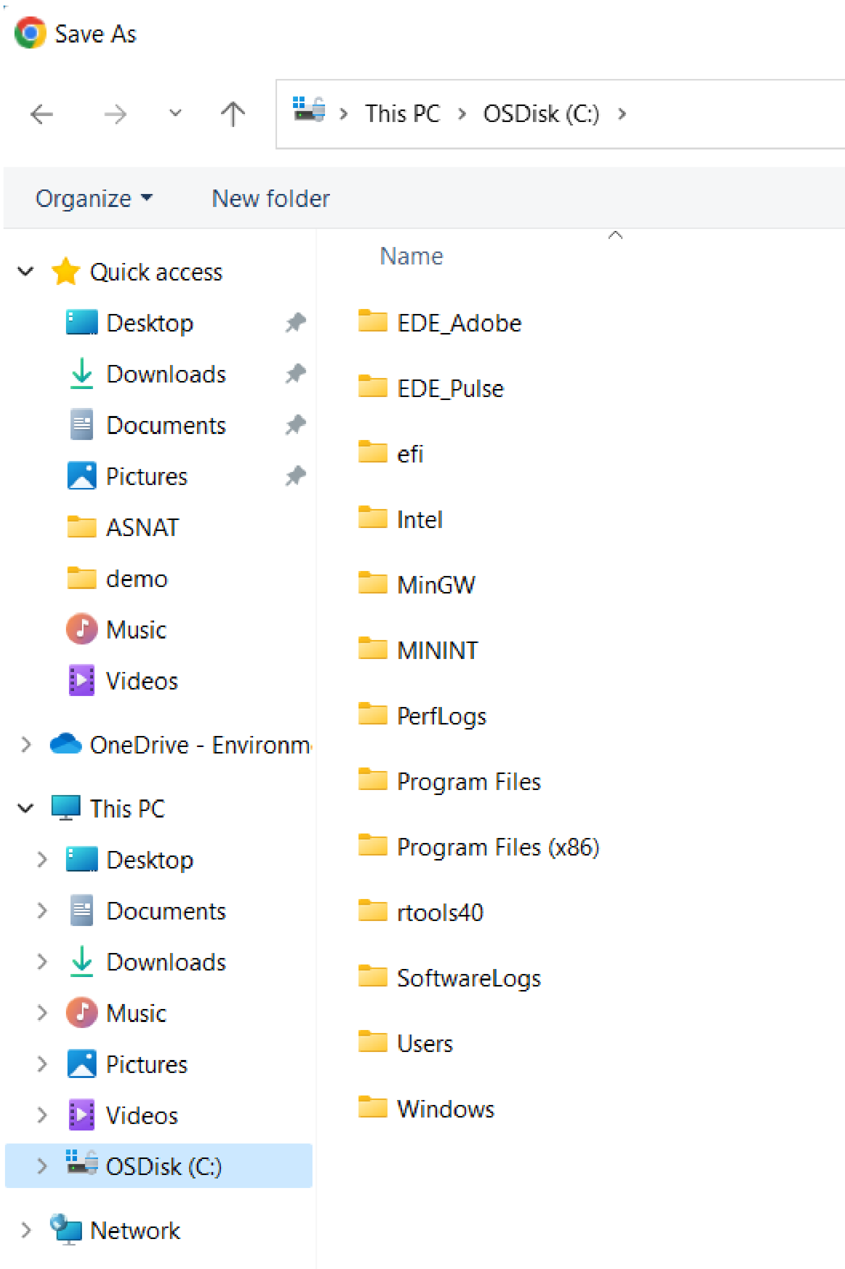
On Mac and Linux: Enter the following command in the shell:

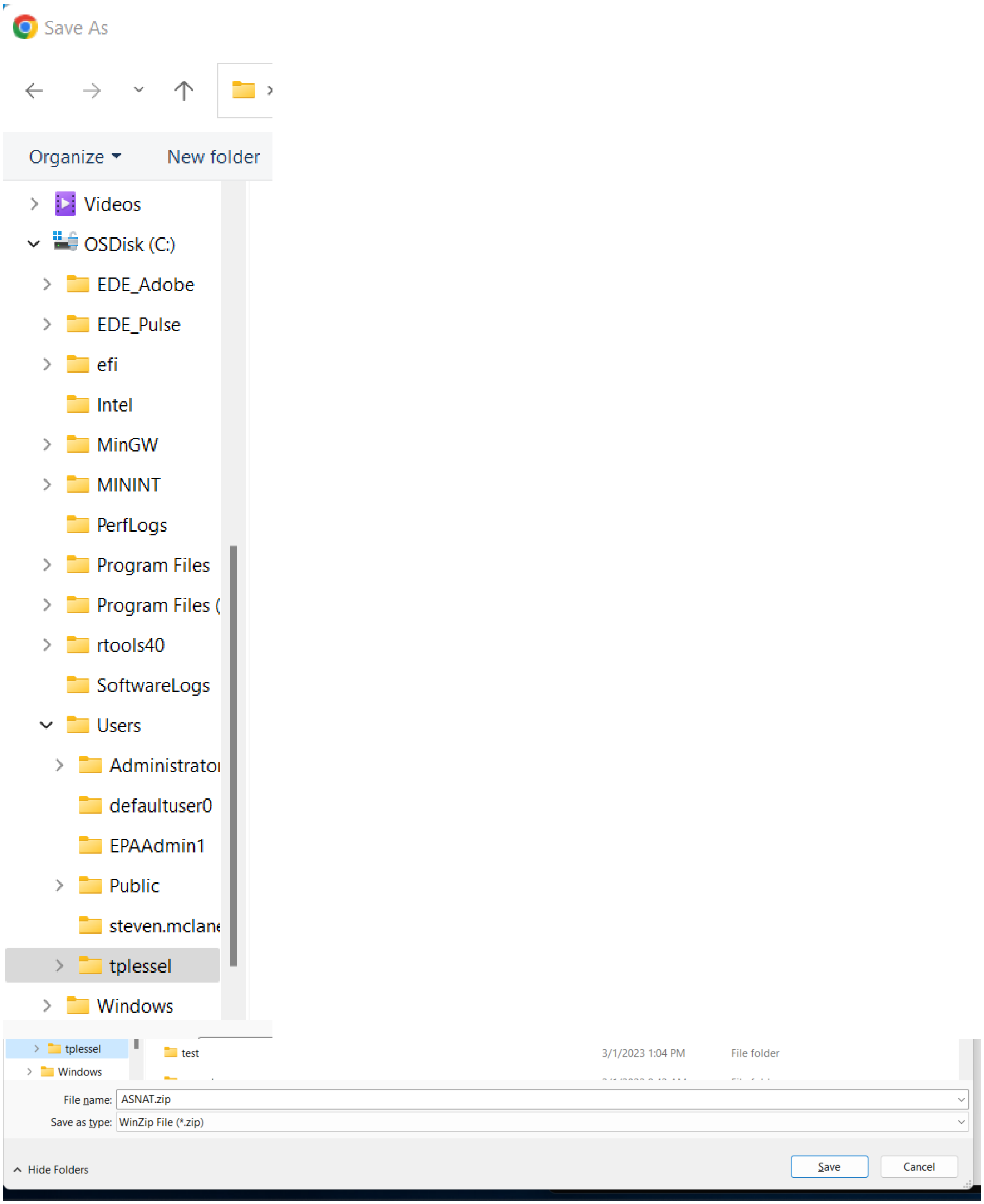
```
cd; mv ~/Downloads/ASNAT.zip .
```

(Remove the old version of ASNAT directory, if it exists.)

```
\rm -rf ASNAT
```

On Windows your home directory is **`C:\Users\logname`** *(Not OneDrive!)*
If the web browser prompts for a location to save the file, then
a. Click **`OSDisk (C:)`**
b. Double-click **`Users`** folder.
c. Double-click your **`logname`** folder.
d. Click **`Save`** button.


Save As
This PC › OSDisk (C:) ›
Organize
New folder
Quick access
Desktop
Downloads
Documents
Pictures
ASNAT
demo
Music
Videos
OneDrive - Environm
This PC
Desktop
Documents
Downloads
Music
Pictures
Videos
OSDisk (C:)
Network
Name
EDE_Adobe
EDE_Pulse
efi
Intel
MinGW
MININT
PerfLogs
Program Files
Program Files (x86)
rtools40
SoftwareLogs
Users
Windows



2. Unzip the `ASNAT.zip` file into `ASNAT` sub-directory/folder in your home directory.

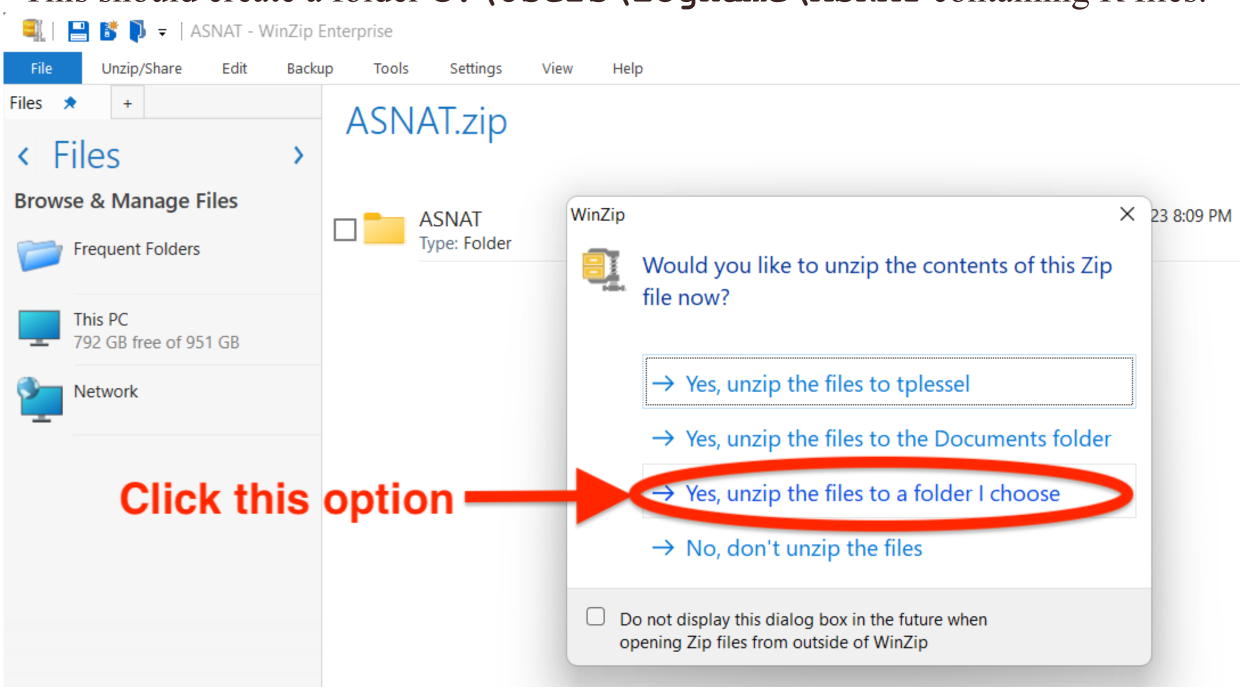
a. On Mac in the shell enter (**required for macOS users**):
`unzip ASNAT.zip`
`xattr -cr .`

b. On Linux:
`unzip ASNAT.zip`

c. On Windows:
Enter the following command in the shell
`start ASNAT.zip`

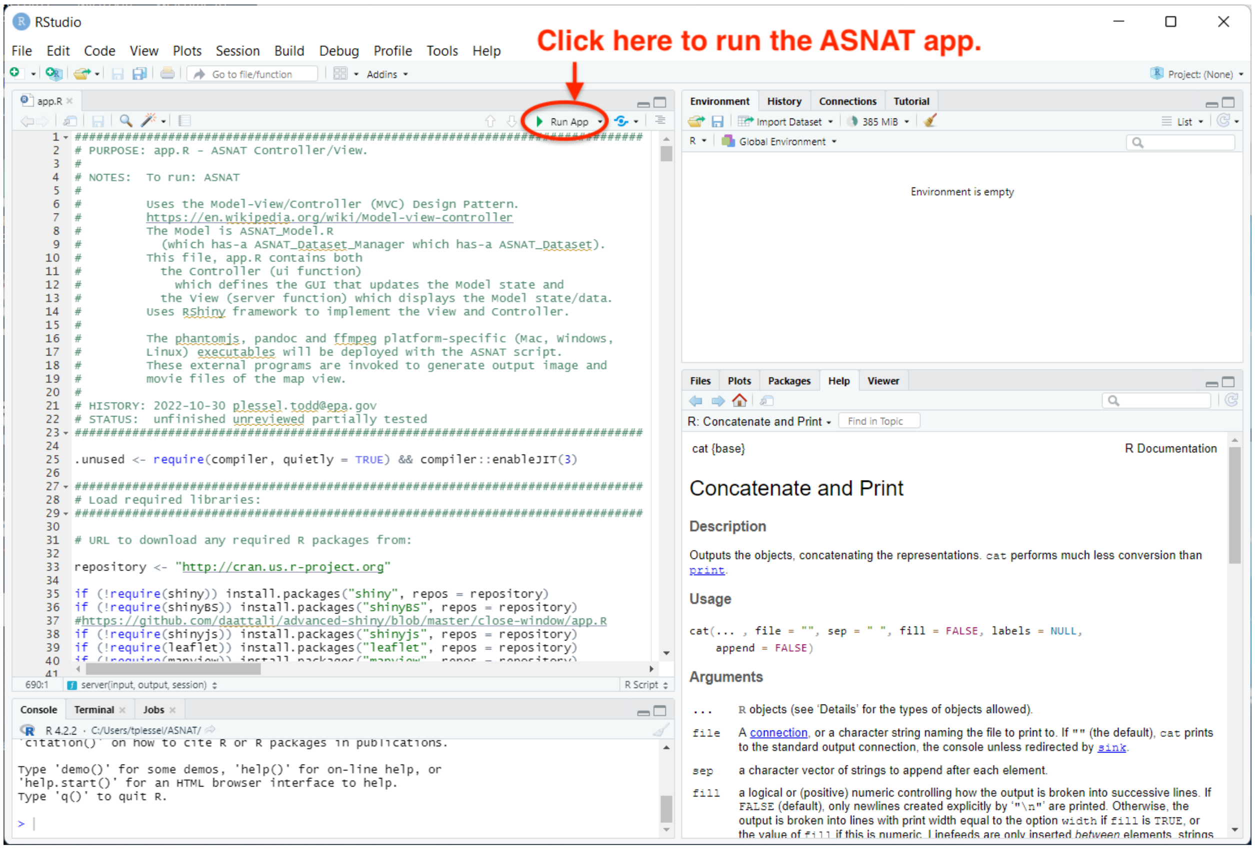
```
C:\Users\tplessel>start ASNAT.zip
```

(or use the File Explorer to find and double-click `ASNAT.zip`)
In the popup choose
`Yes, unzip the files to a folder I choose.`
Press return or Click the **`Unzip`** button.
Click the **`X`** in the upper-right corner to close the File Explorer/WinZip windows.
This should create a folder **`C:\Users\logname\ASNAT`** containing R files.



3. Running ASNAT:

You can run `ASNAT` from within `RStudio` IDE. The first time you run the app it will download, and the installation required
Packages then click **Run App**. A disadvantage of doing this is that the `RStudio` process uses more memory (and screen space) than the smaller `Rscript` interpreter used when running an app from a shell.

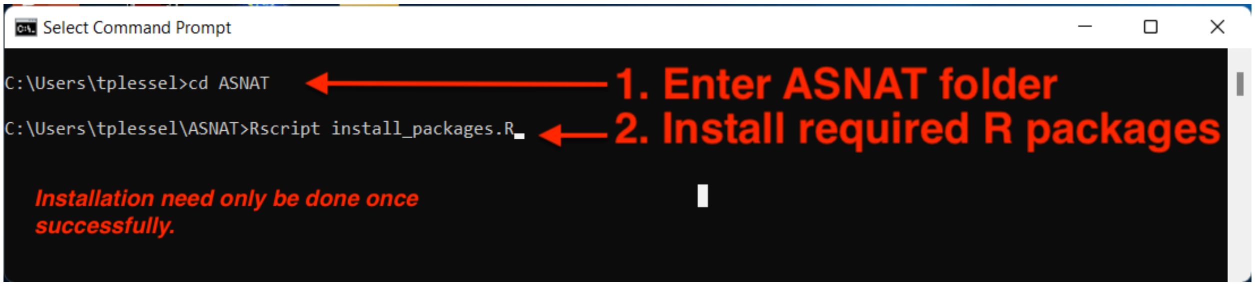


Alternatively, you can run `ASNAT` from the command prompts using the `Rscript` interpreter.

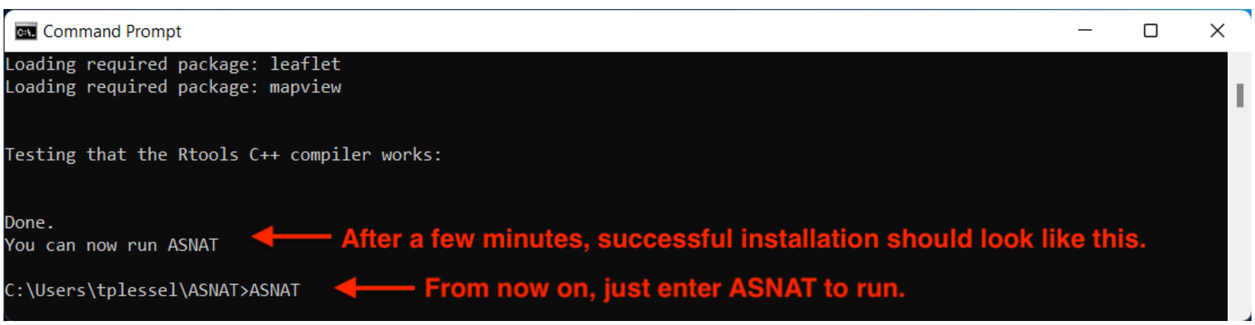
Use `Rscript` to install the required packages *once before you run* `ASNAT` *for the first time*.

Go to the `ASNAT` directory and use `Rscript` to install the various `R` packages that are required by `ASNAT`:

```
cd ASNAT
Rscript install_packages.R
```



It will take several minutes to install the many R packages needed by ASNAT. Check the output messages for any errors such as missing .h files or the message:
`'RTools_is_installed_and_c_plus_plus_compiler_works()' is not TRUE`
The above error message means you must download and install *RTools*.
Ask your system administrator for assistance resolving any problems before attempting to run ASNAT.
This step only needs to be done once successfully. Afterwards, just run ASNAT.



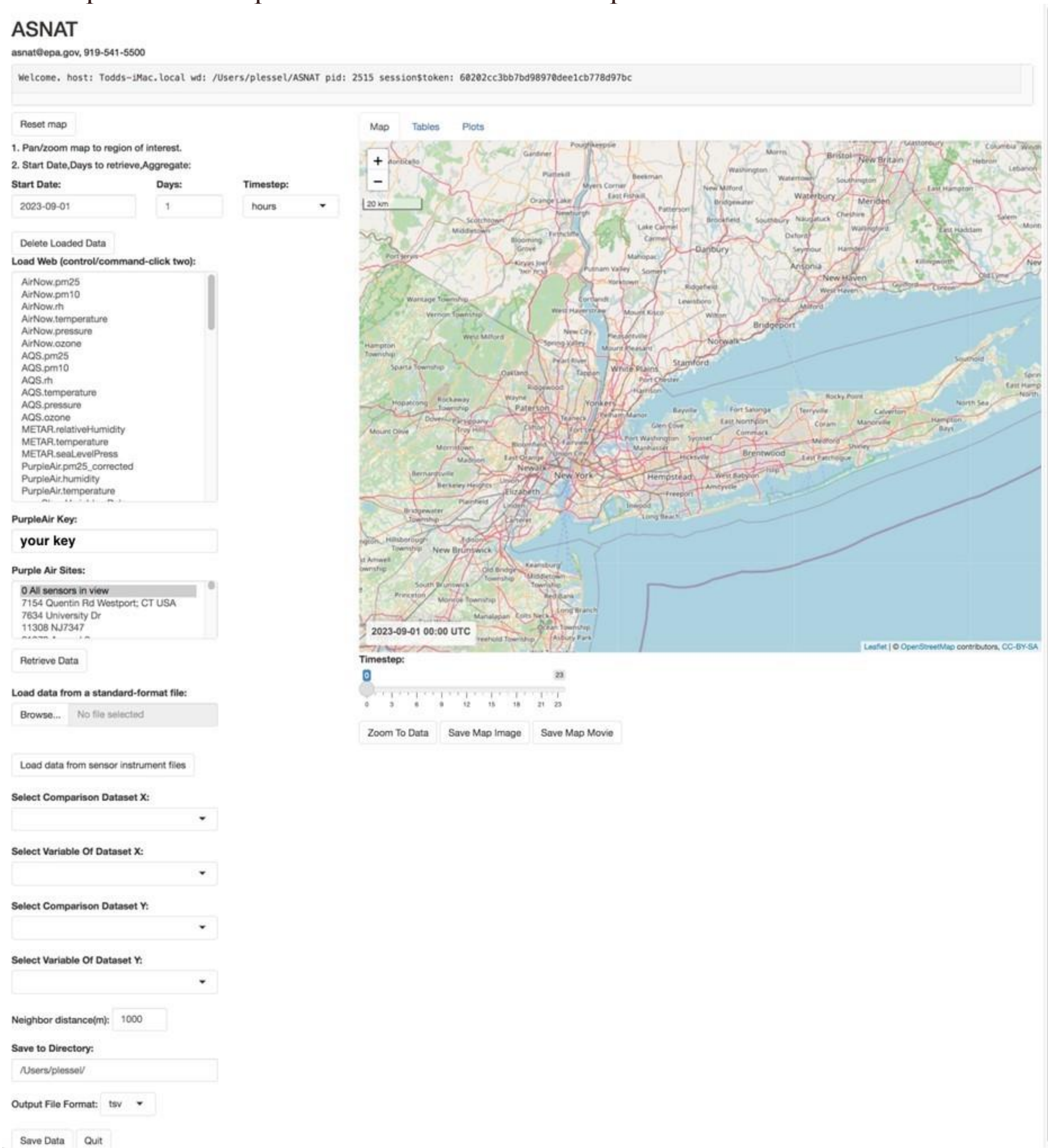
Run ASNAT. This will display the buttons and menus on the left and a map of NYC on the right.
ASNAT

If you are unable to run ASNAT through Command Prompt, please try the following troubleshooting steps:

1. Ensure the folder name of the ASNAT application is 'ASNAT.'
2. Disable or pause any antivirus software.
3. Run Command Prompt as an Administrator.

Here is what `ASNAT` looks like when launched:

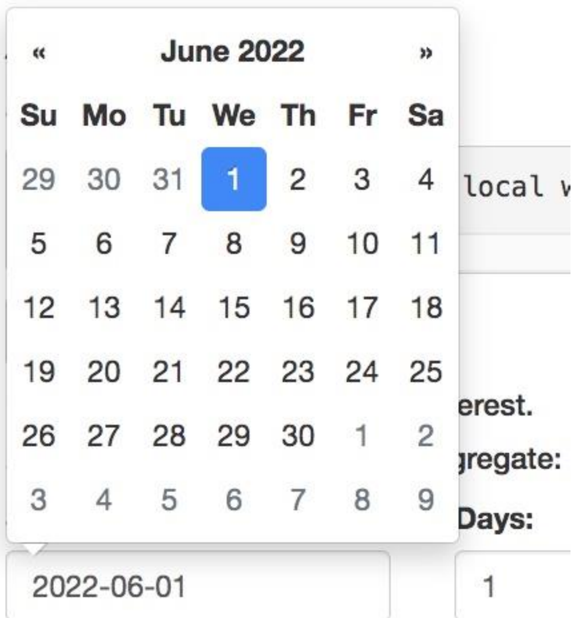
Note: The snippets below were taken from the first version of `ASNAT`. Since then, several new tabs and menus have been added. Users can view all the new developments and updates under the "Demos" options.



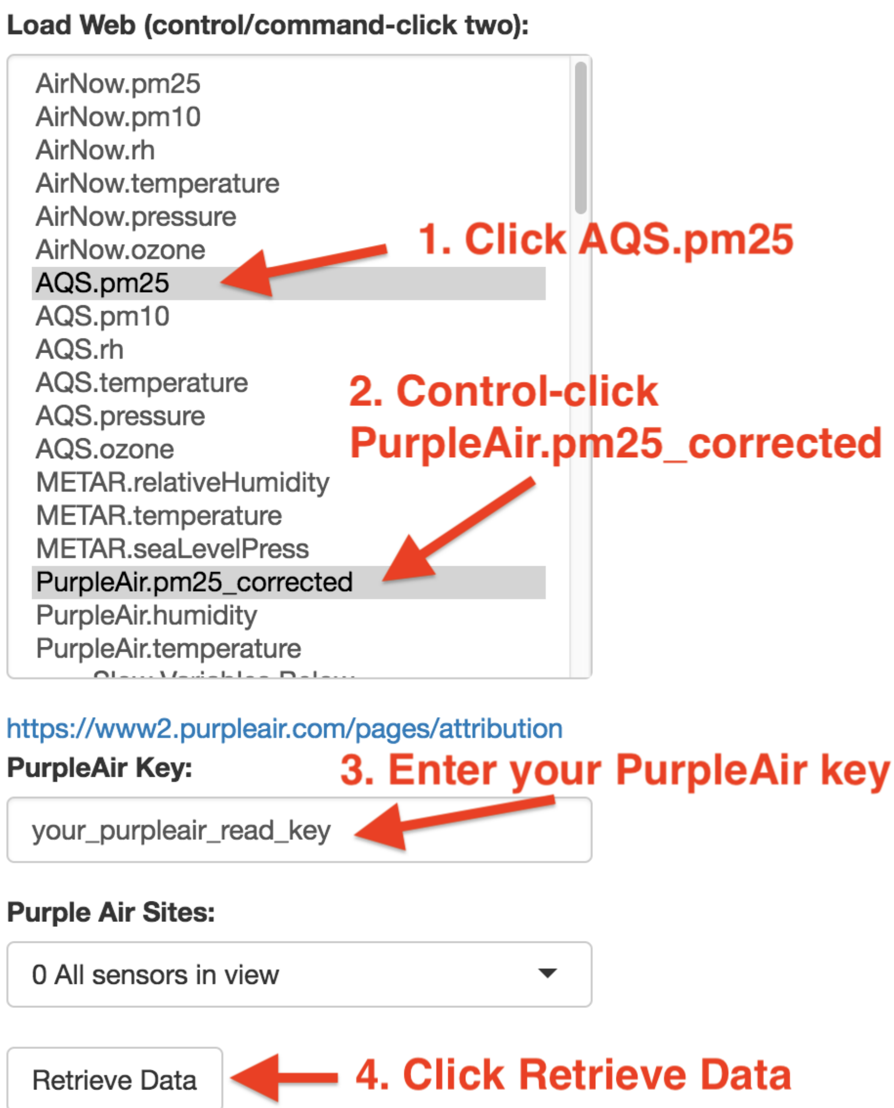
Try the following demos:

Demo 1:

1.1. Use the mouse to select the **Start Date** then type **2022-06-01** and press enter (Make sure to select "hours" for the Timestep dropdown).



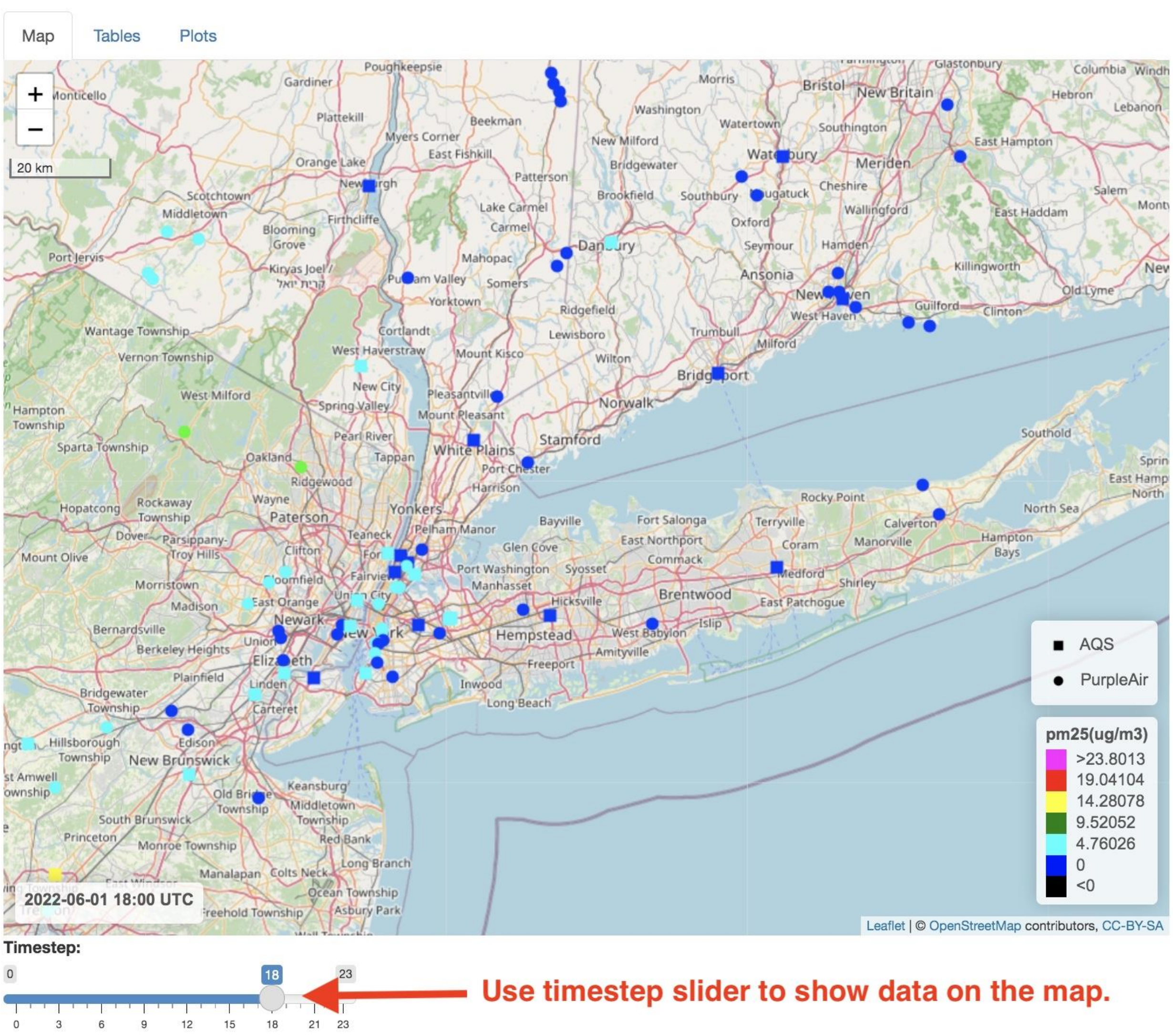
1.2. In the list of variables, **click AQS.pm25** then hold the **control key** down (or **command key on Mac**) and **click PurpleAir.pm25_corrected**.
(This allows you to retrieve multiple datasets. Alternatively, select one dataset at a time and click the Retrieve Data button then repeat for another dataset.)



1.3. Enter your PurpleAir Read Key in the text area below the menu. (Ask for a read key from contact@purpleair.com).
Note: The user will get an error if the user enters/uses an invalid Key.

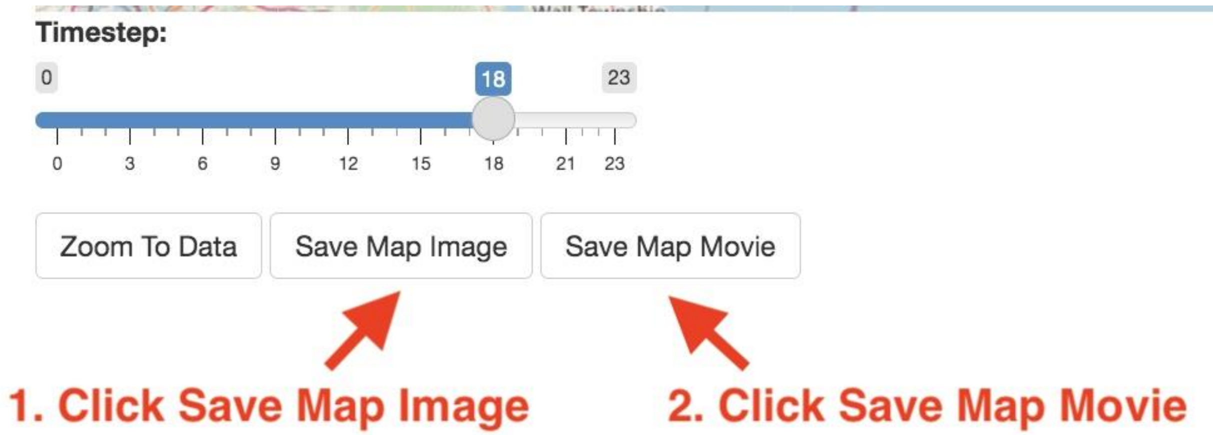
1.4. Click Retrieve Data.
This will retrieve those datasets using the web service commands shown at the top of the window. These commands are also added to the file `ASNAT/log.txt` for possible use in your other scripts.

1.5. Drag the Timestep slider to see mapped data for the given hour.

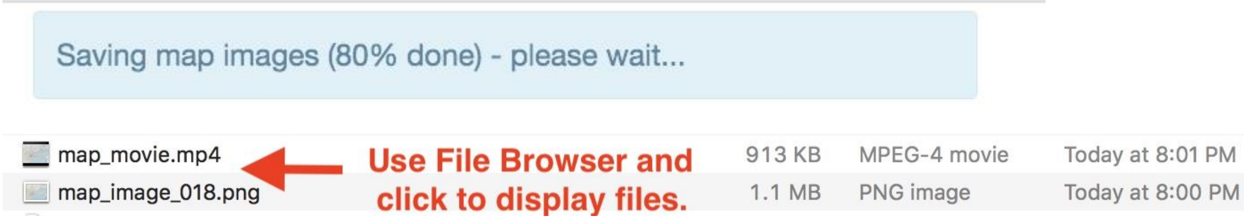


1.6. Click Save Map Image. This will save the displayed map image in the file map_image_ttt.png for timestep 'ttt' in your output directory/folder (your home directory by default).

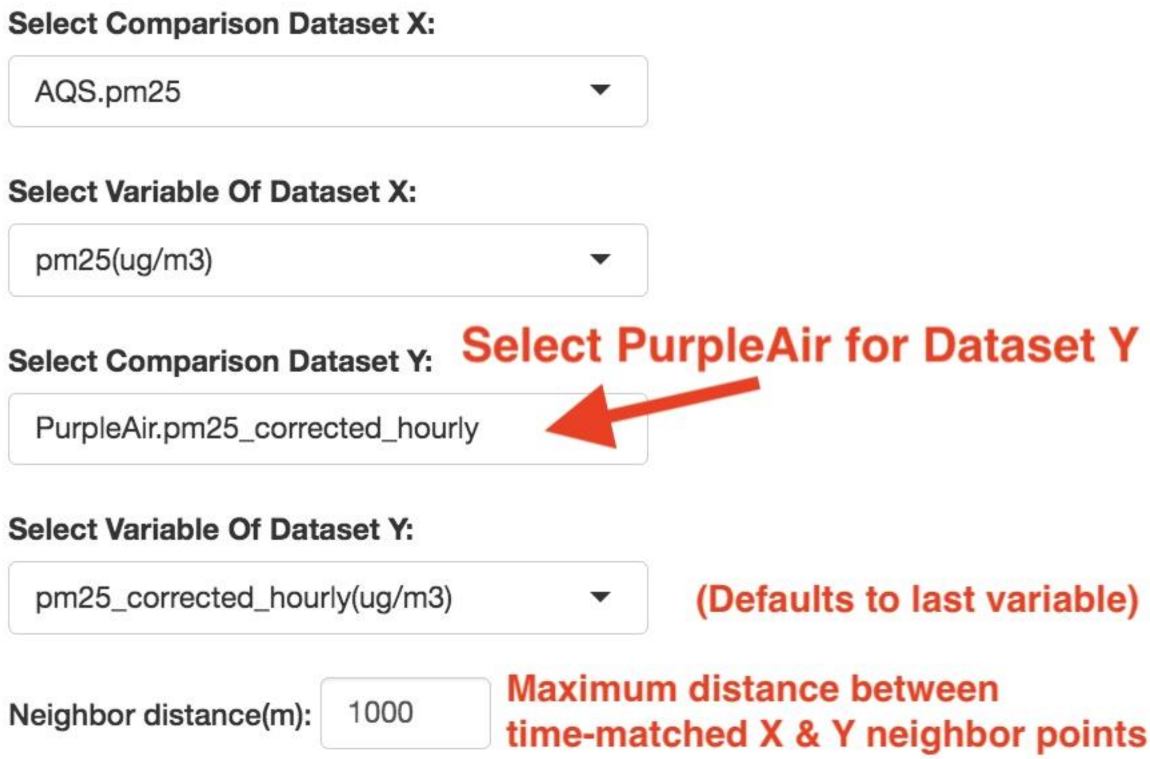
__Note:__ when running *remote-hosted* `ASNAT` instead of *locally* on your computer, There is no **Save To Directory** and **Save** buttons are **Download** buttons.
This is because generated files (data output, tables, plots, images, movie) are created on the remote host, zipped into a zip file, and downloaded to the local computer. In this case, users will be prompted where to save these zip files.
The zip files can then be unzipped to obtain the files.



1.7. Click Save Movie. This will create `map_movie.mp4` containing all timesteps. It will take a couple of minutes to complete.
Later you can use a File Browser / Finder to click on these files to view them.



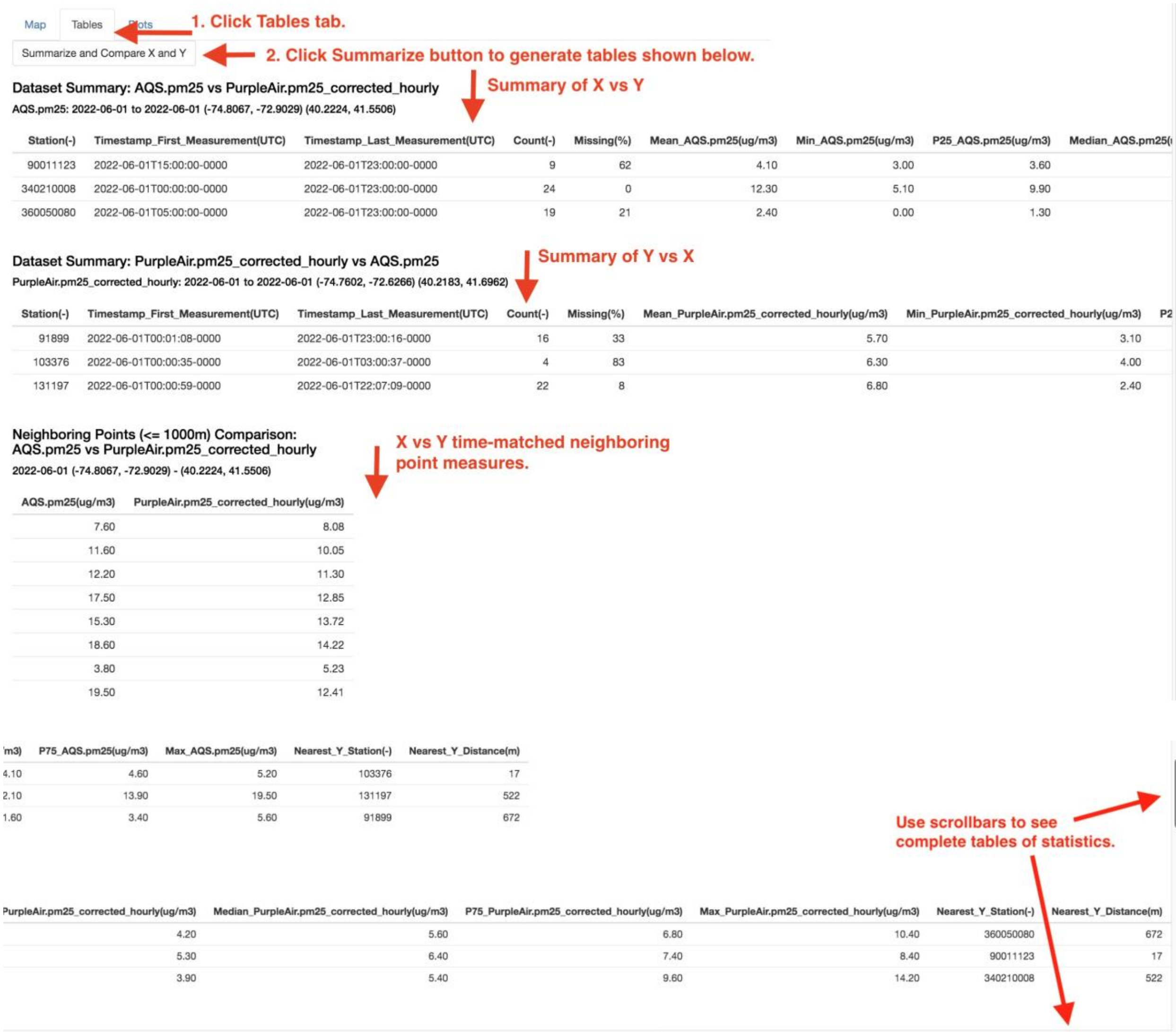
1.8. In the **Select Comparison Dataset Y menu**, select **PurpleAir.pm25_corrected_hourly**.



1.9. Click the Tables tab above the map.

1.10. Click the Summarize and Compare X and Y button to generate tables that summarize and compare AQS and PurpleAir data points.

Use the horizontal and vertical scrollbars to see these three tables.



Dataset Summary: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

AQS.pm25: 2022-06-01 to 2022-06-01 (-74.8067, -72.9029) (40.2224, 41.5506)

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_AQS.pm25(ug/m3)	Min_AQS.pm25(ug/m3)	P25_AQS.pm25(ug/m3)	Median_AQS.pm25(
90011123	2022-06-01T15:00:00-0000	2022-06-01T23:00:00-0000	9	62	4.10	3.00	3.60	
340210008	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	12.30	5.10	9.90	
360050080	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.40	0.00	1.30	

Dataset Summary: PurpleAir.pm25_corrected_hourly vs AQS.pm25

PurpleAir.pm25_corrected_hourly: 2022-06-01 to 2022-06-01 (-74.7602, -72.6266) (40.2183, 41.6962)

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_PurpleAir.pm25_corrected_hourly(ug/m3)	Min_PurpleAir.pm25_corrected_hourly(ug/m3)	P2
91899	2022-06-01T00:01:08-0000	2022-06-01T23:00:16-0000	16	33	5.70	3.10	
103376	2022-06-01T00:00:35-0000	2022-06-01T03:00:37-0000	4	83	6.30	4.00	
131197	2022-06-01T00:00:59-0000	2022-06-01T22:07:09-0000	22	8	6.80	2.40	

Neighboring Points (<= 1000m) Comparison:
AQS.pm25 vs PurpleAir.pm25_corrected_hourly

2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

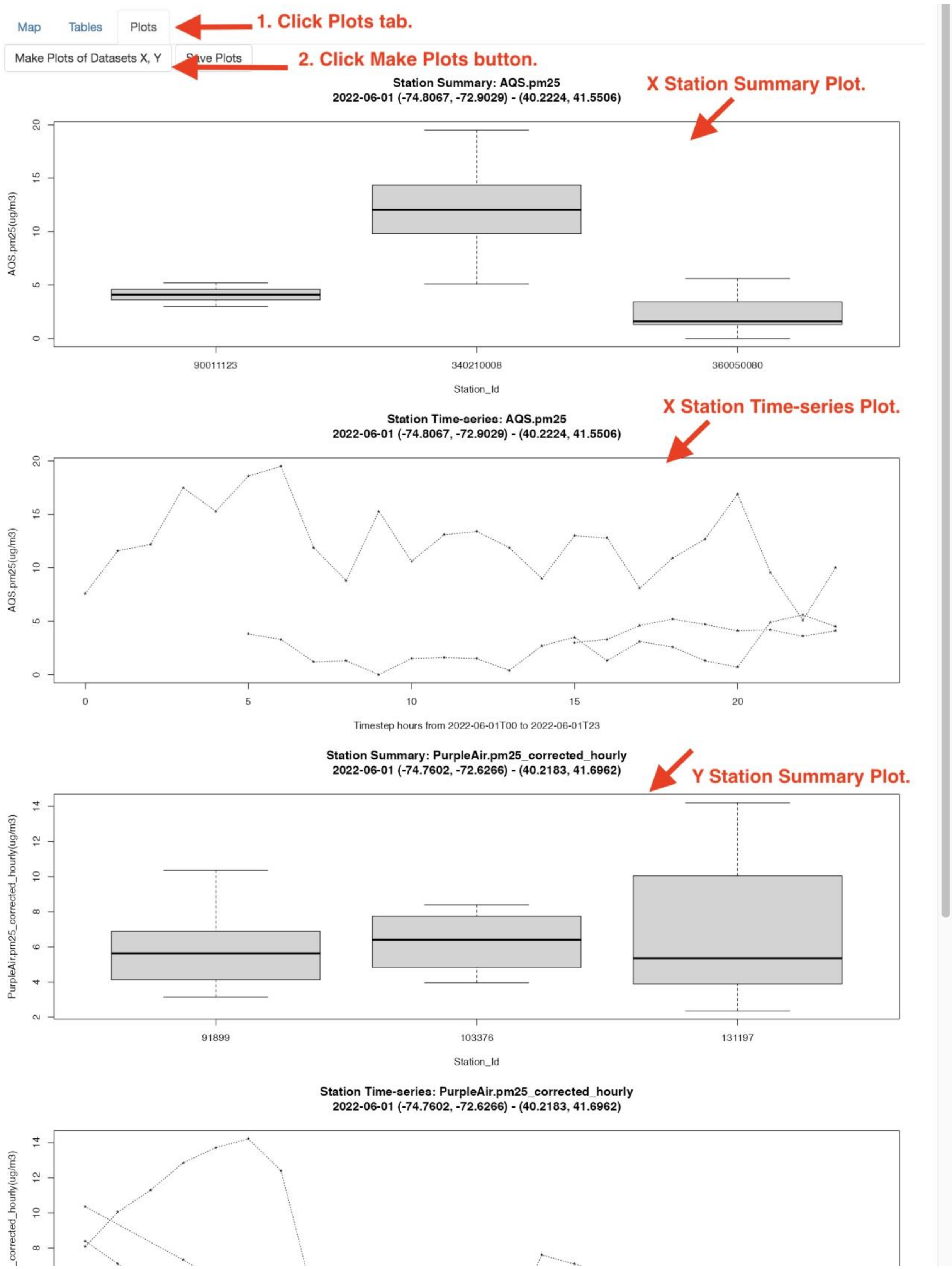
AQS.pm25(ug/m3)	PurpleAir.pm25_corrected_hourly(ug/m3)
7.60	8.08
11.60	10.05
12.20	11.30
17.50	12.85
15.30	13.72
18.60	14.22
3.80	5.23
19.50	12.41

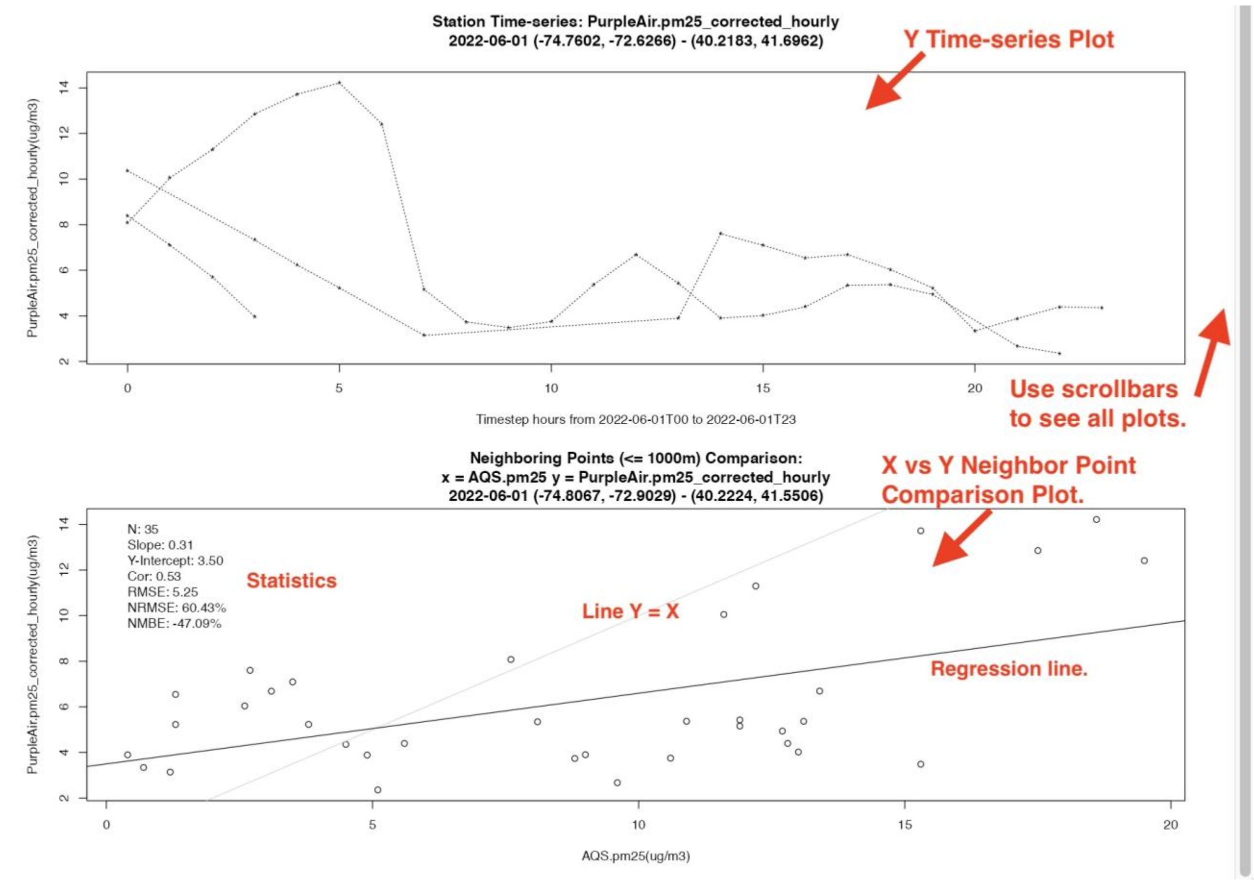
m3)	P75_AQS.pm25(ug/m3)	Max_AQS.pm25(ug/m3)	Nearest_Y_Station(-)	Nearest_Y_Distance(m)
4.10	4.60	5.20	103376	17
2.10	13.90	19.50	131197	522
1.60	3.40	5.60	91899	672

PurpleAir.pm25_corrected_hourly(ug/m3)	Median_PurpleAir.pm25_corrected_hourly(ug/m3)	P75_PurpleAir.pm25_corrected_hourly(ug/m3)	Max_PurpleAir.pm25_corrected_hourly(ug/m3)	Nearest_Y_Station(-)	Nearest_Y_Distance(m)
4.20	5.60	6.80	10.40	360050080	672
5.30	6.40	7.40	8.40	90011123	17
3.90	5.40	9.60	14.20	340210008	522

1.11. Click the Plots tab above the map.

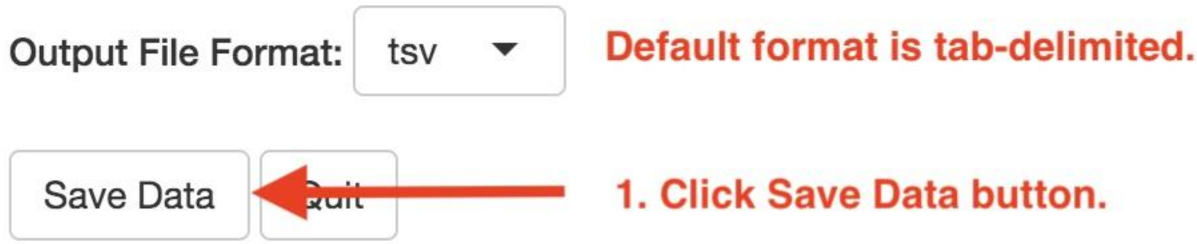
1.12. Click the Make Plots of Datasets X, Y button to generate plots that summarize and compare AQS and PurpleAir data points.
Use the vertical scrollbars to see these five plots.


Map
Tables
Plots
1. Click Plots tab.
Make Plots of Datasets X, Y
Save Plots
2. Click Make Plots button.
Station Summary: AQS.pm25
2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)
X Station Summary Plot.
AQS.pm25(ug/m3)
90011123
340210008
360050080
Station_Id
Station Time-series: AQS.pm25
2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)
X Station Time-series Plot.
AQS.pm25(ug/m3)
Timestep hours from 2022-06-01T00 to 2022-06-01T23
Station Summary: PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.7602, -72.6266) - (40.2183, 41.6962)
Y Station Summary Plot.
PurpleAir.pm25_corrected_hourly(ug/m3)
91899
103376
131197
Station_Id
Station Time-series: PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.7602, -72.6266) - (40.2183, 41.6962)
_corrected_hourly(ug/m3)



1.13. Click the Save Plots button to save pdf files of the plots.
You can use a File Browser / Finder to click on these files to view them.

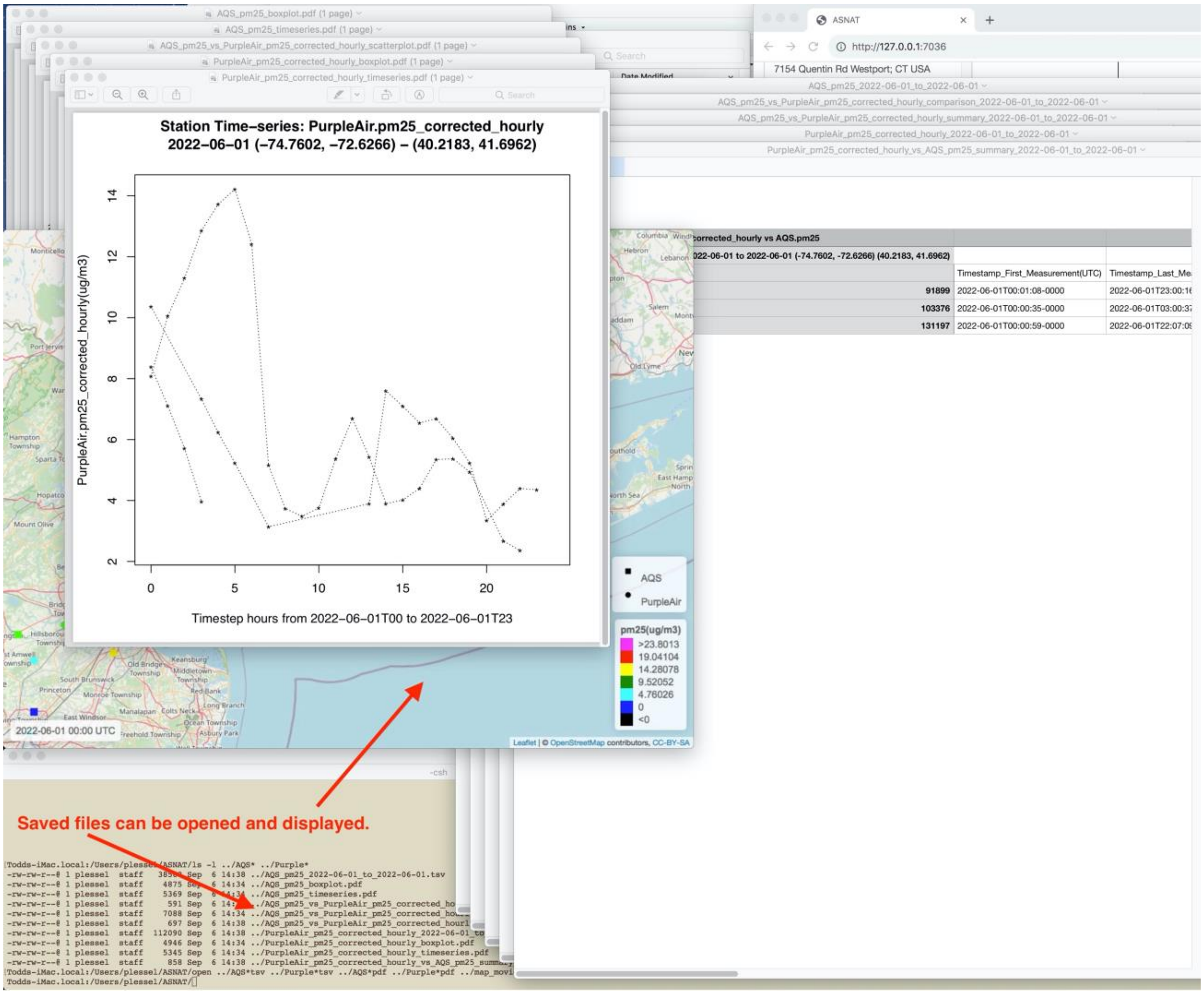
1.14. Scroll to the bottom of the window and **click the Save Data button** (along the left edge of the window).
This will immediately save the datasets and the tables as plain text files in your home directory.
Scroll to the top of the window to see the list of files created.
File formats include tsv (tab-separated values),
csv (comma-separated values), and json (JSON format). The tsv and csv files can be read into a spreadsheet program for further analysis.



Later, they can also be re-loaded into `ASNAT` using the
'Load data from a standard-format file' Browse button.

1.15. Click the Quit button to quit the program and close the web browser tab.

1.16 List the various output files generated:
Windows: **`dir ..\*_to_*sv`**
Mac/Linux: **`ls -l ../*_to_*sv ../*pdf`**
On Mac these files can be displayed using the open command:
`open ../*_to_*sv ../*pdf`

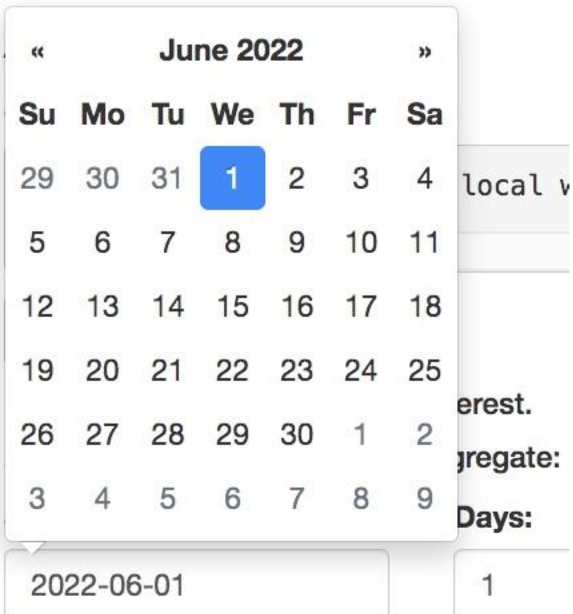


Station Time-series: PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.7602, -72.6266) – (40.2183, 41.6962)
PurpleAir.pm25_corrected_hourly(ug/m3)
Timestep hours from 2022-06-01T00 to 2022-06-01T23
AQS
PurpleAir
pm25(ug/m3)
>23.8013
19.04104
14.28078
9.52052
4.76026
0
<0
2022-06-01 00:00 UTC
ASNAT
http://127.0.0.1:7036
7154 Quentin Rd Westport; CT USA
Saved files can be opened and displayed.

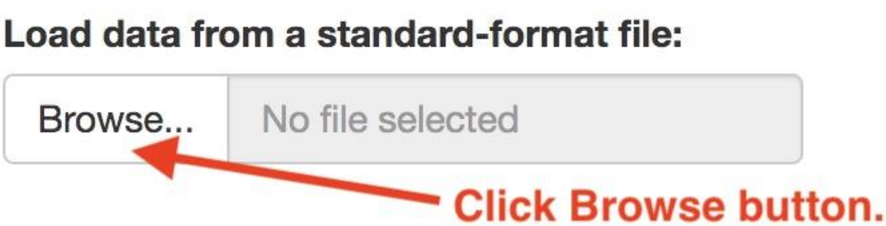
Demo 2: Load data from files:

2.1 Rerun `ASNAT`:
`ASNAT`

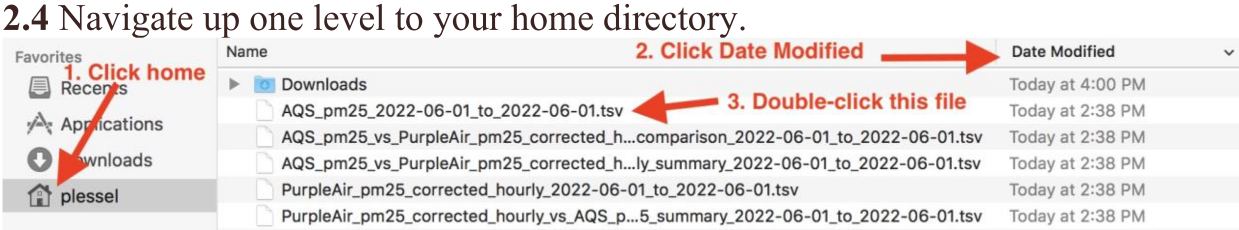
2.2. If the Start Date is not already 2022-06-01 then
use the mouse to select the **Start Date** then type **2022-06-01** and press enter.



2.3 Scroll down to **Load data from a standard-format file**: and **click Browse**.

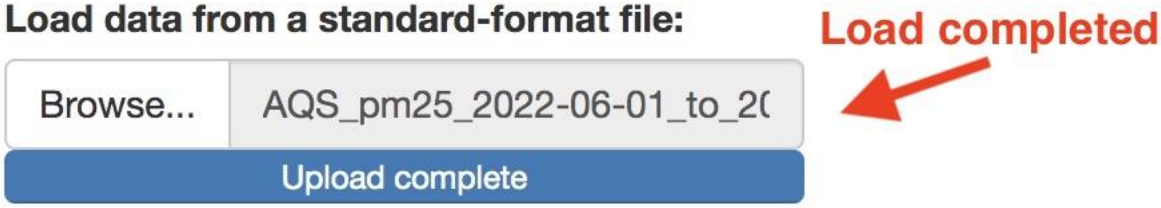


2.4 Navigate up one level to your home directory.



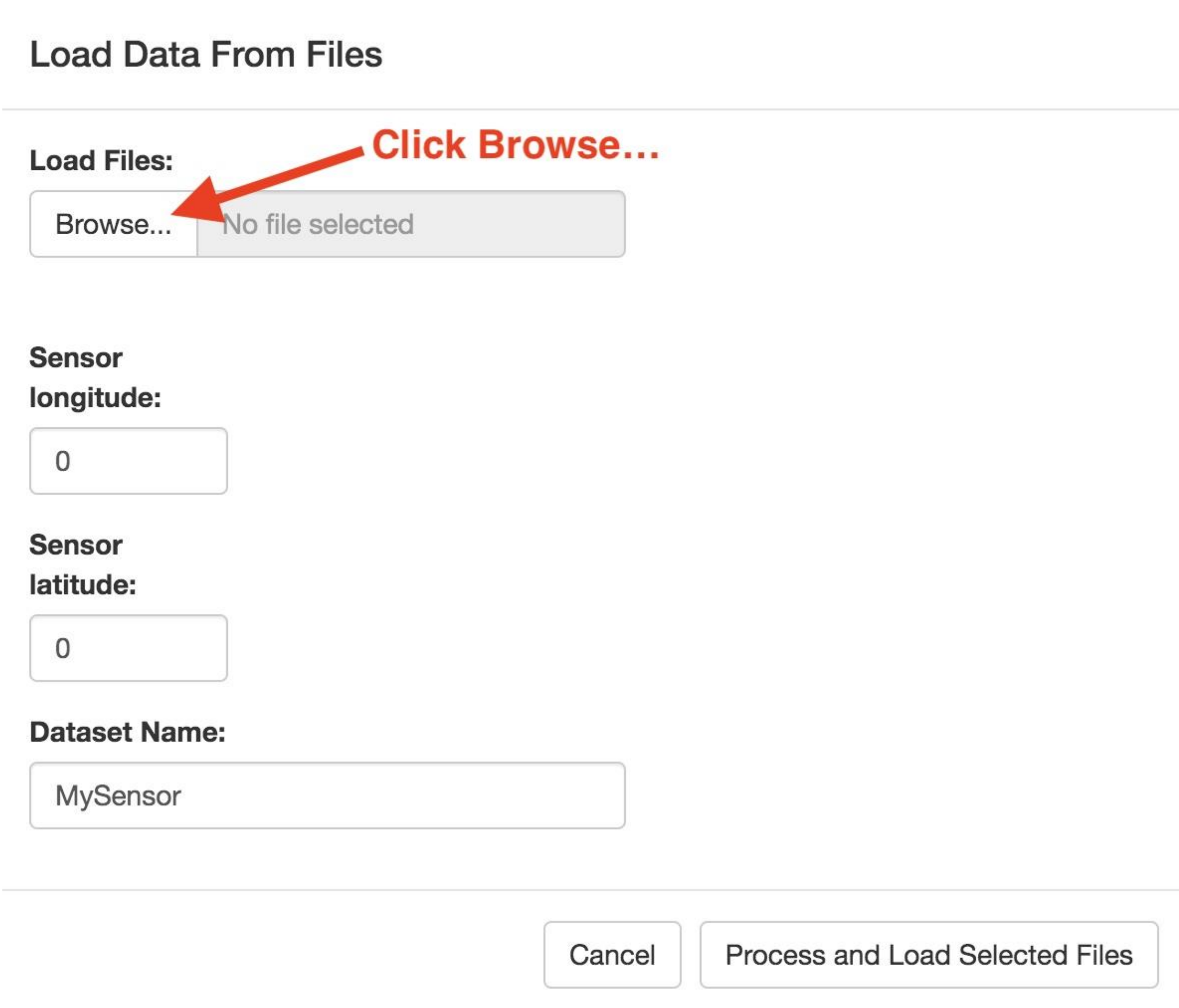
2.5 Select the previously saved (in Demo 1) file

AQS_pm25_2022-06-01_to_2022-06-01.tsv and click Open (or double-click or press return).



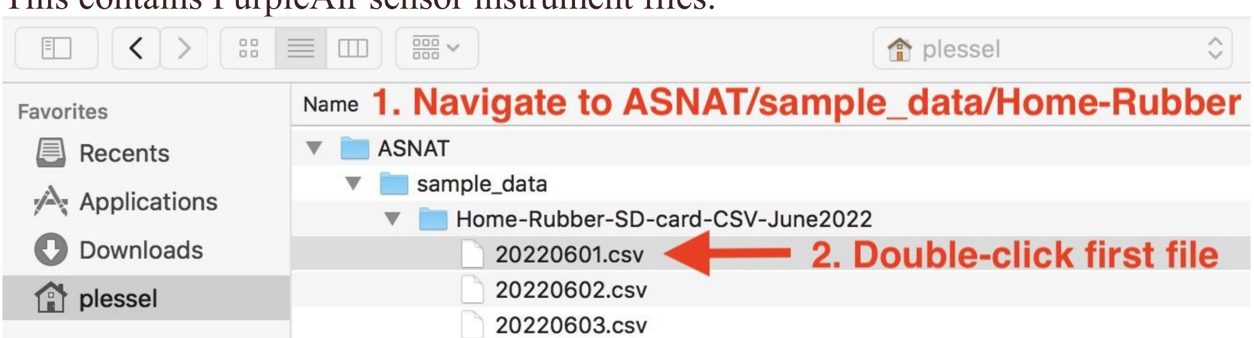
2.6 Click Load data from sensor instrument files.

2.7 Click Browse...



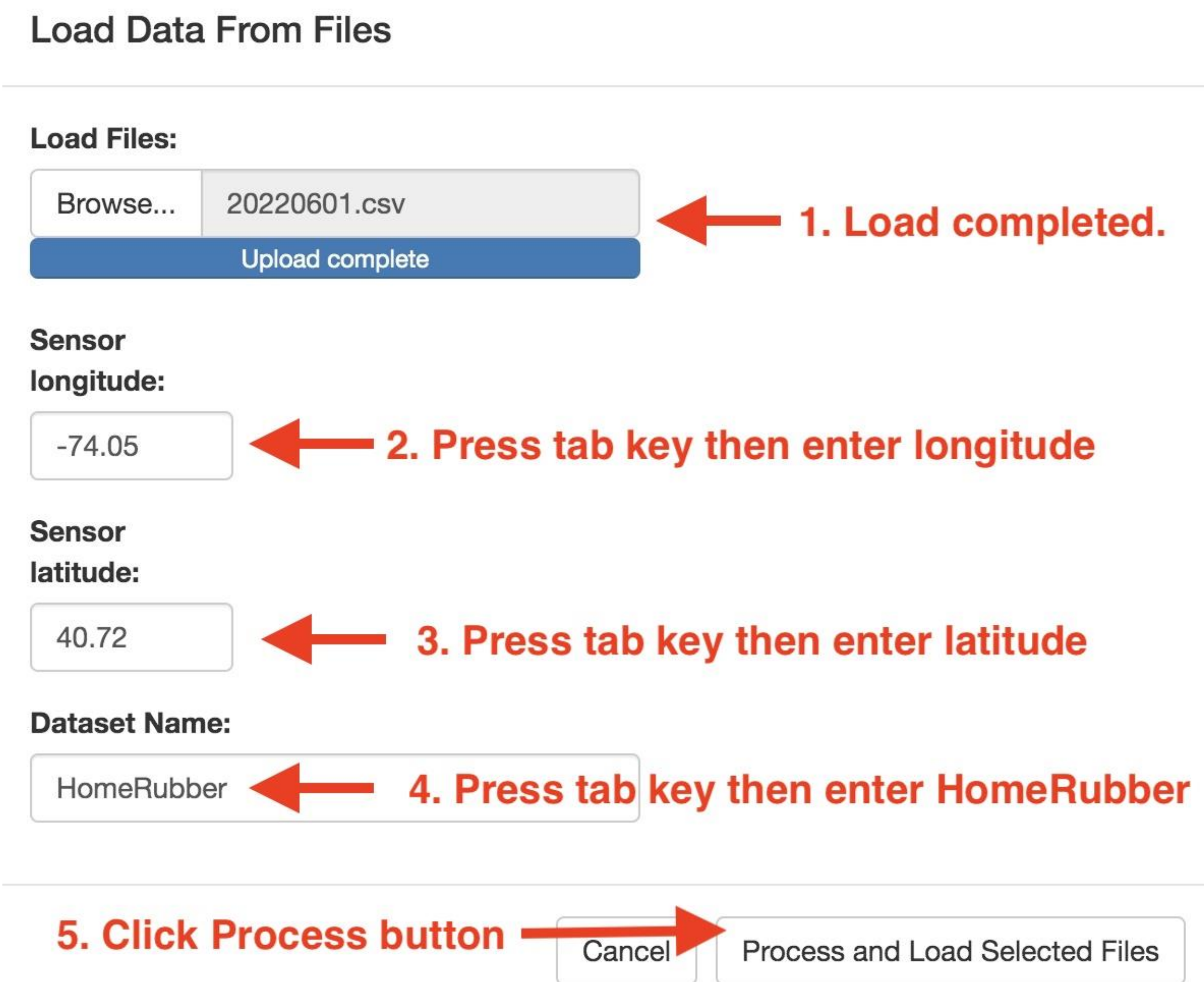
2.8 Navigate into **`sample_data/Home-Rubber-SD-card-CSV-June2022`**.

This contains PurpleAir sensor instrument files.



2.9 Double-click the file 20220601.csv.

2.10 Since the file did not contain the location of the sensor, enter the
Sensor longitude **-74.05**
Sensor latitude **40.72**



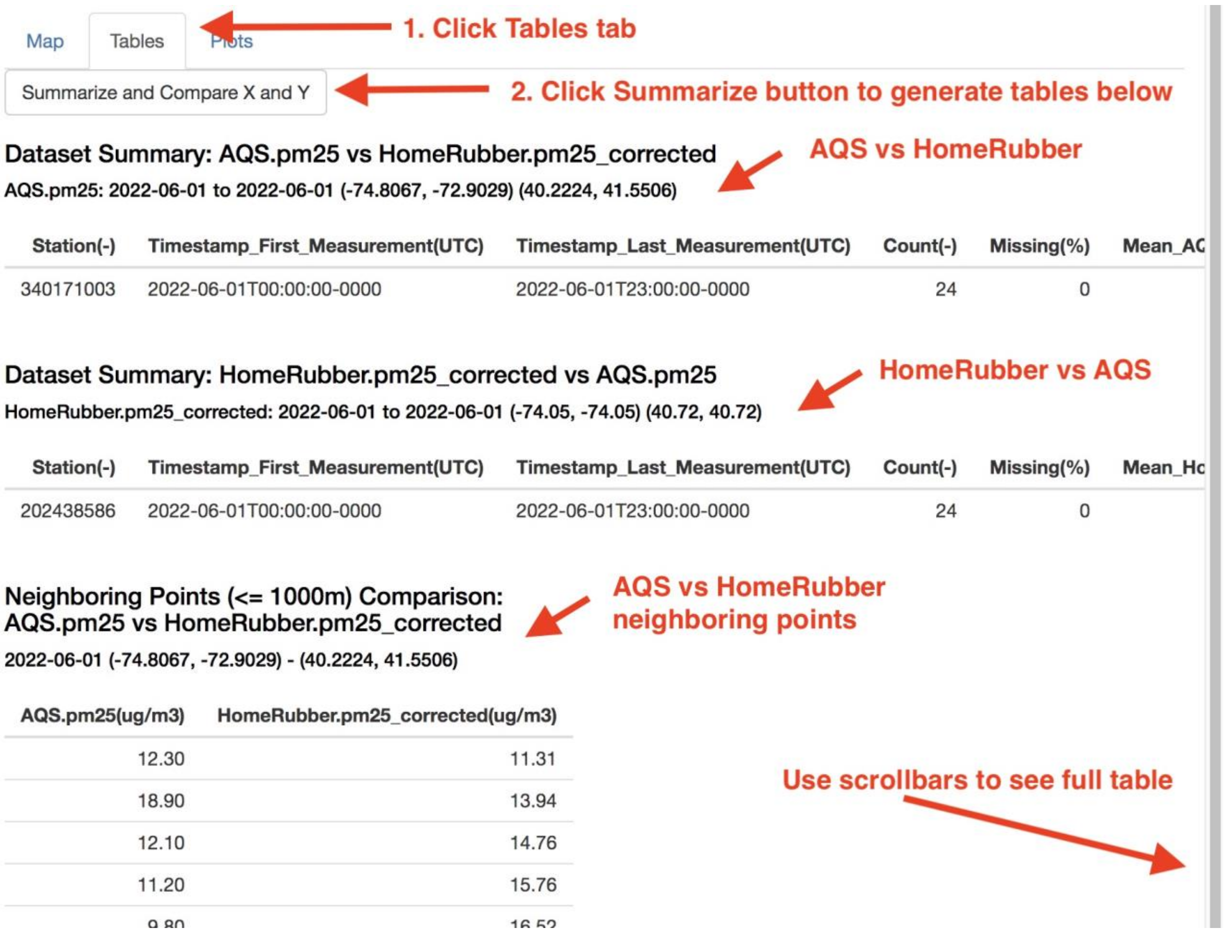
2.11 Change the name of the dataset to **HomeRubber**.

2.12 Click the **Process and Load Selected Files** button (or **tab, tab enter**)

2.13 Change **Select Comparison Dataset Y:** from none to **HomeRubber.hourly_data**.
Note that the 'Select Variable Of Dataset Y' menu automatically selects the last variable **pm25_corrected(ug/m3)** which is *computed by* `ASNAT` *upon import*.

2.14 Click the Tables tab above the map.

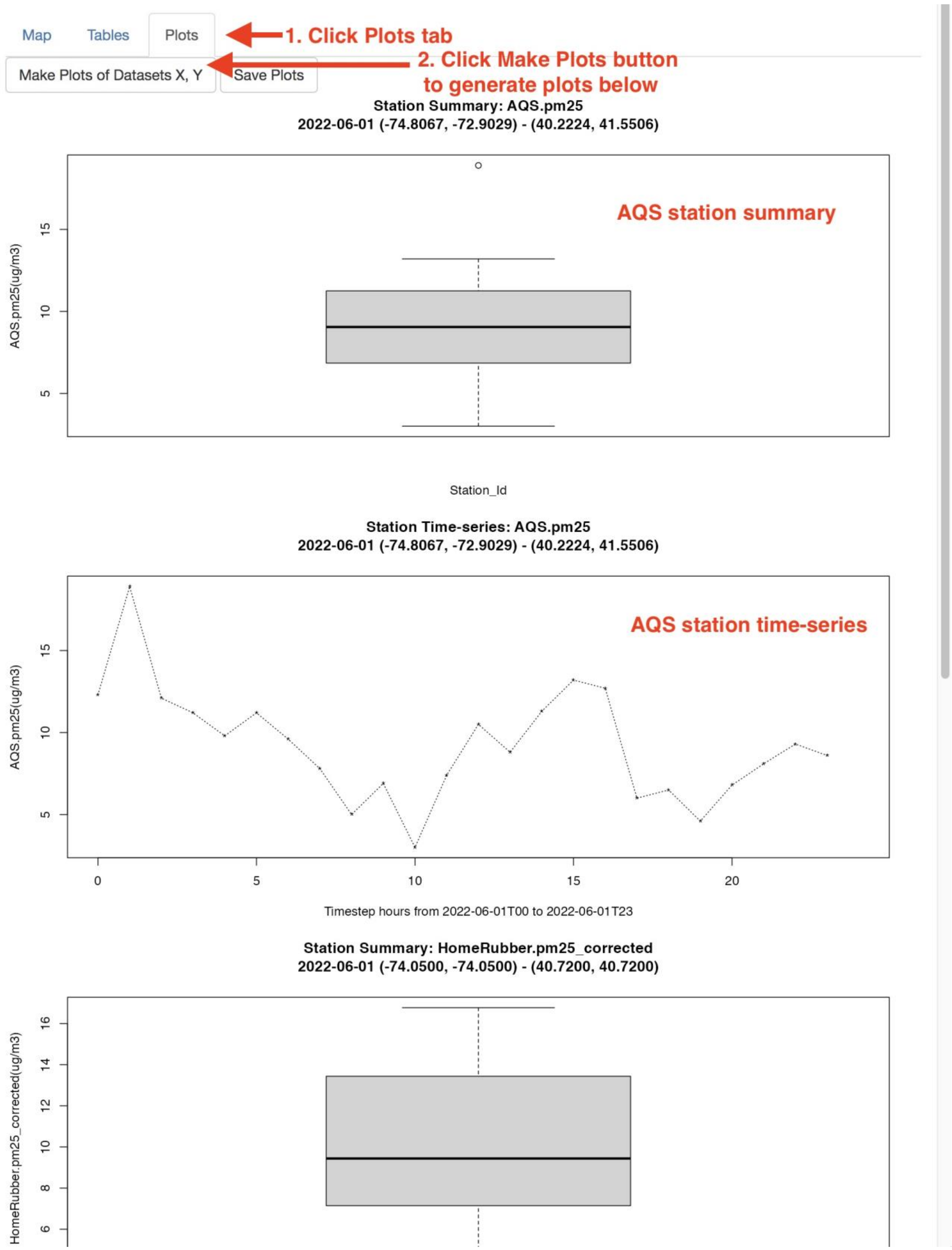
2.15 Click the Summarize and Compare X and Y button.



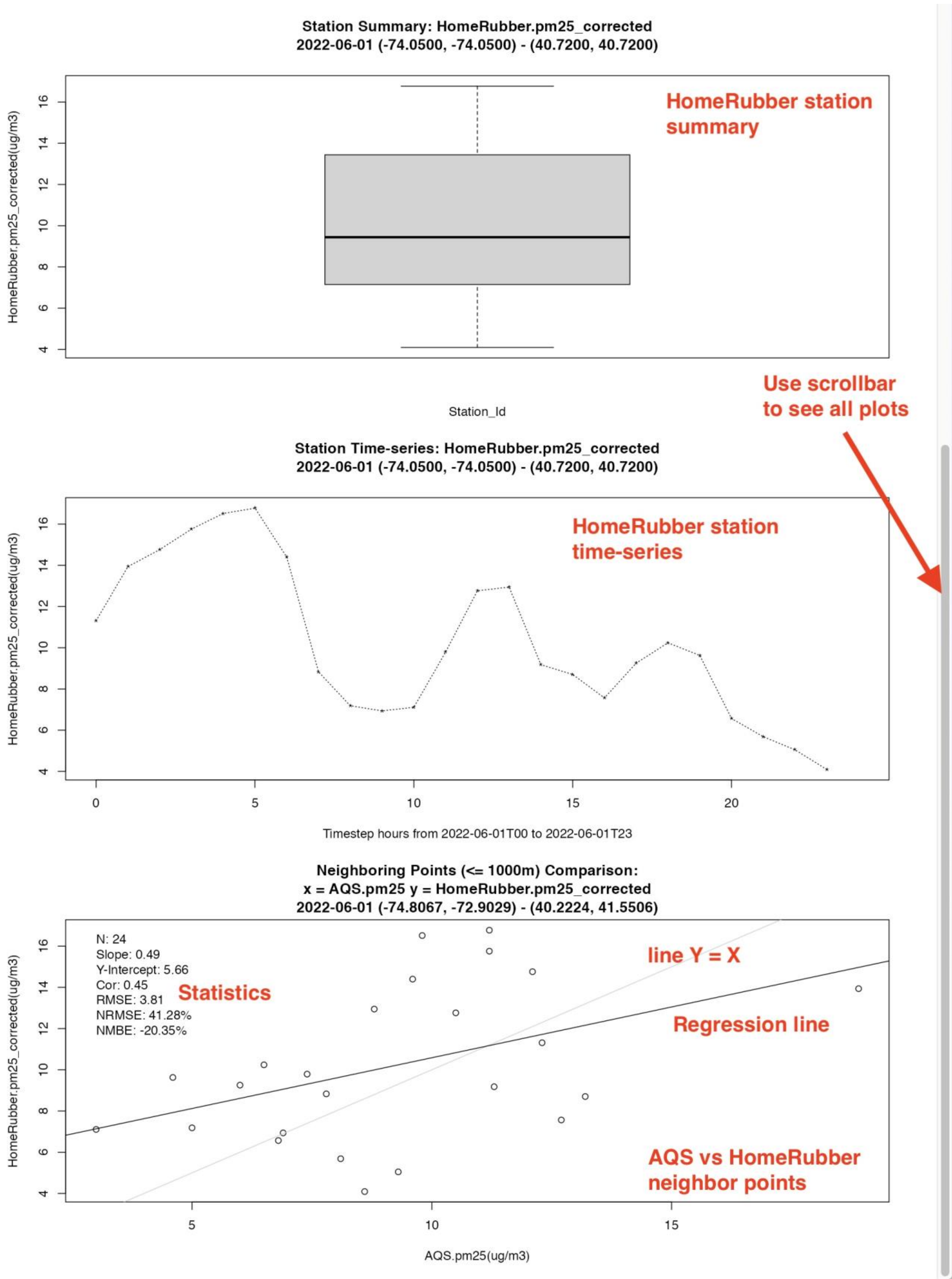
2.16 See that there is one pair of sites within 630 m apart and their paired pm25 measures.

2.17 Click the Plots tab above the map.

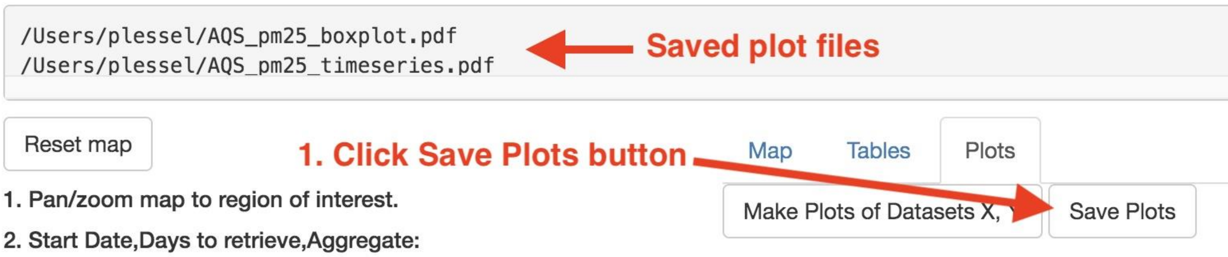
2.18 Click the Make Plots of Datasets X, Y button.



2.19 Scroll to see the five plots.


Station Summary: HomeRubber.pm25_corrected
2022-06-01 (-74.0500, -74.0500) - (40.7200, 40.7200)
HomeRubber.pm25_corrected(ug/m3)
Station_Id
HomeRubber station summary
Use scrollbar to see all plots
Station Time-series: HomeRubber.pm25_corrected
2022-06-01 (-74.0500, -74.0500) - (40.7200, 40.7200)
HomeRubber station time-series
Timestep hours from 2022-06-01T00 to 2022-06-01T23
Neighboring Points (<= 1000m) Comparison:
x = AQS.pm25 y = HomeRubber.pm25_corrected
2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)
N: 24
Slope: 0.49
Y-Intercept: 5.66
Cor: 0.45
RMSE: 3.81
NRMSE: 41.28%
NMBE: -20.35%
Statistics
line Y = X
Regression line
AQS vs HomeRubber neighbor points
AQS.pm25(ug/m3)

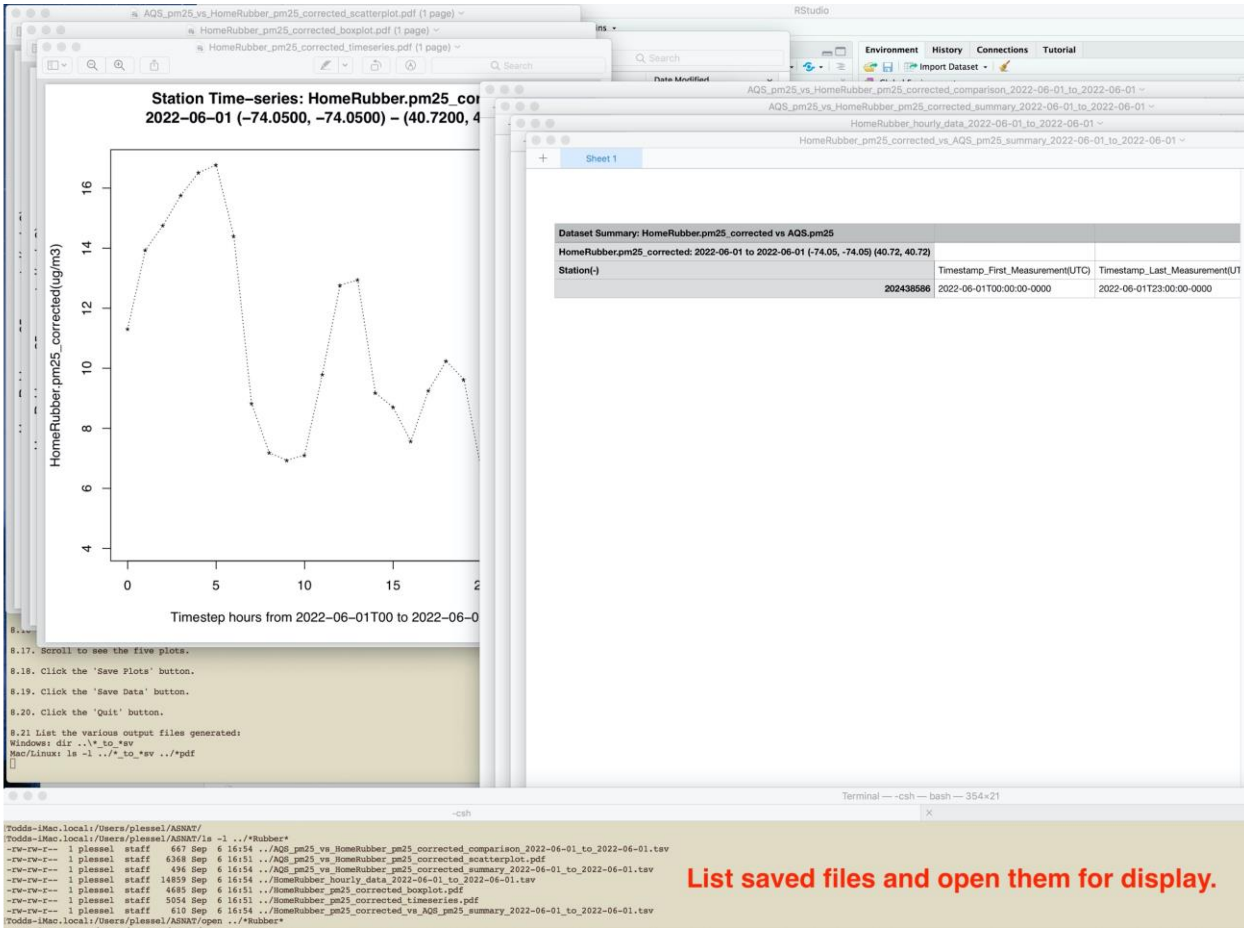
2.20 Click the Save Plots button.



2.21 Click the Save Data button.

2.22 Click the Quit button.

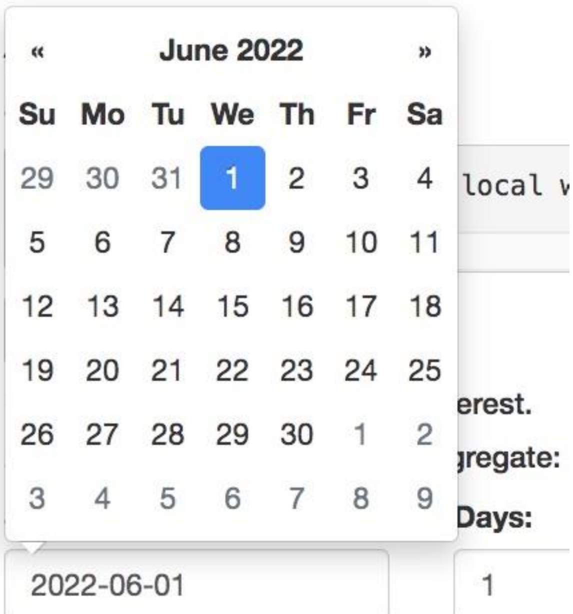
2.23 List the various output files generated:
Windows: **`dir ..\*_to_*sv`**
Mac/Linux: **`ls -l ../*_to_*sv ../*pdf`**



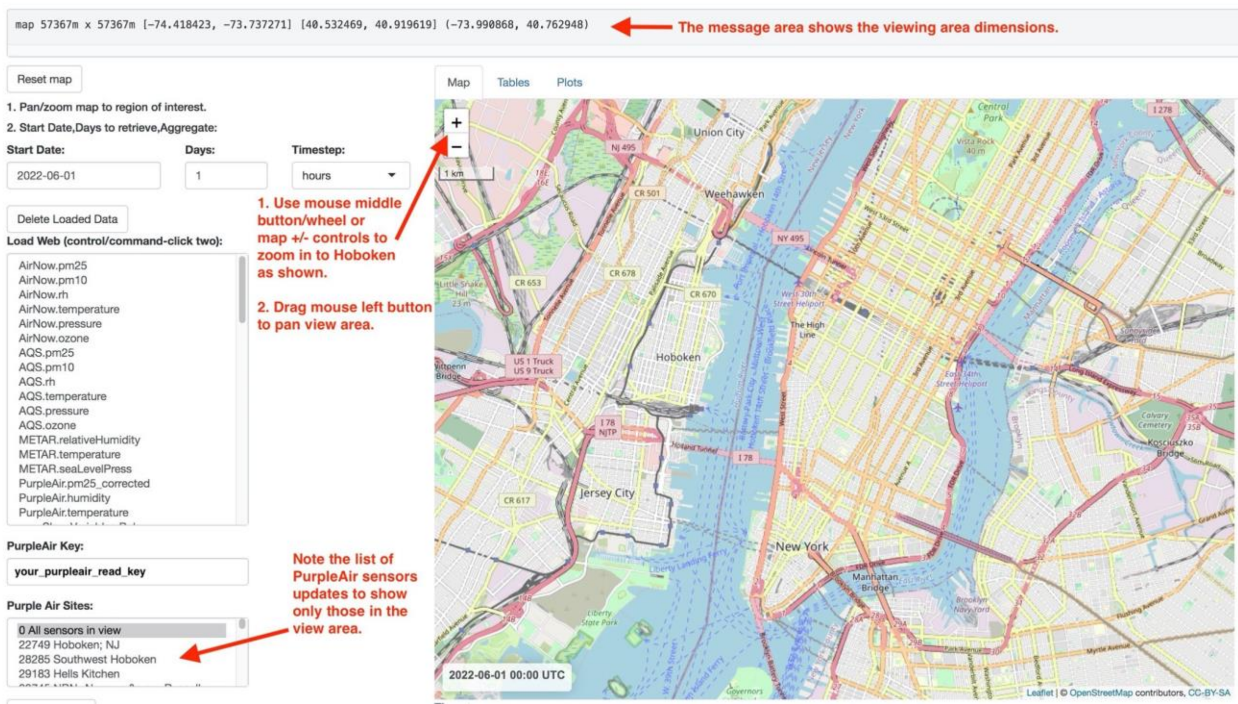
Demo 3: Select single PurpleAir sensor:

3.1 Rerun `ASNAT`:
`ASNAT`

3.2 If the Start Date is not already 2022-06-01 then
use the mouse to select the **Start Date** then type **2022-06-01** and press enter.



3.3 Click **+** in the top-left corner of the map or use the **mouse middle button or wheel to zoom in to Hoboken** and **drag the mouse left button to pan the map view as shown**.
Note the list of Purple Air Sites updates to show only those sites within the view area.



3.4 Click the Delete Loaded Data button to delete previously cached data/files before retrieving data for a new viewing area.

<u>**Note:**</u> temporary data files retrieved from web services are written to
`ASNAT/data/tmp`
and are named by data source, variable and time range. For example:
`/Users/plessel/ASNAT/ls -l data/tmp`

```
plessel  staff  3303 Sep  6 17:22 AQS_pm25_2022-06-01_to_2022-06-01.txt
plessel  staff   878 Sep  6 17:22 PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-01.txt
```

Subsequent web retrievals are avoided if the file is already cached.
However, since the file name does not encode the viewing bounds,
these ***cached files must be deleted before retrievals for a different viewing area***.

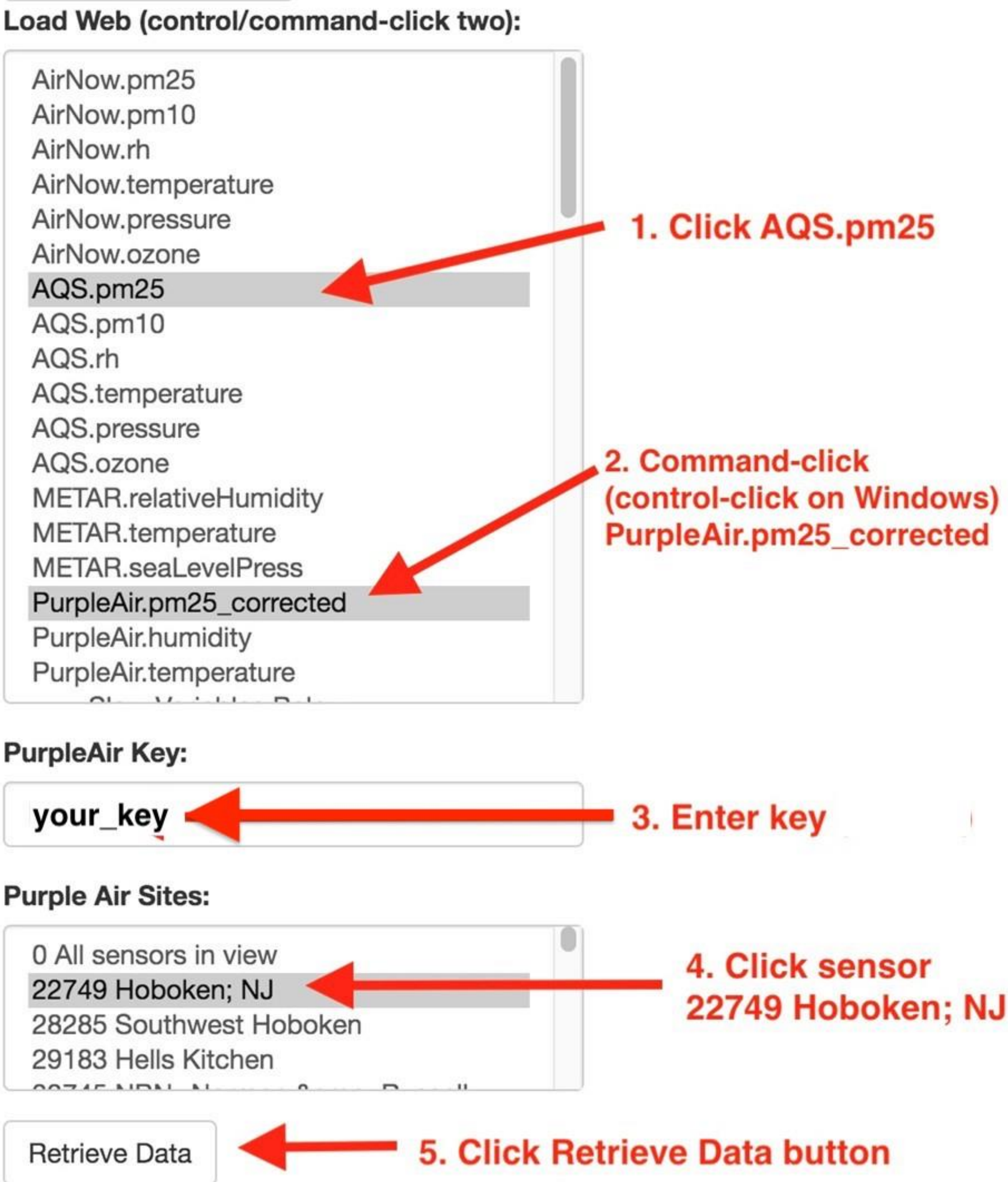
3.5 Select **AQS.pm25** and **control-click** (command-click on `Mac`) on **PurpleAir.pm25_corrected**.

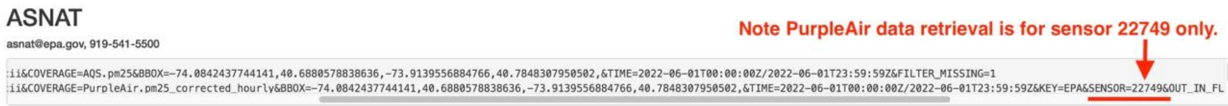
3.6 Enter your PurpleAir read key (obtained from contact@purpleair.com).

3.7 Click on Purple Air site 22749 Hoboken; NJ.

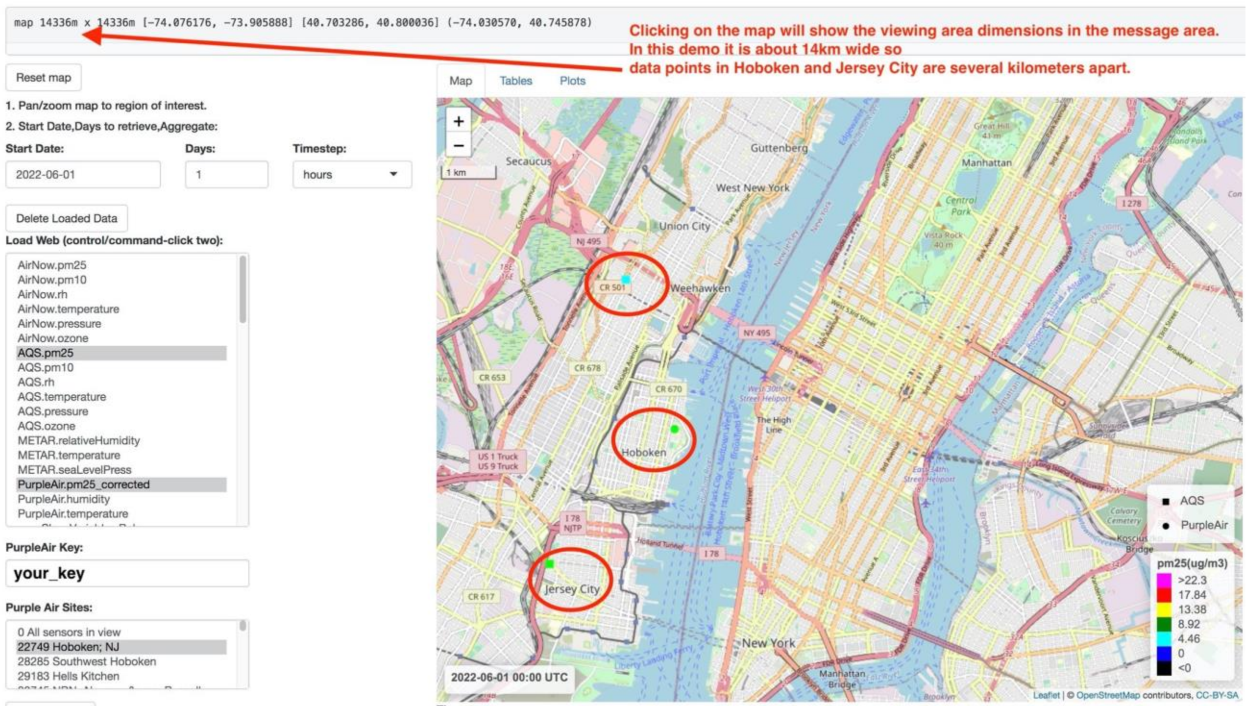
The default is *0 All sensors in view*, but in this demo
we want to compare AQS to a single PurpleAir sensor.

3.8 Click Retrieve Data button to retrieve AQS and PurpleAir data.
Note the message are shows the retrieved PurpleAir data is only for
SENSOR=22749.

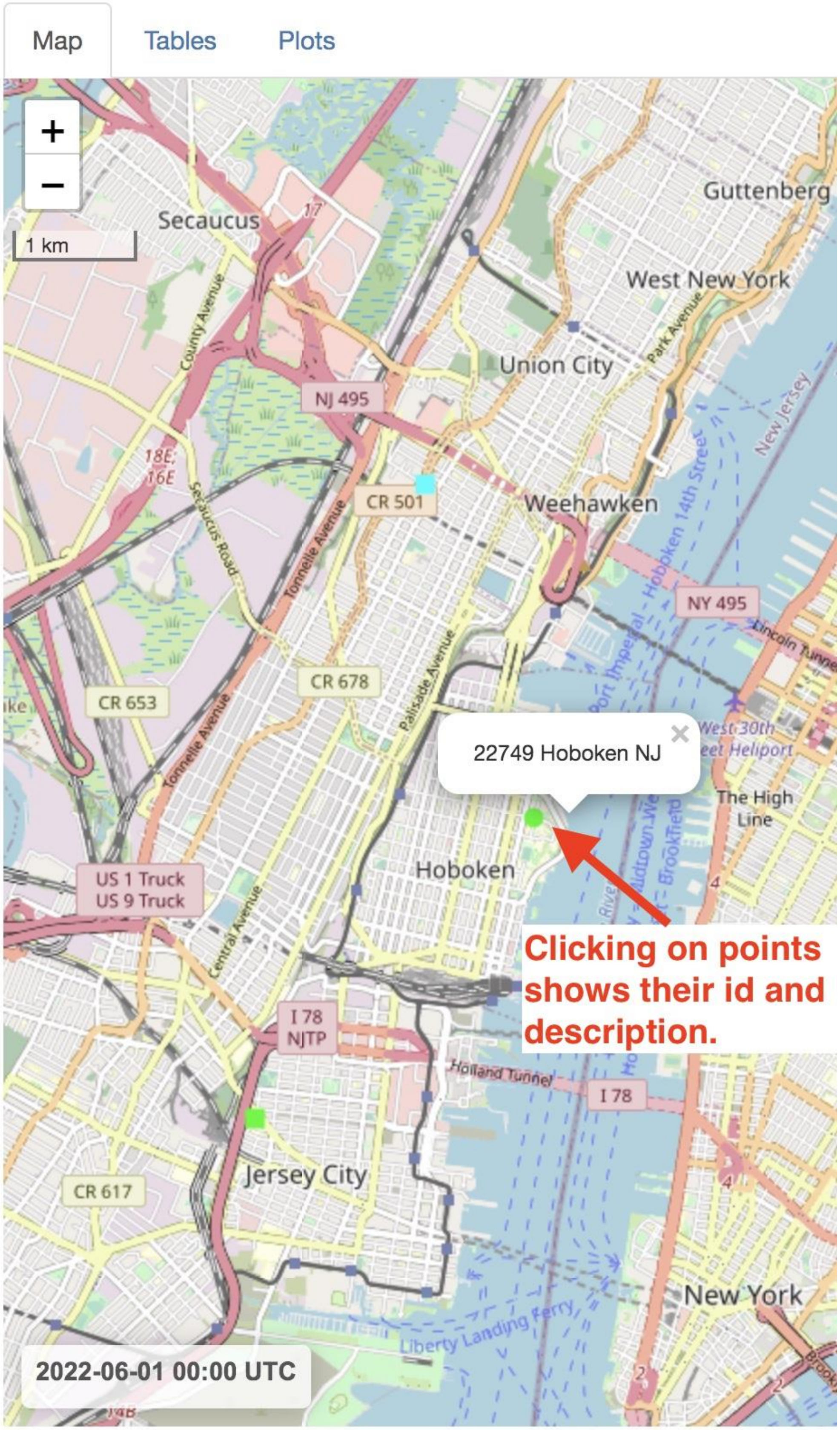




3.9 Clicking on the map shows the viewing are dimensions in the message area. From this we can see that the data points are several kilometers apart.



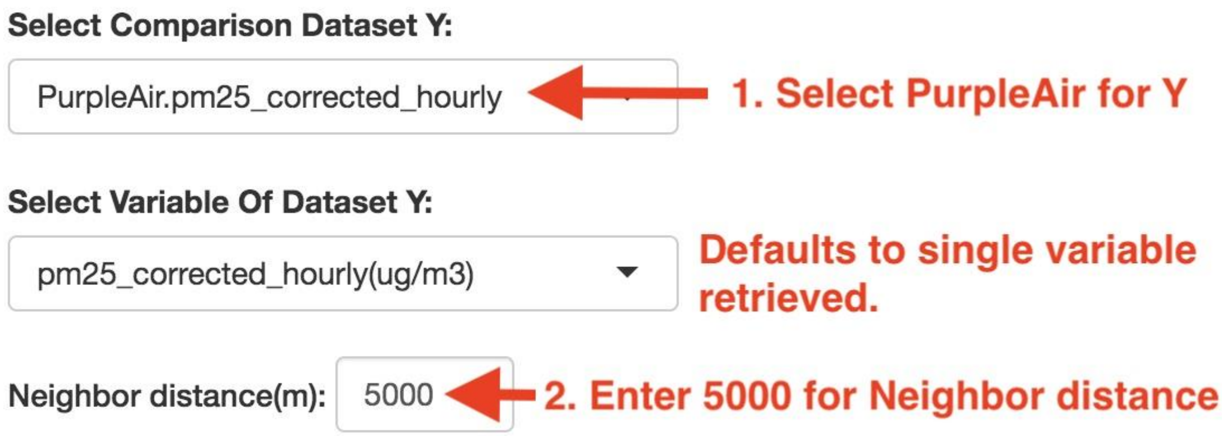
3.10 Clicking on a data point in the map will show its *ID* and *description*. (Caution: AQS data are vetted, but not all PurpleAir descriptions are reasonable.)


Map
Tables
Plots
1 km
Secaucus
Guttenberg
West New York
Union City
NJ 495
Weehawken
CR 501
NY 495
CR 678
CR 653
22749 Hoboken NJ
The High Line
Hoboken
US 1 Truck
US 9 Truck
Clicking on points shows their id and description.
I 78
NJTP
Holland Tunnel
Jersey City
CR 617
New York
Liberty Landing Ferry
2022-06-01 00:00 UTC

Timestep:

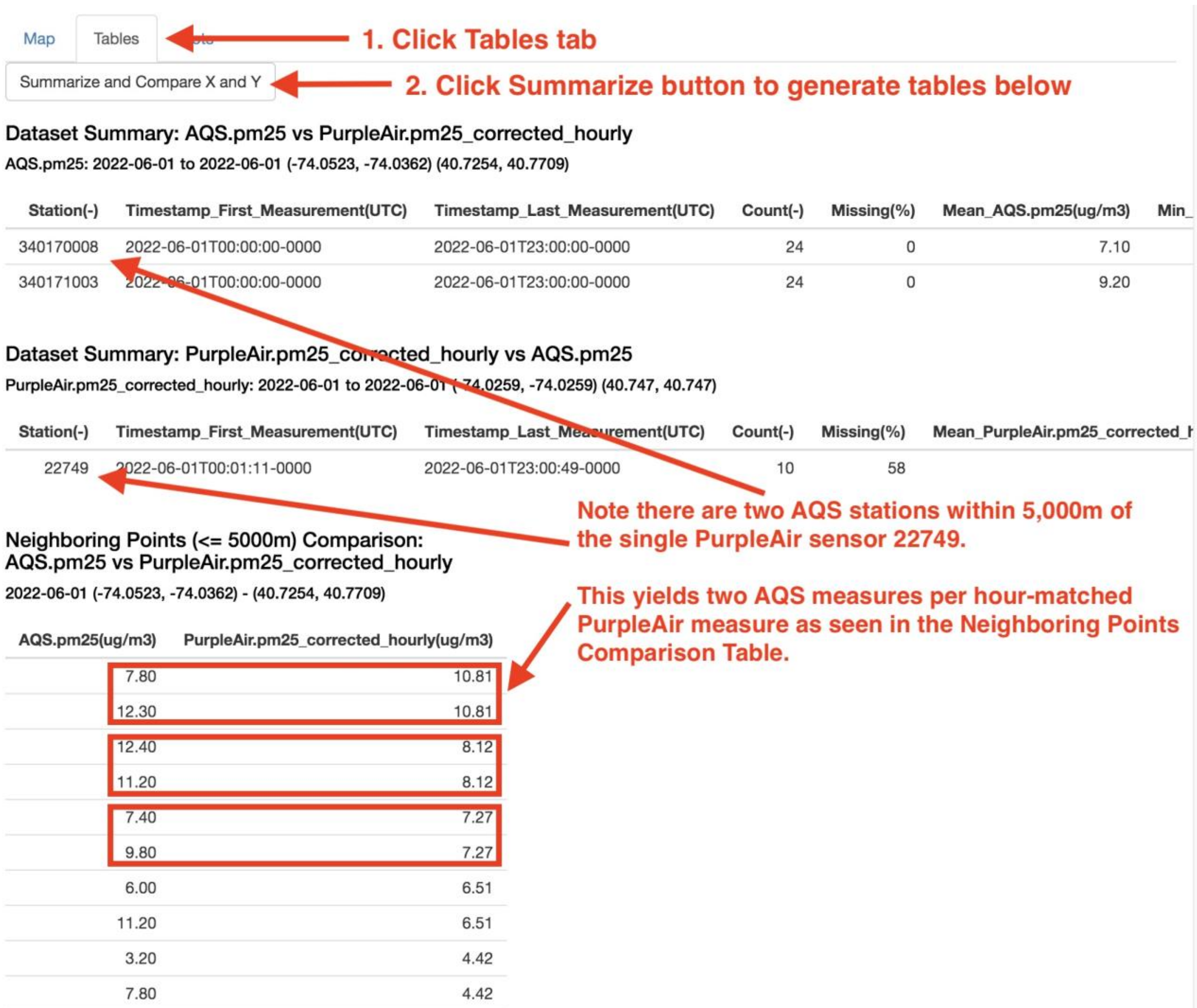
3.11 Select **PurpleAir.pm25_corrected_hourly** for
Select Comparison Dataset Y.
The variable defaults to the single or last variable available.

3.12 Enter 5000 for the **Neighbor distance** since it is clear from the map that the PurpleAir site is several kilometers from the two AQS sites.



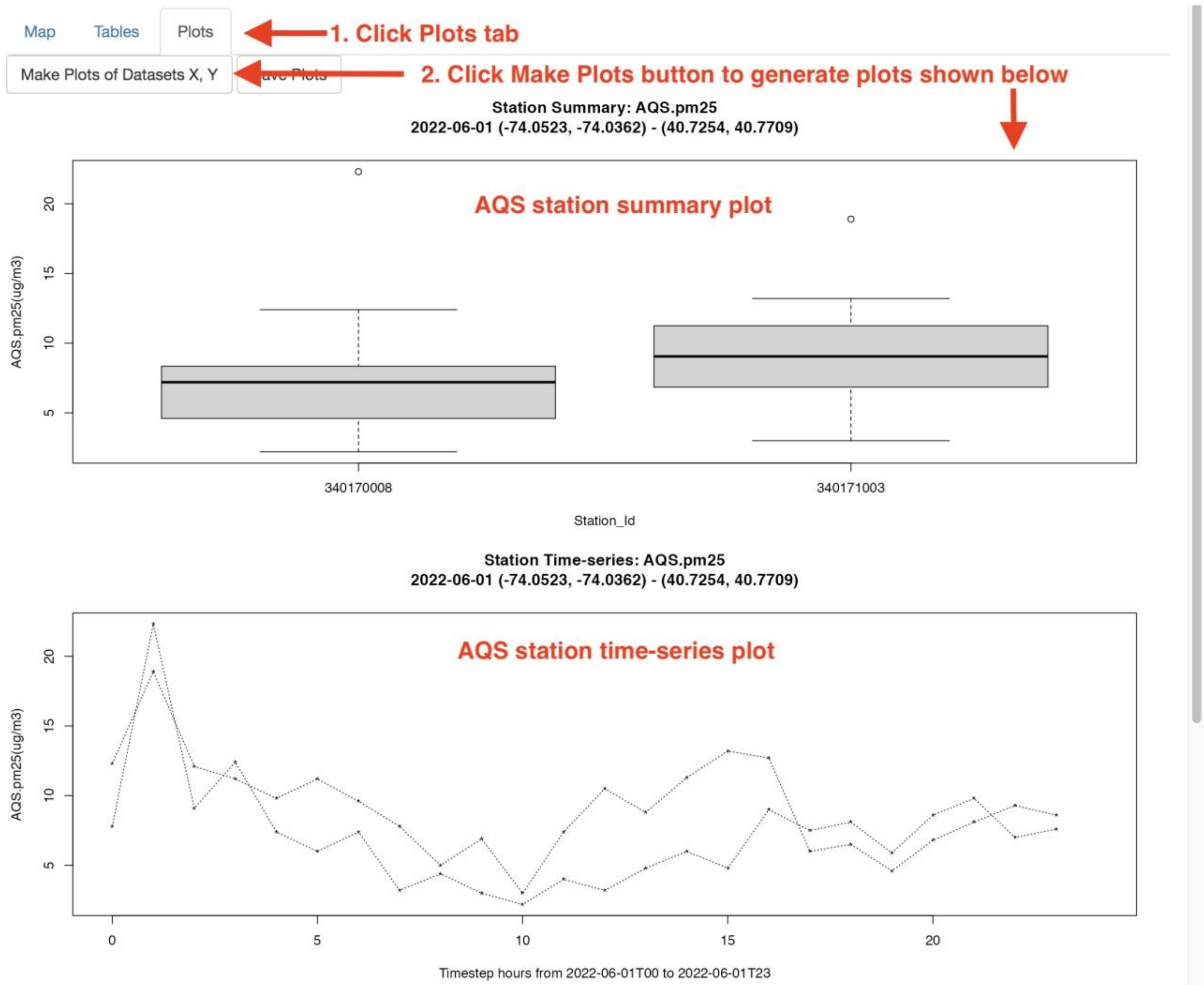
3.13 Click the Tables tab.

3.14 Click the Summarize and Compare X and Y button to generate tables. Note that there are two AQS sites that are neighbors of the single PurpleAir site. This results in two AQS measures for the single hour-matched PurpleAir measure as shown in the Neighboring Points Comparison Table.

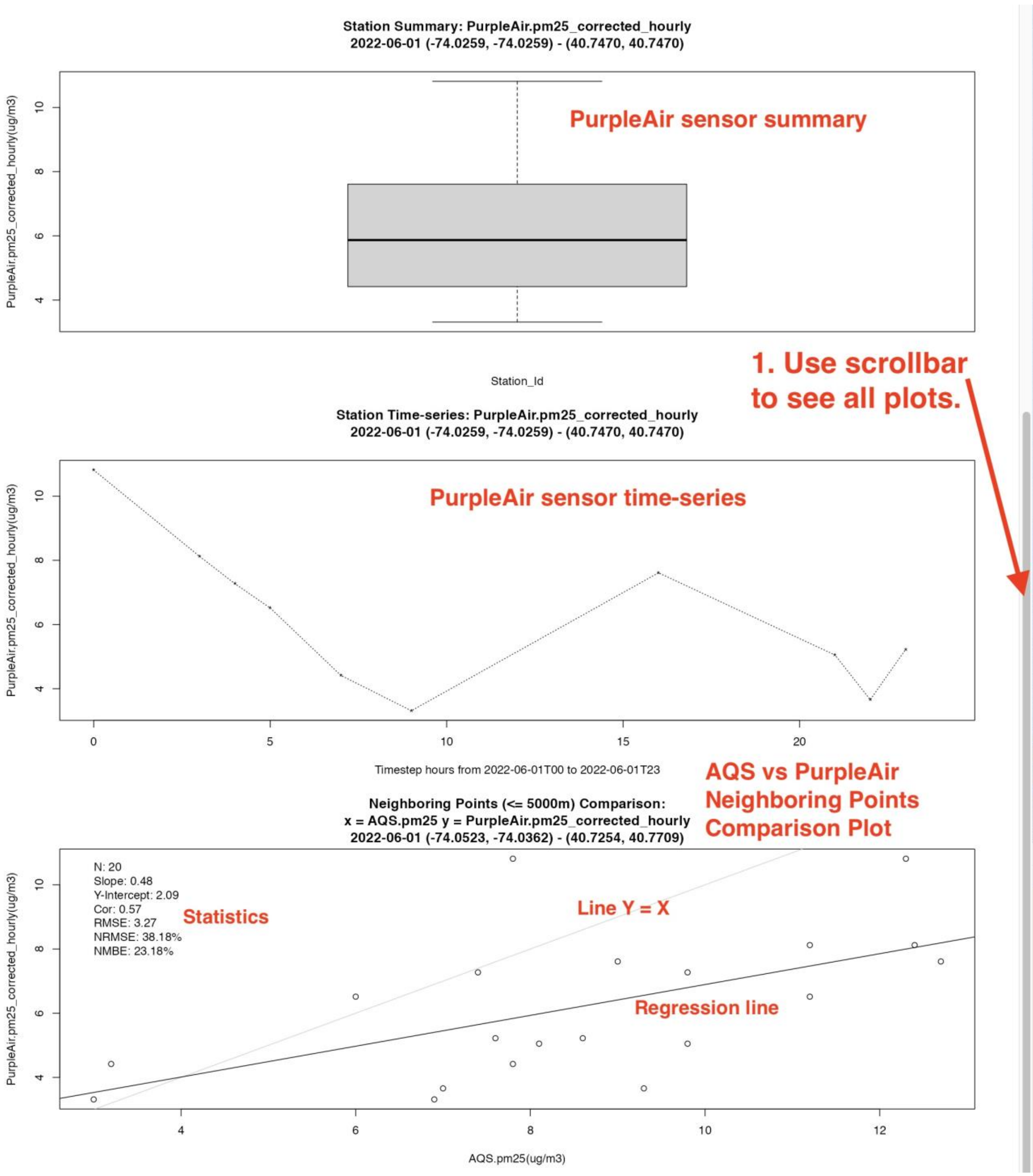


3.15 Click the Plots tab.

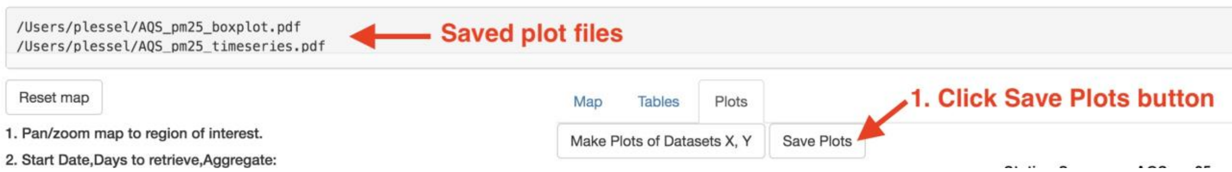
3.16 Click the Make Plots of Datasets X, Y button to generate the plots.



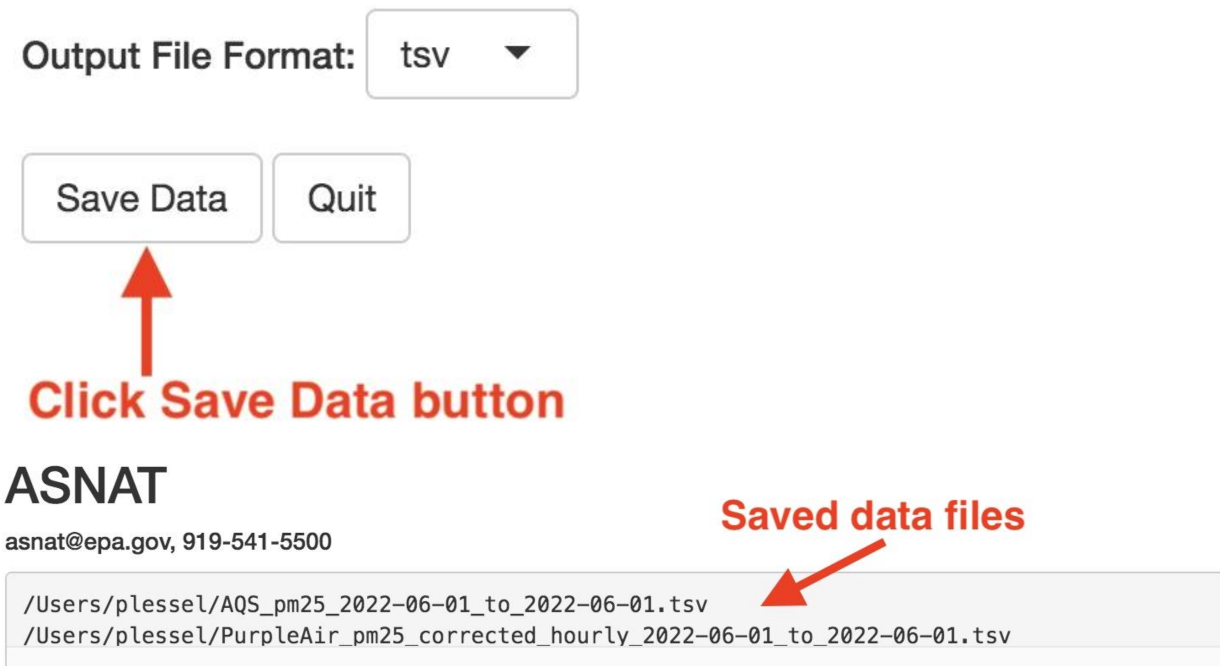
3.17 Use the **scrollbar** to see all five plots.
(Extending to 30 days takes a few minutes to retrieve,
yields many more comparison points and slightly better correlation.)


Station Summary: PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.0259, -74.0259) - (40.7470, 40.7470)
PurpleAir sensor summary
PurpleAir.pm25_corrected_hourly(ug/m3)
Station_Id
1. Use scrollbar to see all plots.
Station Time-series: PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.0259, -74.0259) - (40.7470, 40.7470)
PurpleAir sensor time-series
Timestep hours from 2022-06-01T00 to 2022-06-01T23
AQS vs PurpleAir Neighboring Points Comparison Plot
Neighboring Points (<= 5000m) Comparison:
x = AQS.pm25 y = PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.0523, -74.0362) - (40.7254, 40.7709)
N: 20
Slope: 0.48
Y-Intercept: 2.09
Cor: 0.57
RMSE: 3.27
NRMSE: 38.18%
NMBE: 23.18%
Statistics
Line Y = X
Regression line
AQS.pm25(ug/m3)

3.18 Click Save Plots button.
Note the names of the saved files in the message area.
(Scroll for complete list.)



3.19 Click Save Data button to save the retrieved data files and tables as plain text (tab-delimited) files.



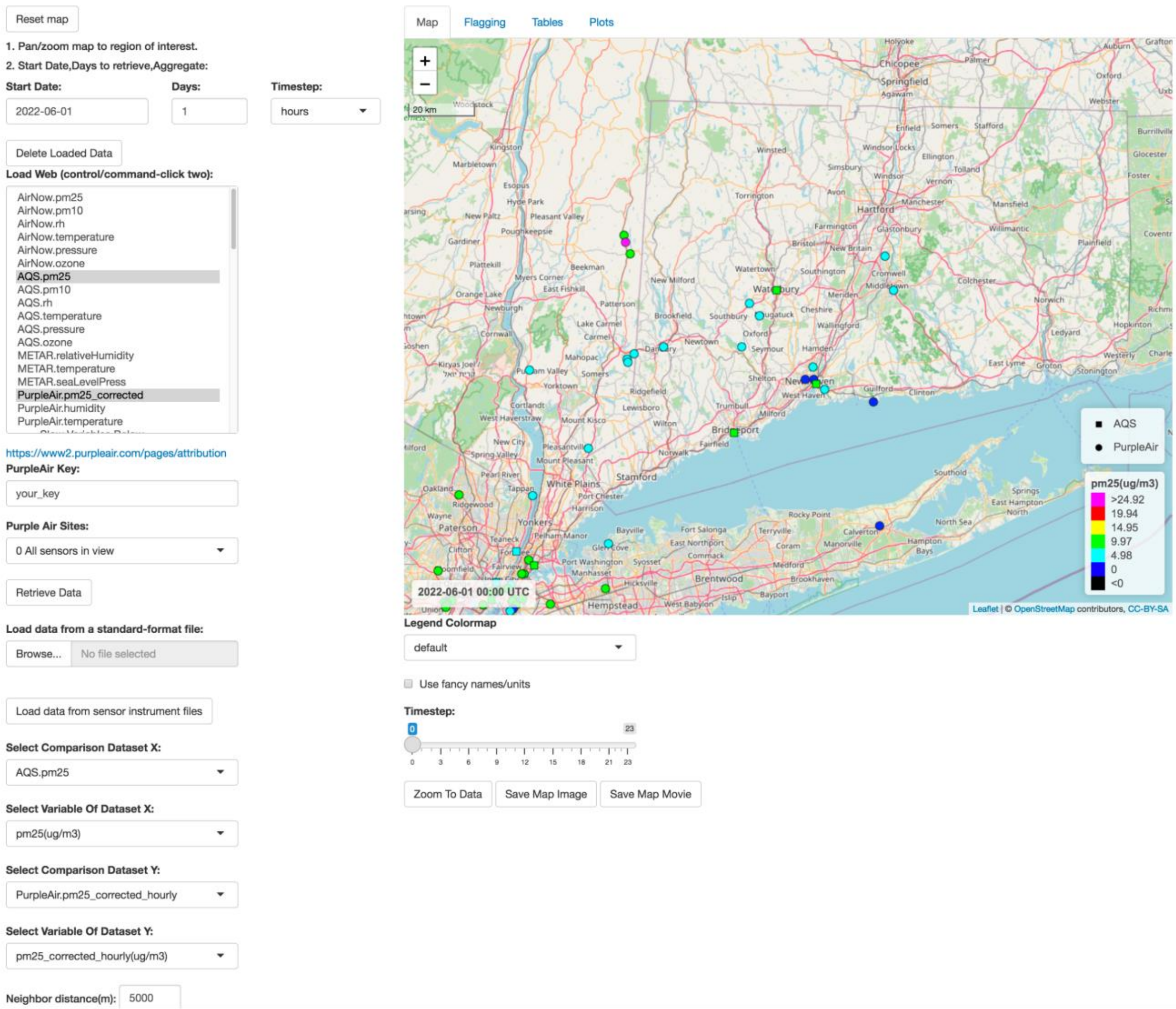
3.20 Click Quit button to quit ASNAT.

Demo 4: Flagging:

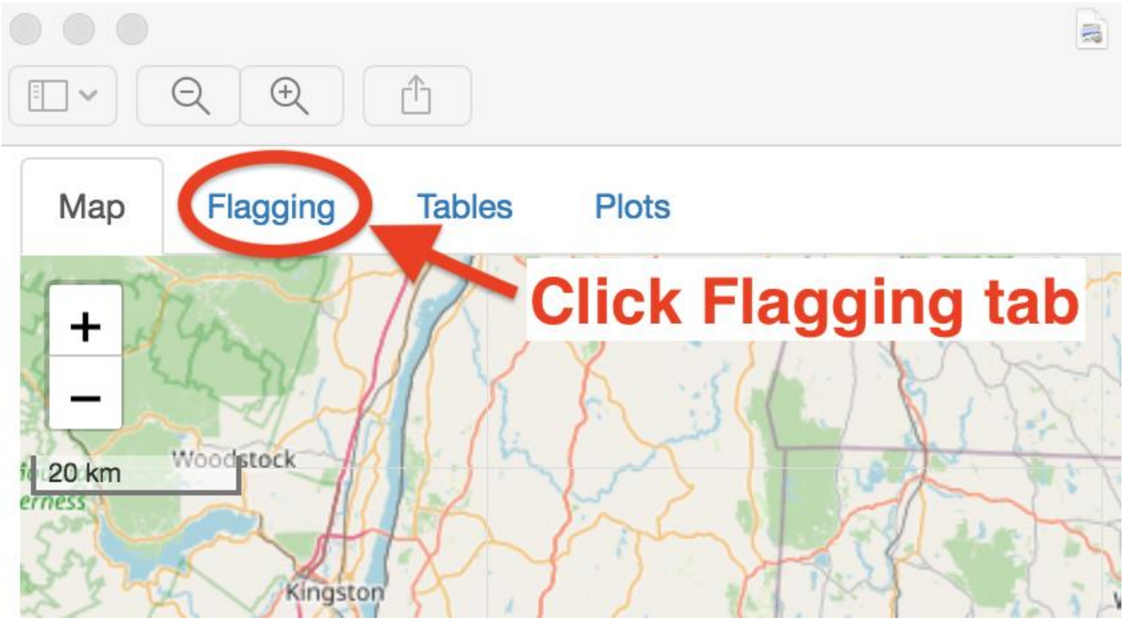
ASNAT allows for a dataset's data lines to be marked with one or more integer codes (separated with semi-colons without spaces) indicating the line is suspect and plotted differently (as gray points) and statistics computed after omitting these flagged lines.
There are three ways to flag data lines:
4.1 Manually selecting lines either in ASNAT or editing/computing the file's flagged column outside ASNAT and loading it into ASNAT.
4.2 Specifying **flag conditions** on the available columns of the dataset.
4.3 Selecting Dataset Y **Neighbor Flagging** which compares time-matched neighboring measures from Dataset X and Dataset Y and flags measures in Dataset Y based on comparison to the measures in Dataset X.

To demo flagging:
Quit/restart ASNAT and load the same datasets as in **Demo 1**:
Reset map
Start Date: 2022-06-01
Days: 1
Timestep: hours
Delete Loaded Data
Load Web:
  AQS.pm25
  PurpleAir.pm25_corrected
Purple Air Key:
  your_key
Retrieve Data
Select Comparison Dataset Y:
  PurpleAir.pm25_corrected_hourly
Neighbor Distance(m): 5000

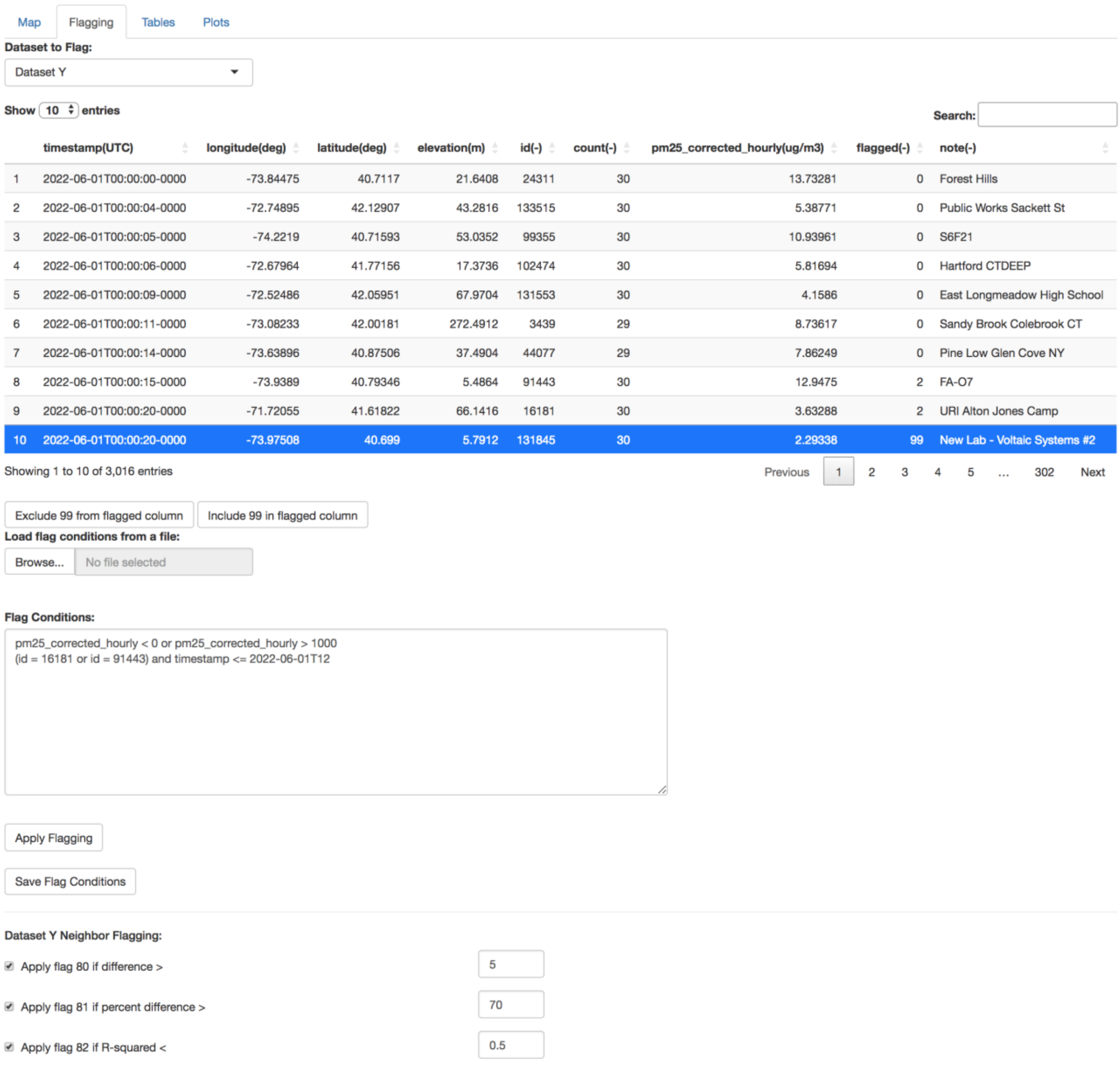
ASNAT should appear as the image below.



Once the data is loaded as shown above,
Click the Flagging tab.

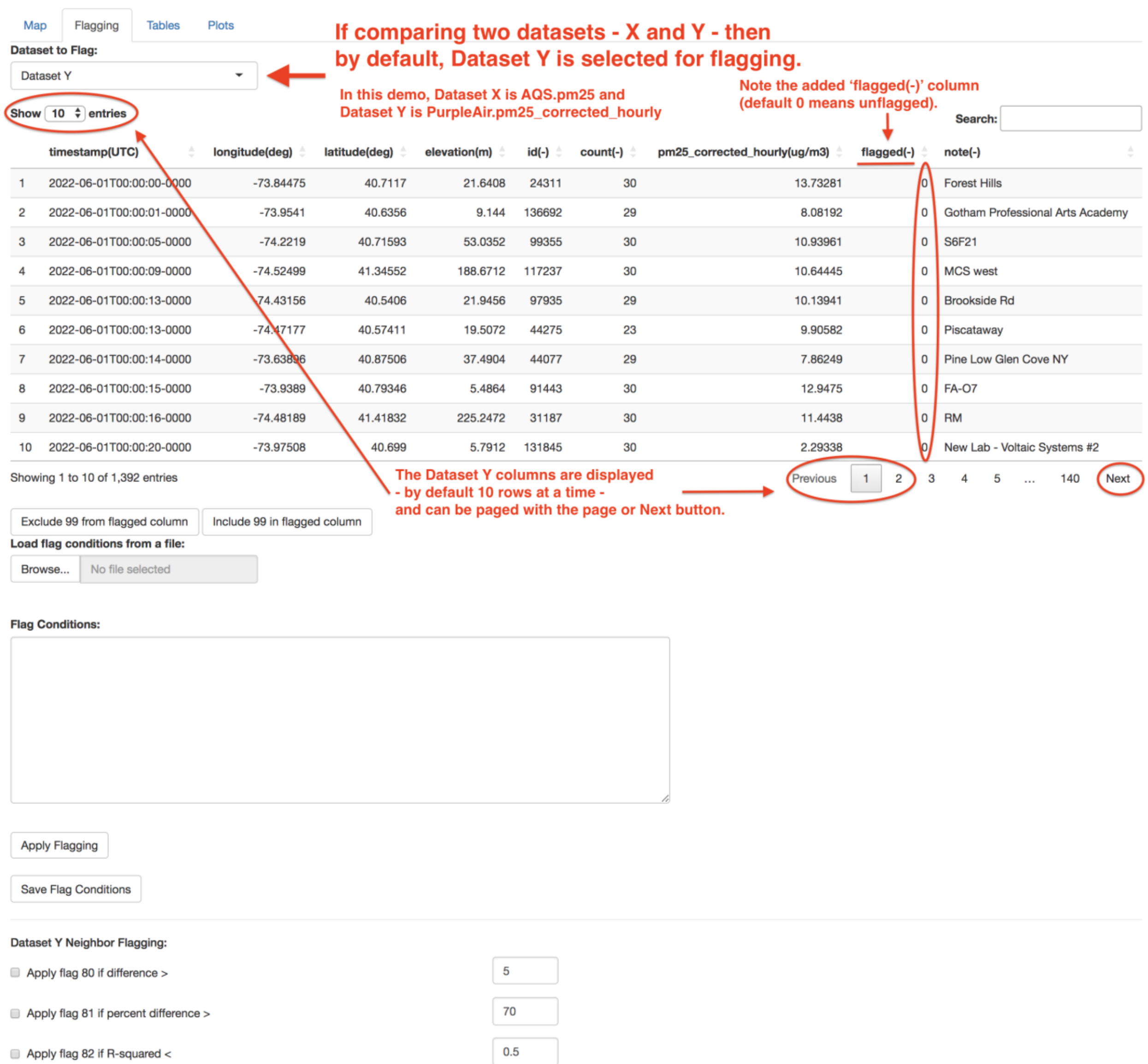


This will display the data lines of Dataset Y – in this example, PurpleAir.pm25_corrected_hourly as shown in the image below.



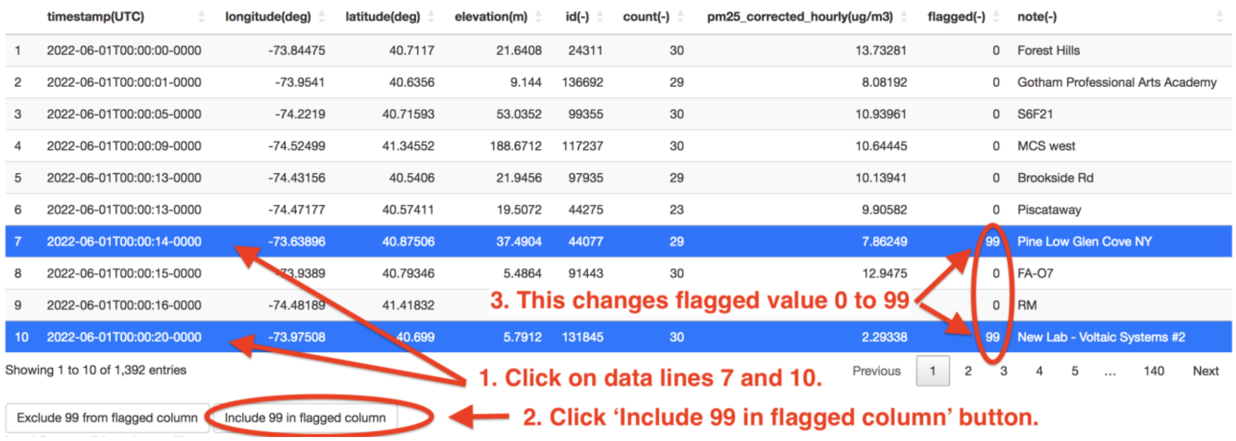
	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
2	2022-06-01T00:00:04-0000	-72.74895	42.12907	43.2816	133515	30	5.38771	0	Public Works Sackett St
3	2022-06-01T00:00:05-0000	-74.2219	40.71593	53.0352	99355	30	10.93961	0	S6F21
4	2022-06-01T00:00:06-0000	-72.67964	41.77156	17.3736	102474	30	5.81694	0	Hartford CTDEEP
5	2022-06-01T00:00:09-0000	-72.52486	42.05951	67.9704	131553	30	4.1586	0	East Longmeadow High School
6	2022-06-01T00:00:11-0000	-73.08233	42.00181	272.4912	3439	29	8.73617	0	Sandy Brook Colebrook CT
7	2022-06-01T00:00:14-0000	-73.63896	40.87506	37.4904	44077	29	7.86249	0	Pine Low Glen Cove NY
8	2022-06-01T00:00:15-0000	-73.9389	40.79346	5.4864	91443	30	12.9475	2	FA-O7
9	2022-06-01T00:00:20-0000	-71.72055	41.61822	66.1416	16181	30	3.63288	2	URI Alton Jones Camp
10	2022-06-01T00:00:20-0000	-73.97508	40.699	5.7912	131845	30	2.29338	99	New Lab - Voltaic Systems #2

Since the dataset can have many data lines, they are paged, by default 10 lines at a time.
This can be increased with the Show entries menu.
Pages can be navigated by clicking on the page numbers or the Next button.
Note the 'flagged (-)' column added before the last 'note(-)' column.
`ASNAT` adds this column when loading a dataset if it does not already exist.
0 indicates the data line is unflagged (the default).



4.1 Manual Flagging:

Click on line **7** and then click on line **10**.
(Re-clicking toggles line selection.
Shift-clicking selects/toggles a range of lines.)
Click the 'Include 99 in flagged column' button.
This will change the flagged column of these lines from 0 to 99.
99 is the integer code used to indicate manually flagged lines.



	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
2	2022-06-01T00:00:01-0000	-73.9541	40.6356	9.144	136692	29	8.08192	0	Gotham Professional Arts Academy
3	2022-06-01T00:00:05-0000	-74.2219	40.71593	53.0352	99355	30	10.93961	0	S6F21
4	2022-06-01T00:00:09-0000	-74.52499	41.34552	188.6712	117237	30	10.64445	0	MCS west
5	2022-06-01T00:00:13-0000	-74.43156	40.5406	21.9456	97935	29	10.13941	0	Brookside Rd
6	2022-06-01T00:00:13-0000	-74.47177	40.57411	19.5072	44275	23	9.90582	0	Piscataway
7	2022-06-01T00:00:14-0000	-73.63896	40.87506	37.4904	44077	29	7.86249	99	Pine Low Glen Cove NY
8	2022-06-01T00:00:15-0000	73.9389	40.79346	5.4864	91443	30	12.9475	0	FA-O7
9	2022-06-01T00:00:16-0000	-74.48189	41.41832					0	RM
10	2022-06-01T00:00:20-0000	-73.97508	40.699	5.7912	131845	30	2.29338	99	New Lab - Voltaic Systems #2

Showing 1 to 10 of 1,392 entries

Previous 1 2 3 4 5 ... 140 Next

Exclude 99 from flagged column | Include 99 in flagged column

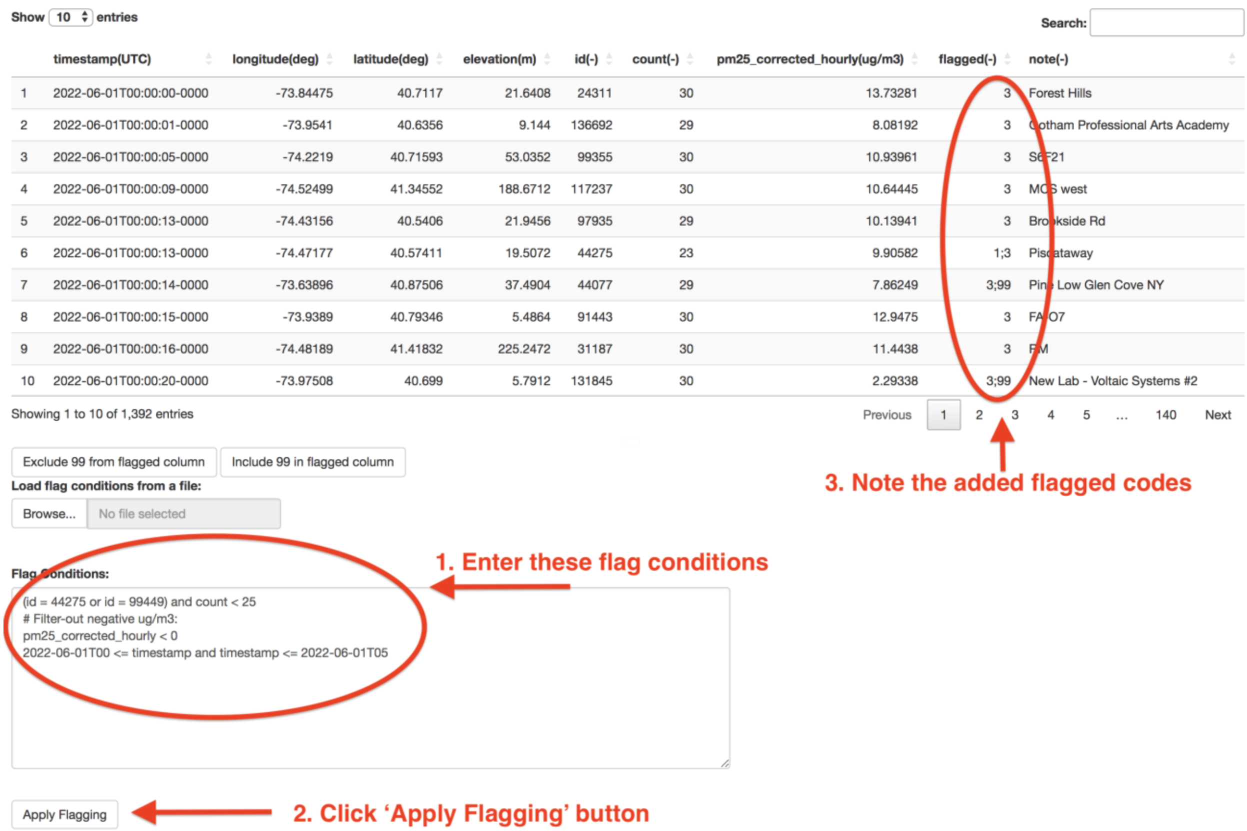
4.2 Flag Conditions:

Flag conditions are Boolean expressions of the available columns of a dataset.
These expressions must be evaluated as either true or false.
They are evaluated on each data line and, if true,
they include an integer code for each true flag condition.
The numbers are 1 for the first condition, 2 for the second condition, and so on.

Let’s specify a few flag conditions on Dataset Y using the named columns shown in the table.
These are timestamp, longitude, latitude, elevation, id, count, pm25_corrected_hourly and note.
The column names are operands.
Available relational operators are: <, <=, >, >=, =, !=, or, and, (,).
One may also use R operators such as: ||, &&, -, +, *, /, ^, etc.
Note that * and 'and' have higher precedence than + and 'or' and ^ is right-associative.
(E.g., 2 + 3 * 4 = 14 and 2^3^2 = 512.)
Use parentheses only when necessary.
can be used for comment lines.

Click in the text editing area labeled 'Flag Conditions:'
Enter the following lines:
(id = 44275 or id = 99449) and count < 25
 # Filter-out negative ug/m3:
pm25_corrected_hourly < 0
2022-06-01T00 <= timestamp and timestamp <= 2022-06-01T05

Click the 'Apply Flagging' button.
ASNAT should appear as shown in the next image.

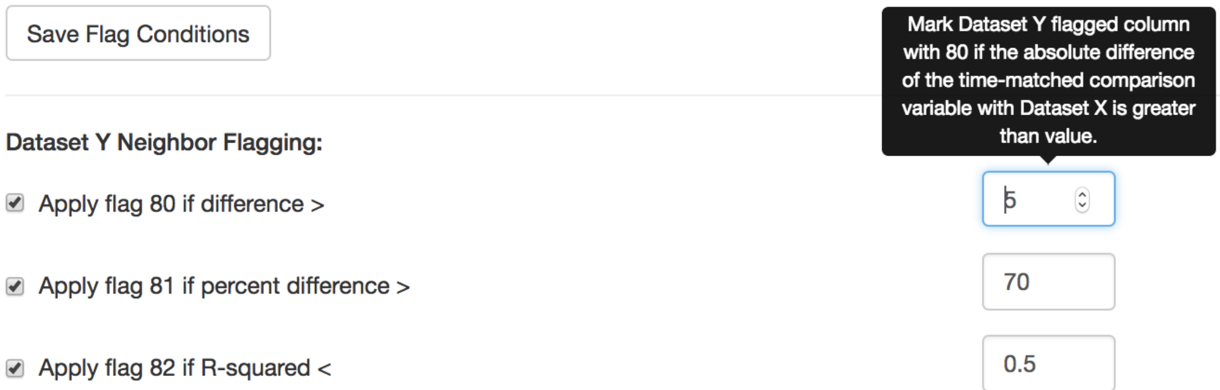


Note the integer codes included in the flagged column.
Regarding timestamps, the literal comparison values can be any starting sub-length of the timestamp.
In the example here, all data lines between hours 00 and 05 are flagged.
For multi-day datasets, one may omit the hour to omit entire days.
E.g., for a 30-day dataset, omit the first 3 and last 6 days with:
timestamp <= 2022-06-03 or timestamp >= 2022-06-25
ASNAT's unique ability to evaluate arbitrary expressions of possibly multiple columns is more powerful than simple range specifications on single columns.

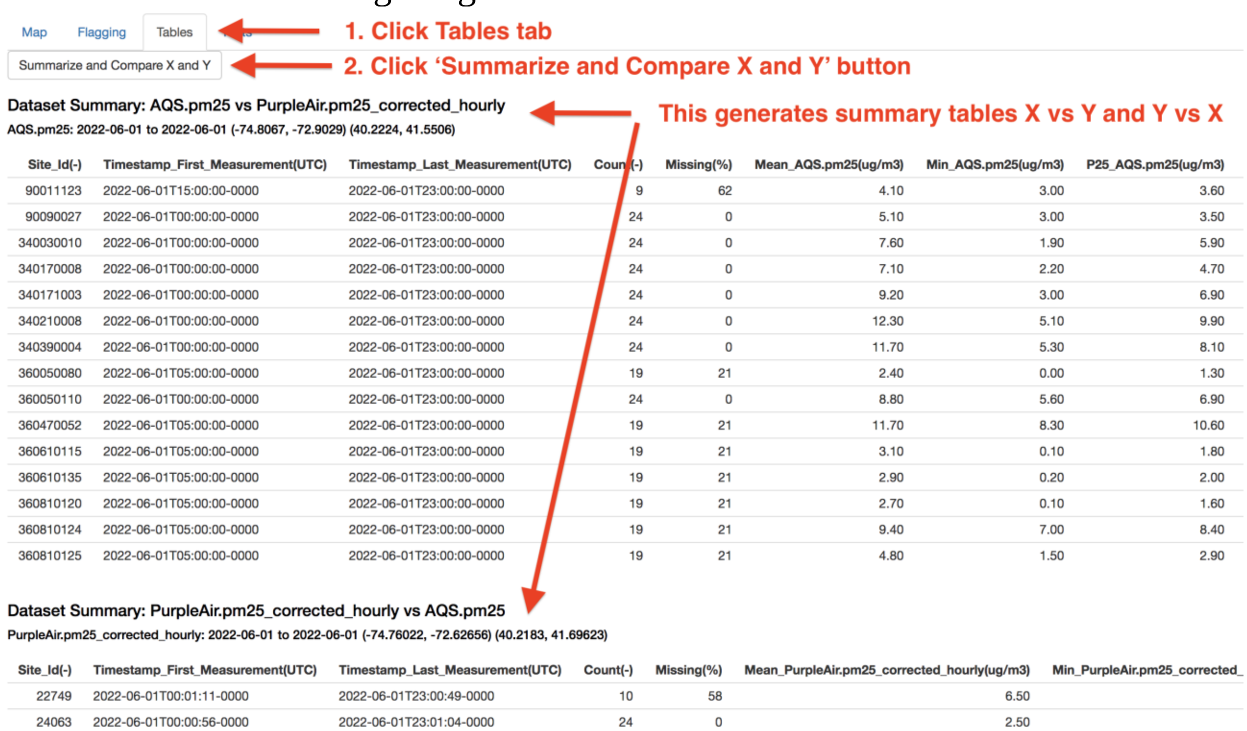
These flag conditions can be saved to plain text files and reloaded into ASNAT later.
Click the 'Save Flag Conditions' button to save them to the file 'flags.txt' in the 'Save to Directory'. Rename this file if desired.
Click the 'Browse...' button to reload them later.
Flag conditions can only be applied to a dataset containing the columns named in the flag conditions.

4.3 Neighbor Flagging:

Manual flagging and condition-based flagging operate only on single lines of the 'Dataset to Flag'.
`ASNAT` also includes flagging based on differences between matched Dataset X and Dataset Y measures.
In this example, **'Select Variable of Dataset** Y' is pm25_corrected_hourly.
This value is compared to Dataset X (AQS.pm25).
For each Dataset Y time-matched point within the specified '**Neighbor Distance**' of a Dataset X point, one can flag based on the absolute difference, percent difference, or R-squared value of the matched time series of X and Y measures.
(R-squared is called the coefficient of determination and equals the square of r, the Pearson sample correlation coefficient.
https://en.wikipedia.org/wiki/Pearson_correlation_coefficient)
These are the buttons shown at the bottom and the default values are shown but may be changed.
One can select/deselect these additional flagging options which are applied after the Summary tables are computed.
Select all three buttons and use the default values.
As shown in the tooltip for these values, the flagged integer codes are 80, 81, 82 respectively.



To apply Neighbor Flagging, compute the Summary and Comparison tables.
Click the Tables tab.
Click the 'Summarize and Compare X and Y' button.
This generates four tables:
Dataset Summary of X vs Y
Dataset Summary of Y vs X
Neighboring Points of Y for each point in X
Neighboring Points of Y for each point in X R-squared values
As shown in the following images.



Dataset Summary: AQS.pm25 vs PurpleAir.pm25_corrected_hourly
AQS.pm25: 2022-06-01 to 2022-06-01 (-74.8067, -72.9029) (40.2224, 41.5506)

Site_Id(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_AQS.pm25(ug/m3)	Min_AQS.pm25(ug/m3)	P25_AQS.pm25(ug/m3)
90011123	2022-06-01T15:00:00-0000	2022-06-01T23:00:00-0000	9	62	4.10	3.00	3.60
90090027	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	5.10	3.00	3.50
340030010	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	7.60	1.90	5.90
340170008	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	7.10	2.20	4.70
340171003	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	9.20	3.00	6.90
340210008	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	12.30	5.10	9.90
340390004	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	11.70	5.30	8.10
360050080	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.40	0.00	1.30
360050110	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	8.80	5.60	6.90
360470052	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	11.70	8.30	10.60
360610115	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	3.10	0.10	1.80
360610135	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.90	0.20	2.00
360810120	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.70	0.10	1.60
360810124	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	9.40	7.00	8.40
360810125	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	4.80	1.50	2.90

Dataset Summary: PurpleAir.pm25_corrected_hourly vs AQS.pm25
PurpleAir.pm25_corrected_hourly: 2022-06-01 to 2022-06-01 (-74.76022, -72.62656) (40.2183, 41.69623)

Site_Id(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_PurpleAir.pm25_corrected_hourly(ug/m3)	Min_PurpleAir.pm25_corrected_
22749	2022-06-01T00:01:11-0000	2022-06-01T23:00:49-0000	10	58	6.50	
24063	2022-06-01T00:00:56-0000	2022-06-01T23:01:04-0000	24	0	2.50	

Neighboring Points (<= 5000m) Comparison: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

And Neighbor Points table

timestamp(UTC)	AQS.id(-)	AQS.pm25(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly(ug/m3)	PurpleAir.flagged(-)
2022-06-01T00:00:00-0000	90090027	10.30	39689	7.12	3
2022-06-01T00:00:00-0000	90090027	10.30	24063	4.06	3;80;81
2022-06-01T00:00:00-0000	90090027	10.30	106862	4.54	3;80;81
2022-06-01T00:00:00-0000	340030010	6.90	91899	10.79	3
2022-06-01T00:00:00-0000	340170008	7.80	98723	11.23	3
2022-06-01T00:00:00-0000	340170008	7.80	22749	11.24	3
2022-06-01T00:00:00-0000	340170008	7.80	101655	8.60	3

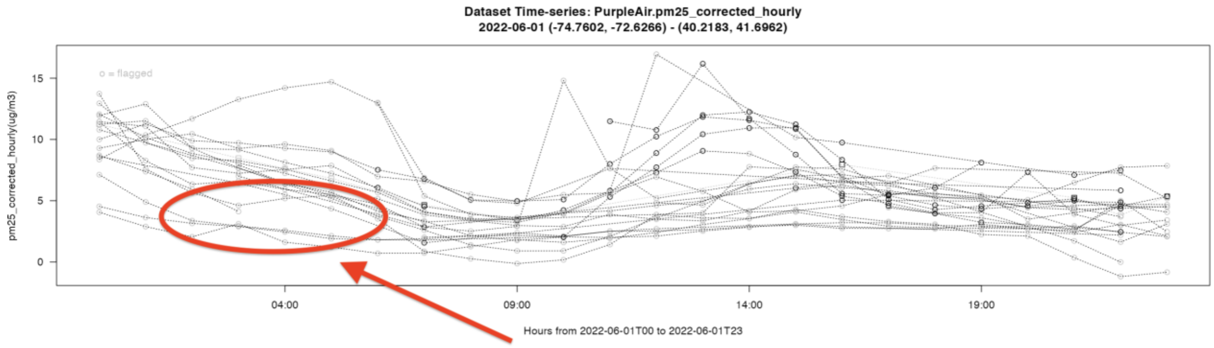
Neighboring Points (<= 5000m) Comparision R2: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

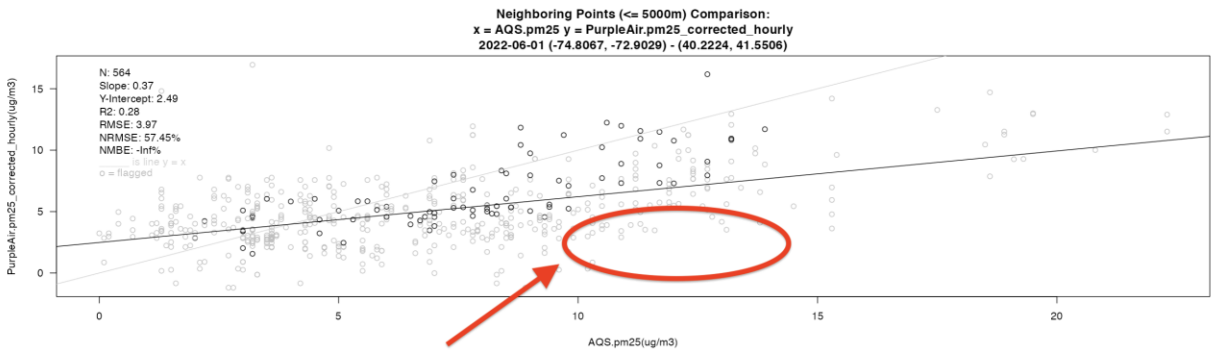
And R-squared table

AQS.id(-)	AQS.longitude(deg)	AQS.latitude(deg)	PurpleAir.id(-)	PurpleAir.longitude(deg)	PurpleAir.latitude(deg)	distance(m)	R2(-)
340170008	-74.0362	40.7709	22749	-74.0259	40.7470	2793.1682	0.3943
340171003	-74.0523	40.7254	22749	-74.0259	40.7470	3269.6087	0.9564
340170008	-74.0362	40.7709	58069	-73.9949	40.7544	3932.7059	0.7242
360050080	-73.9201	40.8361	91423	-73.8916	40.8611	3671.8532	0.0000
360610115	-73.9356	40.8496	91423	-73.8916	40.8611	3917.0500	0.9051
340390004	-74.2084	40.6414	96741	-74.2089	40.6635	2455.6155	0.7617
360470052	-74.0187	40.6418	97641	-73.9894	40.6596	3159.8462	0.9179
340170008	-74.0362	40.7709	101655	-73.9836	40.7643	4484.1806	0.9104
340171003	-74.0523	40.7254	110490	-74.0845	40.7104	3178.8728	0.5163
340171003	-74.0523	40.7254	114593	-74.0727	40.7269	1730.5630	0.8508
360470052	-74.0187	40.6418	129163	-73.9962	40.6760	4248.7797	0.6746
340210008	-74.7632	40.2224	131197	-74.7602	40.2183	520.7753	1.0000

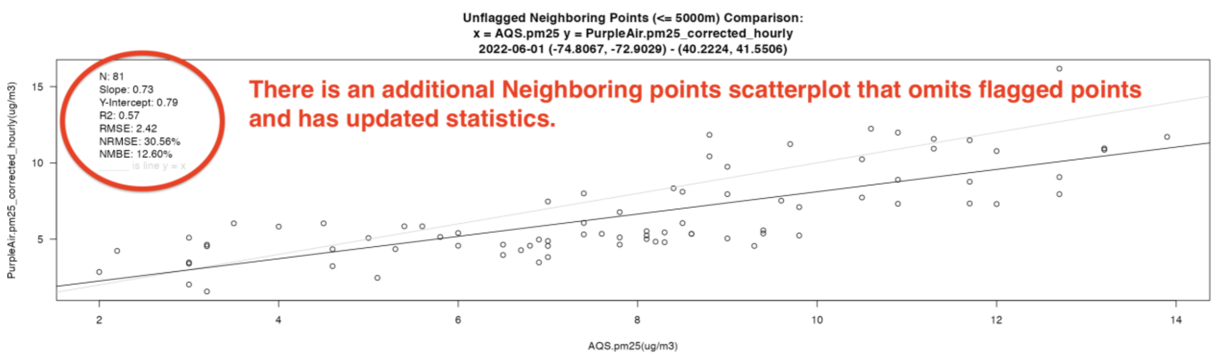
Finally, plot the results from these tables.
Click the 'Plots' tab.
Click the 'Make Points of Datasets X, Y' button.
Note in the time-series plot of Dataset Y the flagged points are gray.
The scatter plot flagged points are also gray.
There is an additional scatter plot of just the non-flagged points (N = 81).



The Time-series plot of Dataset Y shows flagged points in gray.



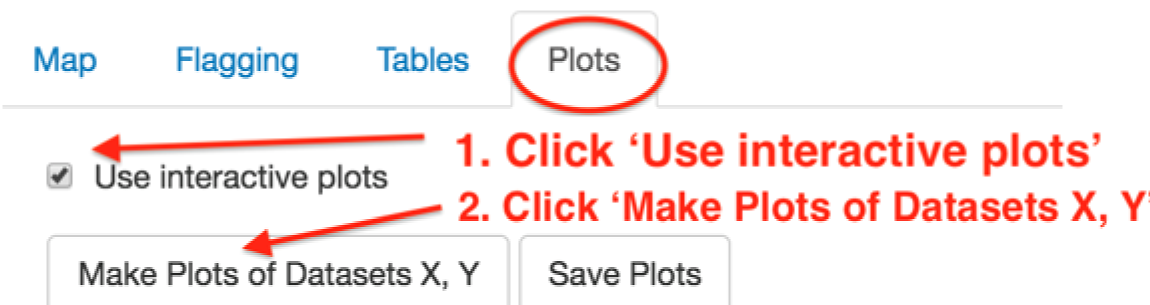
The Neighboring points scatterplot shows flagged points in gray.



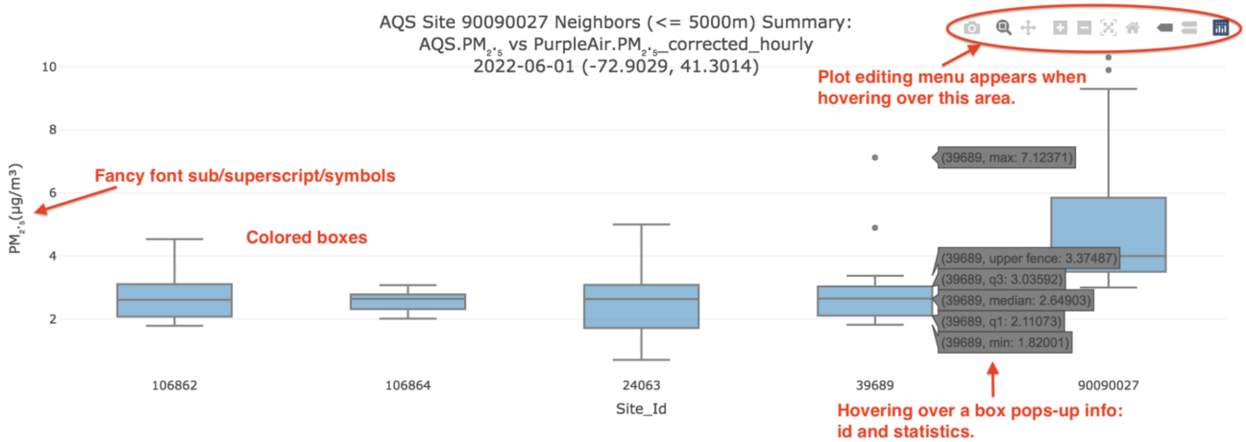
Demo 5: Interactive Plots:

By default, `ASNAT` uses the basic plots which are grayscale, fast, clear, and best for use in publication-quality PDF documents. Lacking color, they are also understandable by those with color vision impairment. PDF files are printed at the highest resolution of the printer rather than the screen resolution they were captured at (as is the case with pixel image formats).
However, `ASNAT` also provides the option of colored 'interactive plots'.
These allow the user to zoom, probe, select, and hide parts of plots and save the modified plot as a PNG image file.
Continuing from the previous demo, you can explore interactive plots.

On the **Plots tab**,
Click the 'Use interactive plots' button.
Click the 'Make Plots of Datasets X, Y' button.
(You may have to click twice for all plots to appear/update.)



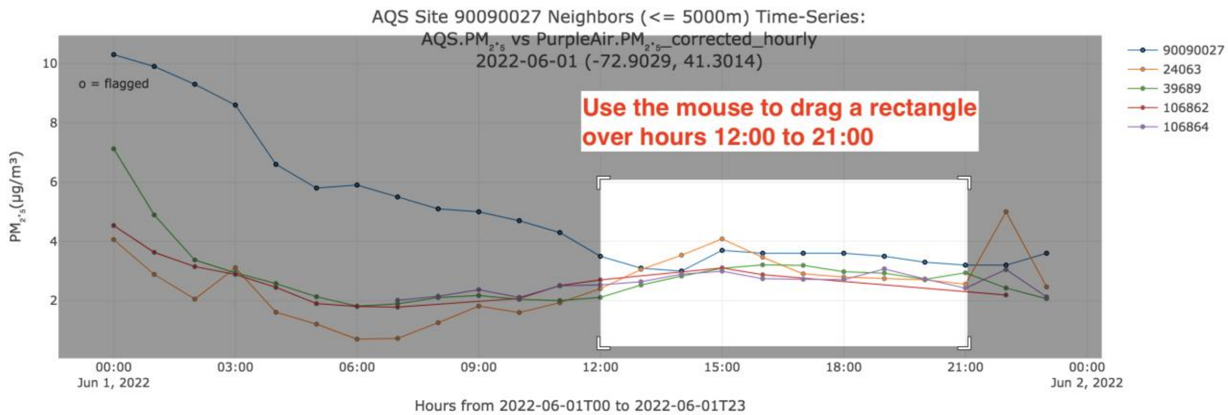
The summary boxplot now has blue boxes, fancy fonts (with sub/superscripts and symbols), and hovering over a box pops-up ID and statistics for that plot element. Hovering over the top right corner of an interactive plot reveals a plot editing menu.



Explore these plot editing features with the Time-series plot.

5.1 Zoom:

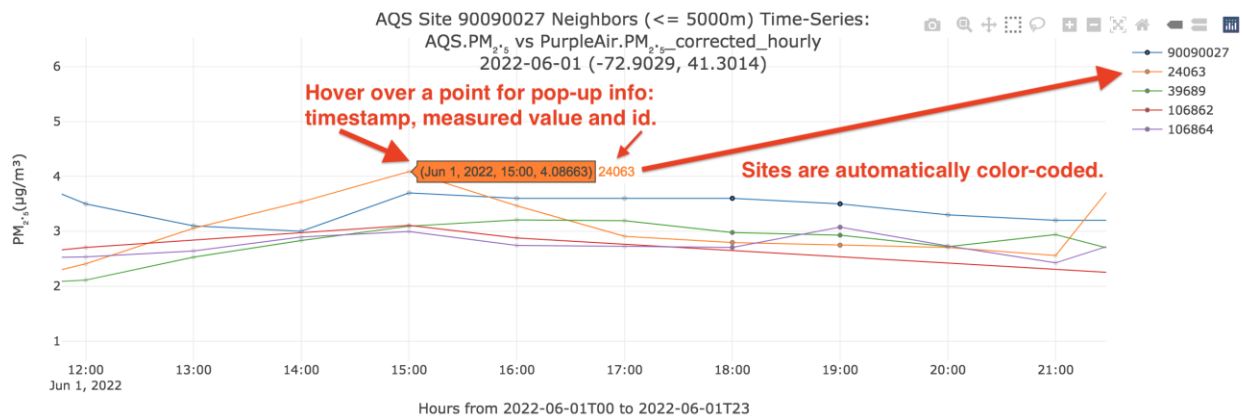
The first feature is zooming.
Use the mouse to drag a rectangular area on the plot from hours 12:00 to 21:00.



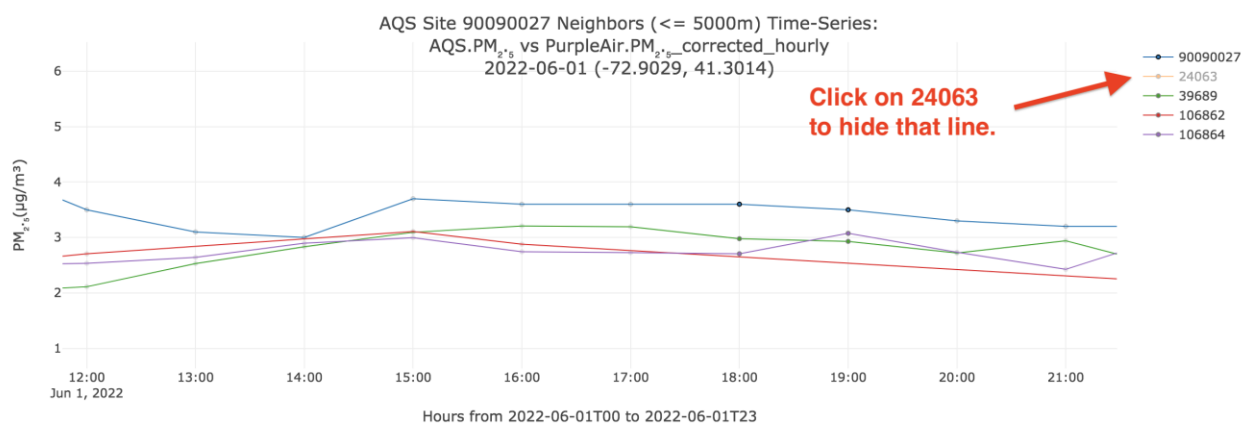
Releasing the mouse button causes the plot to be zoomed in to show just these hours.
In a dense plot, zooming in can help you visualize the data for small areas at a time.

If you mess up, just click the 'home' symbol to reset everything back to the default.

Hovering pops up info about a data point: timestamp, measured value, and ID. Note that each site is automatically assigned a unique color as shown in the legend on the right.

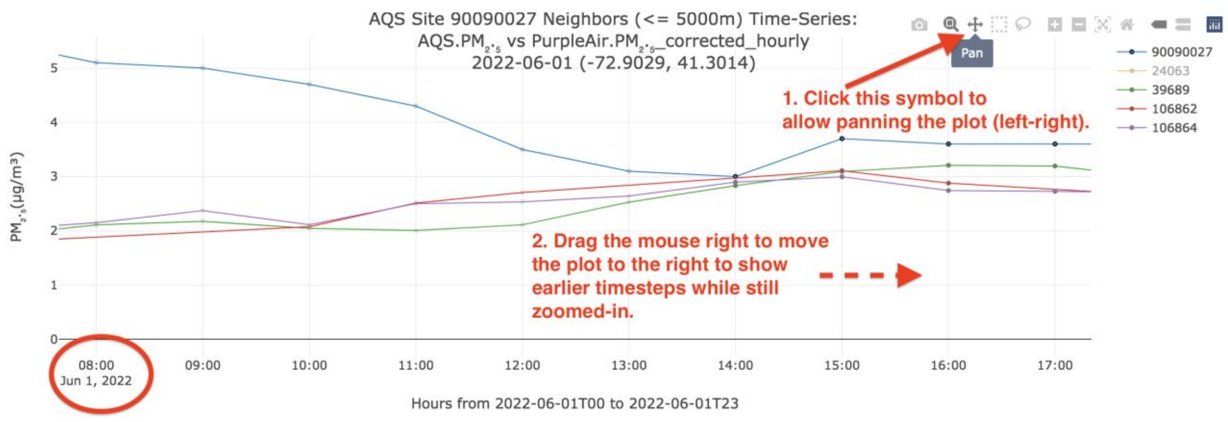


Plot elements, in this case, a time-series line for each site, can be toggled visible or invisible.
Click on the legend for site 24063 to hide the time-series line for that site.



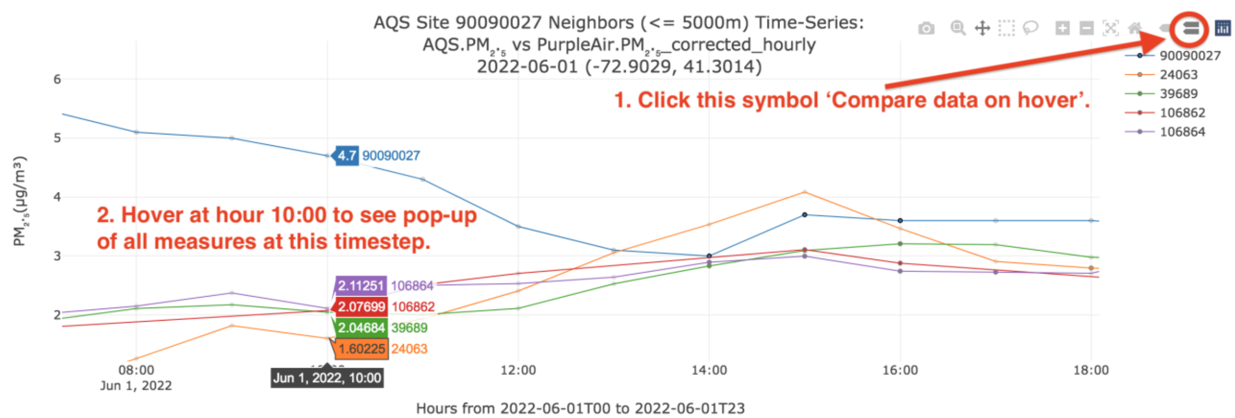
You may click it again to make it re-appear.
Several time-series lines may be toggled at a time.
Click the '+' symbol to allow panning the plot.

Then **drag the mouse to the right** to show earlier timesteps while still zooming in.

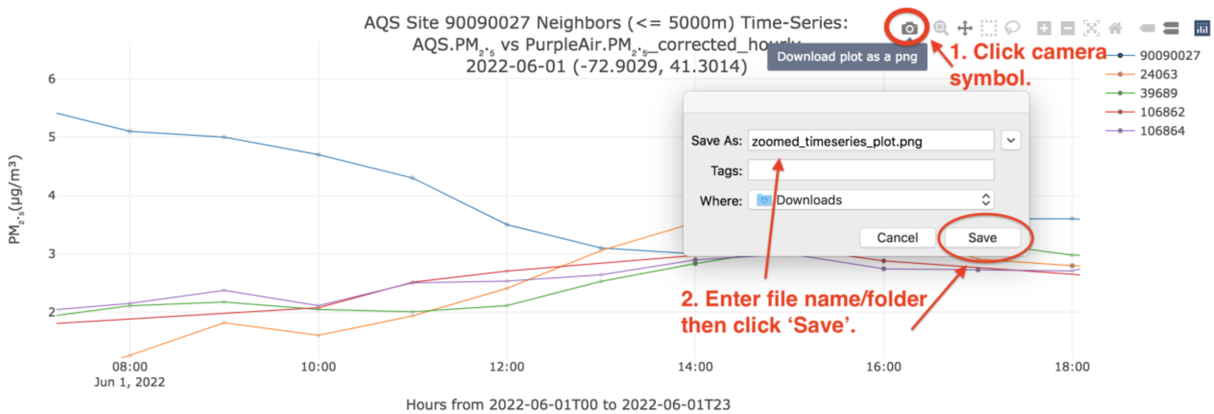


5.2 Compare data on hover:

Click the second to last symbol in the plot edit menu.
Then **hover over hour 10:00** to see a pop-up showing the values of all measurements for that timestep.



Finally, **click the camera symbol** to save this plot as a PNG image file.



The last plot - a scatterplot - is similarly interactive but the time-series plot is most appropriate for interaction.

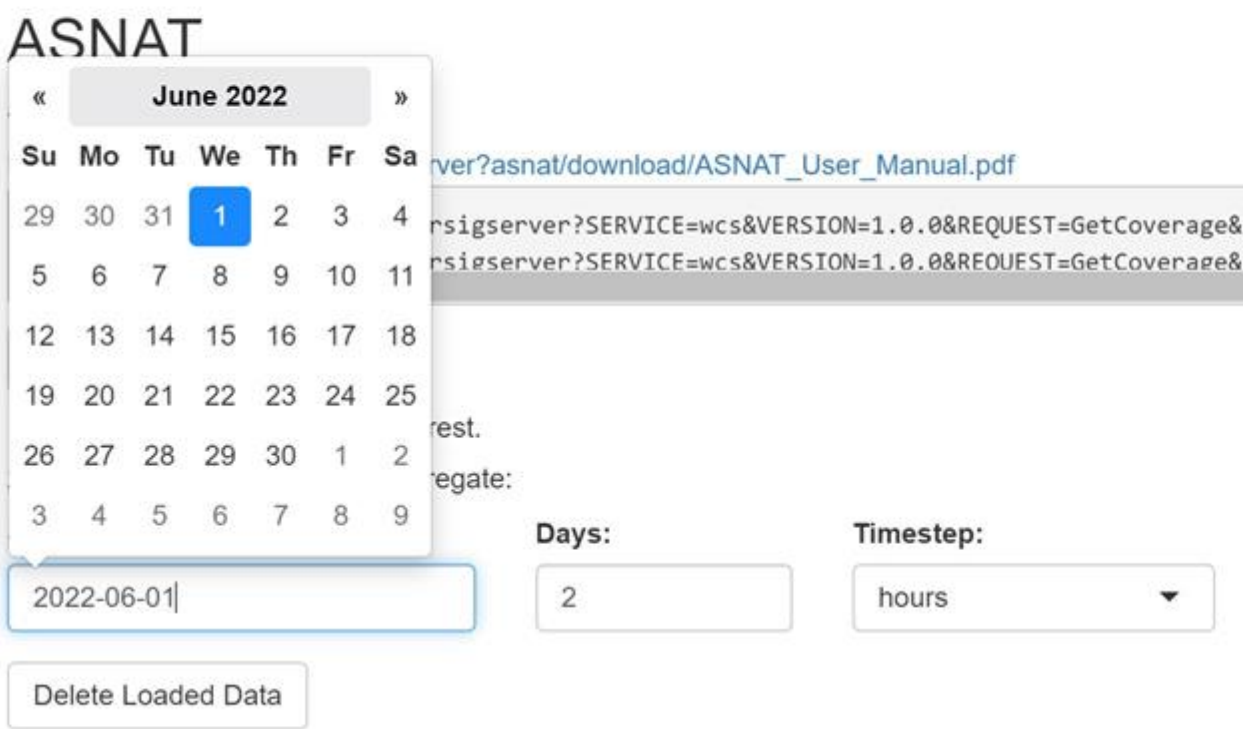
Demo 6: Data Correction:

ASNAT allows for data correction between two datasets. The user can select the device ID in the dataset and perform corrections based on the selected models (currently, only linear regression is supported).
The correction results will be displayed in both static and interactive plots, as well as in a table.

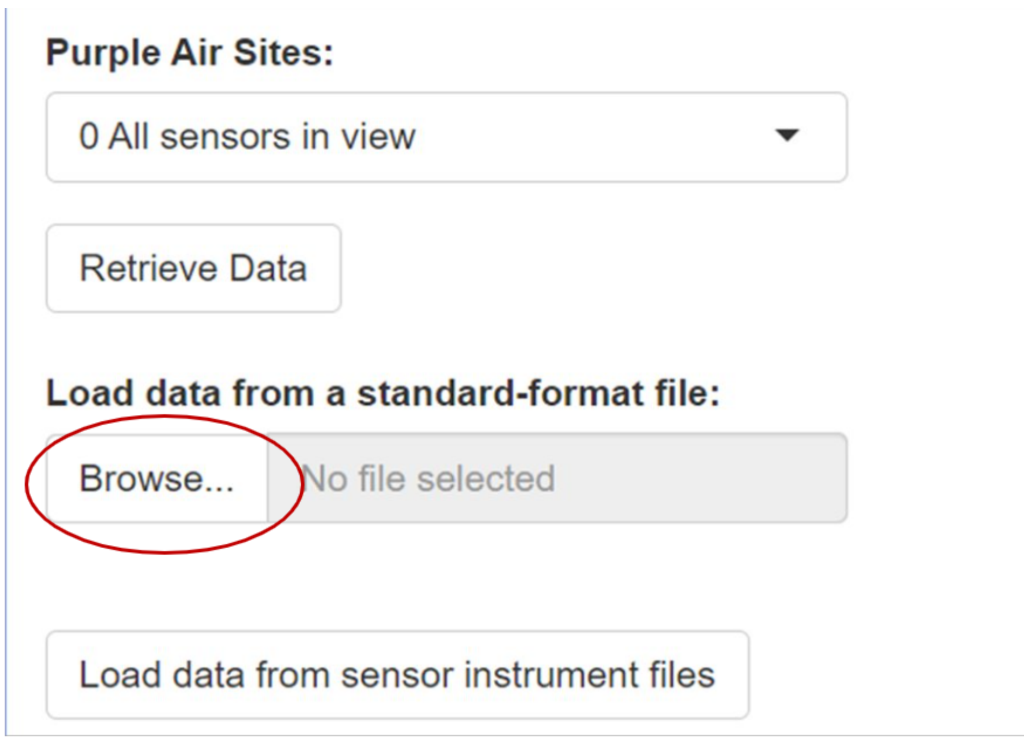
6.1 Relaunch **RStudio** and **ASNAT**.

6.2 Use the mouse to select the **Start Date**, then type **2022-06-01** and press Enter.

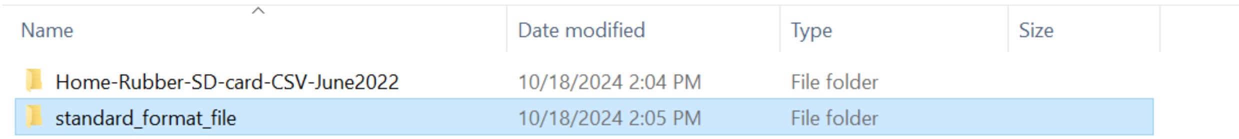
6.3 Use the mouse to select **Days**, then enter **2**.

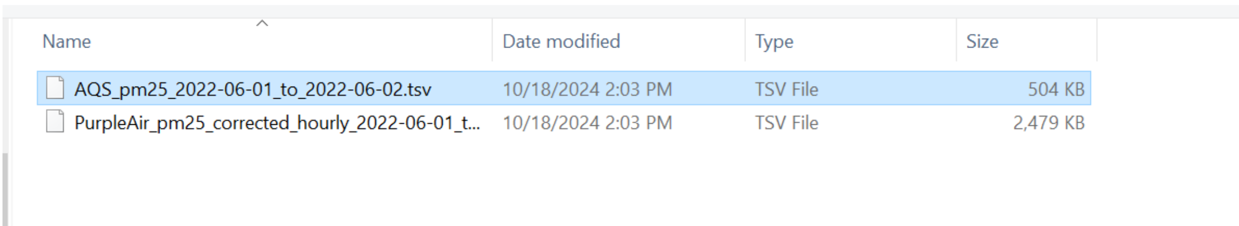


6.4 Click **Browse** under **Load data from a standard format file**.

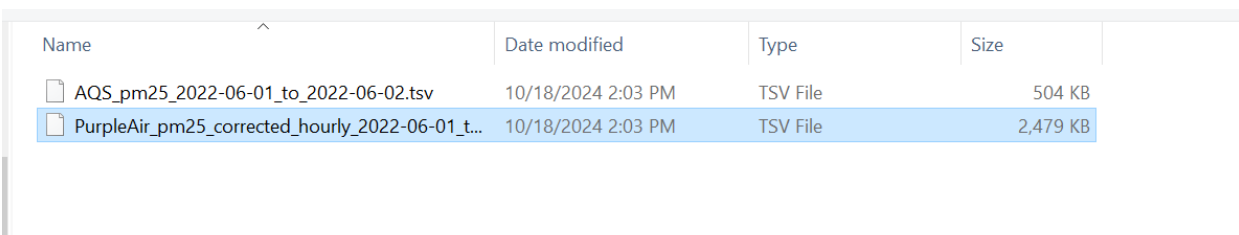


6.5 In the newly opened window, navigate to the \ASNAT\ sample_data\ standard_format_file, and select **AQS_pm25_2022-06-01_to_2022-06-02.tsv**.

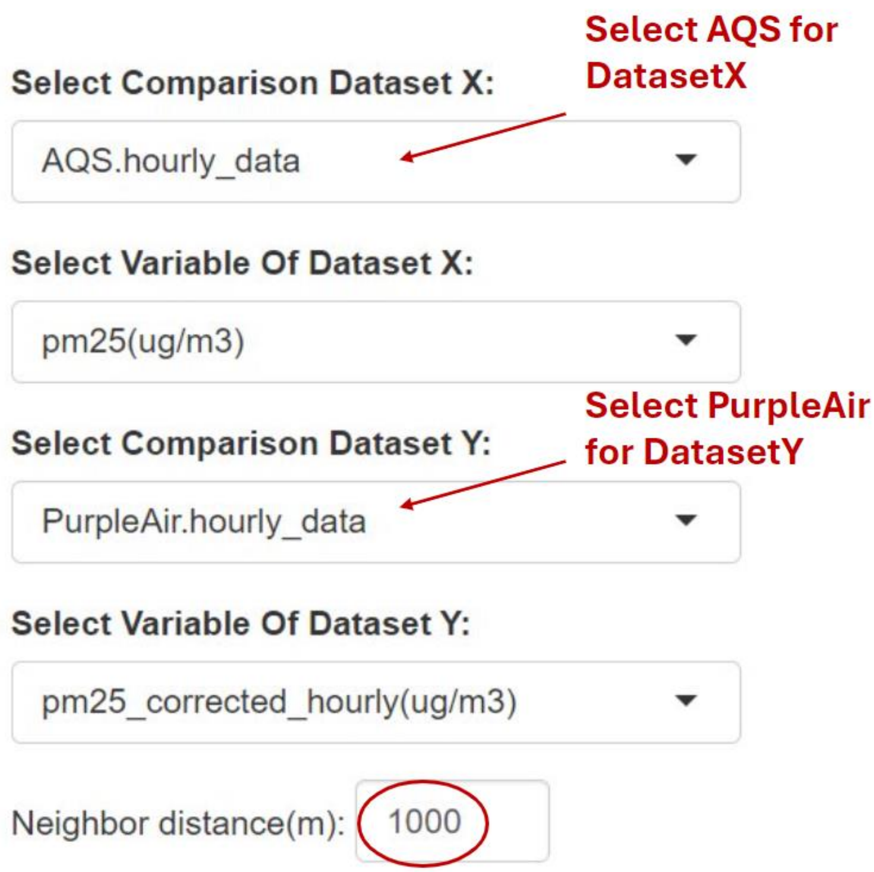




6.6 Repeat the process and select **PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-02.tsv**.



6.7 In the **Select Comparison Dataset Y menu**, select **PurpleAir.hourly_data**, and change the **Neighbor distance(m**) to 1000.

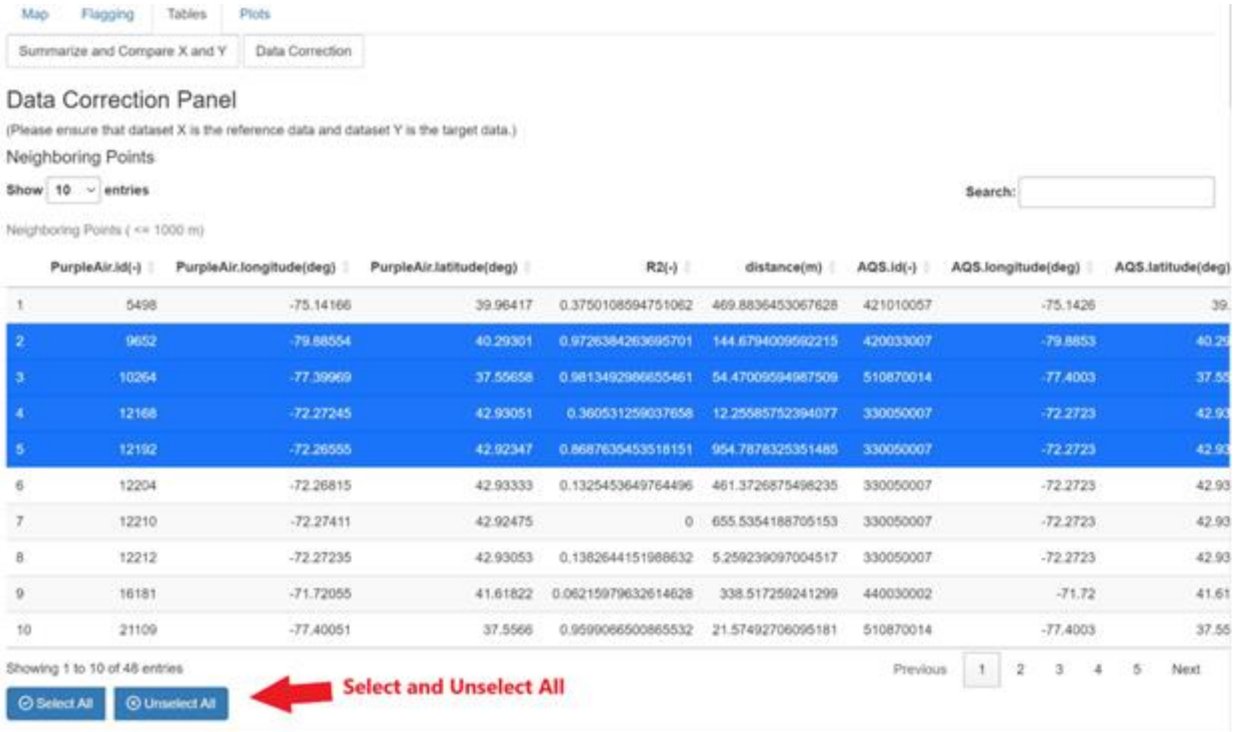


6.8 Click the **Corrections** tab.

6.9 Click the **Regression Correction** tab.

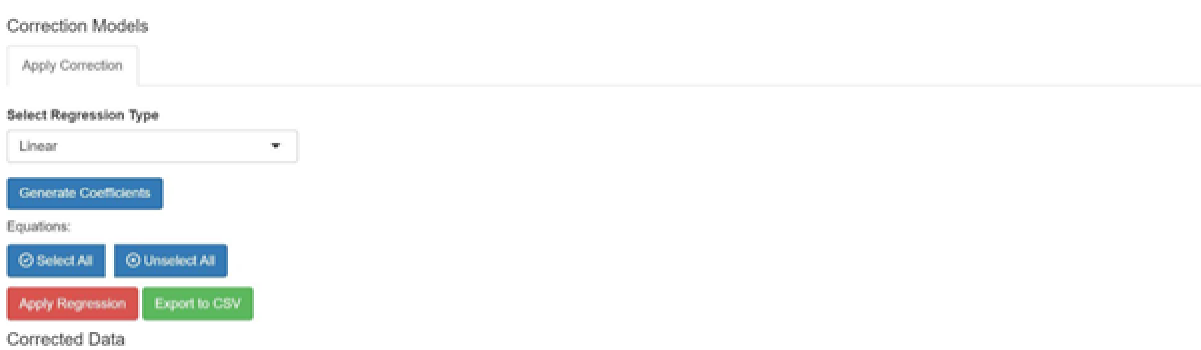
6.10 Use the mouse to click the **Select All** and **Unselect All** buttons to select or unselect all devices.

6.11 Use the mouse to select **rows 2 to 5**.



Scroll down to review the selected devices.

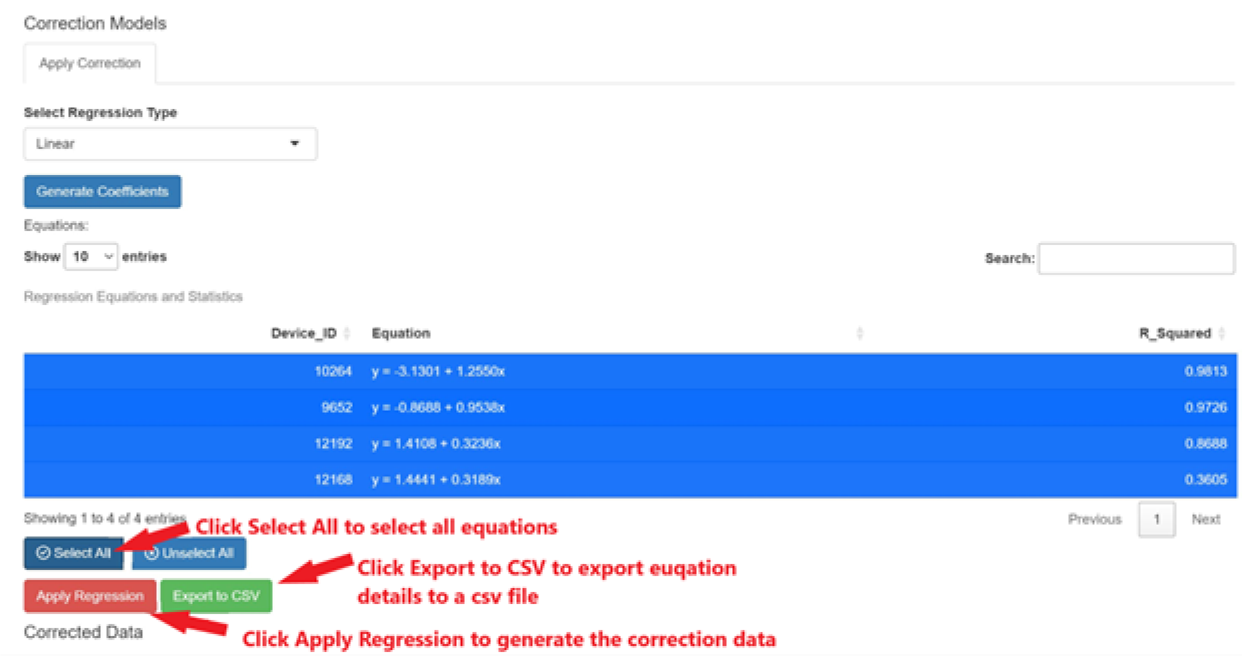
6.12 Select **Linear** from the **Select Regression Type** dropdown menu, then click **Generate Coefficients**.

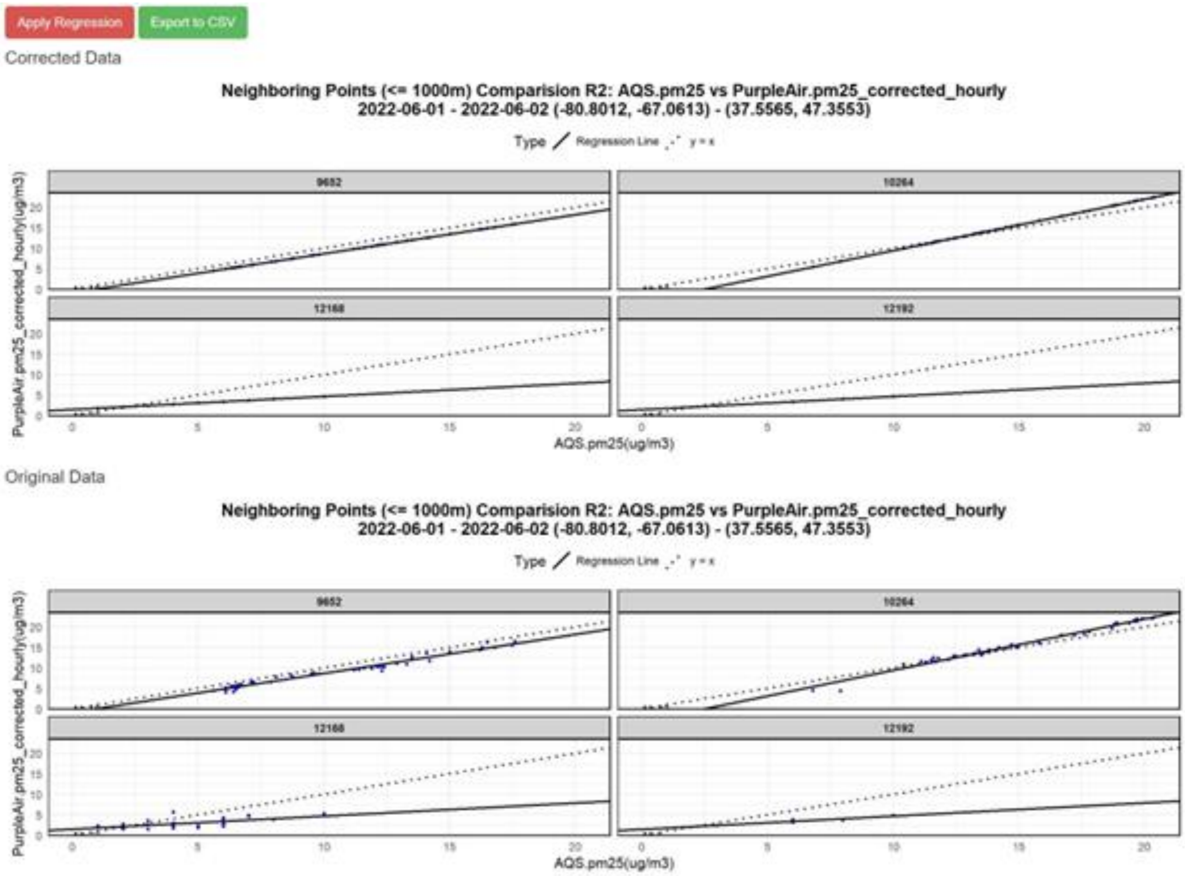


6.13 Click the **Select All** button to select all newly generated equations.

6.14 Click **Export to CSV** to export the details of the selected equations to a CSV file under \ASNAT\data\equation

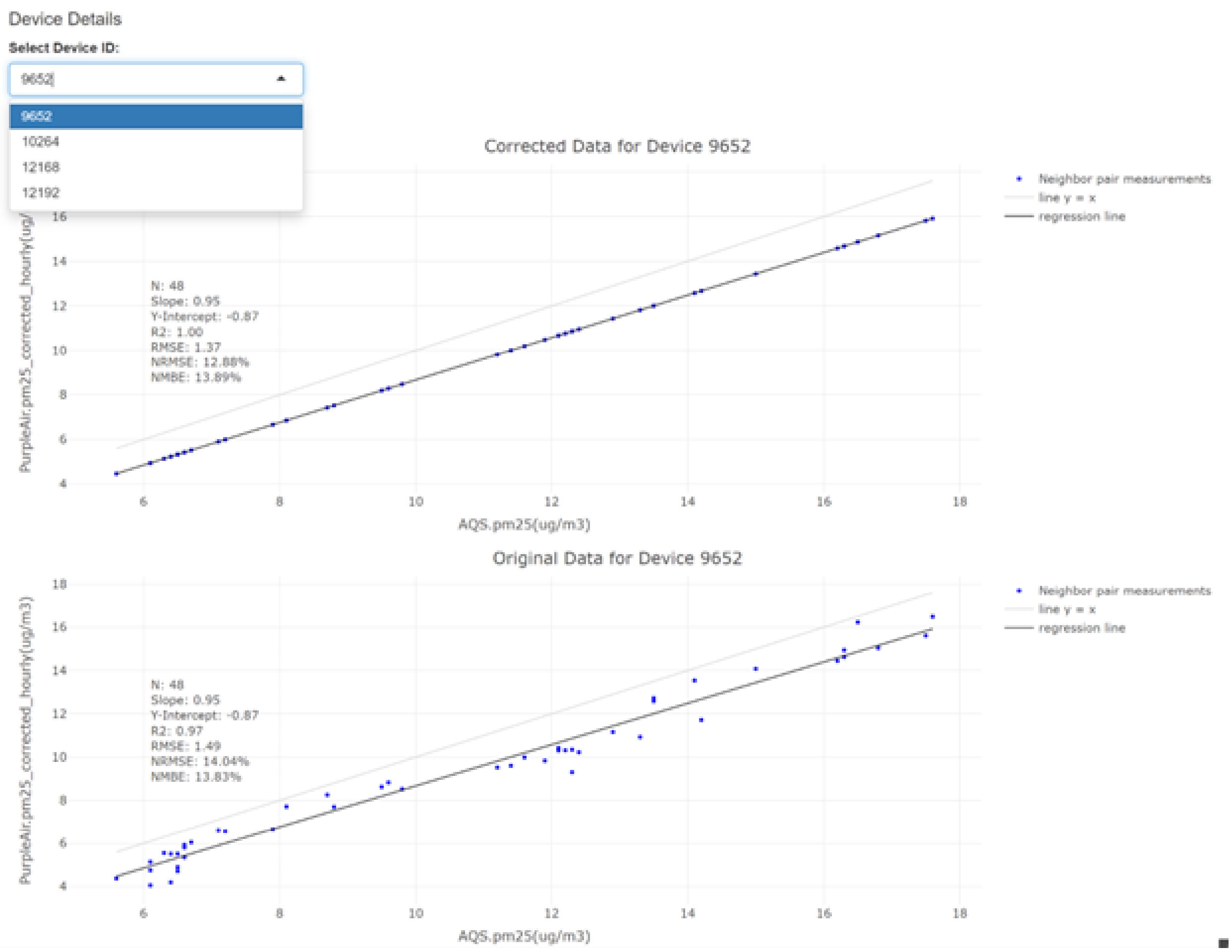
6.15 Click **Apply Regression** to generate correction plots and a table.



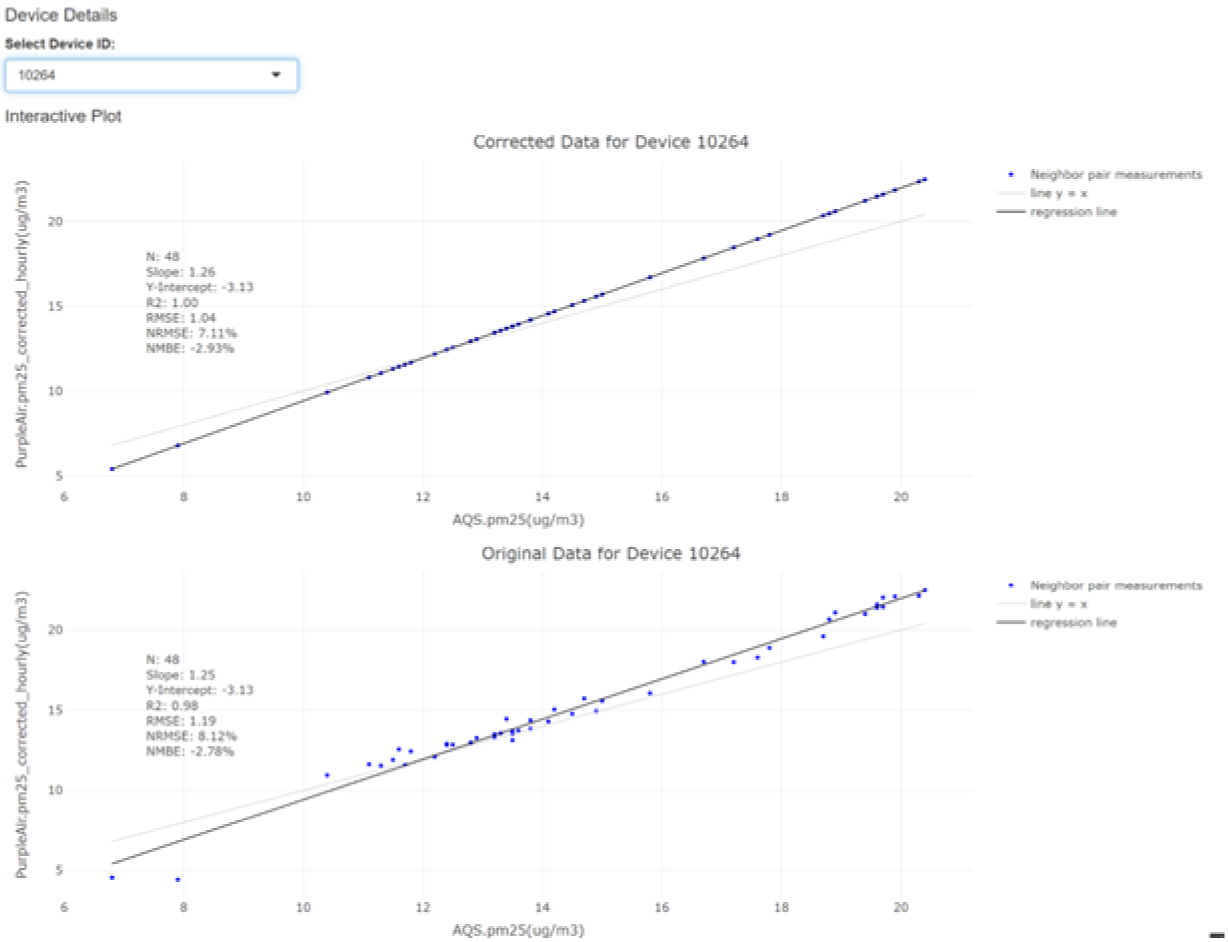


6.16 Scroll down to **Device Details**, and click the **Select Device ID** dropdown menu to choose a device ID to view detailed interactive plots and the data table.

Static plots (before and after correction) will be generated for each device, while interactive plots and the details table will only be displayed for the selected device in the **Select Device ID** menu.



Device Details
Select Device ID:
9652
9652
10264
12168
12192
Corrected Data for Device 9652
Neighbor pair measurements
line y = x
regression line
N: 48
Slope: 0.95
Y-Intercept: -0.87
R2: 1.00
RMSE: 1.37
NRMSE: 12.88%
NMBE: 13.89%
PurpleAir.pm25_corrected_hourly(ug/
AQS.pm25(ug/m3)
Original Data for Device 9652
Neighbor pair measurements
line y = x
regression line
N: 48
Slope: 0.95
Y-Intercept: -0.87
R2: 0.97
RMSE: 1.49
NRMSE: 14.04%
NMBE: 13.83%
PurpleAir.pm25_corrected_hourly(ug/m3)
AQS.pm25(ug/m3)


Device Details
Select Device ID:
10264
Interactive Plot
Corrected Data for Device 10264
N: 48
Slope: 1.26
Y-Intercept: -3.13
R2: 1.00
RMSE: 1.04
NRMSE: 7.11%
NMBE: -2.93%
Neighbor pair measurements
line y = x
regression line
PurpleAir.pm25_corrected_hourly(ug/m3)
AQS.pm25(ug/m3)
Original Data for Device 10264
N: 48
Slope: 1.25
Y-Intercept: -3.13
R2: 0.98
RMSE: 1.19
NRMSE: 8.12%
NMBE: -2.78%
Neighbor pair measurements
line y = x
regression line
PurpleAir.pm25_corrected_hourly(ug/m3)
AQS.pm25(ug/m3)

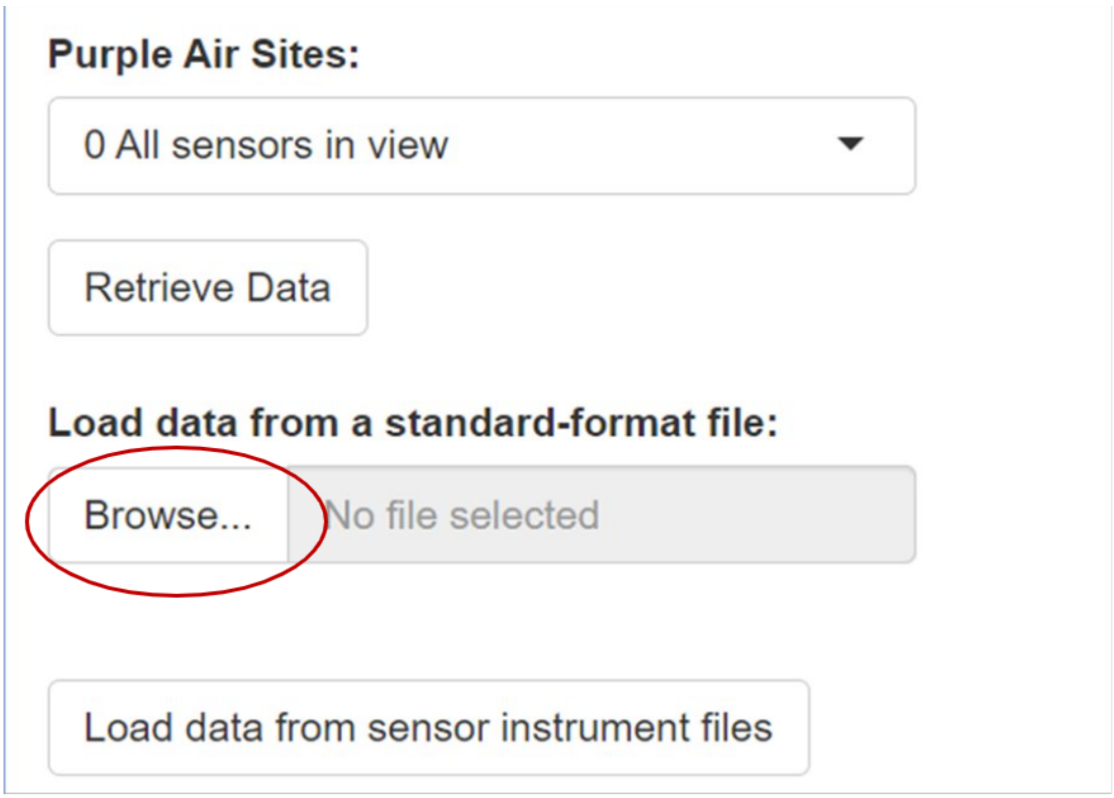
timestamp(UTC)	AQS.id(-)	AQS.pm25(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly(ug/m3)	PurpleAir.flagged(-)	PurpleAir.pm25_corrected_hourly(ug/m3)_corrected
2022-06-01T00:00:00-0000	420033007	11.40	9652	9.60	0	10.00
2022-06-01T01:00:00-0000	420033007	12.30	9652	10.35	0	10.86
2022-06-01T02:00:00-0000	420033007	12.90	9652	11.16	0	11.44
2022-06-01T03:00:00-0000	420033007	13.50	9652	12.58	0	12.01
2022-06-01T04:00:00-0000	420033007	13.50	9652	12.73	0	12.01
2022-06-01T05:00:00-0000	420033007	14.10	9652	13.54	0	12.58
2022-06-01T06:00:00-0000	420033007	15.00	9652	14.08	0	13.44
2022-06-01T07:00:00-0000	420033007	16.30	9652	14.63	0	14.68
2022-06-01T08:00:00-0000	420033007	16.20	9652	14.44	0	14.58
2022-06-01T09:00:00-0000	420033007	16.30	9652	14.94	0	14.68
2022-06-01T10:00:00-0000	420033007	16.80	9652	15.04	0	15.16
2022-06-01T11:00:00-	420033007	17.60	9652	16.49	0	15.92

timestamp(UTC)	AQS.id(-)	AQS.pm25(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly(ug/m3)	PurpleAir.flagged(-)	PurpleAir.pm25_corrected_hourly(ug/m3)_corrected
2022-06-01T00:00:00-0000	510870014	17.60	10264	18.29	0	18.96
2022-06-01T01:00:00-0000	510870014	18.70	10264	19.60	0	20.34
2022-06-01T02:00:00-0000	510870014	19.60	10264	21.38	0	21.47
2022-06-01T03:00:00-0000	510870014	19.60	10264	21.59	0	21.47
2022-06-01T04:00:00-0000	510870014	19.70	10264	21.43	0	21.59
2022-06-01T05:00:00-0000	510870014	20.30	10264	22.14	0	22.35
2022-06-01T06:00:00-0000	510870014	20.40	10264	22.49	0	22.47
2022-06-01T07:00:00-0000	510870014	19.90	10264	22.10	0	21.84
2022-06-01T08:00:00-0000	510870014	19.70	10264	22.03	0	21.59
2022-06-01T09:00:00-0000	510870014	19.40	10264	21.00	0	21.22
2022-06-01T10:00:00-0000	510870014	18.90	10264	21.09	0	20.59
2022-06-01T11:00:00-	510870014	18.80	10264	20.67	0	20.46

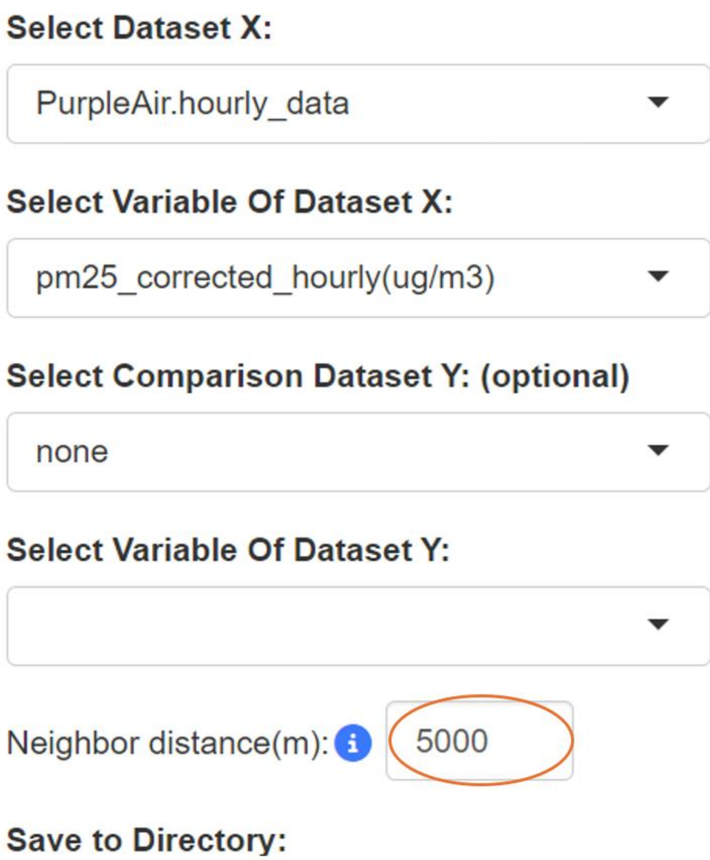
Demo 7: Additional Flagging:

7.1 Rerun `ASNAT`:
`ASNAT`

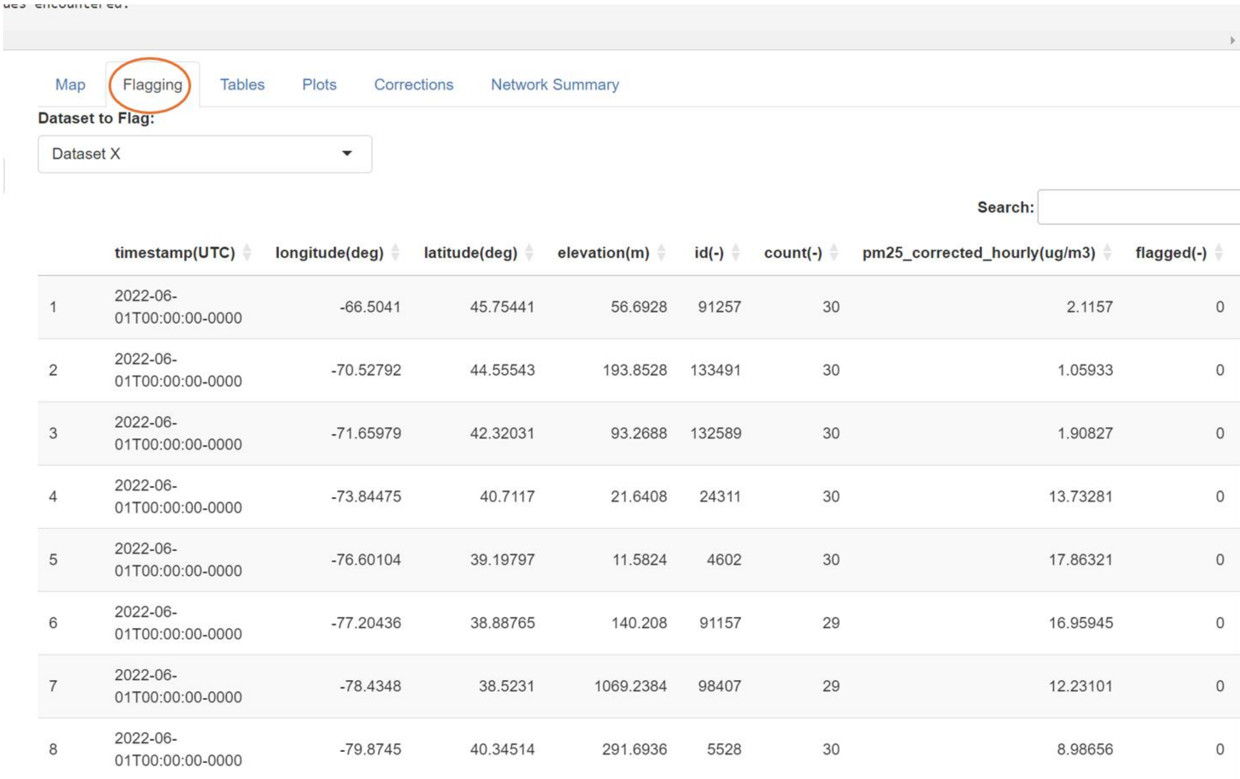
7.2 Load the sample data PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-02.tsv through "**Load data from a standard-format file**" in \ASNAT\ sample_data\ standard_format_file.



Scroll down to the **Dataset Y Neighbor Flagging** section and change the **Neighbor distance(m)** to 5000.

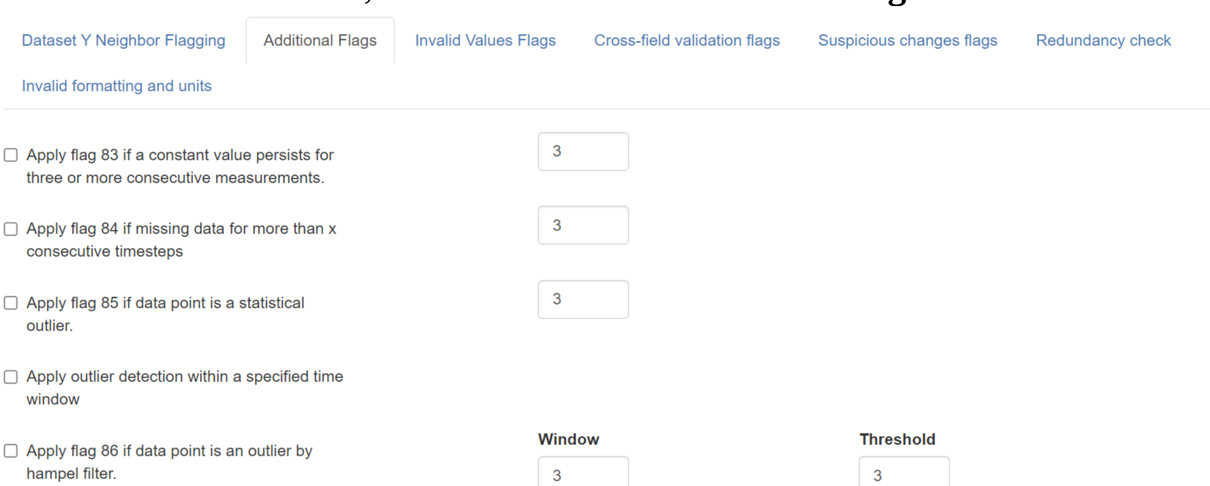


7.3 Use the mouse to click the **Flagging** tab.

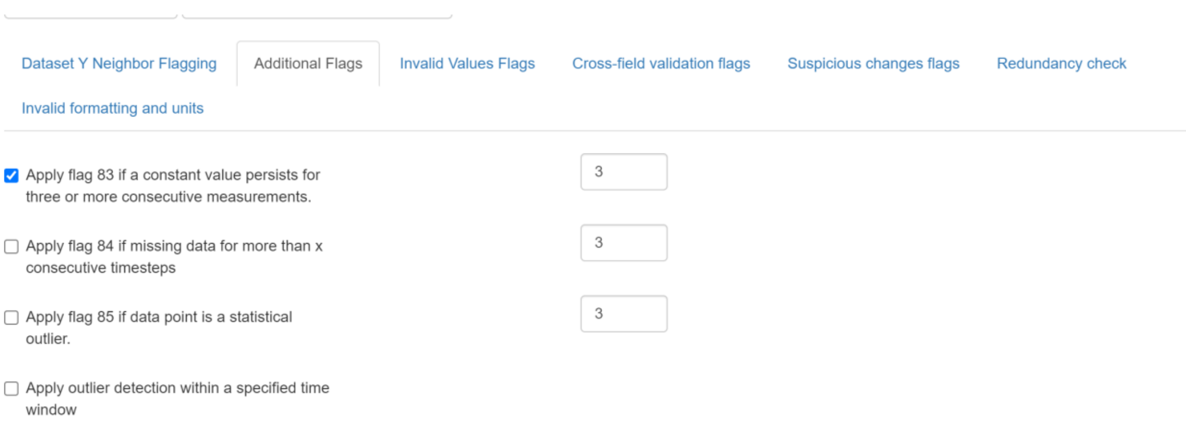


	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)
1	2022-06-01T00:00:00-0000	-66.5041	45.75441	56.6928	91257	30	2.1157	0
2	2022-06-01T00:00:00-0000	-70.52792	44.55543	193.8528	133491	30	1.05933	0
3	2022-06-01T00:00:00-0000	-71.65979	42.32031	93.2688	132589	30	1.90827	0
4	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0
5	2022-06-01T00:00:00-0000	-76.60104	39.19797	11.5824	4602	30	17.86321	0
6	2022-06-01T00:00:00-0000	-77.20436	38.88765	140.208	91157	29	16.95945	0
7	2022-06-01T00:00:00-0000	-78.4348	38.5231	1069.2384	98407	29	12.23101	0
8	2022-06-01T00:00:00-0000	-79.8745	40.34514	291.6936	5528	30	8.98656	0

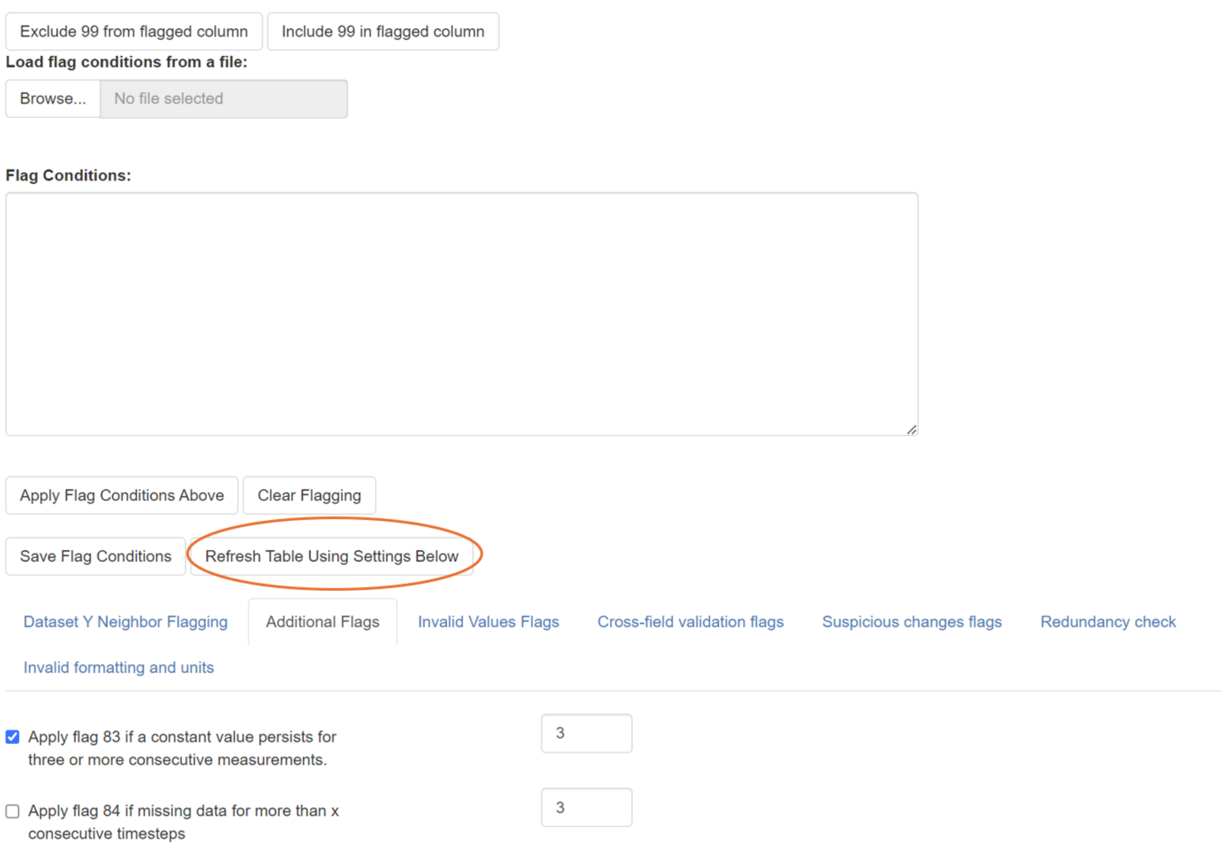
Scroll down to the bottom, and then click the **Additional Flags** tab.



7.4 Click **Apply Flag 83** to enable the flag, which checks if a constant value persists for three or more consecutive measurements.



7.5 Scroll up and click the **Refresh Table Using Settings Below** button to apply the preset flag conditions listed below to the flag table.



7.6 Scroll up and review the new flags in the table.

Show 10 entries

Search:

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-66.5041	45.75441	56.6928	91257	30	2.1157	83	AQSU-A8D6
2	2022-06-01T00:00:00-0000	-70.52792	44.55543	193.8528	133491	30	1.05933	0	Mountain Valley Middle School
3	2022-06-01T00:00:00-0000	-71.65979	42.32031	93.2688	132589	30	1.90827	0	Melican Middle School
4	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
5	2022-06-01T00:00:00-0000	-76.60104	39.19797	11.5824	4602	30	17.86321	0	ex
6	2022-06-01T00:00:00-0000	-77.20436	38.88765	140.208	91157	29	16.95945	0	Falls Church - Westwood Park
7	2022-06-01T00:00:00-0000	-78.4348	38.5231	1069.2384	98407	29	12.23101	0	SHEN- Big Meadows
8	2022-06-01T00:00:00-0000	-79.8745	40.34514	291.6936	5528	30	8.98656	0	Port Vue PA
9	2022-06-01T00:00:00-0000	-80.0476	40.30428	358.7496	9036	30	9.02467	0	ROCIS_h5w7_O
10	2022-06-01T00:00:00-0000	-80.30667	40.69105	240.1824	94491	30	8.58554	0	CMP11 - Beaver

Showing 1 to 10 of 28,674 entries

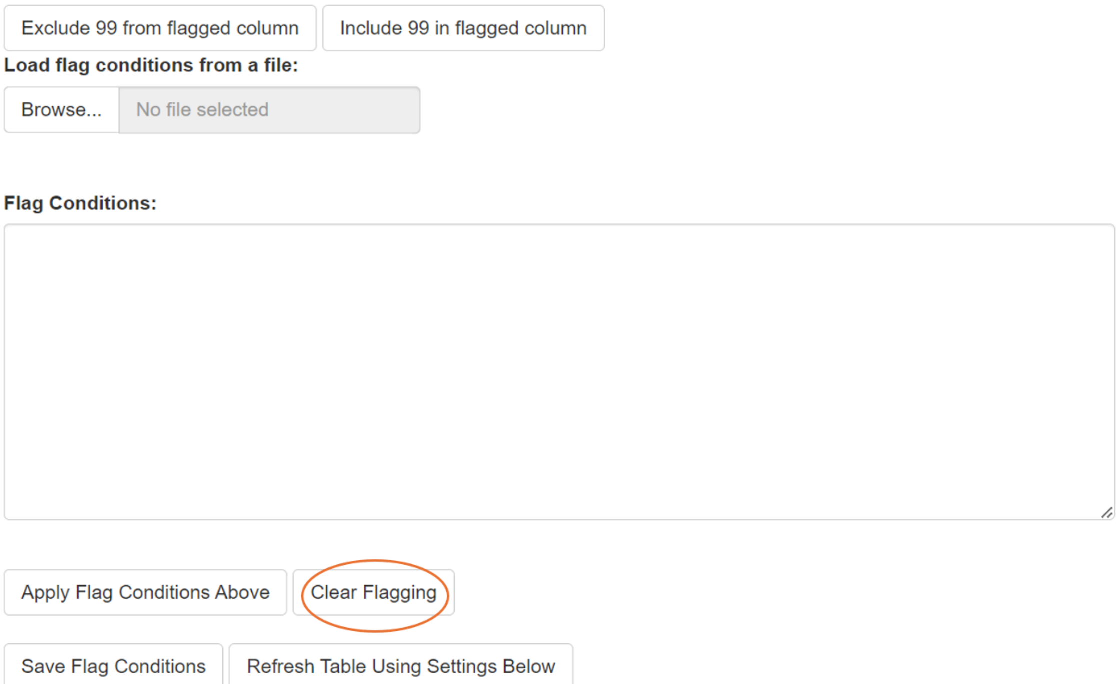
Previous 1 2 3 4 5 … 2,868 Next

Exclude 99 from flagged column | Include 99 in flagged column

Load flag conditions from a file:

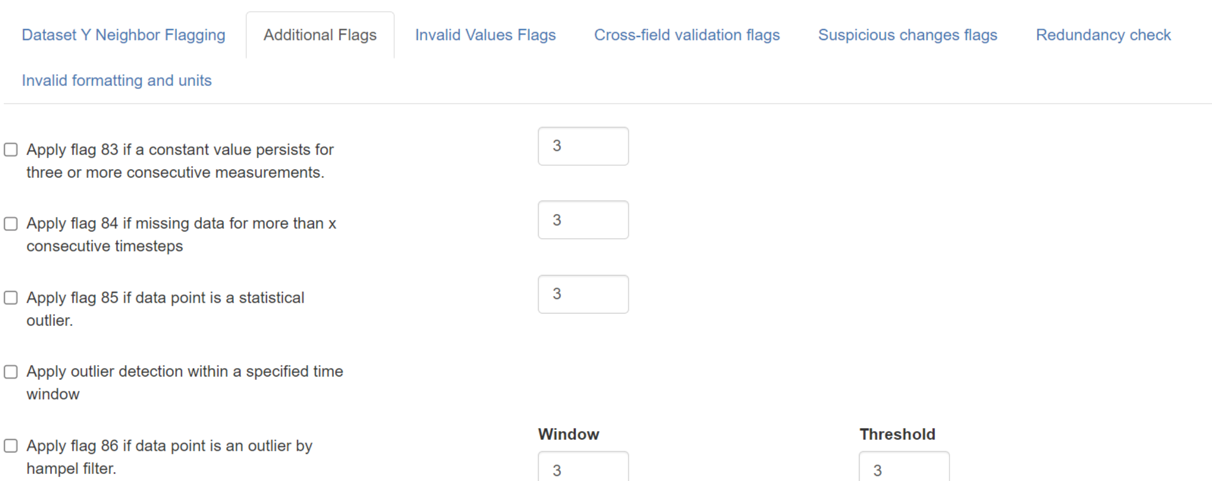
Browse... No file selected

7.7 Scroll down to Flag Conditions and click the **Clear Flagging** button to reset the flags in the table.

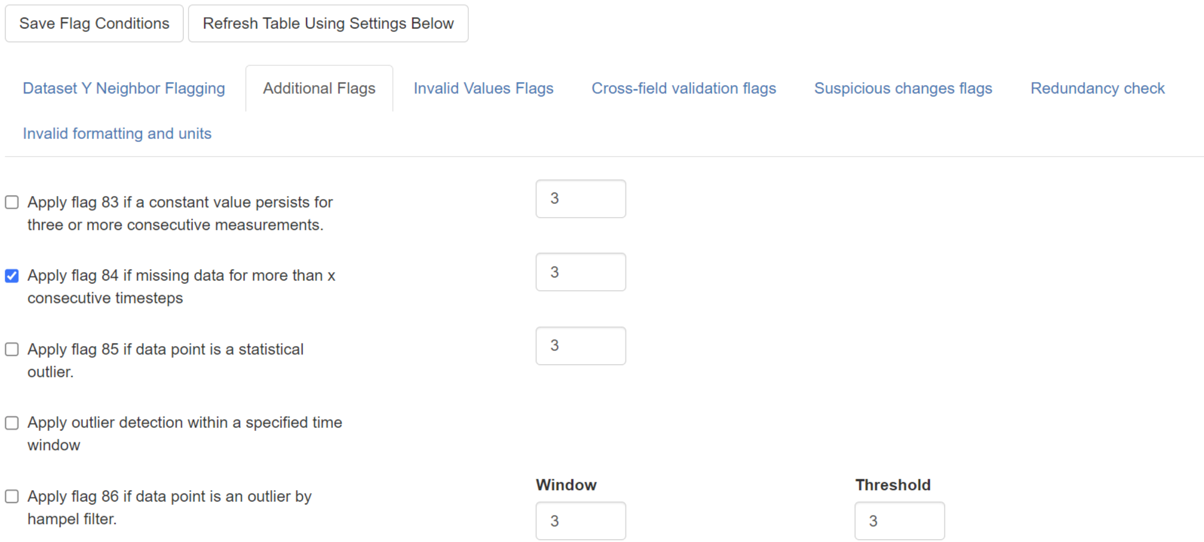


7.8 Scroll down to the **Dataset Y Neighbor Flagging** section.

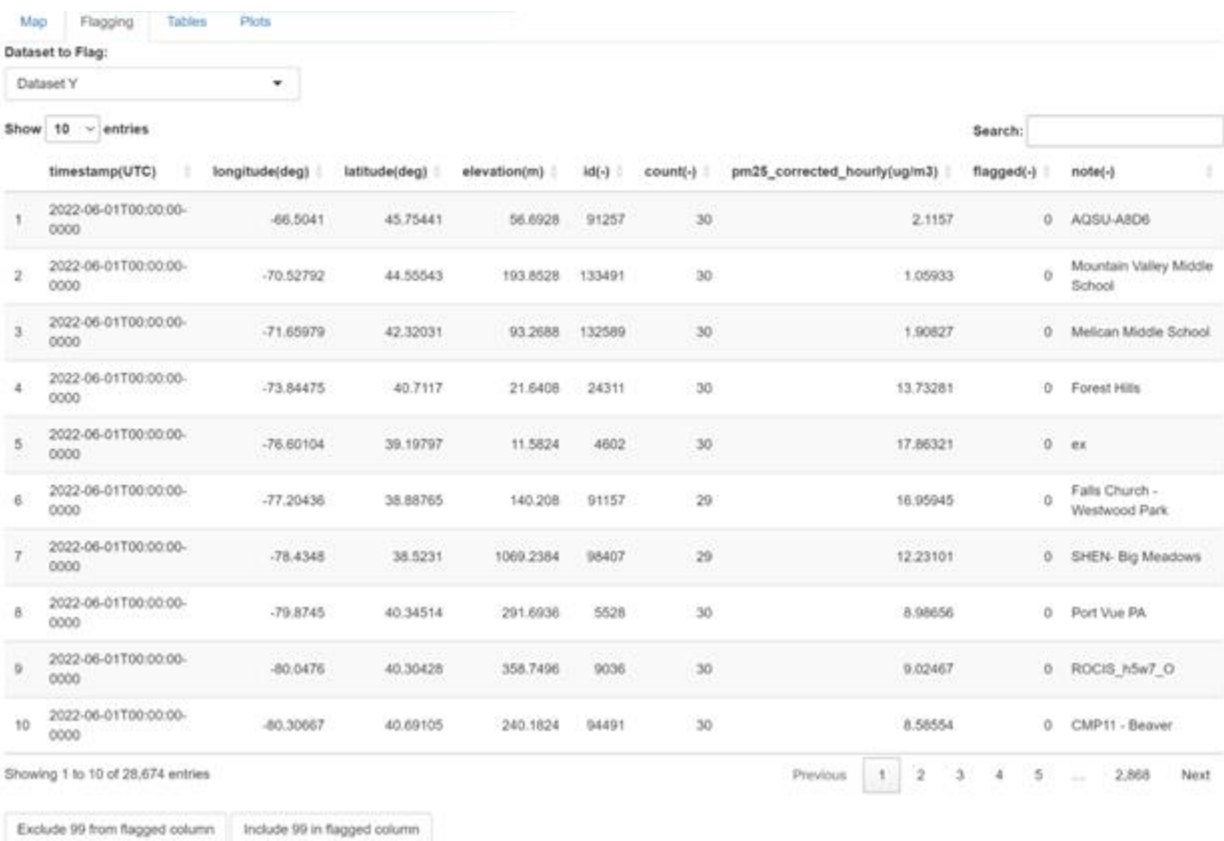
7.9 Click **Apply Flag 83** again to disable the flag.



7.10 Click **Apply Flag 84** to enable the, which checks if a missing value persists for x or more consecutive measurements. The value for **x** can be adjusted by entering a new number in the text input field.



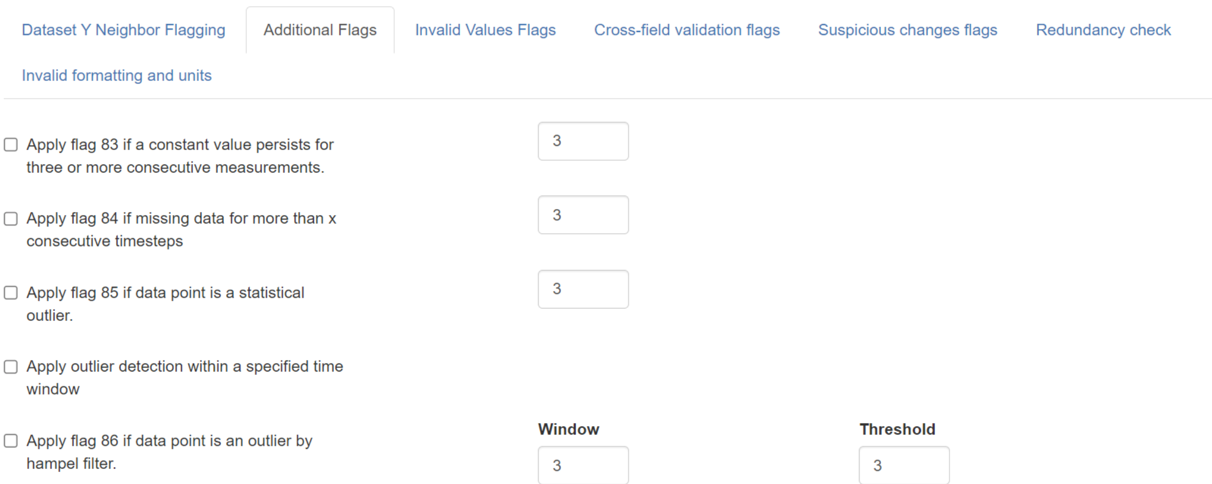
7.11 Scroll up and click the **Refresh Table Using Settings Below** button. Review the new flags in the table. (Since ASNAT currently does not load missing data into the dataset, no changes will be displayed.)



	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-66.5041	45.75441	56.6928	91257	30	2.1157	0	AQSU-A8D6
2	2022-06-01T00:00:00-0000	-70.52792	44.55543	193.8528	133491	30	1.05933	0	Mountain Valley Middle School
3	2022-06-01T00:00:00-0000	-71.65979	42.32031	93.2688	132589	30	1.90827	0	Melican Middle School
4	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
5	2022-06-01T00:00:00-0000	-76.60104	39.19797	11.5824	4602	30	17.86321	0	ex
6	2022-06-01T00:00:00-0000	-77.20436	38.88765	140.208	91157	29	16.95945	0	Falls Church - Westwood Park
7	2022-06-01T00:00:00-0000	-78.4348	38.5231	1069.2384	98407	29	12.23101	0	SHEN- Big Meadows
8	2022-06-01T00:00:00-0000	-79.8745	40.34514	291.6936	5528	30	8.98656	0	Port Vue PA
9	2022-06-01T00:00:00-0000	-80.0476	40.30428	358.7496	9036	30	9.02467	0	ROCIS_h5w7_O
10	2022-06-01T00:00:00-0000	-80.30667	40.69105	240.1824	94491	30	8.58554	0	CMP11 - Beaver

7.12 Scroll down to the **Dataset Y Neighbor Flagging** section.

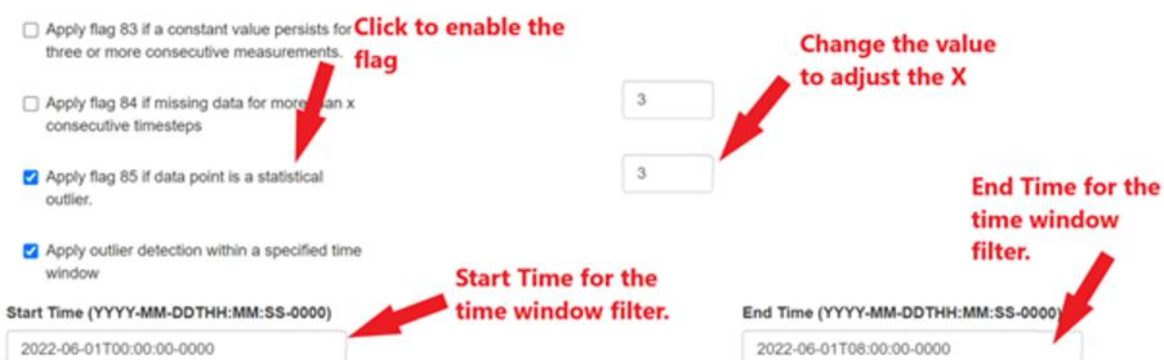
7.13 Click **Apply Flag 84** again to disable the flag.



7.14 Click **Apply Flag 85** to enable the flag, which identifies data as a statistical outlier if the standard deviation is a factor of **X** greater than the mean.

7.15 Change the **corresponding** value **X** to 1.

7.16 Click **Apply Outlier Detection within a Specified Time Window** to enable the time window filter. In the **Start Time** field (YYYY-MM-DDTHH:MM), type **2022-06-01T00:00:00-0000**, and in the **End Time** field (YYYY-MM-DDTHH:MM), type **2022-06-01T08:00:00-0000**.



- [x] Apply flag 85 if data point is a statistical outlier.

1

- [x] Apply outlier detection within a specified time window

Start Time (YYYY-MM-DDTHH:MM:SS-0000)

2022-06-01T00:00:00-0000

End Time (YYYY-MM-DDTHH:MM:SS-0000)

2022-06-01T08:00:00-0000

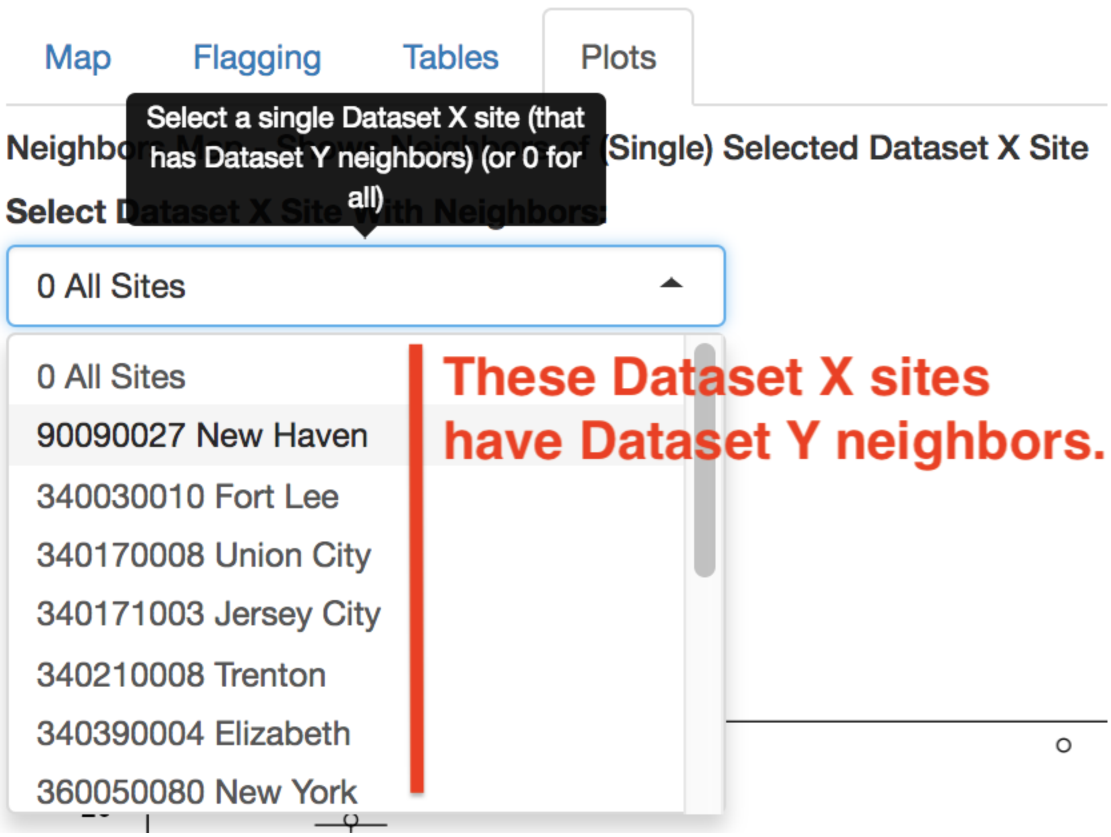
7.17 Scroll up and click the **Refresh Table Using Settings Below** button to apply **Flag 85**.

7.18 Click the **Flagged (-)** column to sort the table and display 85 flagged entries.

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
564	2022-06-01T00:01:12-0000	-71.10436	42.38097	6.096	3642	25	-1.63504	85	Dane St & Village St.
1807	2022-06-01T03:00:08-0000	-72.2593	42.93803	241.4016	1224	23	0.95097	85	KSC 17
2337	2022-06-01T03:03:14-0000	-71.10436	42.38097	6.096	3642	26	-1.91622	85	Dane St & Village St.
2499	2022-06-01T04:00:09-0000	-72.2593	42.93803	241.4016	1224	30	0.32927	85	KSC 17
3000	2022-06-01T04:01:15-0000	-71.10436	42.38097	6.096	3642	24	-2.06457	85	Dane St & Village St.
3158	2022-06-01T05:00:10-0000	-72.2593	42.93803	241.4016	1224	30	-0.36902	85	KSC 17
3656	2022-06-01T05:01:15-0000	-71.10436	42.38097	6.096	3642	23	-1.9219	85	Dane St & Village St.
3764	2022-06-01T06:00:10-0000	-72.2593	42.93803	241.4016	1224	30	-0.50745	85	KSC 17
4357	2022-06-01T07:00:11-0000	-72.2593	42.93803	241.4016	1224	30	-0.44966	85	KSC 17
4871	2022-06-01T07:03:16-0000	-71.10436	42.38097	6.096	3642	25	-2.03527	85	Dane St & Village St.

Demo 8: Site Neighbor Plots:

ASNAT has three modes of plotting:
8.1 Single entire dataset
8.2 Compared pair of datasets X, Y - i.e., only points that are neighbors.
8.3 Single selected Dataset X site and its Dataset Y neighbors.
This demo is the last mode.
It will plot a single AQS.pm25 site with its Dataset Y neighbors.
Continuing on from Demo 4 on the Plots tab:
Note the menu 'Select Dataset X Site With Neighbors'.
It defaults to '0 - All Sites' which is used in the first two modes.

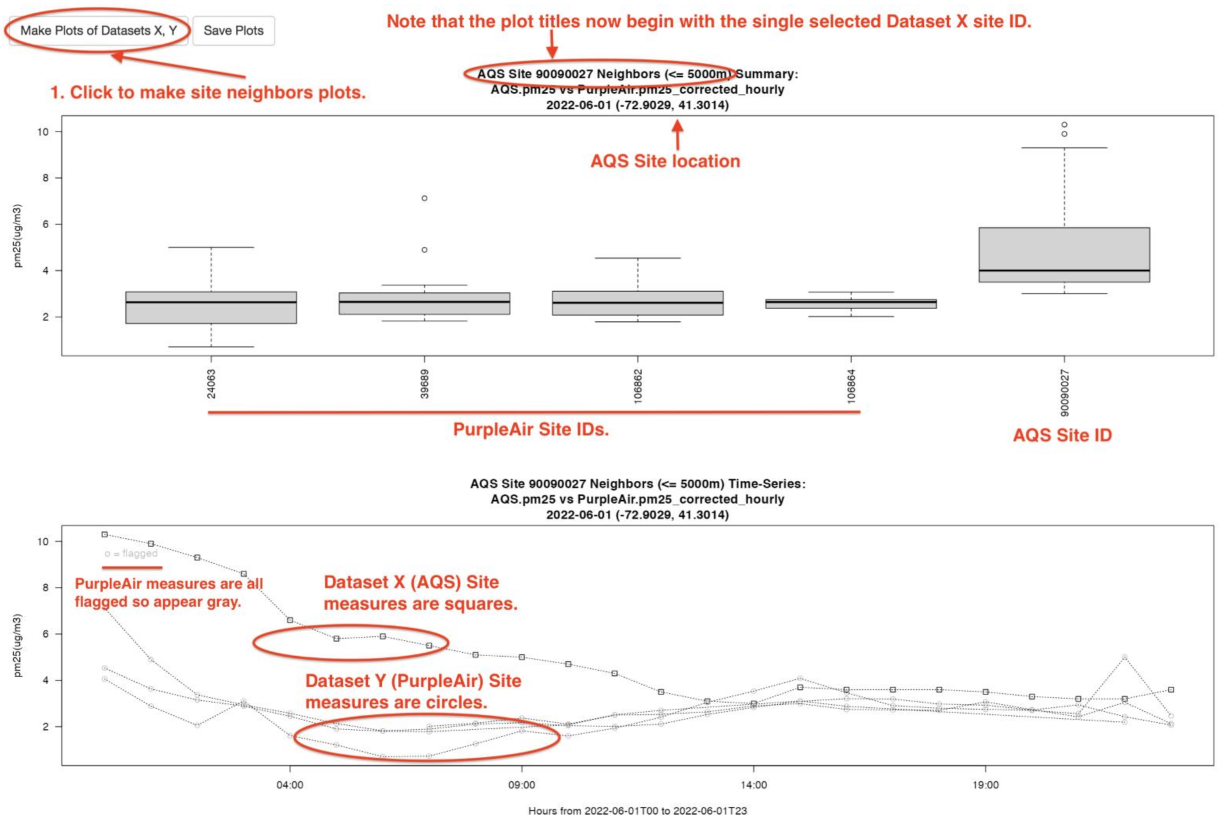


Click the menu to see a list of Dataset X sites that have Dataset Y neighbors.
Click on '90090027 New Haven' to select this AQS site (and its PurpleAir neighbors).

The Neighbors Map shows the location of the single selected Dataset X site and its Dataset Y neighbors. In this example, since the Dataset X is AQS, its site appears as a square glyph on the map. And since Dataset Y is PurpleAir, its sites appear as circles as shown by the map legend.
The 'Show site labels' button adds the site ID to each site displayed.
Clicking on a site pops up a larger label that shows the source, id, and note for that site.

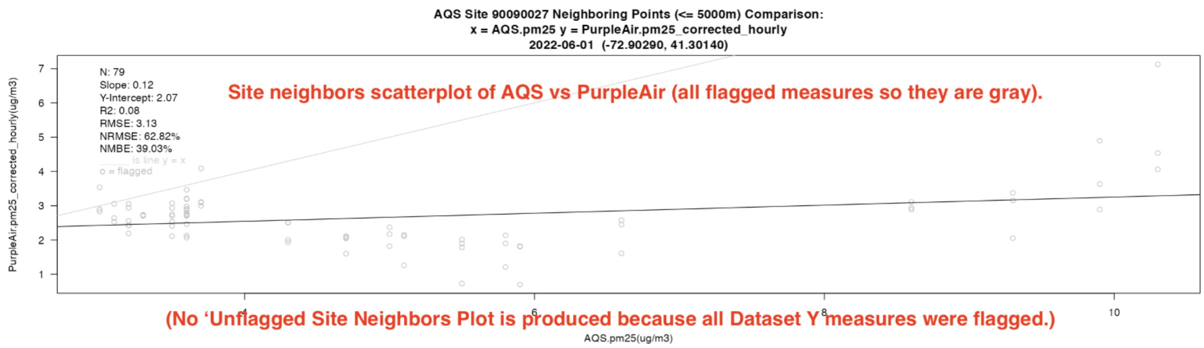
Click the 'Make Plots of Dataset X, Y' button.

This changes the plots from mode 2 to mode 3 - i.e., from Compared Dataset to a single site plus its neighbors.



The Summary boxplot now shows the single selected Dataset X site (AQS 90090027)
along with its four Dataset Y (PurpleAir) neighbors - IDs 24063, 39689, 106882, 106864).
The plot titles now begin with the single selected Dataset X site and show its longitude, latitude location, and maximum neighbor distance.
The second plot is the time series of these five sites.
Note that the Dataset X site measures use square points while the Dataset Y measures use circles.
Also, in this example, due to extensive flagging, all Dataset Y measures were flagged so they
appear gray instead of black.
Because there are no unflagged measures, there is no 'Unflagged' scatterplot.

(No Dataset Y Summary Boxplot or Time-series plots are produced since we are doing Site Neighbor Plots.)

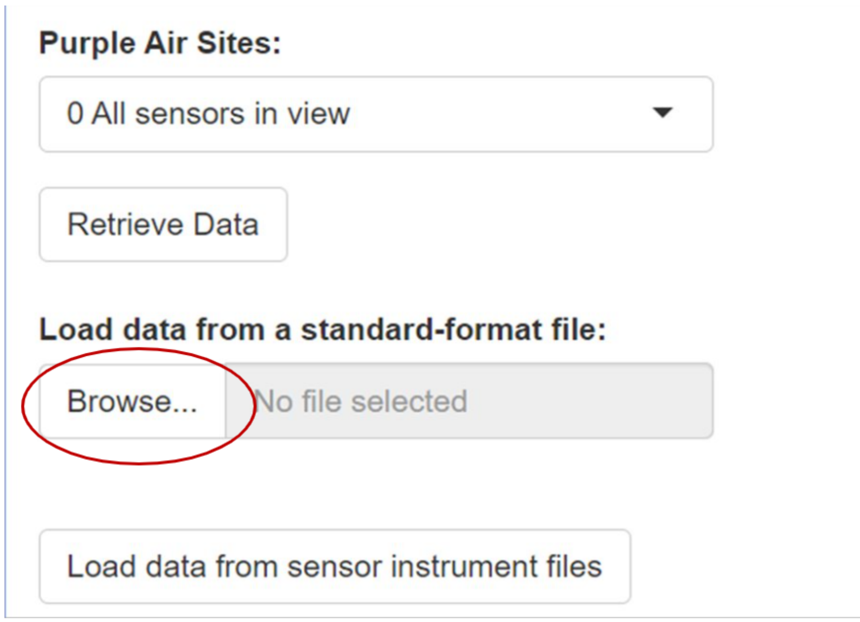


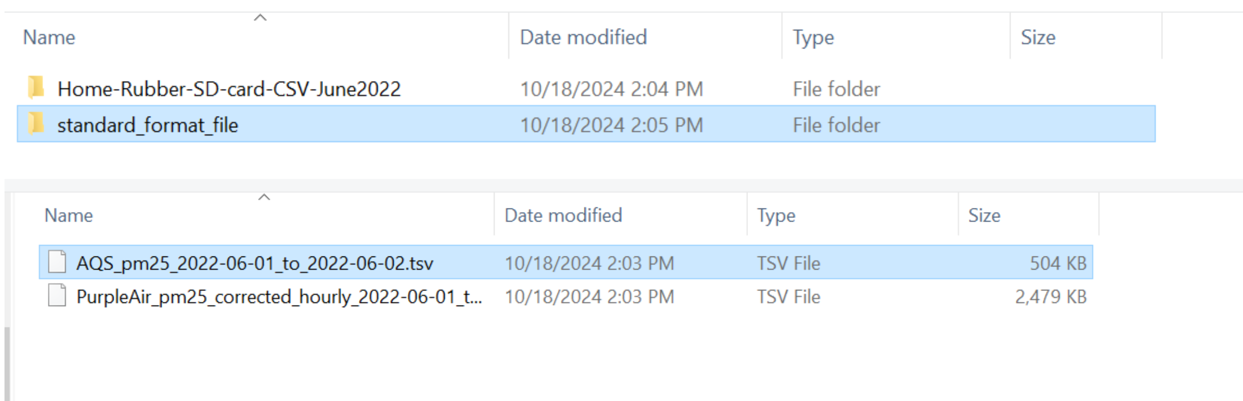
Demo 9: Moving Average:

`ASNAT` can correct repetitive/constant values with a moving average.
To use this feature, please refer to Demo 7: Additional Flagging, and ensure that flag 83 is enabled and applied (by clicking the "**Refresh Table Using Settings Below**" button).

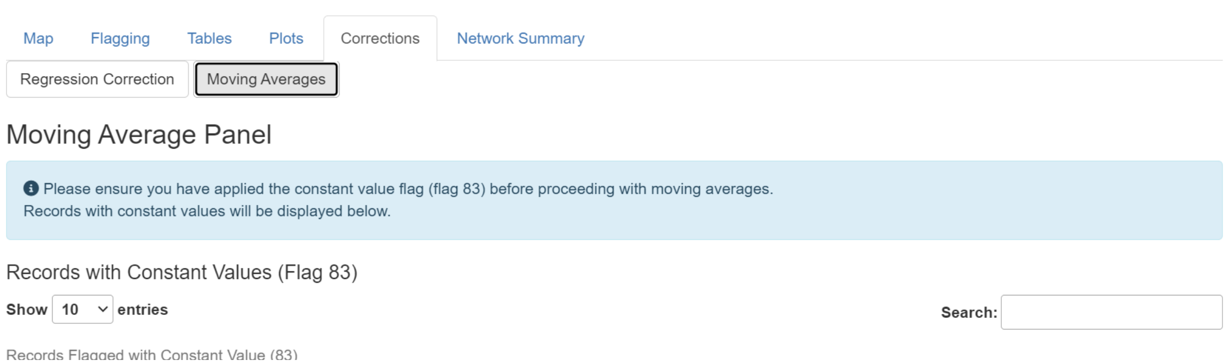
Relaunch `ASNAT`

Load the sample data AQS_pm25_2022-06-01_to_2022-06-02.tsv through "load data from a standard-format file" in \ASNAT\ sample_data\ standard_format_file





Go to the **"Correction" tab.**
Click the **"Moving Average"** button.



Click the **“id(-)”** to sort the repetitive values by ID.
Select all entries from ID 90010010.

Records with Constant Values (Flag 83)

Show 10 **entries** **Search:**

Records Flagged with Constant Value (83)

	timestamp(UTC)	longitude(deg)	latitude(deg)	id(-)	pm25(ug/m3)	flagged(-)	note(-)
827	2022-06-01T06:00:00-0000	-73.1947	41.1708	90010010	6.5	83	Bridgeport
985	2022-06-01T07:00:00-0000	-73.1947	41.1708	90010010	6.5	83	Bridgeport
1142	2022-06-01T08:00:00-0000	-73.1947	41.1708	90010010	6.5	83	Bridgeport
2543	2022-06-01T17:00:00-0000	-73.1947	41.1708	90010010	4.3	83	Bridgeport
2700	2022-06-01T18:00:00-0000	-73.1947	41.1708	90010010	4.3	83	Bridgeport
2859	2022-06-01T19:00:00-0000	-73.1947	41.1708	90010010	4.3	83	Bridgeport
6959	2022-06-02T21:00:00-0000	-72.6799	41.7714	90030025	7.2	83	Hartford
7118	2022-06-02T22:00:00-0000	-72.6799	41.7714	90030025	7.2	83	Hartford
7276	2022-06-02T23:00:00-0000	-72.6799	41.7714	90030025	7.2	83	Hartford
2395	2022-06-01T16:00:00-0000	-72.9029	41.3014	90090027	3.6	83	New Haven

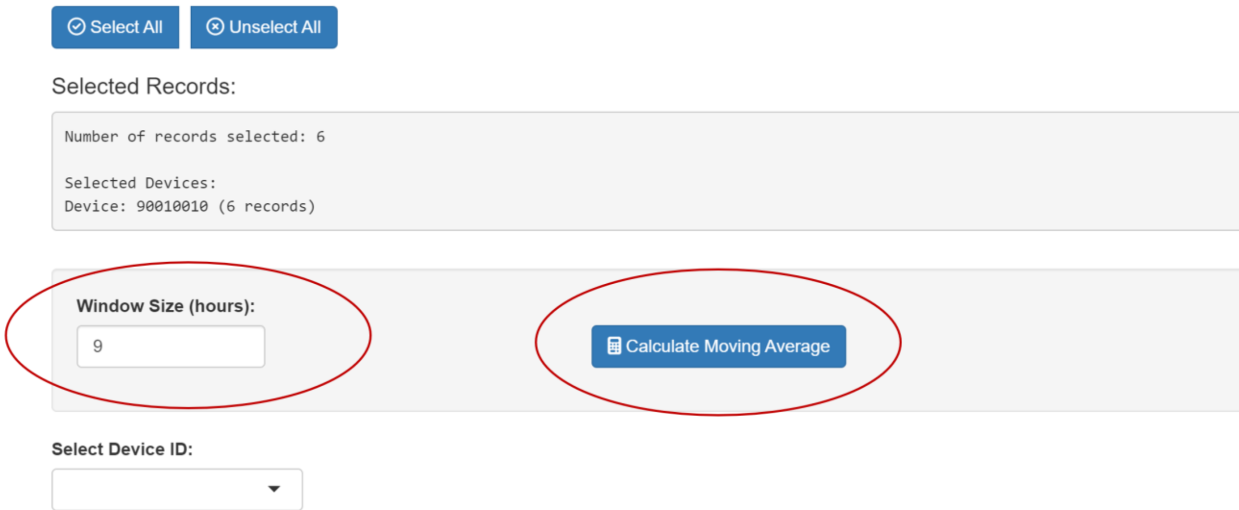
Showing 1 to 10 of 484 entries   Previous 1 2 3 4 5 … 49 Next

Select All   Unselect All

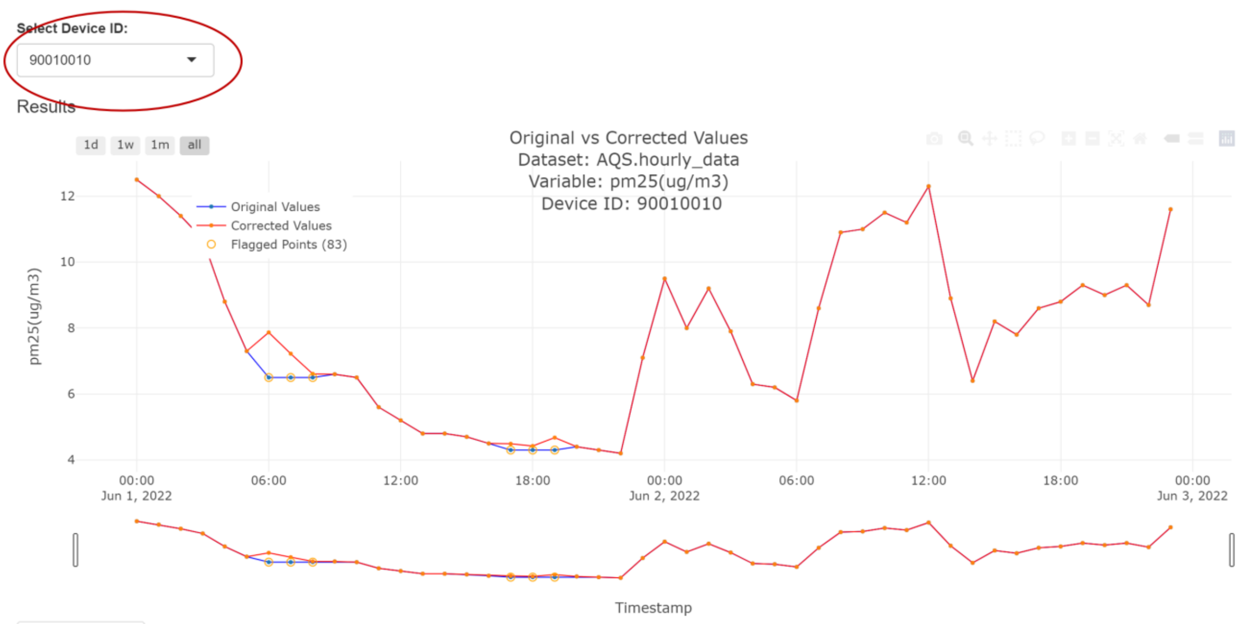
Under “**Number of records selected**” you can confirm the number of records and devices selected.

Enter 9 in the Window Size (the window size should be greater than the selected records to achieve the best moving average result).

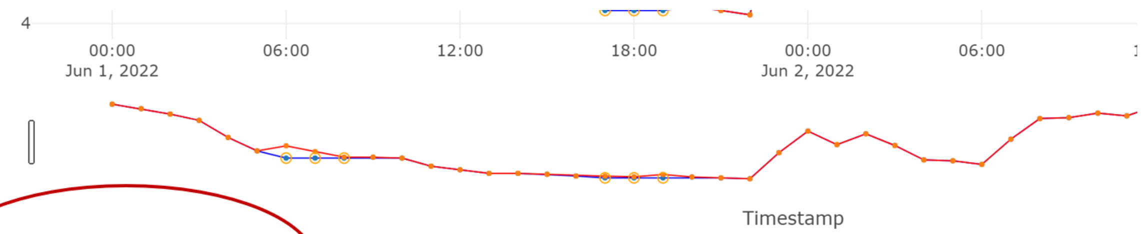
Click **“Calculate Moving Average”** to generate the result.



From the drop-down menu, select "Device ID" to switch to the results of different devices (if any records are selected in the prior steps).

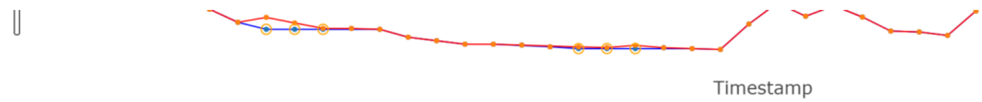


Click **"After Correction"** and **"Original Data"** tabs to switch between different table views for corrected and original data.



After Correction | Original Data

timestamp(UTC)	longitude(deg)	latitude(deg)	id(-)	pm25(ug/m3)	flagged(-)	note(-)
2022-06-01T00:00:00-0000	-73.19	41.17	90010010	12.50	0	Bridgeport
2022-06-01T01:00:00-0000	-73.19	41.17	90010010	12.00	0	Bridgeport
2022-06-01T02:00:00-0000	-73.19	41.17	90010010	11.40	0	Bridgeport
2022-06-01T03:00:00-0000	-73.19	41.17	90010010	10.70	0	Bridgeport
2022-06-01T04:00:00-0000	-73.19	41.17	90010010	8.80	0	Bridgeport
2022-06-01T05:00:00-0000	-73.19	41.17	90010010	7.30	0	Bridgeport
2022-06-01T06:00:00-0000	-73.19	41.17	90010010	7.87	83	Bridgeport
2022-06-01T07:00:00-0000	-73.19	41.17	90010010	7.22	83	Bridgeport
2022-06-01T08:00:00-0000	-73.19	41.17	90010010	6.61	83	Bridgeport
2022-06-01T09:00:00-0000	-73.19	41.17	90010010	6.60	0	Bridgeport
2022-06-01T10:00:00-0000	-73.19	41.17	90010010	6.50	0	Bridgeport
2022-06-01T11:00:00-0000	-73.19	41.17	90010010	5.60	0	Bridgeport
2022-06-01T12:00:00-0000	-73.19	41.17	90010010	5.20	0	Bridgeport
2022-06-01T13:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T14:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T15:00:00-0000	-73.19	41.17	90010010	4.70	0	Bridgeport
2022-06-01T16:00:00-0000	-73.19	41.17	90010010	4.50	0	Bridgeport
2022-06-01T17:00:00-0000	-73.19	41.17	90010010	4.49	83	Bridgeport
2022-06-01T18:00:00-0000	-73.19	41.17	90010010	4.42	83	Bridgeport
2022-06-01T19:00:00-0000	-73.19	41.17	90010010	4.68	83	Bridgeport
2022-06-01T20:00:00-0000	-73.19	41.17	90010010	4.40	0	Bridgeport
2022-06-01T21:00:00-0000	-73.19	41.17	90010010	4.30	0	Bridgeport



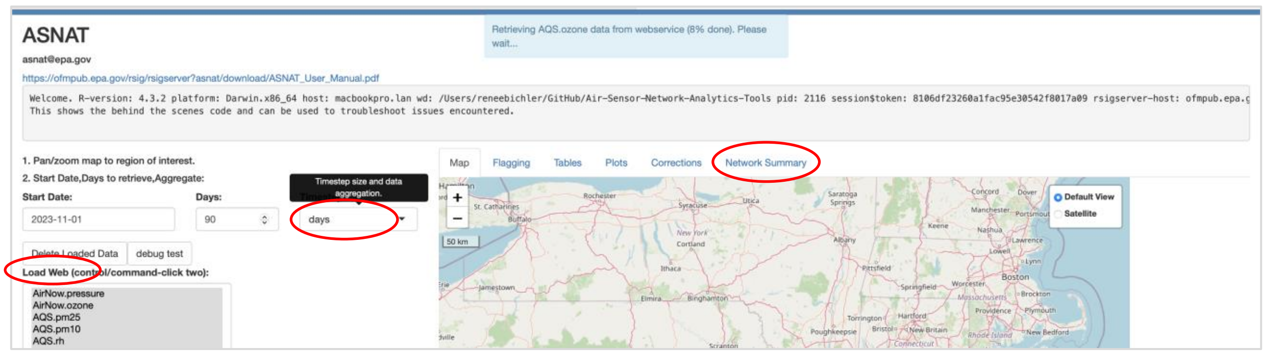
After Correction | Original Data

timestamp(UTC)	longitude(deg)	latitude(deg)	id(-)	pm25(ug/m3)	flagged(-)	note(-)
2022-06-01T00:00:00-0000	-73.19	41.17	90010010	12.50	0	Bridgeport
2022-06-01T01:00:00-0000	-73.19	41.17	90010010	12.00	0	Bridgeport
2022-06-01T02:00:00-0000	-73.19	41.17	90010010	11.40	0	Bridgeport
2022-06-01T03:00:00-0000	-73.19	41.17	90010010	10.70	0	Bridgeport
2022-06-01T04:00:00-0000	-73.19	41.17	90010010	8.80	0	Bridgeport
2022-06-01T05:00:00-0000	-73.19	41.17	90010010	7.30	0	Bridgeport
2022-06-01T06:00:00-0000	-73.19	41.17	90010010	6.50	83	Bridgeport
2022-06-01T07:00:00-0000	-73.19	41.17	90010010	6.50	83	Bridgeport
2022-06-01T08:00:00-0000	-73.19	41.17	90010010	6.50	83	Bridgeport
2022-06-01T09:00:00-0000	-73.19	41.17	90010010	6.60	0	Bridgeport
2022-06-01T10:00:00-0000	-73.19	41.17	90010010	6.50	0	Bridgeport
2022-06-01T11:00:00-0000	-73.19	41.17	90010010	5.60	0	Bridgeport
2022-06-01T12:00:00-0000	-73.19	41.17	90010010	5.20	0	Bridgeport
2022-06-01T13:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T14:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T15:00:00-0000	-73.19	41.17	90010010	4.70	0	Bridgeport
2022-06-01T16:00:00-0000	-73.19	41.17	90010010	4.50	0	Bridgeport
2022-06-01T17:00:00-0000	-73.19	41.17	90010010	4.30	83	Bridgeport
2022-06-01T18:00:00-0000	-73.19	41.17	90010010	4.30	83	Bridgeport
2022-06-01T19:00:00-0000	-73.19	41.17	90010010	4.30	83	Bridgeport
2022-06-01T20:00:00-0000	-73.19	41.17	90010010	4.40	0	Bridgeport
2022-06-01T21:00:00-0000	-73.19	41.17	90010010	4.30	0	Bridgeport

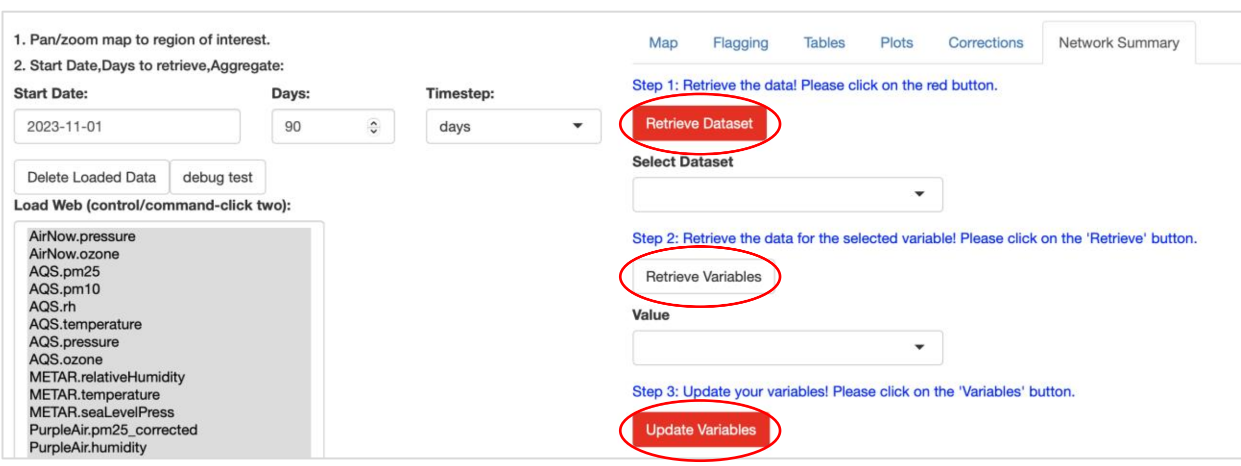
Demo 10: Network Summary:

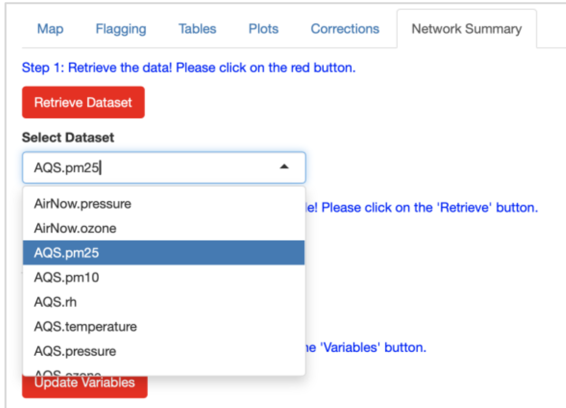
The Network Summary Panel will give insights about basic statistics regarding selected data from the **"Load Web"** panel. Some plots depend on the selected timestep size which means hourly or daily data.

After loading the data frames via **"Load Web"** move over to the "Network Summary" panel.

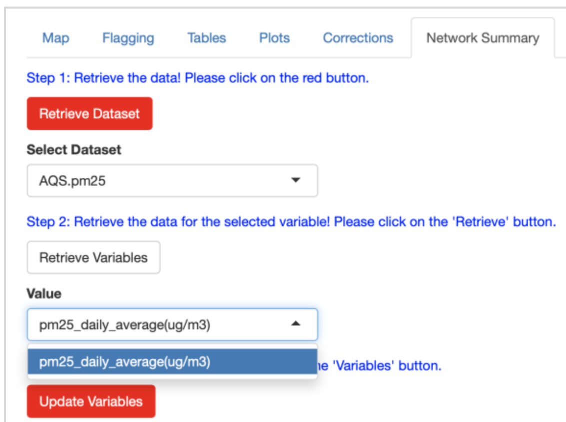


After clicking on **"Retrieve Dataset"**, select a dataset from the dropdown menu for example AQS.pm25. The dropdown menu will offer different selection options depending on the loaded data from **"Load Web"**.

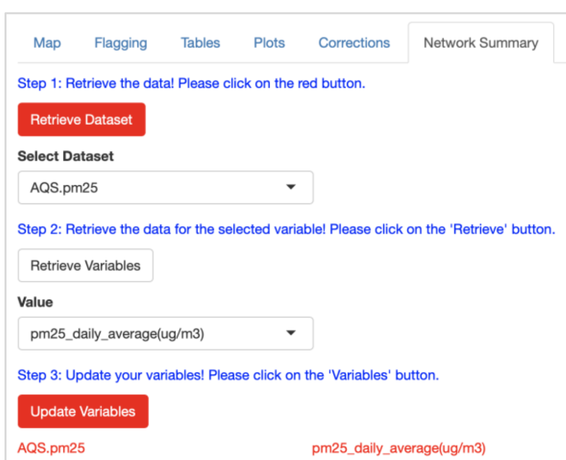




In the next step click on the **"Retrieve Variables"** button. This triggers the retrieval of the column names of the selected data frame. If own datasets were uploaded the selection will show different options.



With **"Update Variables"** the selected input variables will be used to process the data frame.

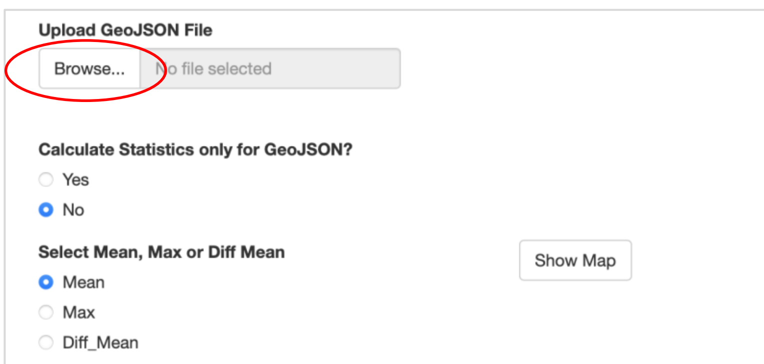


It is also possible to upload a GeoJSON file to either visually check if there are measurement stations in the polygon or to process the network summary statistics only for observations within the GeoJSON file. **This is an optional feature and can be skipped if not needed**.

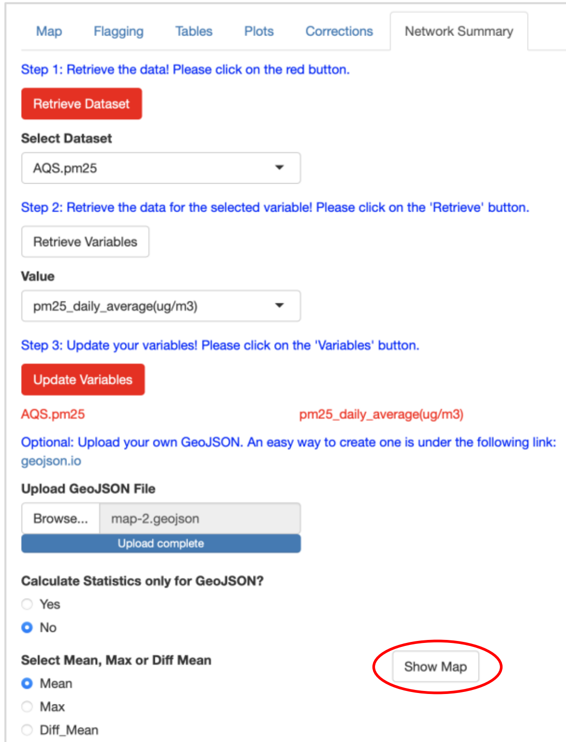
To create a GeoJSON it is possible to use ArcGIS, QGIS, or simply visit www.geojson.io to create and download a polygon file.

Afterward, select if the statistics should be carried out with measurement sites inside the polygon or not. The default setting is set to **"No"** and can always be changed later on.

The selection below will show the mean or maximum value over the selected period for each measurement site. It is also possible to visualize how big the difference of the mean value for each site differs from the overall mean of all selected sites or only for sites within the polygon.



After checking the selected dataset and variables click on **“Show Map”**.

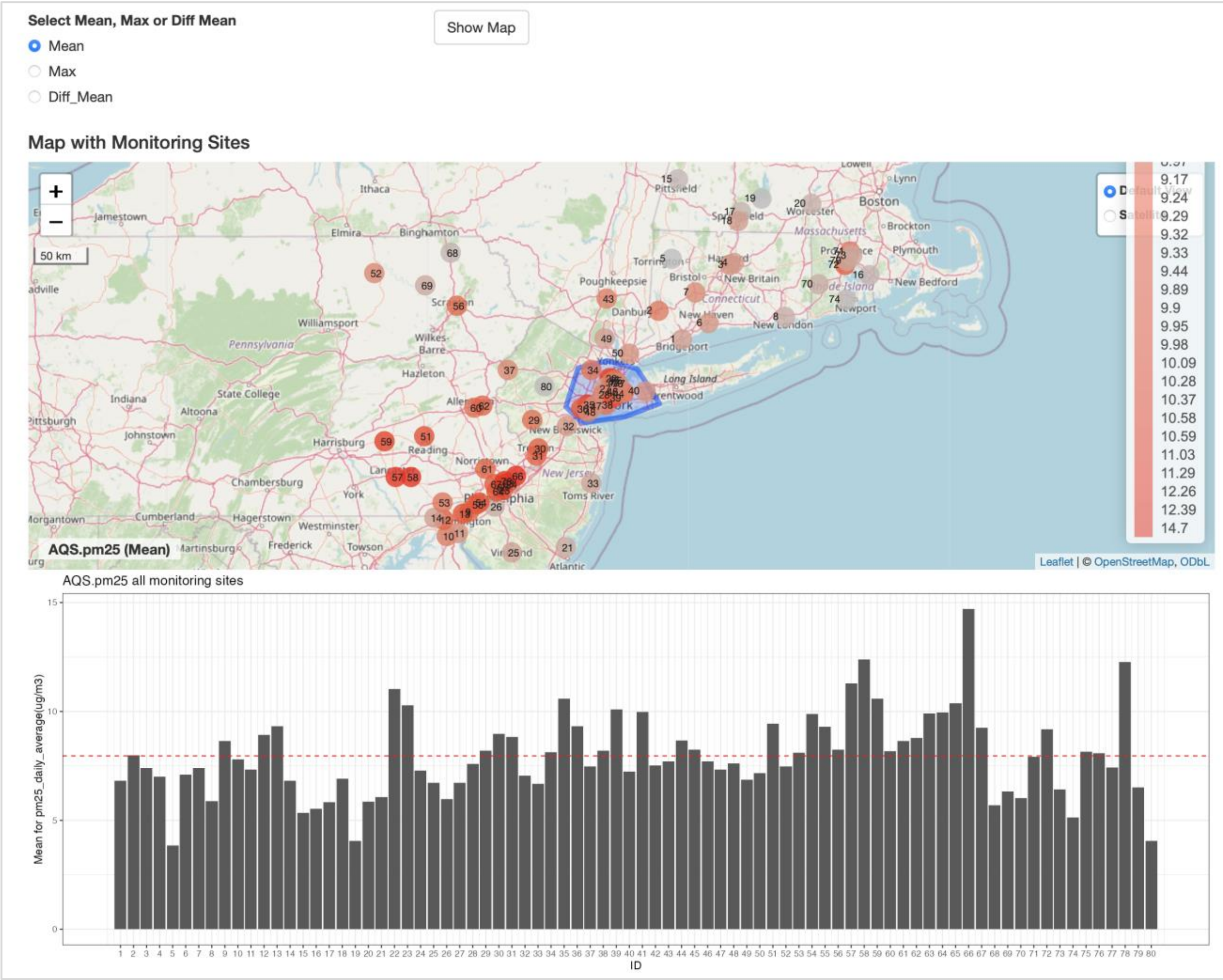


A leaflet map with the uploaded GeoJSON should be displayed. The map title will appear in the bottom left corner. The values in the circle show the new ID number that is connected to the bar chart below.

Note: The new ID is not the same as the original measurement site ID.

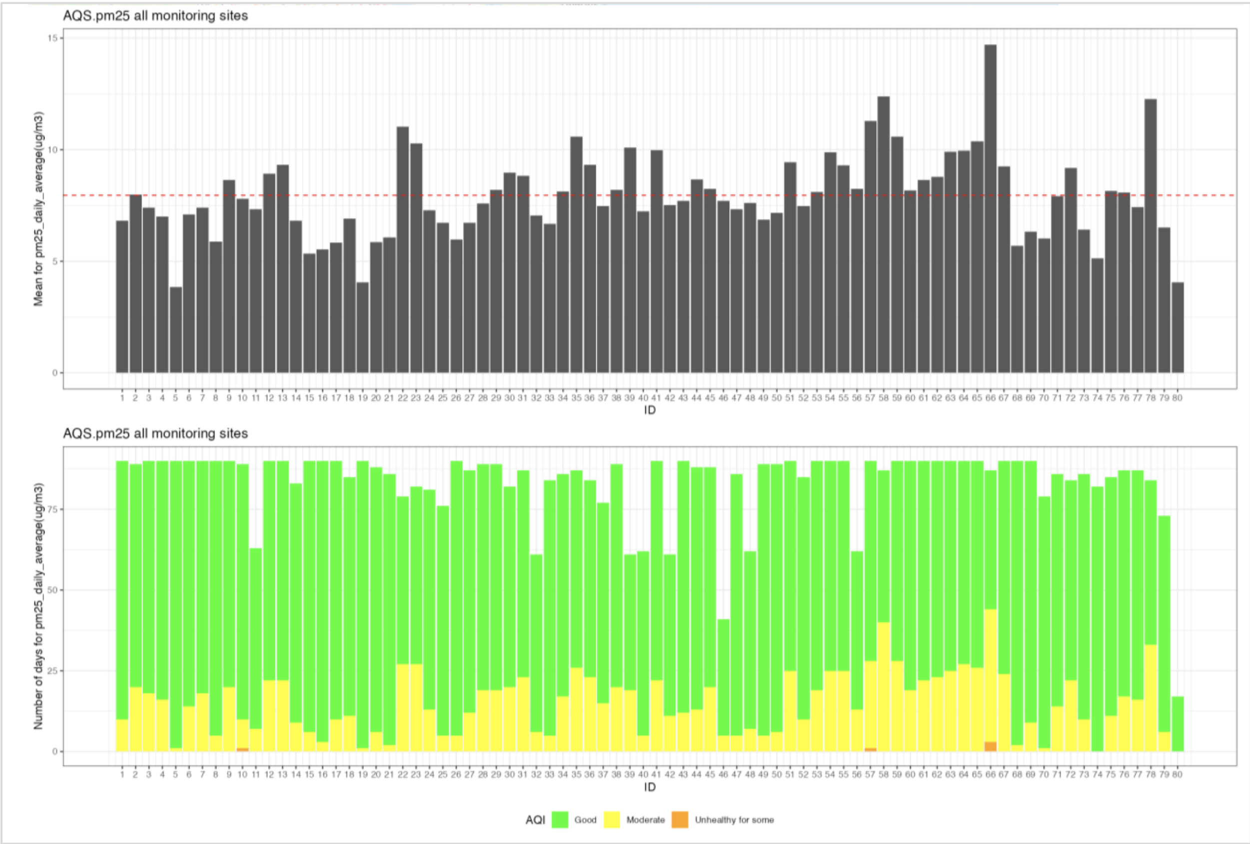
When clicking on one of the circle markers a popup feature will appear.

The red dotted line in the bar chart below presents the overall mean value for all monitoring sites displayed in the leaflet map. If the maximum values or the difference values for the mean value shall be displayed, simply select the option and click on **“Show Map”** again.

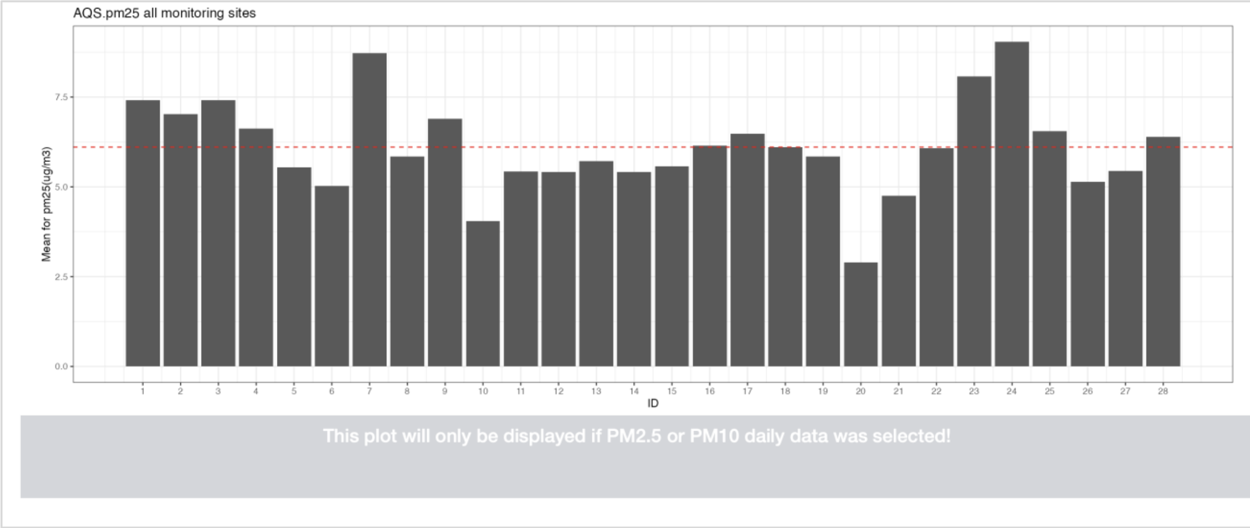


In this example, AQS.pm25 daily data was loaded. As a result, a second bar chart will be displayed that counts the number of days with a certain air quality during the selected time range.

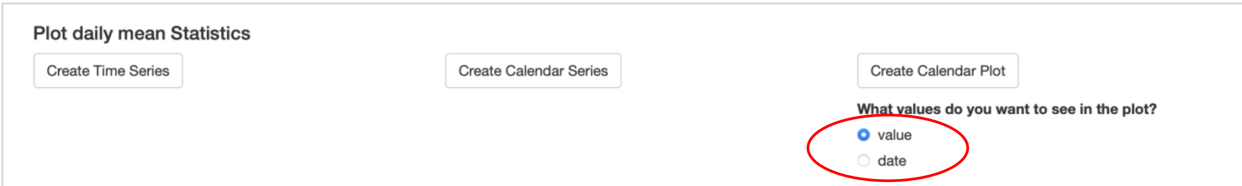
Note: This plot only appears if PM2.5 or PM10 daily data was selected! Therefore, the plot will not appear if PM2.5 hourly data is loaded.



When hourly PM2.5 dataset was selected for example, the plot which represents the number of days with a certain air quality will not appear in the dashboard.

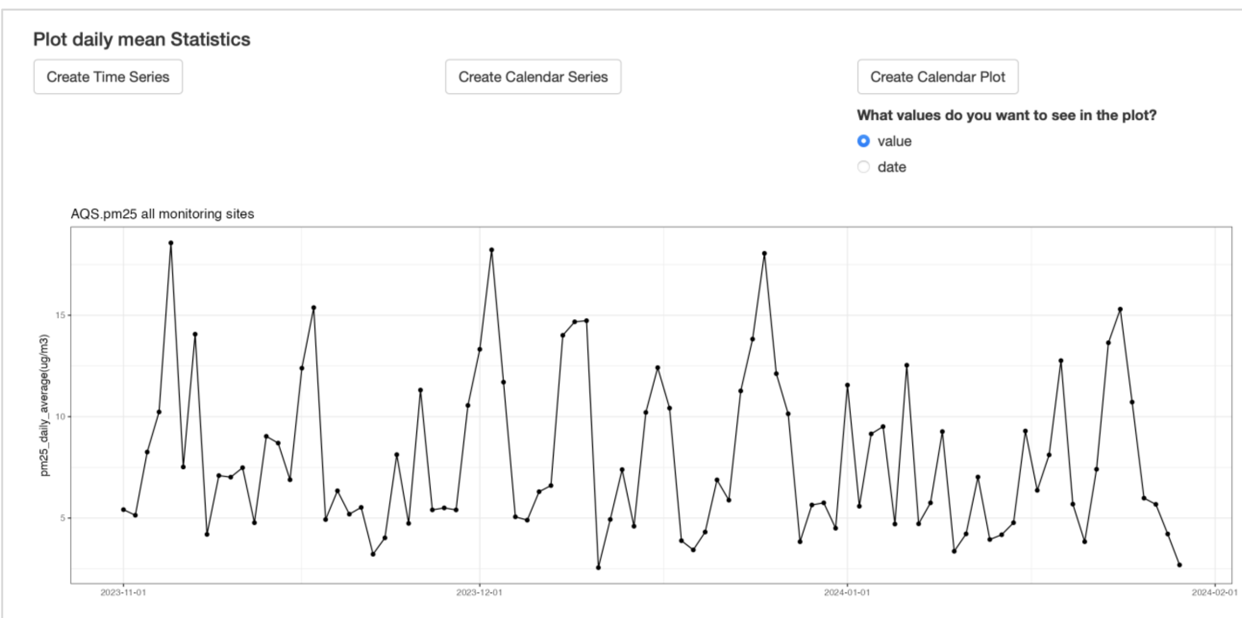


The network summary offers the possibility to produce a time series plot, a calendar series plot as well as a normal calendar plot. On top of that, the calendar plot offers to option to show the mean values of a day in the calendar or the number of days.

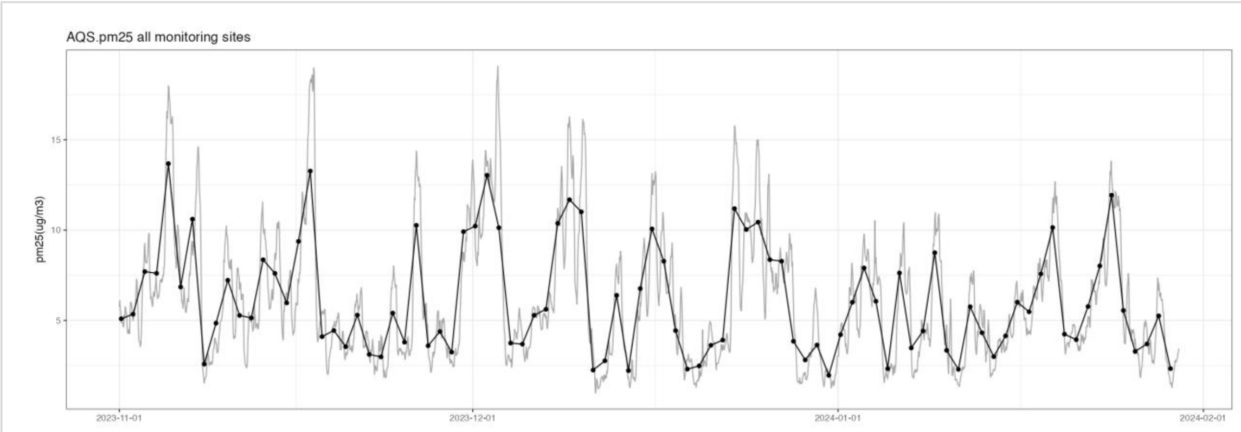


When clicking on **"Create Time Series"** a time series of the daily mean of the selected data frame is processed.

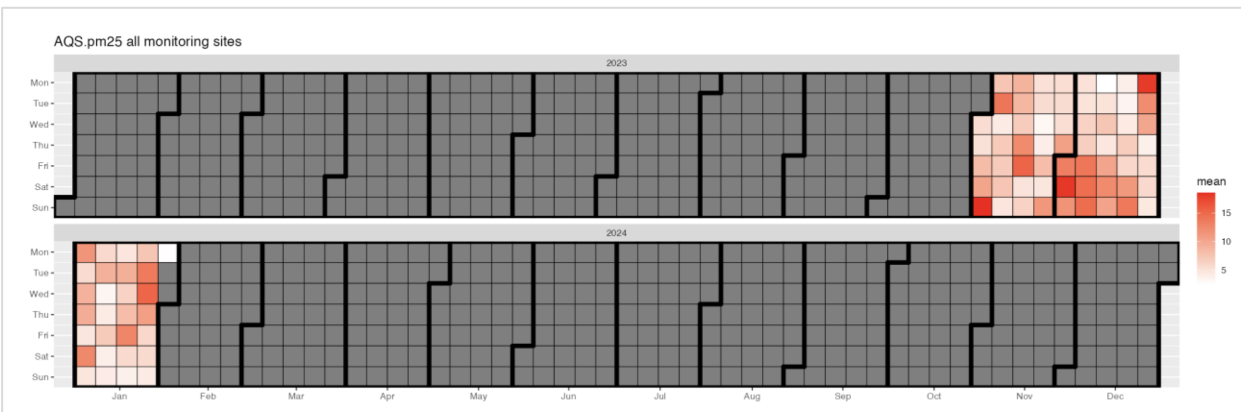
Note: When a GeoJSON was uploaded and only measurements of monitoring sites within the polygon are analyzed, the title of the plots will change. This indicates what area of interest was used, either **"all monitoring sites"** or **"within GeoJSON".**



When a data frame with hourly data is loaded the time series plot will look like the following plot. The gray line represents the hourly data and the black line represents the daily mean based on the hourly input data.

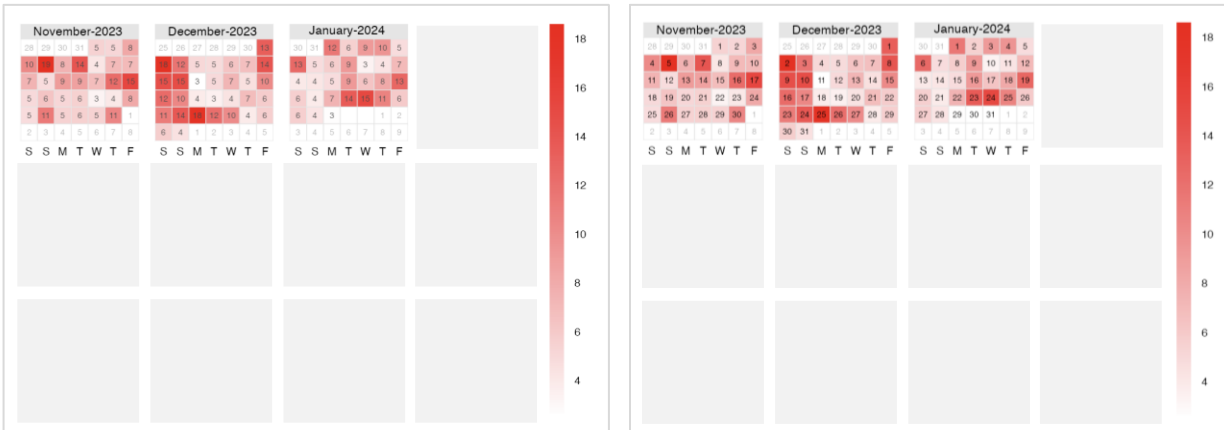


The **“Create Calendar Series”** displays the daily mean of a time series. The y-axis shows the name of the day from Monday to Sunday, and the x-axis represents the month. Darker red colors imply higher mean values, whereas lighter red colors represent lower mean values.

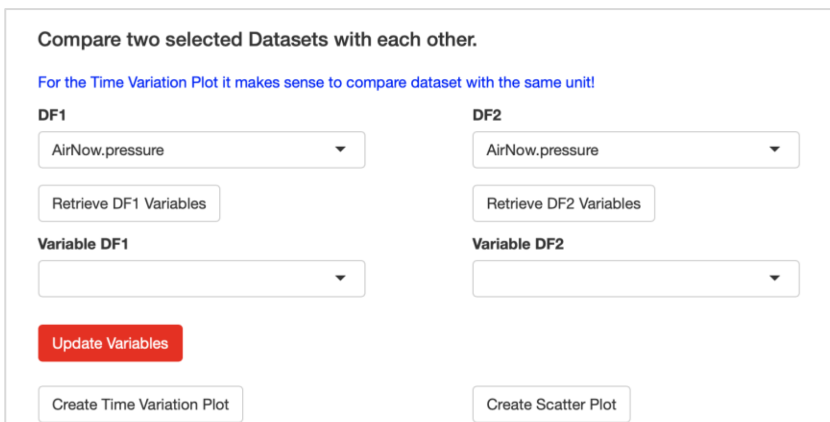


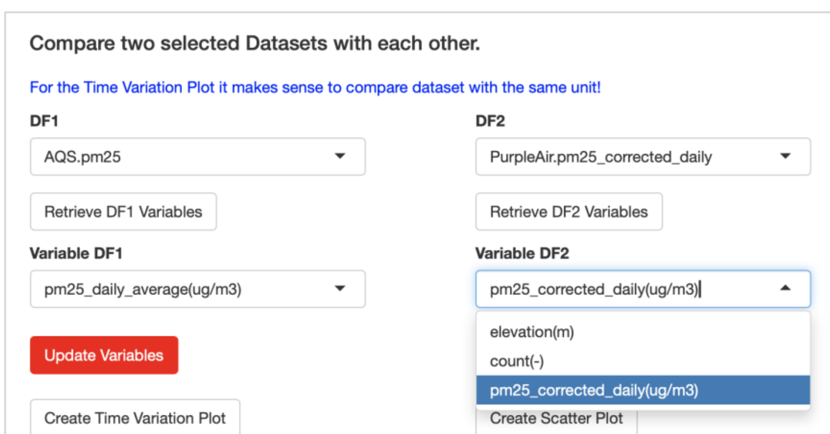
The **“Create Calendar Plot”** presents the daily mean of a time series and comes in a 3x4 months matrix. Thereby, it is no problem to show a time range that starts in 2023 and ends in 2024 for example. However, this plot cannot plot more than 12 months in total.

The darker red color means higher values, and light red color presents lower values. The values in the calendar plot can either display the daily mean value or represent the day of a month.

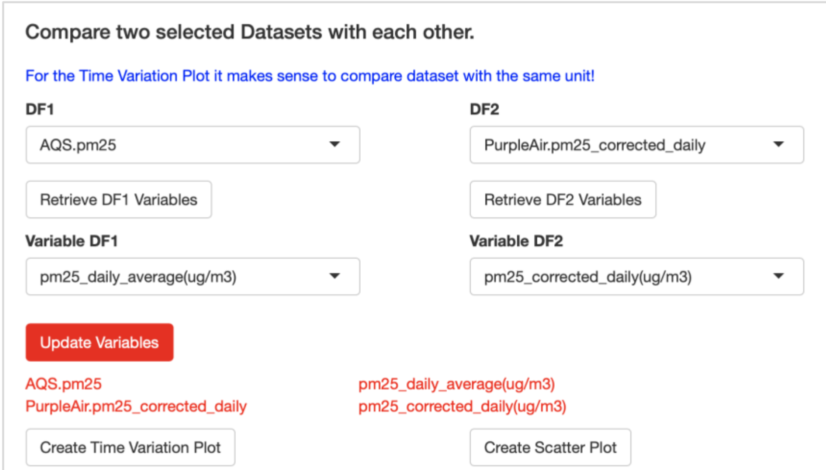


The **“Network Summary”** panel also offers the possibility to compare two datasets with each other. To do so select the two data frames and retrieve the variables/column names that should be used.

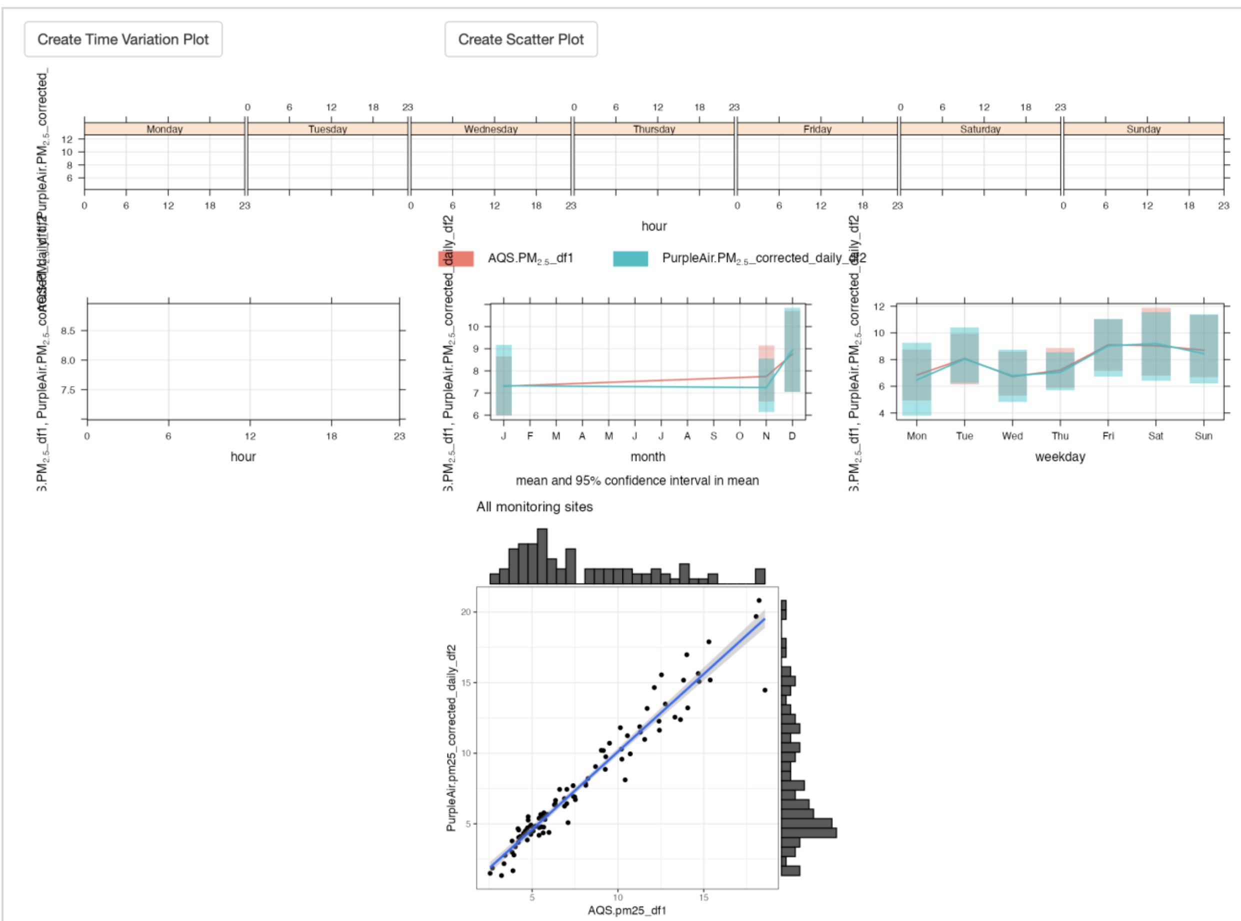




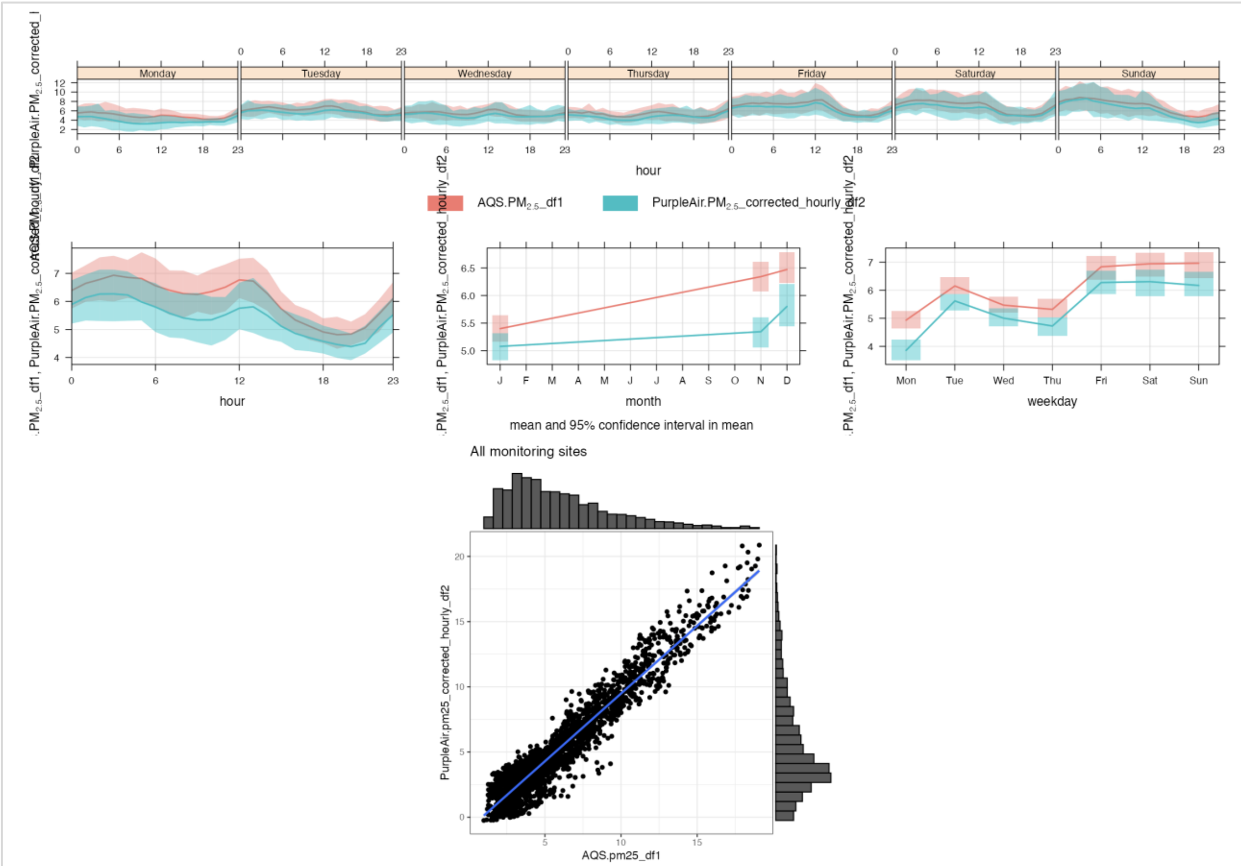
Do not forget to click on **“Update Variables”** to confirm the choices.



The data frames used to create those plots are daily observations. In that case, some parts of the time variation plot cannot be displayed since hourly observations are needed.



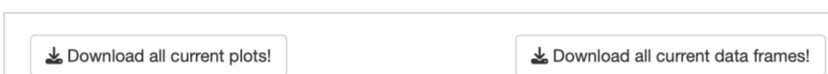
Here is an example of how the plot looks like with hourly PM2.5 observations.

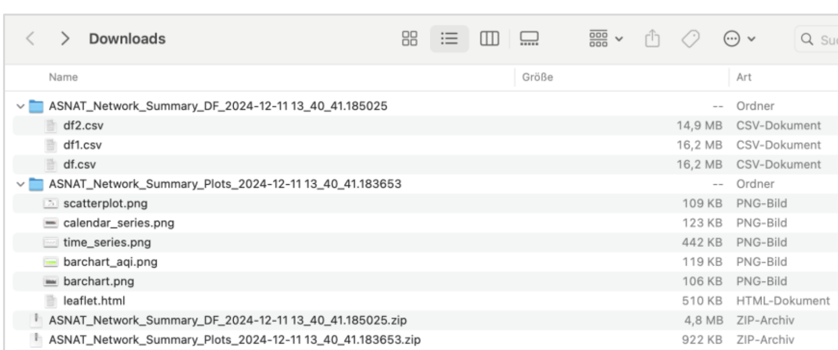


Finally, download the plots and the original data frames that were used to create them in the first place.

`ASNAT` **"Network Summary"** plots and original data frames will be downloaded as zip files and include only visible plots from the **"Network Summary"** panel. If the process of the time series plot is skipped, for example, the download will not include the time series plot.

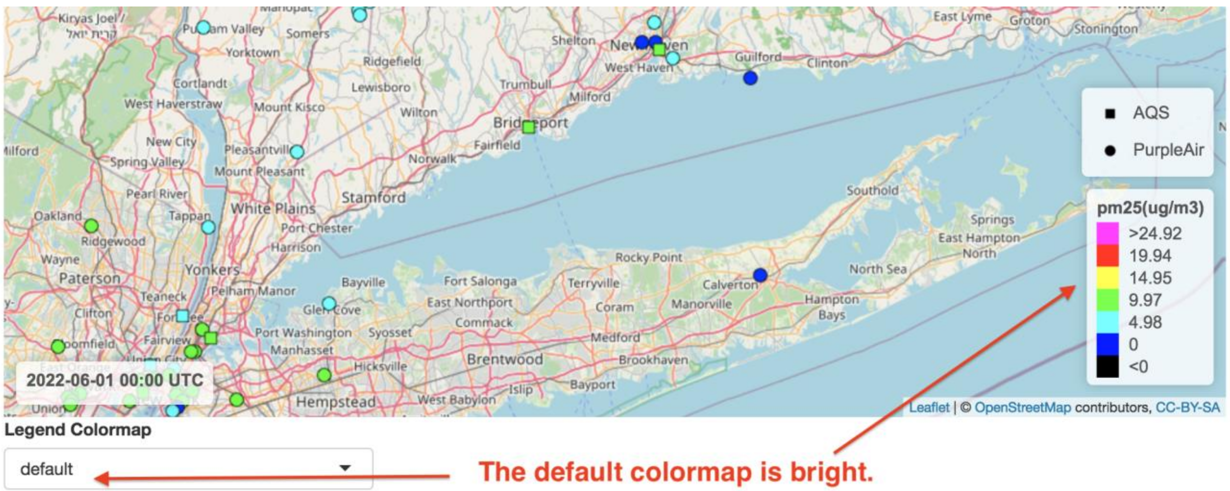
<u>Note:</u> The leaflet map will be downloaded as an HTML file.



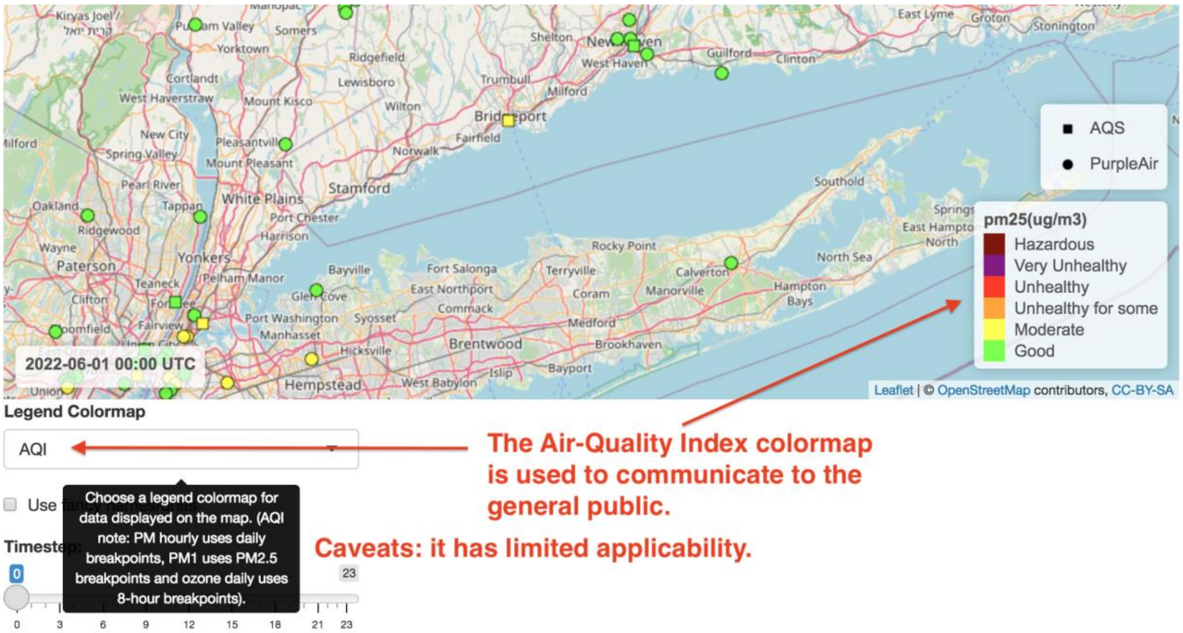


Colormaps/legends:

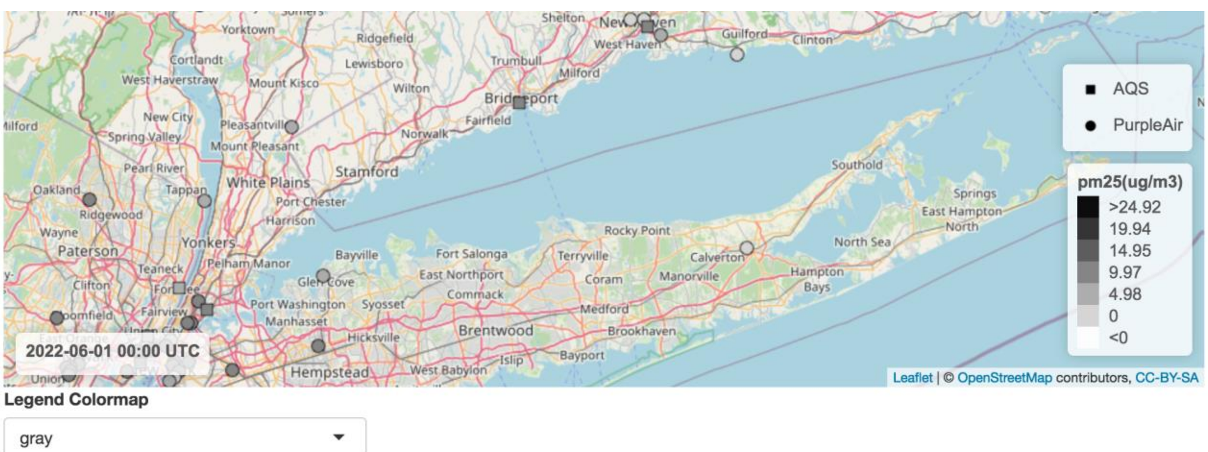
ASNAT has a few colormaps/legends.
Each has a half-dozen hues since human vision cannot distinguish many subtly differing hues.
The default colormap contains bright most easily discernable hues.



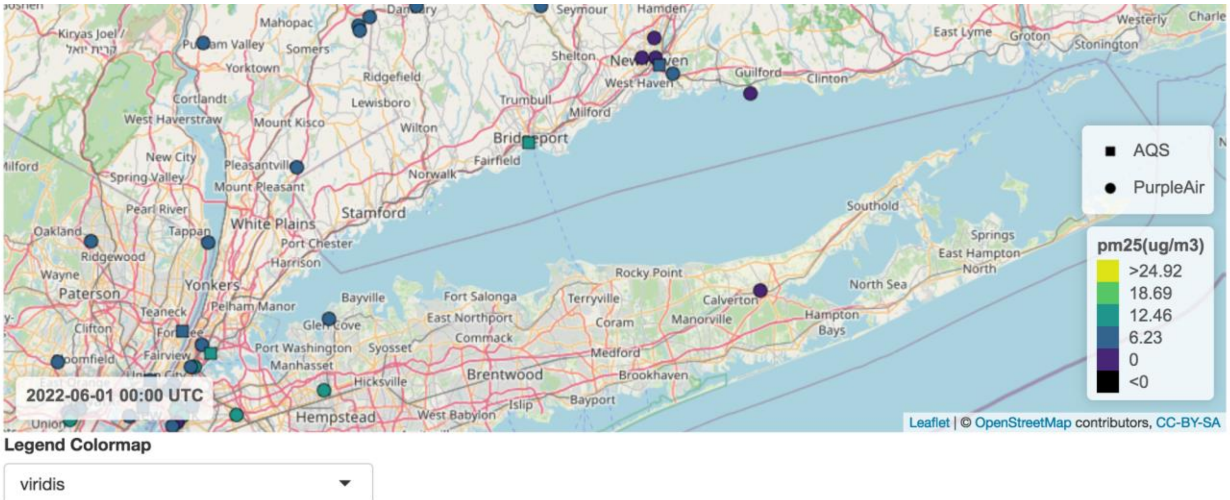
The AQI colormap is familiar to the general public but the standard is not defined for all measures available in ASNAT.
https://aqs.epa.gov/aqsweb/documents/codetables/aqi_breakpoints.html
For hourly PM2.5 and PM10, ASNAT uses daily breakpoints.
For PM1 ASNAT uses PM2.5 breakpoints.
For daily ozone, ASNAT uses 8-hour average breakpoints.
If a variable is selected that is not in the AQI standard, then the default colormap will be used instead.



The grayscale colormap can work well for saved map images for publication and color-impaired viewers.



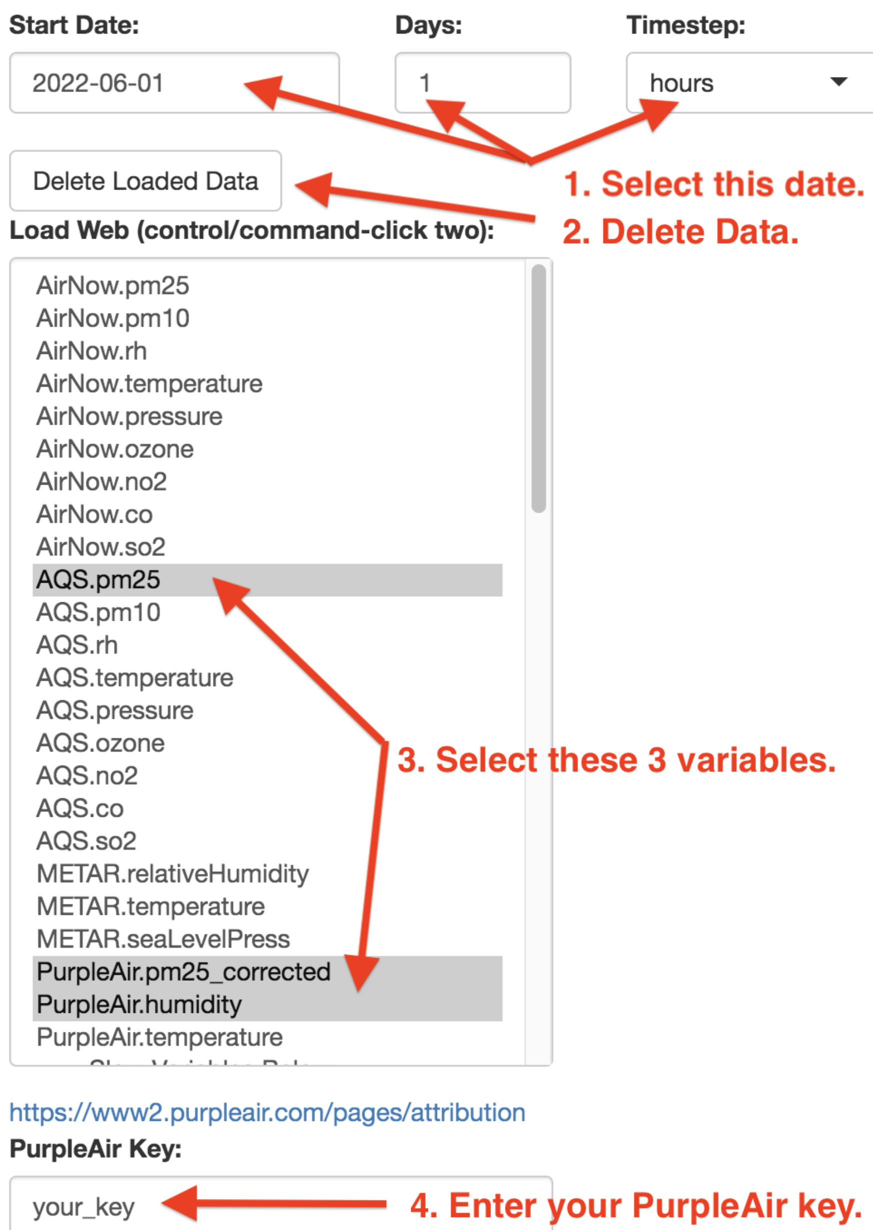
The last colormap, viridis (Latin for green) is supposedly also suitable.
for color-impaired viewers.


AQS
PurpleAir
pm25(ug/m3)
>24.92
18.69
12.46
6.23
0
<0
2022-06-01 00:00 UTC
Leaflet | © OpenStreetMap contributors, CC-BY-SA
Legend Colormap
viridis

Demo 11: PM2.5 Parameter Codes, Neighbor Statistics and AQI Plots

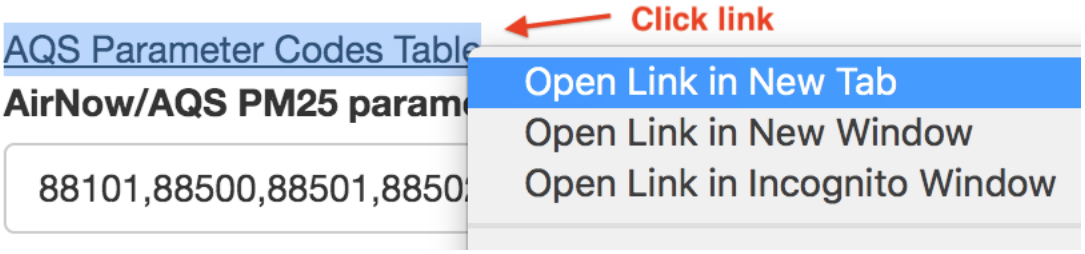
This demo will show retrieval by PM2.5 parameter codes, Neighbor Statistics and AQI plots.

11.1 To start, **Quit** ASNAT and **restart** it and make the selections shown in the following figures.

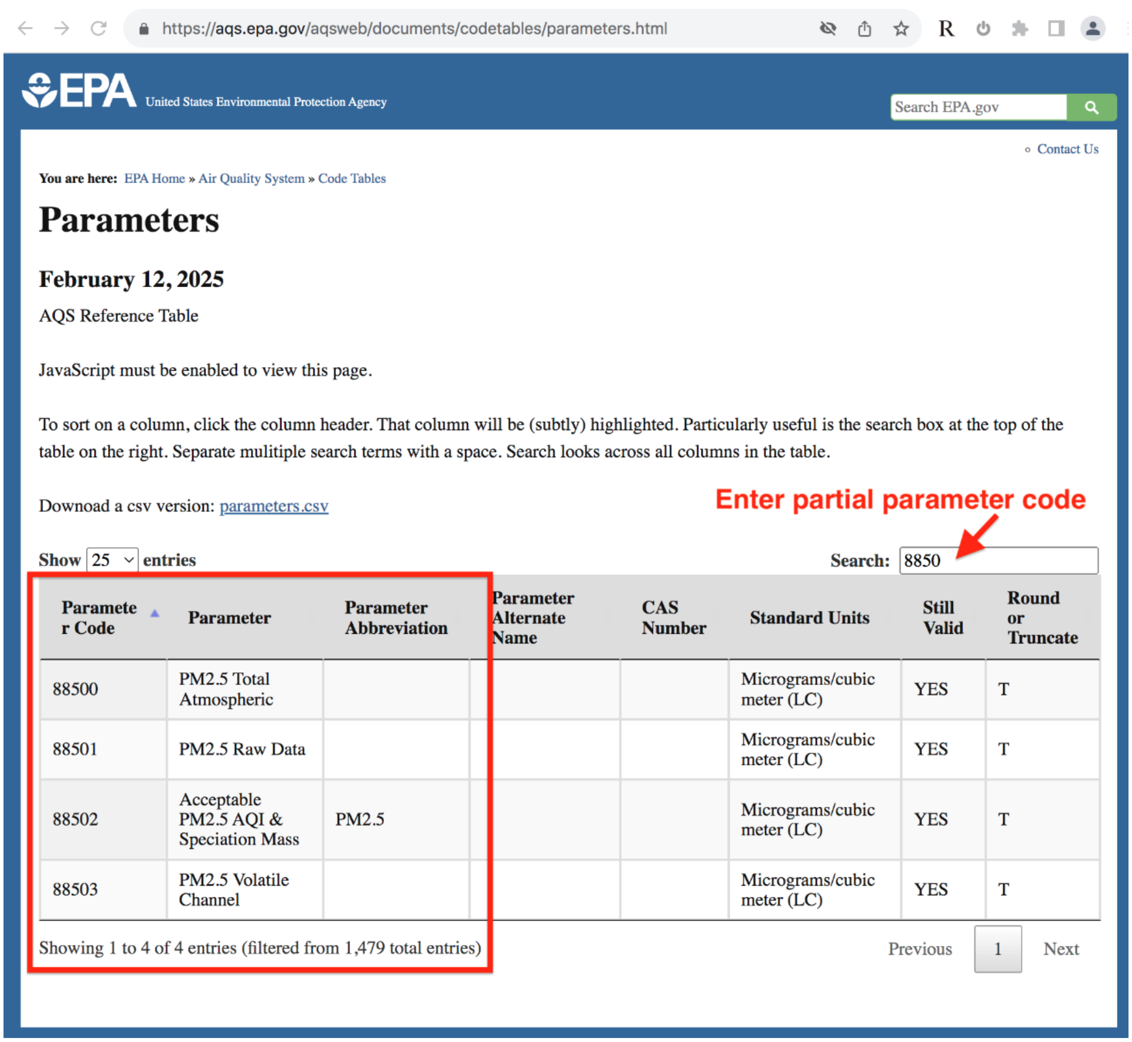


11.2 For AQS.pm25, we will specify only data designated with a particular parameter code will be retrieved.

Click on the AQS Parameter Codes Table hyperlink.



This will display the web page below. **Enter 8850 partial code** and the table will show PM2.5 parameters that start with those digits:



https://aqs.epa.gov/aqsweb/documents/codetables/parameters.html

EPA United States Environmental Protection Agency

Search EPA.gov

Contact Us

You are here: EPA Home » Air Quality System » Code Tables

Parameters

February 12, 2025

AQS Reference Table

JavaScript must be enabled to view this page.

To sort on a column, click the column header. That column will be (subtly) highlighted. Particularly useful is the search box at the top of the table on the right. Separate mulitiple search terms with a space. Search looks across all columns in the table.

Download a csv version: parameters.csv

Show 25 entries

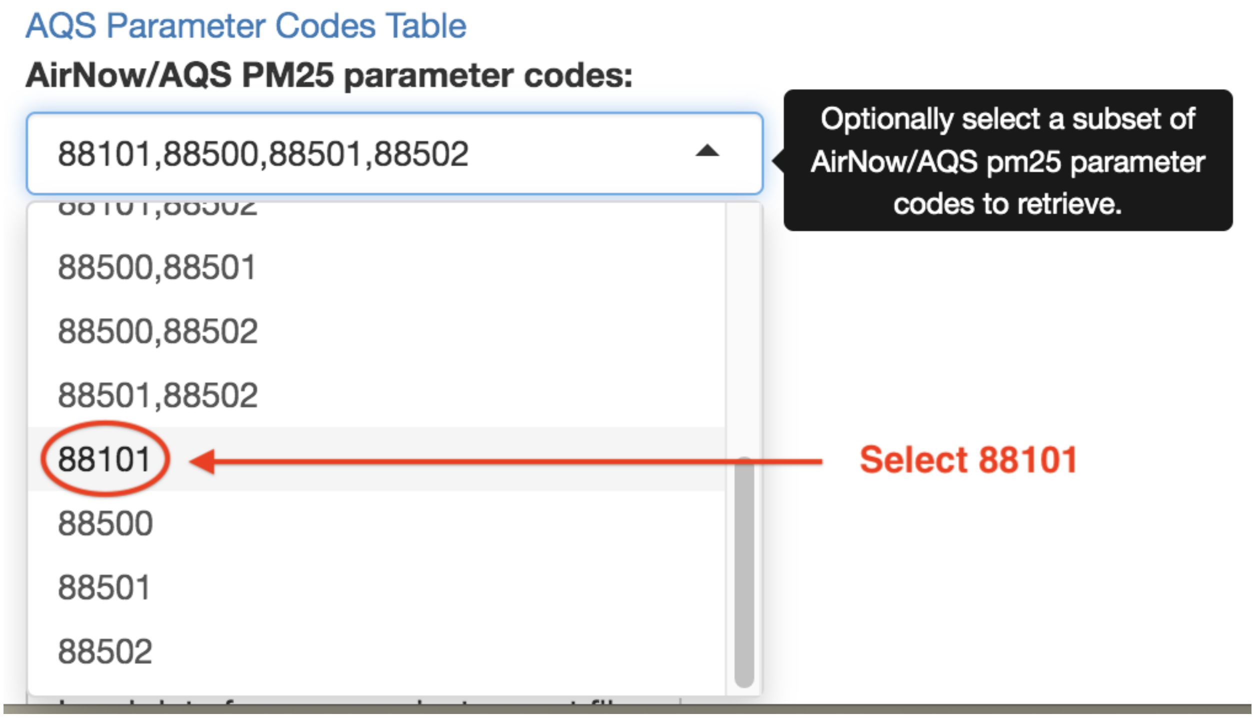
Search: 8850

Parameter Code	Parameter	Parameter Abbreviation	Parameter Alternate Name	CAS Number	Standard Units	Still Valid	Round or Truncate
88500	PM2.5 Total Atmospheric				Micrograms/cubic meter (LC)	YES	T
88501	PM2.5 Raw Data				Micrograms/cubic meter (LC)	YES	T
88502	Acceptable PM2.5 AQI & Speciation Mass	PM2.5			Micrograms/cubic meter (LC)	YES	T
88503	PM2.5 Volatile Channel				Micrograms/cubic meter (LC)	YES	T

Showing 1 to 4 of 4 entries (filtered from 1,479 total entries)

Previous 1 Next

11.3 **In ASNAT, Select 88101** in the **AirNow/AQS PM25 parameter codes menu**.
This specifies that AirNow/AQS PM25 data retrieved will only match code 88101. Note the menu lists all possible subsets of the four PM25 parameter codes.

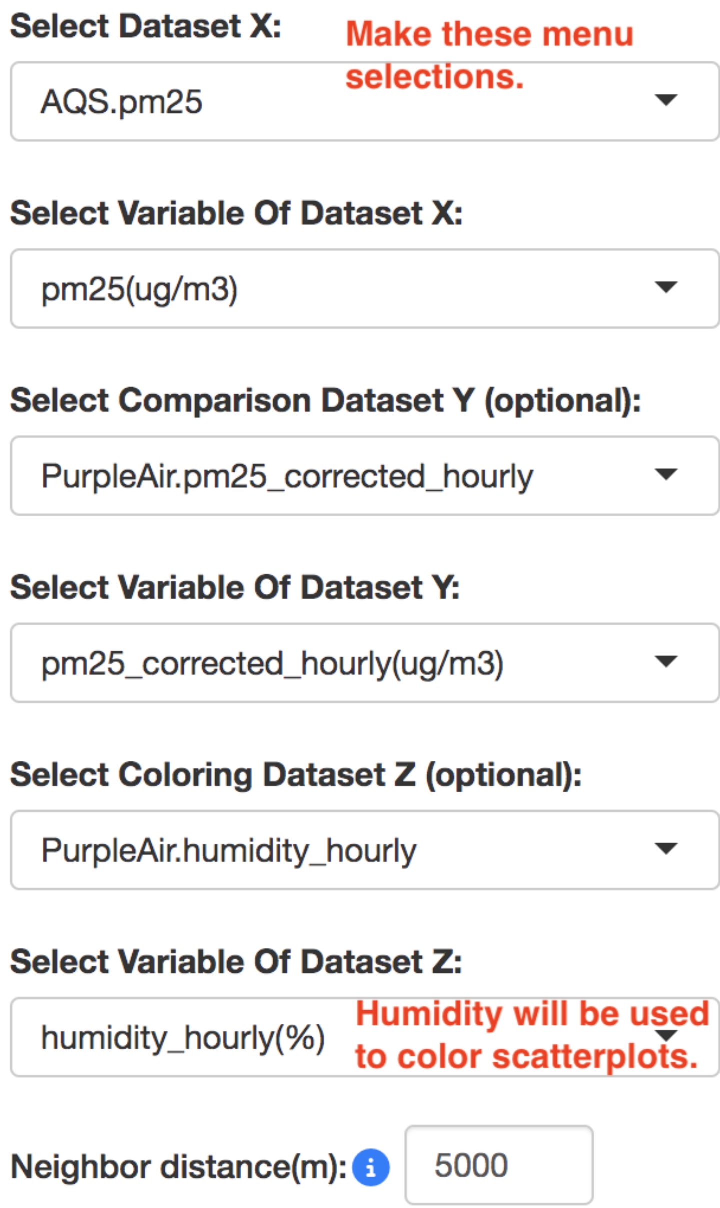


11.4 Click the **Reset map** button (below the map image).

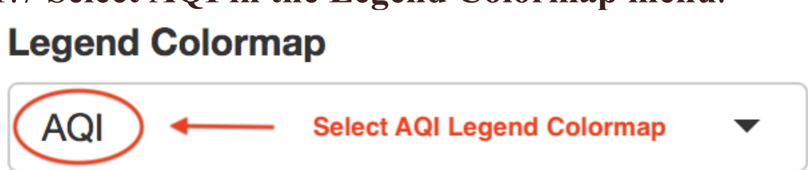
11.5 Click the **Retrieve Data** button.

11.6 Select the following Datasets X, Y and Z and Neighbor distance shown below. Dataset Z specifies and optional humidity variable to be used to color scatterplot points. ***The humidity dataset must match either Dataset X or Dataset Y***

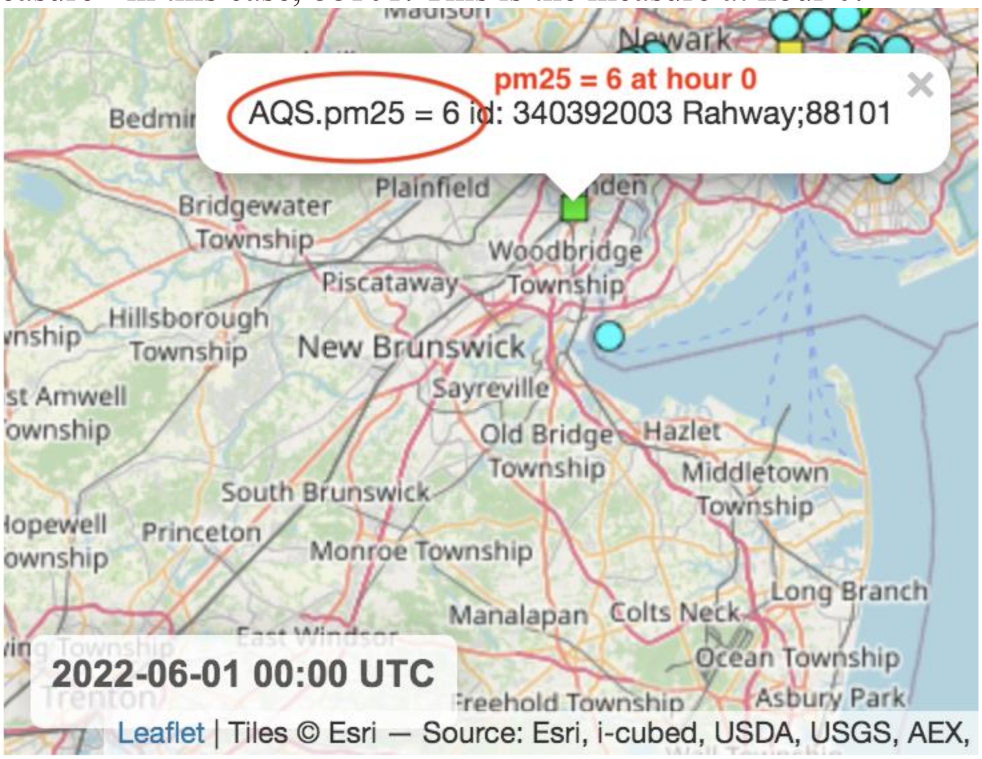
- in this case, either AQS.rh or PurpleAir.humidity. We will choose the latter since it contains more data points.



11.7 **Select AQI in the Legend Colormap menu**.

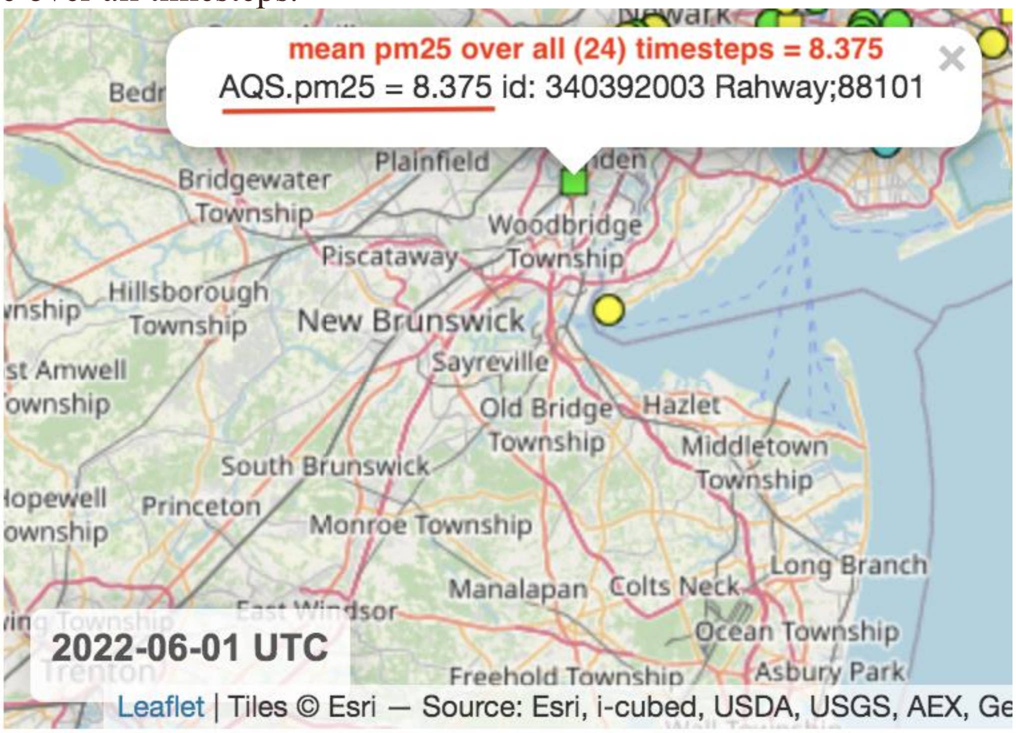


11.8 Look at the map and **find and click on the AQS site shown below**. The pop-up shows the AQS site #340392003 labeled *Rahway;88101*. Note that site numbers are unique but site labels are not. The label shows the parameter code of that measure - in this case, 88101. This is the measure at hour 0.



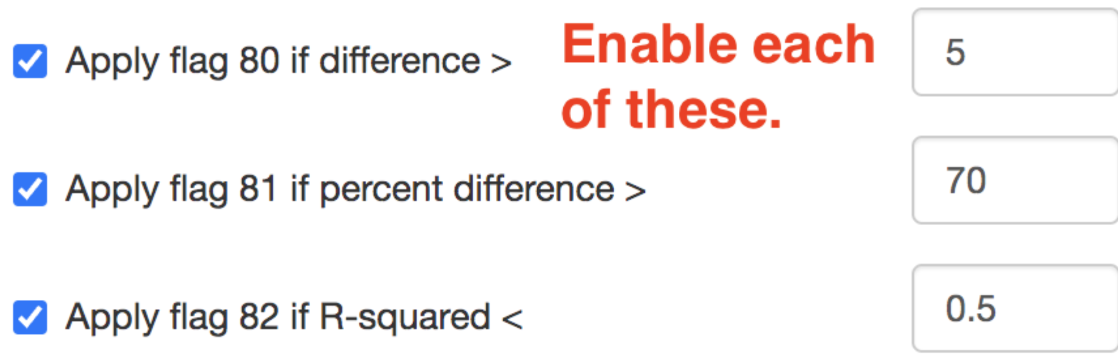
11.8 ASNAT can quickly display the mean of each site's measures - over all loaded timesteps -by simply clicking the **Show mean values button**.

11.9 Now **click on the AQS site again** to see the mean value (8.375 ug/m3) of that site over all timesteps.

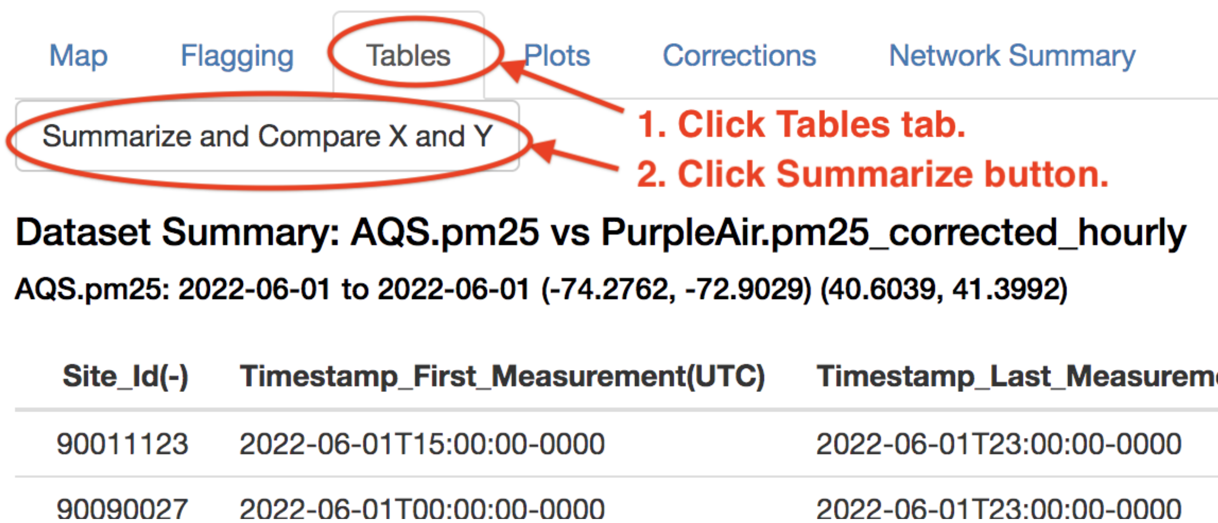


11.10 **Select the Flagging tab**.

11.11 Select the following flags to be applied to Dataset Y (PurpleAir.pm25_corrected_hourly).

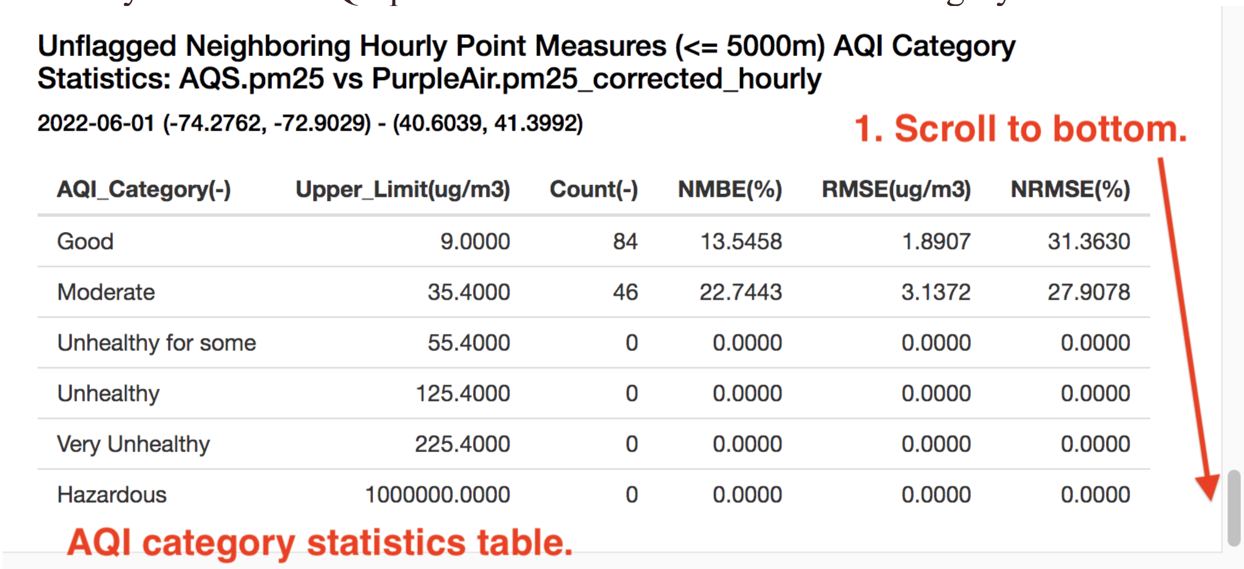


11.12 **Select Tables tab** and click the **Summarize** button.



11.13 **Scroll to the bottom of the page** to see the **AQI Category Statistics** summary that compares each AQS measure with its *unflagged* PurpleAir neighbor measures. Looking at the first row, we see there are 84 AQS paired measures not exceeding 9 ug/m3 which is *Good* category and PurpleAir neighboring measures differences are summarized by the statistics: Normalized Mean Bias Error, etc.

Similarly for the 46 AQS paired measures in the Moderate category.

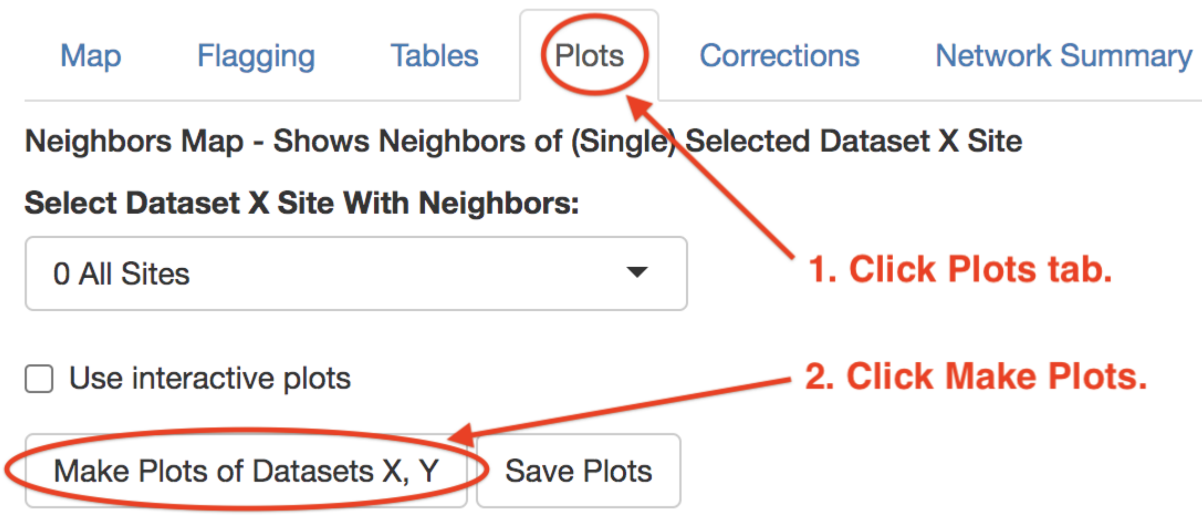


Unflagged Neighboring Hourly Point Measures (<= 5000m) AQI Category Statistics: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

2022-06-01 (-74.2762, -72.9029) - (40.6039, 41.3992)

AQI_Category(-)	Upper_Limit(ug/m3)	Count(-)	NMBE(%)	RMSE(ug/m3)	NRMSE(%)
Good	9.0000	84	13.5458	1.8907	31.3630
Moderate	35.4000	46	22.7443	3.1372	27.9078
Unhealthy for some	55.4000	0	0.0000	0.0000	0.0000
Unhealthy	125.4000	0	0.0000	0.0000	0.0000
Very Unhealthy	225.4000	0	0.0000	0.0000	0.0000
Hazardous	1000000.0000	0	0.0000	0.0000	0.0000

11.14 **Select the Plots tab and Make Plots button**.

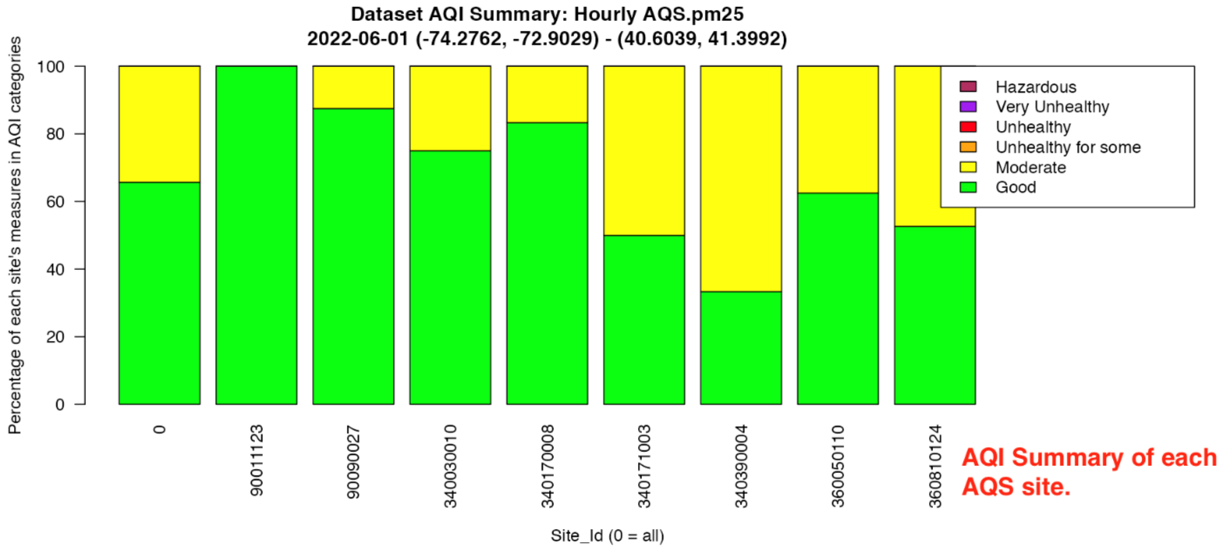


11.15 **Scroll to the 2nd plot to see the AQI Summary Plot of AQS PM25**.

This shows the percentage each site's measures in each AQI category.

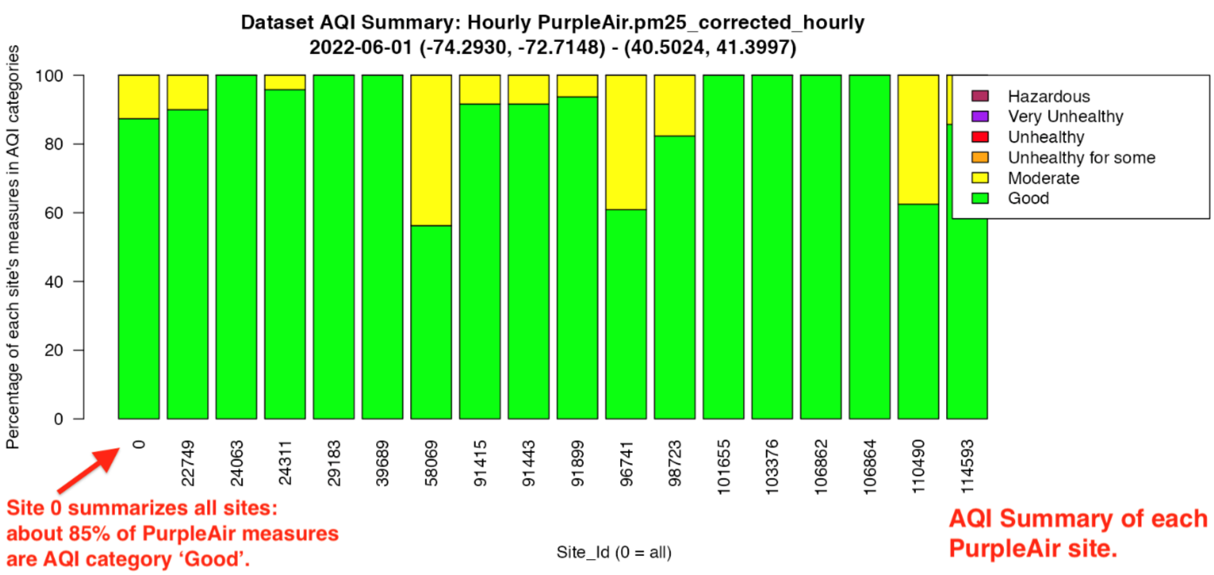
Site 0 is a summary of all sites. We see that site # 90011123 measures are all in the Good category, while overall (labeled site 0) about 65% of all AQS measures are in the Good category with the remainder in the Moderate category.

.



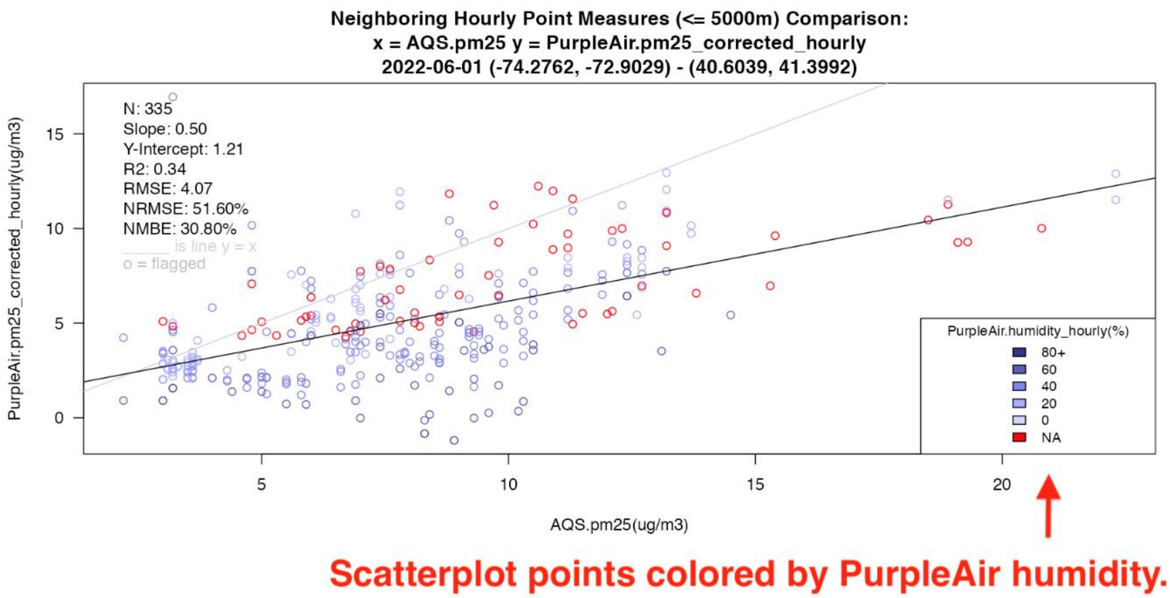
11.16 **Scroll to the 5th plot** to see the AQI Summary Plot of PurpleAir PM25.

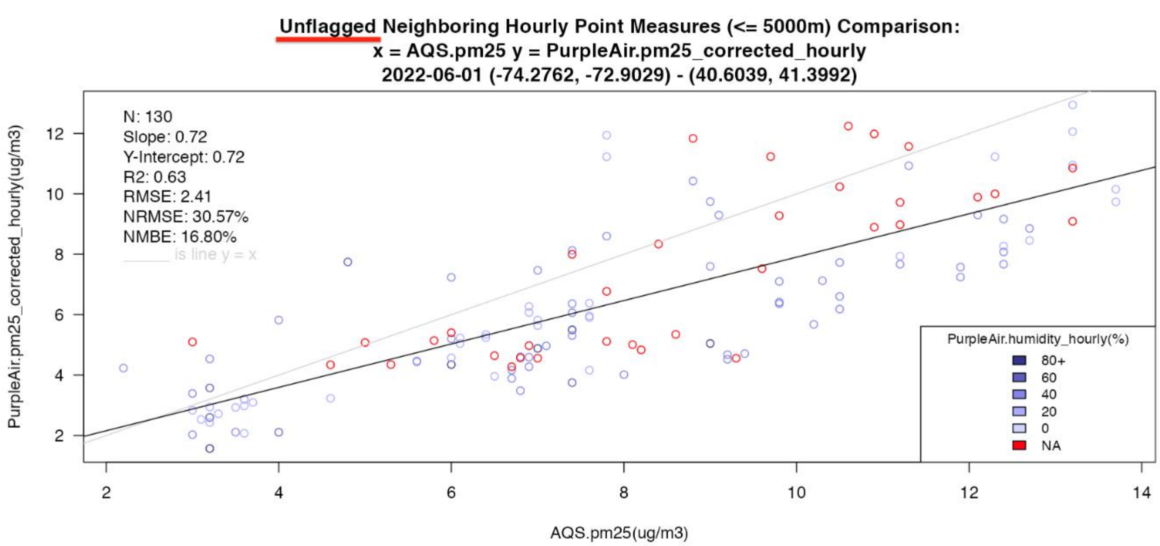
This shows that, overall, about 85% of PurpleAir unflagged measures are in the Good category with the remainder in the Moderate category.



11.17 **Scroll down to the scatterplots.** The top one shows all neighbor pair measures AQS vs PurpleAir) and the bottom one shows only unflagged neighbors.

These points are *colored by PurpleAir humidity* when there is a coincident PurpleAir humidity measure. Not all PurpleAir points had a humidity measure and in those cases the point is red to indicate NA (not available).





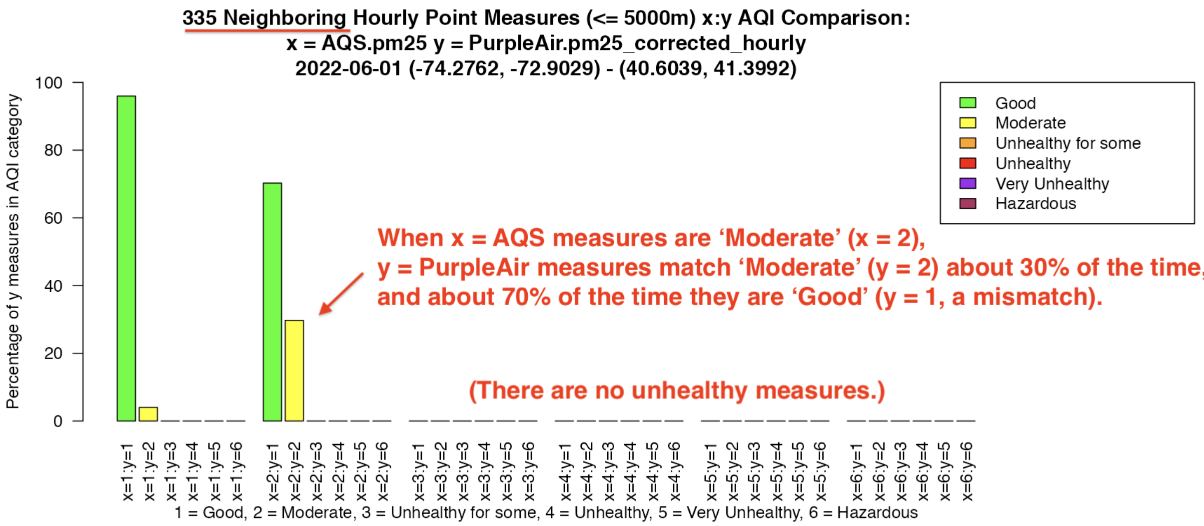
11.18 The next pair of plots compare the AQI categories of neighboring measurement pairs - in this case, x = AQS PM2.5 vs y = PurpleAir PM2.5 - with the bottom plot showing unflagged PurpleAir measures.

There are 6 AQI categories numbered from 1=Good, 2=Moderate ... 6=Hazardous.

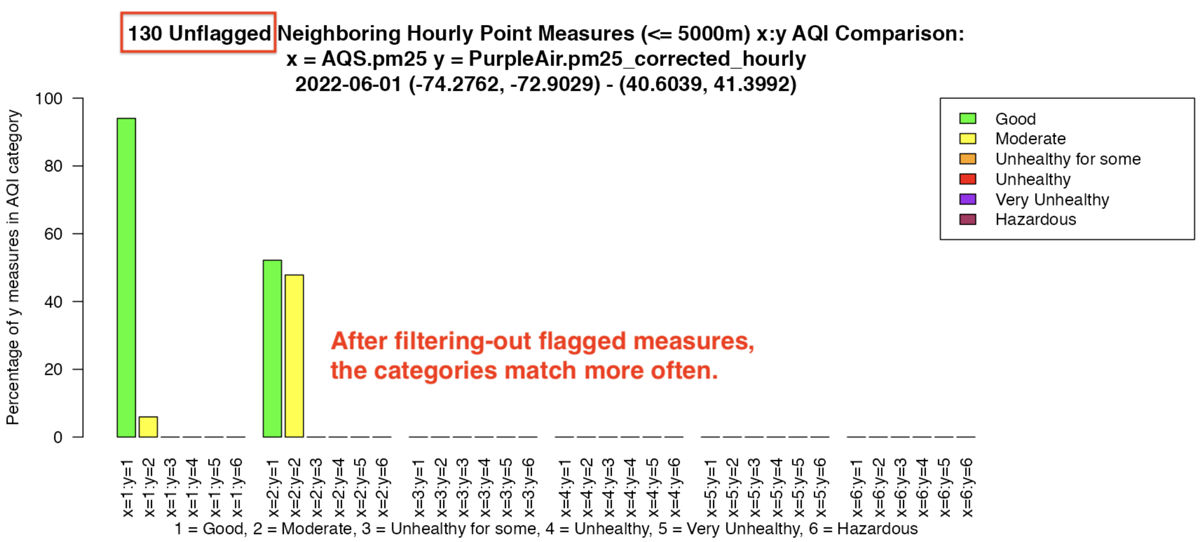
The left-most green bar labelled *x=1:y=1* indicates that when AQS measures are *1=Good*, 95% of PurpleAir measures are *1=Good* (matched) and about 5% of PurpleAir measures are *2=Moderate* (2nd yellow bar labelled x=1:y=2).

Next, *x=2:y=1* means when AQS is *2=Moderate*, about 70% of PurpleAir measures are *1=Good* (unmatched) with the remaining 30% *2=Moderate* (matched). There are no measures in the unhealthy categories.

The bottom plot shows the *unflagged* neighboring measure comparisons.
Of course, after filtering-out various mismatches, the remaining paired measures match categories better. In this example, there are no instances of mismatches differing by more than one category.

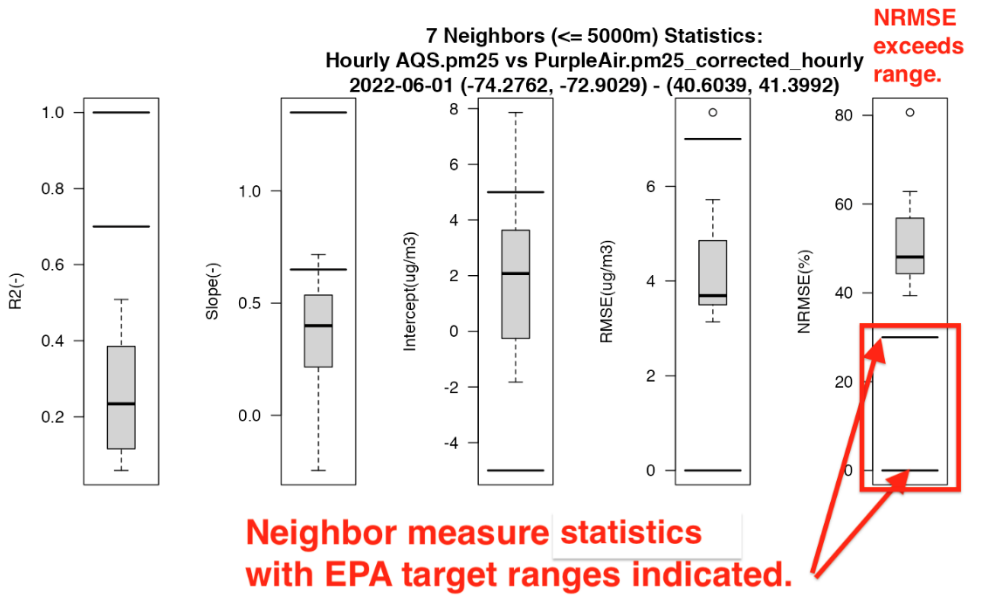


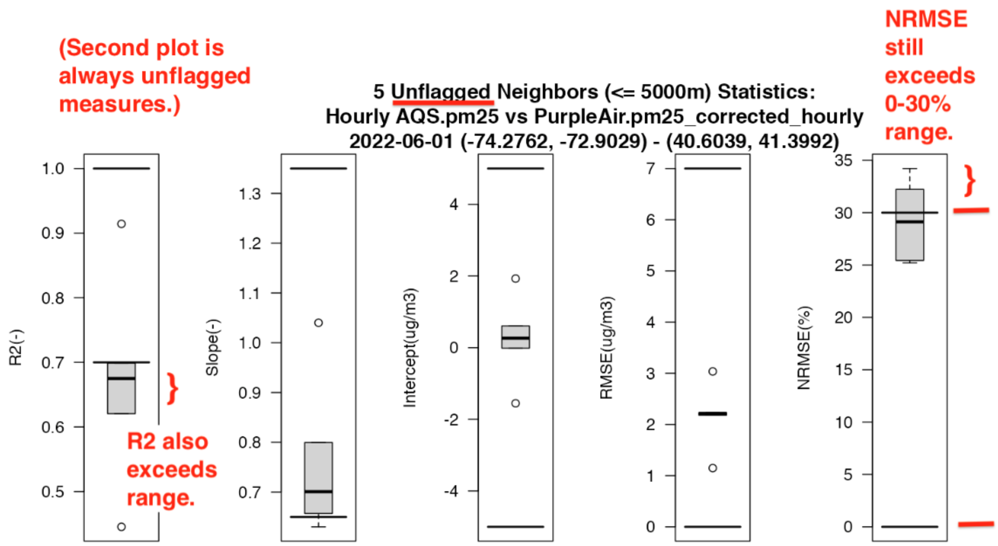
Neighboring measures AQI Comparison



11.19 **Scroll to the bottom of the page** to see the Neighbor Statistics Plots.

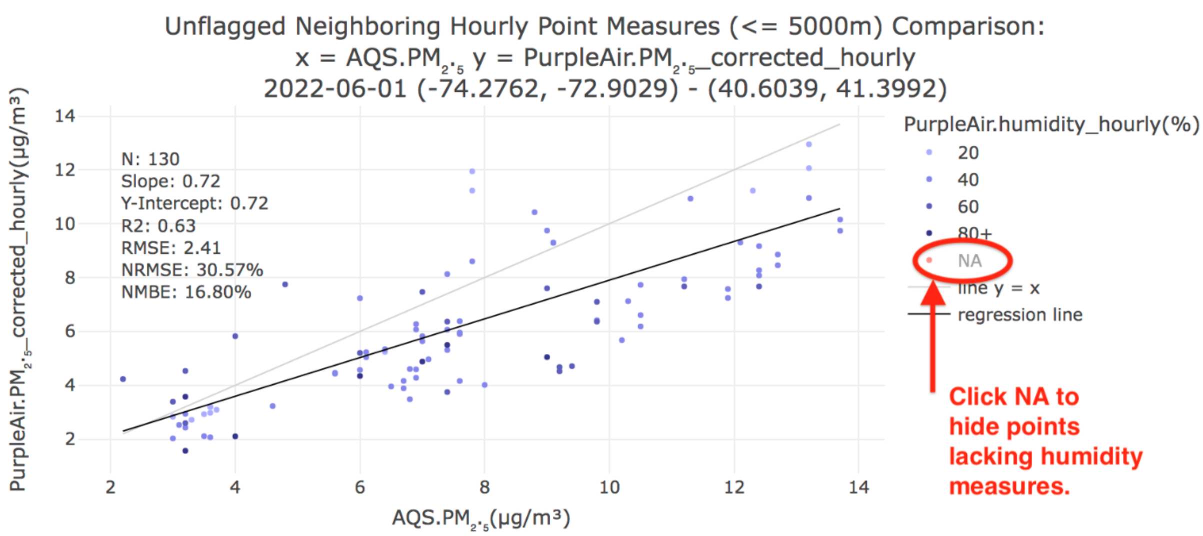
These plots show several statistics for each set of neighbor measures.
As before, the bottom plot shows only unflagged neighbor measures.
If the variable has an EPA target range (currently PM1, PM2.5, PM10, O3, NO2, SO2, CO) for each statistic then that range is indicated with the bold lines.
For example, The Normalized Root Mean Square Error target range for PM2.5 is [0, 30%] and both plots show exceedance. They both show exceedance of the R-squared statistic target range [0.7, 1].


7 Neighbors (<= 5000m) Statistics:
Hourly AQS.pm25 vs PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.2762, -72.9029) - (40.6039, 41.3992)
NRMSE exceeds range.
R2(-)
Slope(-)
Intercept(ug/m3)
RMSE(ug/m3)
NRMSE(%)
Neighbor measure statistics with EPA target ranges indicated.

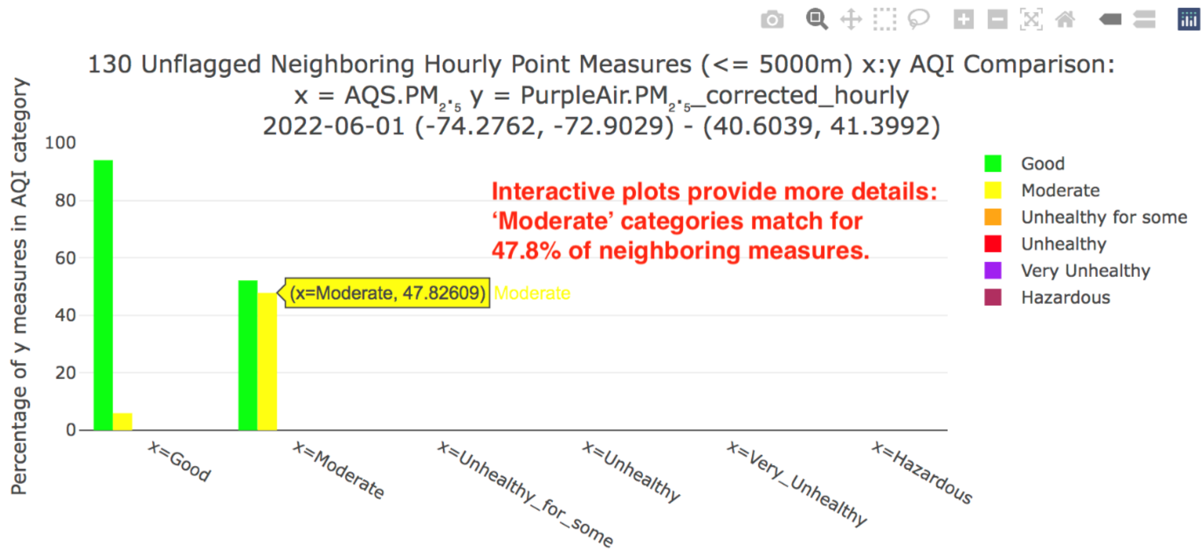

(Second plot is always unflagged measures.)
5 Unflagged Neighbors (<= 5000m) Statistics:
Hourly AQS.pm25 vs PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.2762, -72.9029) - (40.6039, 41.3992)
NRMSE still exceeds 0-30% range.
R2(-)
Slope(-)
Intercept(ug/m3)
RMSE(ug/m3)
NRMSE(%)
R2 also exceeds range.

11.20 The Interactive Plots provide more detail.
Click the Use interactive plots button and then **click the Make Plots button**.

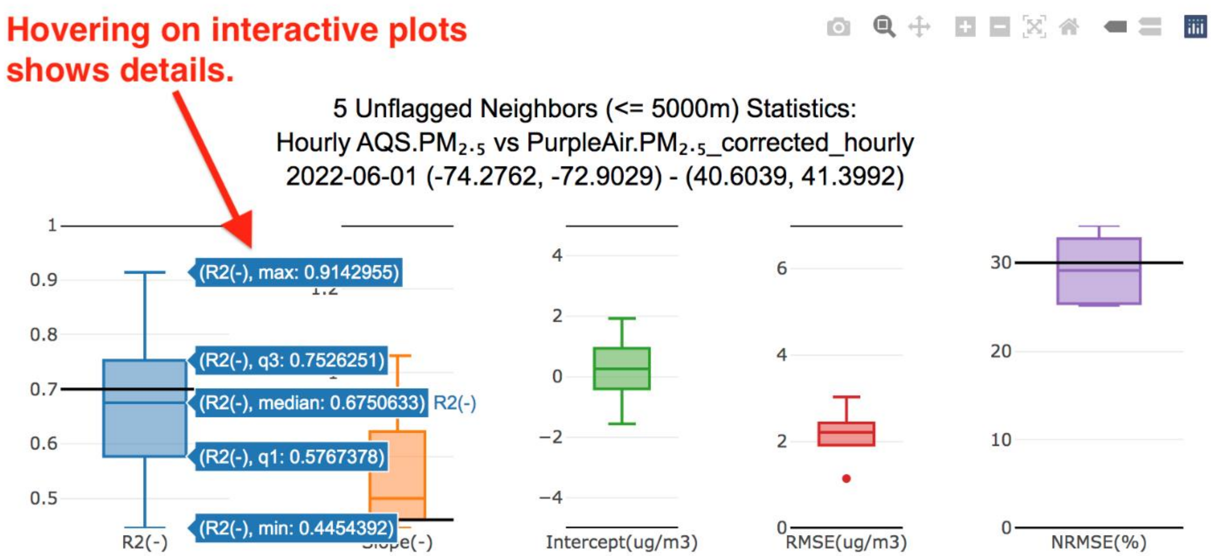
11.21 **Scroll down to the 2nd scatterplot**. Then **click on NA in the legend** to hide the red points that indicate missing humidity. Click again to toggle it or click other values such as 80+ to hide points with humidity greater than 80%. Moving the mouse over points will display their measures as (AQS, PurpleAir), humidity.



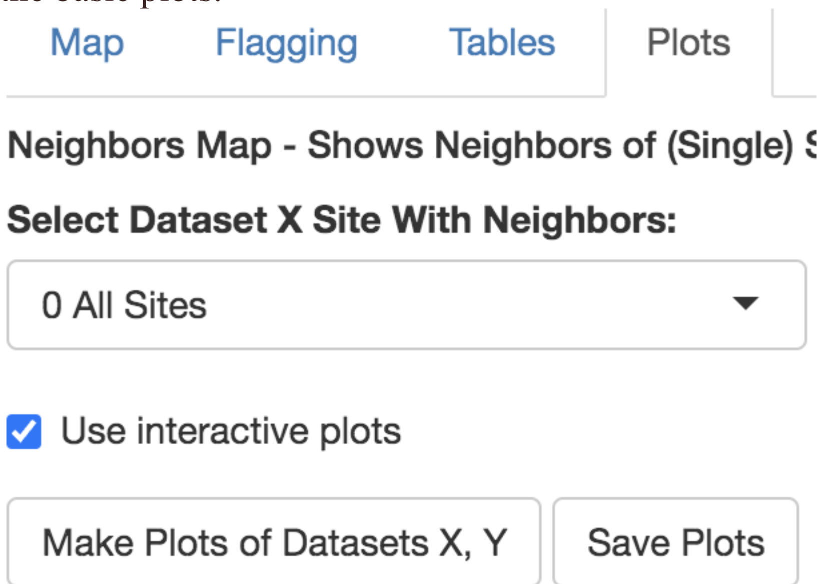
11.22 Below the scatterplots are the **Interactive AQI Comparison** plots. These are labelled more clearly and moving the mouse pointer over the bars shows numeric values. For example, in the bottom plot of *unflagged* neighboring measures, when AQS is *Moderate*, 47.8% of PurpleAir measures are *Moderate* (matched) and 52.2% are *Good* (mismatched by one category).



11.23 **Scroll to the bottom of the page** to see the **Interactive Neighbor Statistics Plots**. Moving the mouse pointer over a boxplot will show the numeric values of the quartiles.



11.24 **Scroll to the top of the page** and click **Save Plots** if desired. Note these will be the basic plots.



To save the interactive plots **click the camera icon** in each plot's menu.

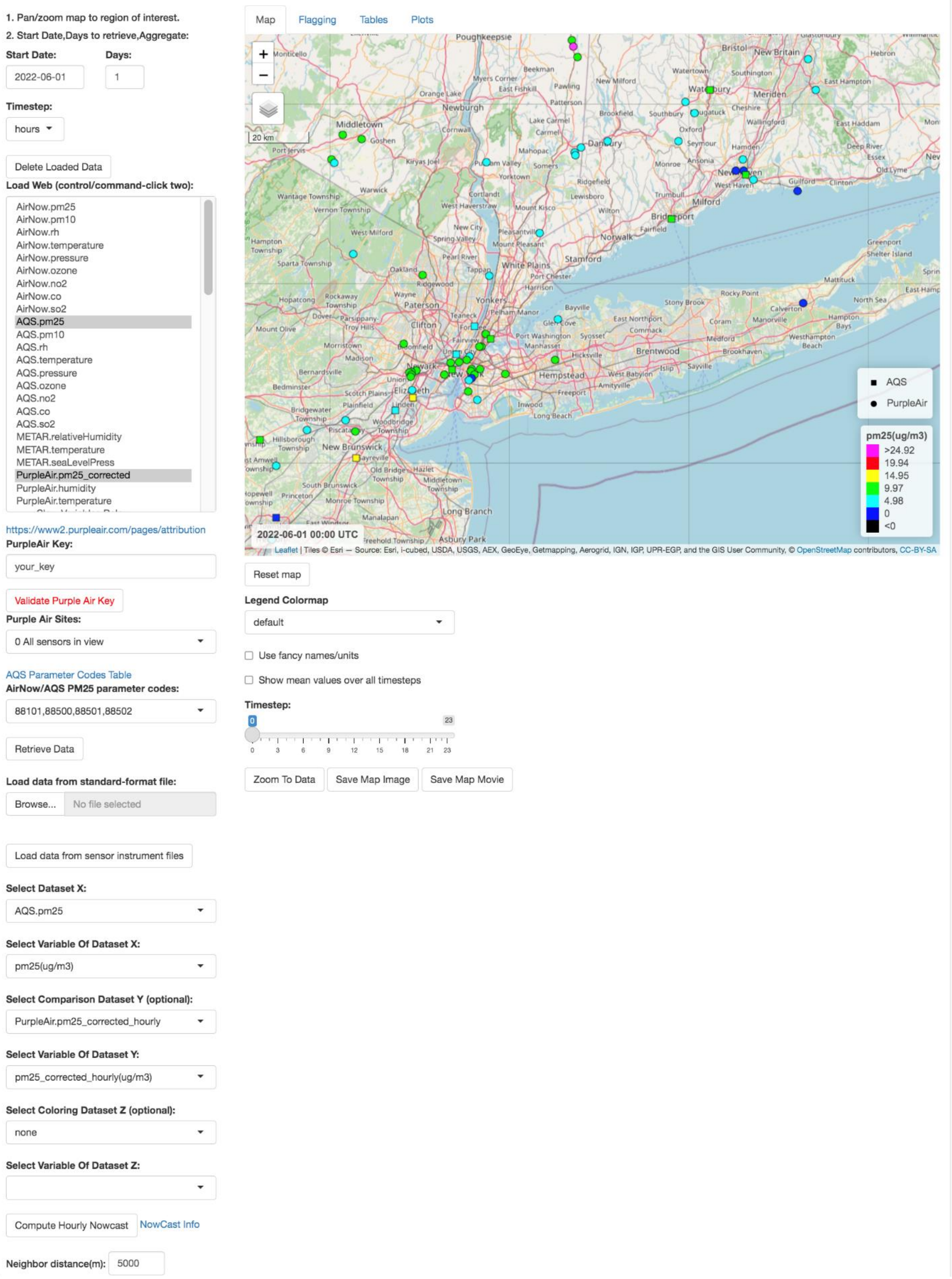
11.25 **Click the Quit button** to exit ASNAT.

Demo 12: Nowcast

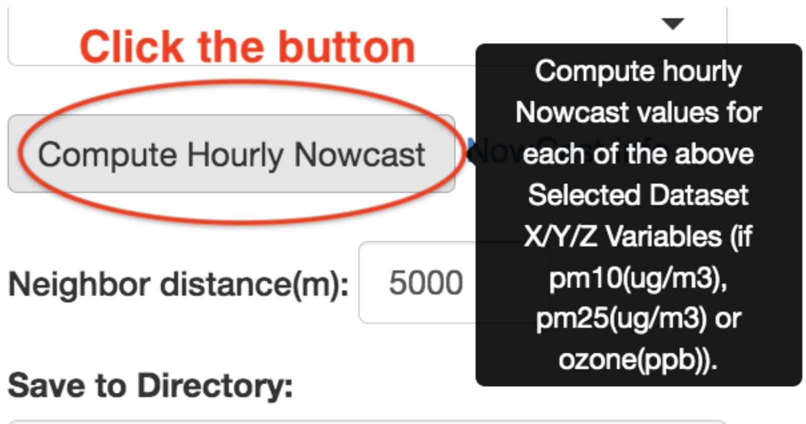
EPA's Nowcast algorithm computes an hourly ozone, PM10 or PM2.5 concentration estimate based on a number of preceding hourly measures (window).
Mostly used to forecast a daily AQI based on yesterday's hourly measures, it can be used for any sequence of hourly values for a dataset loaded into ASNAT, provided the selected variable is one of: ozone(ppb), pm10(ug/m3), pm25(ug/m3). For ozone the window is 8 hours with no minimum weight factor. For PM the window is 12 hours with a minimum weight factor of 0.5. Note that the Nowcast hourly concentrations cannot be computed until the window of preceding measures are available therefore the value NA will be used for the first window hours.
Detailed descriptions of the Nowcast algorithm are available from these websites:
https://en.wikipedia.org/wiki/NowCast_(air_quality_index)
https://www.epa.gov/sites/default/files/2018-01/documents/nowcastfactsheet.pdf
There are different implementations of Nowcast code. ASNAT's implementation is based on code from jonathan.s.callahan@gmail.com downloaded on 2025-02-07 from:
https://rdrr.io/cran/AirMonitor/src/R/monitor_nowcast.R

To demo Nowcast:
12.1 **Quit/restart** ASNAT and load the same datasets as in Demo 1:
Reset map
Start Date: 2022-06-01
Days: 1
Timestep: hours
Delete Loaded Data
Load Web:
AQS.pm25
PurpleAir.pm25_corrected
Purple Air Key:
your_key
Retrieve Data
Select Comparison Dataset Y:
PurpleAir.pm25_corrected_hourly
Neighbor Distance(m): 5000

ASNAT should appear as the image below.



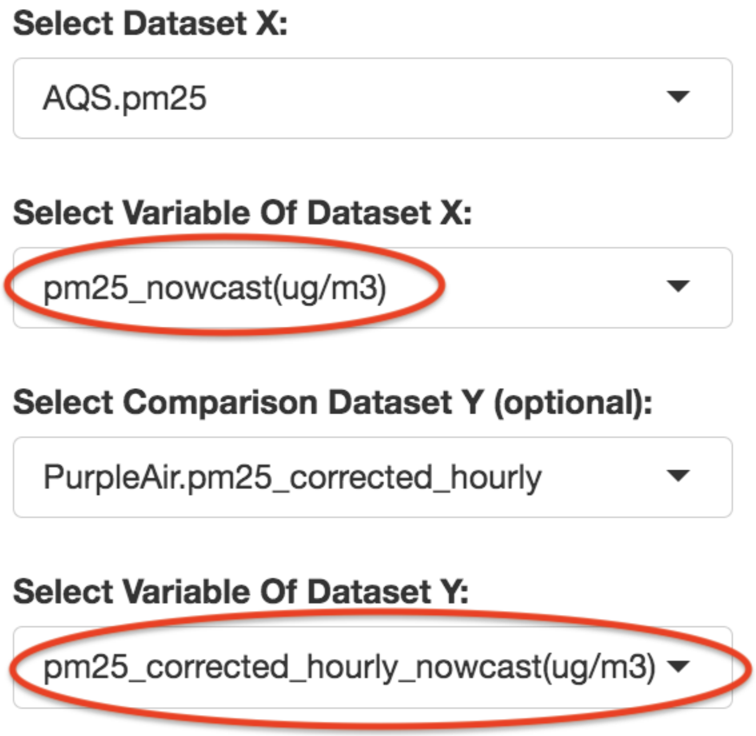
12.2 **Click the Compute Hourly Nowcast button**.
Hovering over the button will pop-up a tooltip explaining that hourly Nowcast values will be computed for each of the selected Dataset variables - provided they are compatible with the Nowcast algorithm (including units).



12.3 The message area will report that the new **_nowcast** variables have been added to their datasets:

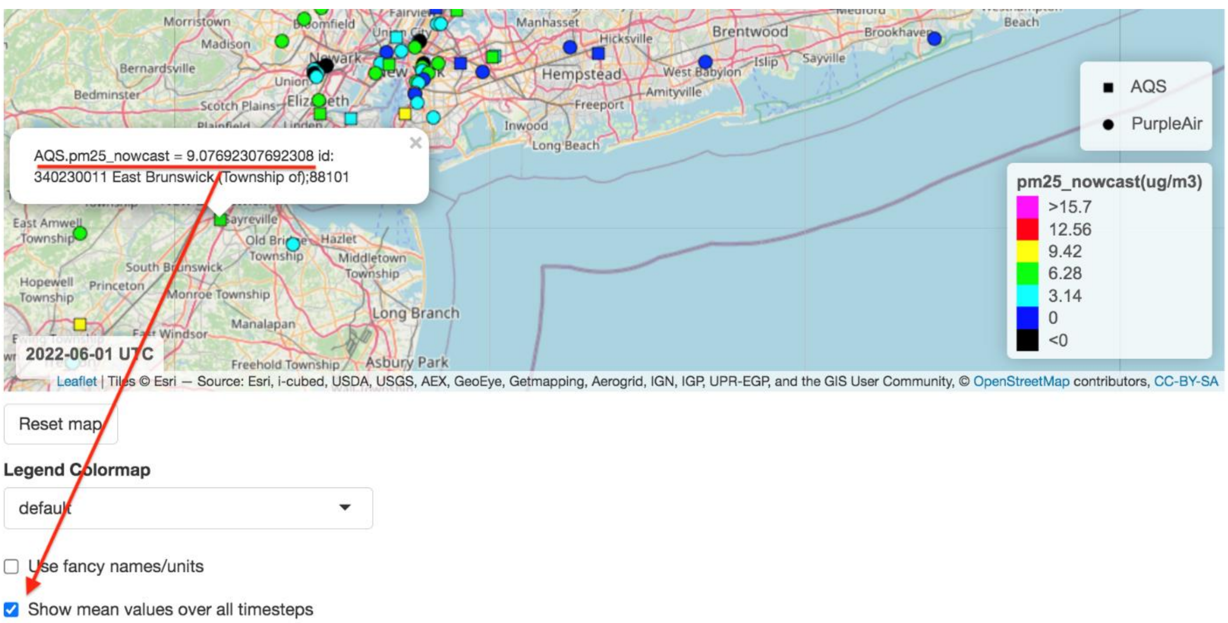
```
Added variable AQS.pm25_nowcast(ug/m3)
Added variable PurpleAir.pm25_corrected_hourly_nowcast(ug/m3)
```

12.4 Select these **_nowcast** variables from the menus:

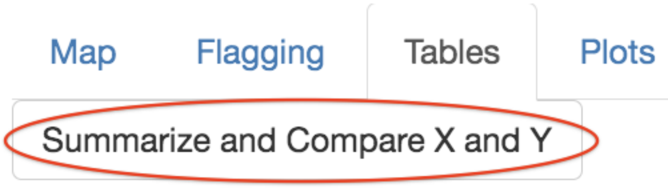


12.5 Note that the map shows the nowcast measures as **black indicating NA** since the Nowcast hourly value is not available for the first window = 12 hours.

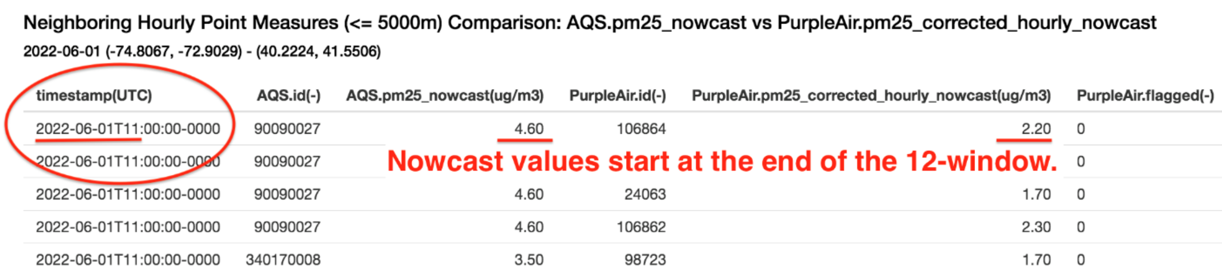
12.6 You may **click the checkbox *Show mean values over all timesteps*** to see the means instead.



12.7 **Click the *Summarize and Compare X and Y*** button to compute tables.



12.8 Note that in the Comparison Table, Nowcast values start at hour 1, the end of the 12-hour window for PM.

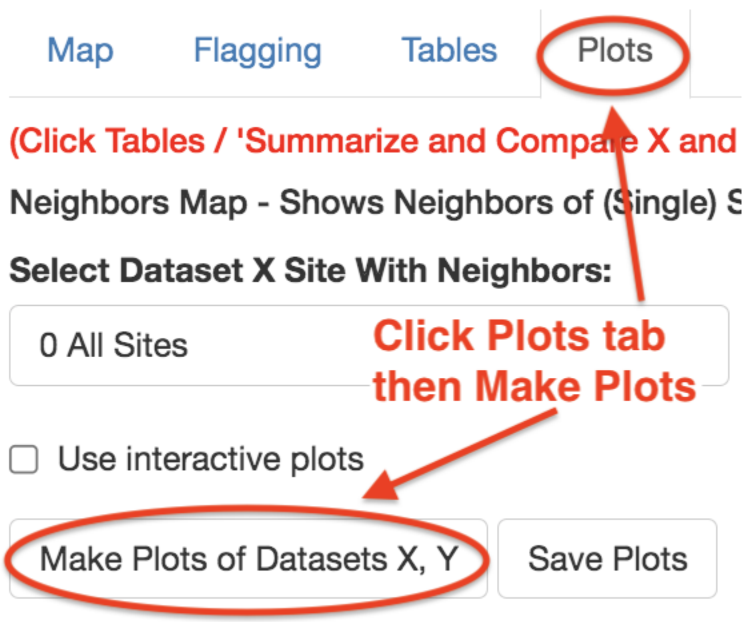


Neighboring Hourly Point Measures (<= 5000m) Comparison: AQS.pm25_nowcast vs PurpleAir.pm25_corrected_hourly_nowcast
2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

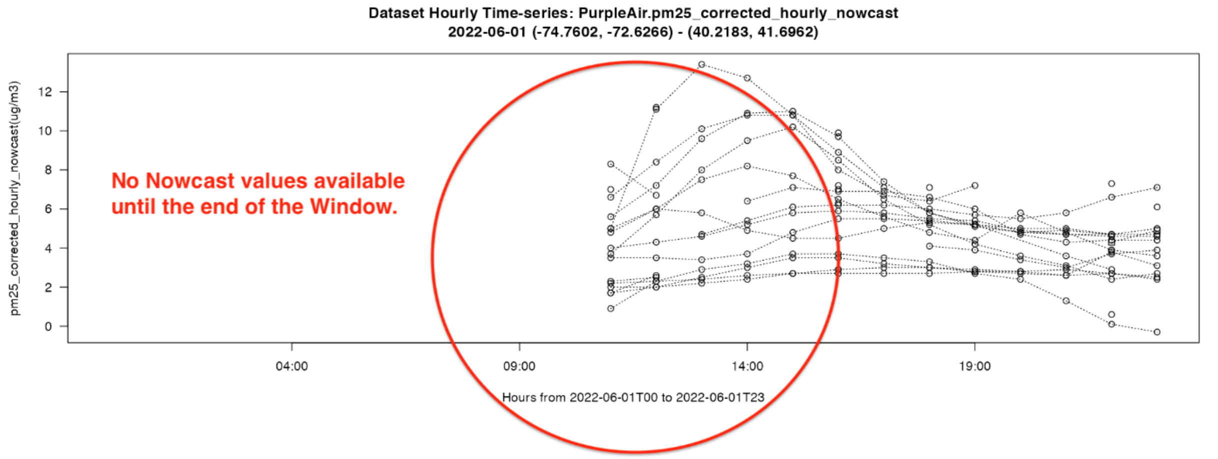
timestamp(UTC)	AQS.id(-)	AQS.pm25_nowcast(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly_nowcast(ug/m3)	PurpleAir.flagged(-)
2022-06-01T11:00:00-0000	90090027	4.60	106864	2.20	0
2022-06-01T11:00:00-0000	90090027				0
2022-06-01T11:00:00-0000	90090027	4.60	24063	1.70	0
2022-06-01T11:00:00-0000	90090027	4.60	106862	2.30	0
2022-06-01T11:00:00-0000	340170008	3.50	98723	1.70	0

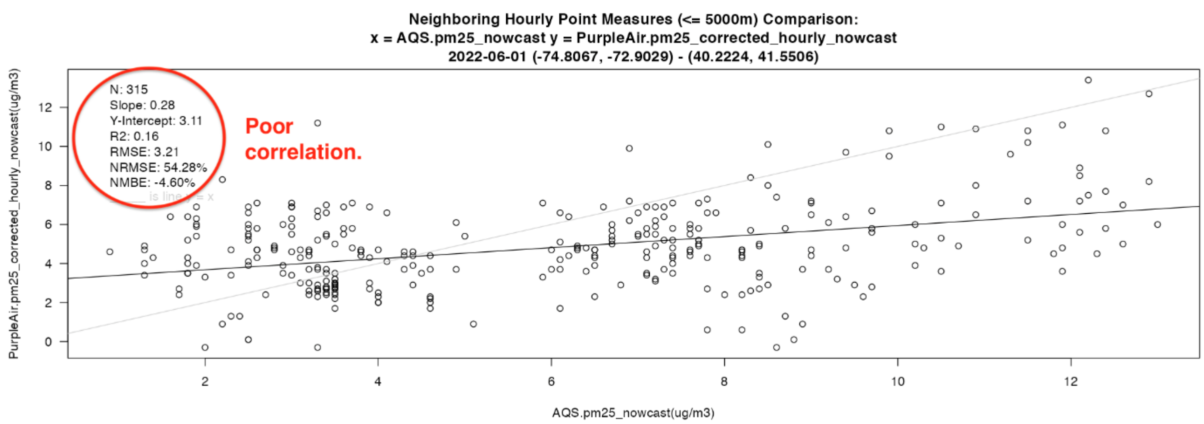
Nowcast values start at the end of the 12-window.

12.9 **Click Plots tab** then **click Make Plots** button:

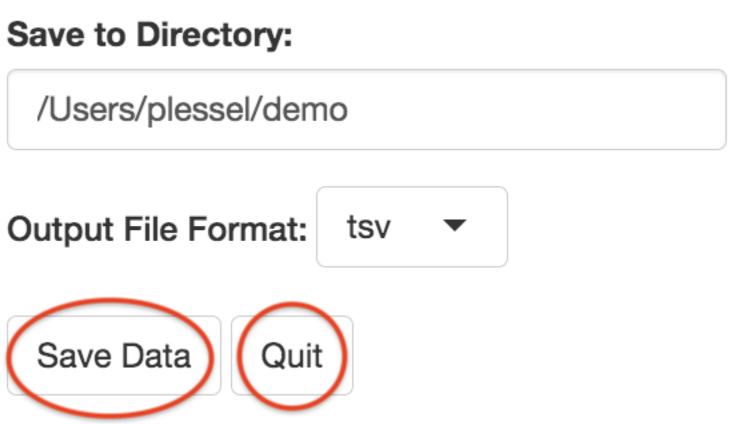


12.10: **Scroll down to the PurpleAir time-series.** Note that there are no Nowcast values available until hour 11 (the end of the window). Also, note the poor correlation.





12.11: **Click the Save Data button** to save the datasets with the new _nowcast variables and also save the tables.
Then **click the Quit button**.

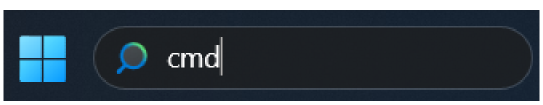


Demo 13: Spatial Filtering

Data retrieved from webservices will be for the current date range and viewing area shown on the map. Data loaded from files will be at whatever their time and locations are. (Click the Zoom To Data button to see data loaded from files.) ASNAT can also filter data *that has already been loading into memory* by selecting one or more US states or counties and click the 'Filter Data To Match Above Places' button. The *filtering will be applied to all datasets in memory* and such filtered data can be plotted and saved to files as usual.

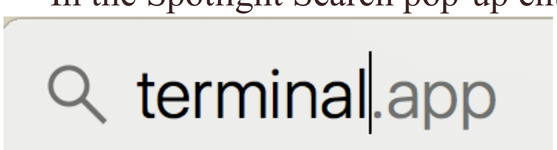
This feature relies on externally developed software and data files containing place names and polygon boundaries that must be downloaded into data/tmp and loaded into ASNAT when it starts. This can take a few minutes the first time ASNAT is run. It is best to re-run the install program to be sure that all needed software and data is installed before running ASNAT. *This only needs to be done once successfully* - after that simply run ASNAT.
Optional but recommended: Recall the earlier instructions for installing:

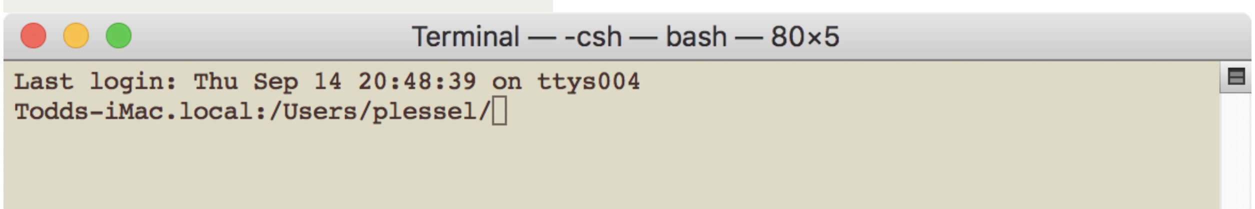
13.1. Open a shell window.
a. On Windows: this is called `Command Prompt (CMD)`.
Enter **`CMD`** in the Search area at the bottom of the screen.



b. On Mac it is called `Terminal`.
Click the magnifying glass at the upper-right corner of the screen.

In the Spotlight Search pop-up enter **`terminal`**

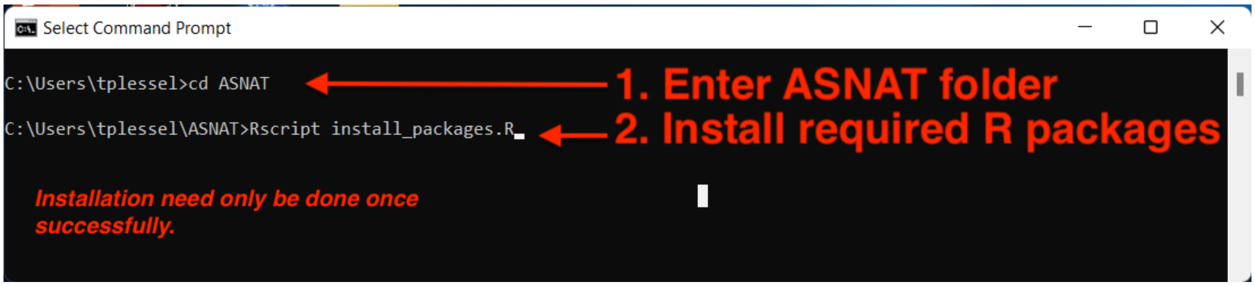




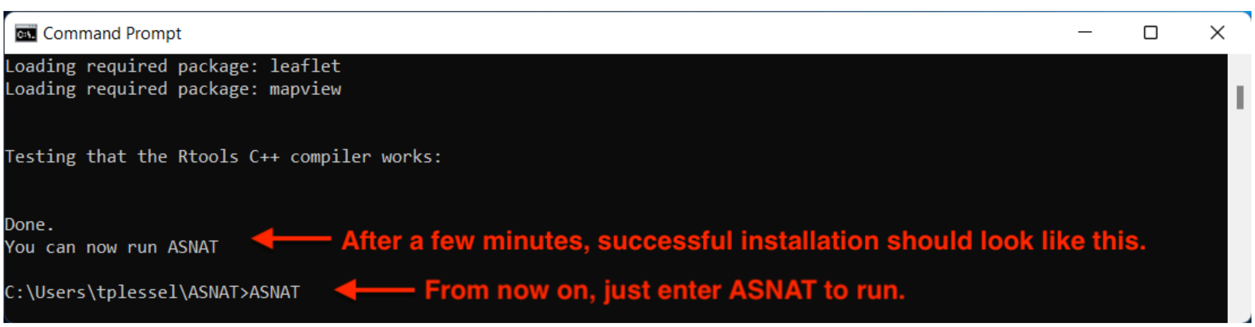
c. On Linux it is called `xterm`. Click right-mouse button then select Shell.

13.2 Go to the ASNAT directory and use `Rscript` to install the various R packages that are required by ASNAT:

```
cd ASNAT
Rscript install_packages.R
```



It will take several minutes to install the many R packages needed by ASNAT. Check the output messages for any errors such as missing .h files or the message:
`'RTools_is_installed_and_c_plus_plus_compiler_works()' is not TRUE`
The above error message means you must download and install *RTools*.
Ask your system administrator for assistance resolving any problems before attempting to run ASNAT.
This step only needs to be done once successfully. Afterwards just run ASNAT.



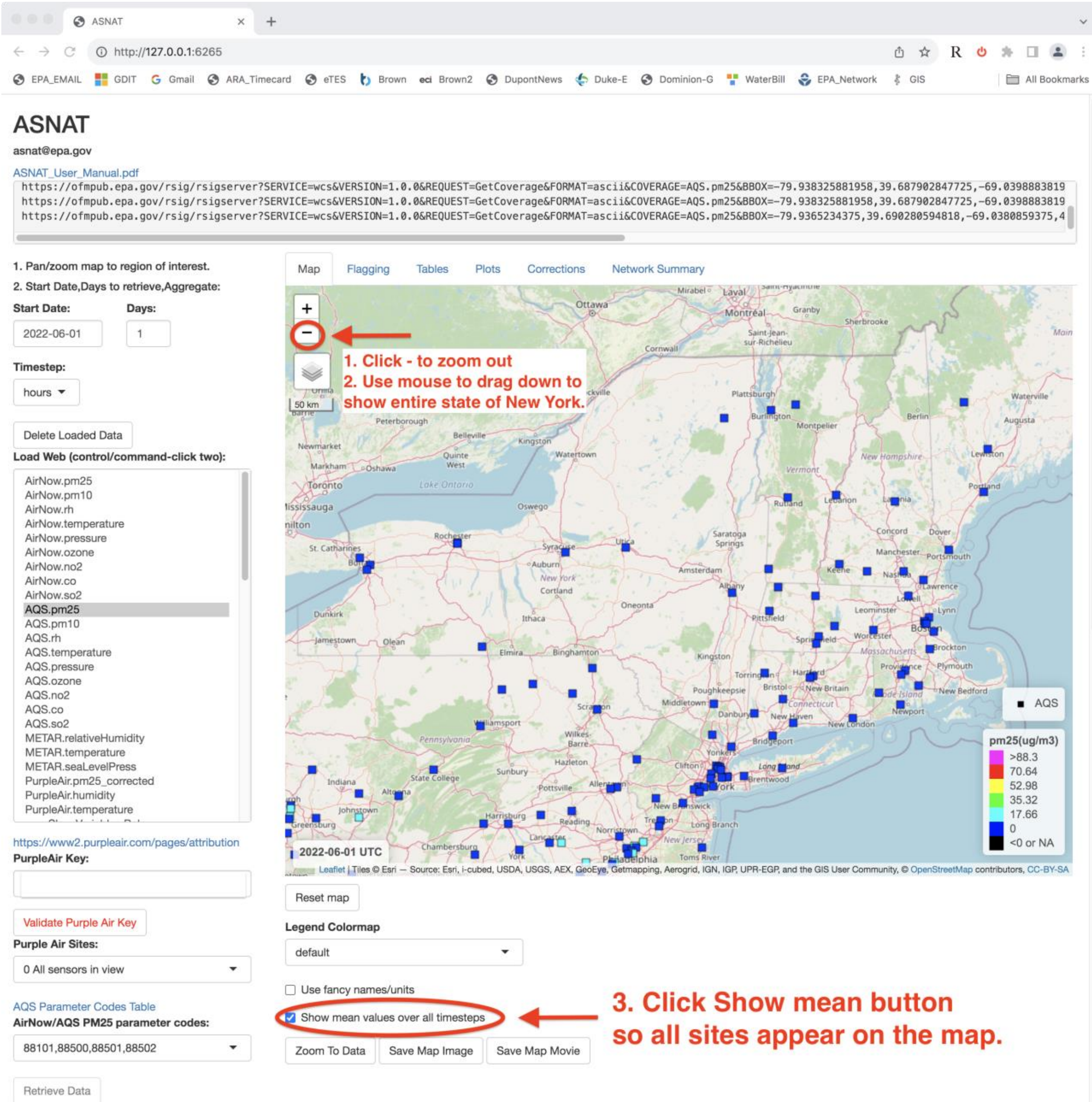
13.3 Run ASNAT. This will display the buttons and menus on the left and a map of NYC on the right.

```
ASNAT
```

13.4 To demo spatial filtering we'll load AQS.pm25 data for an area that includes the entire state of New York then filter it by NY state and then NY counties.

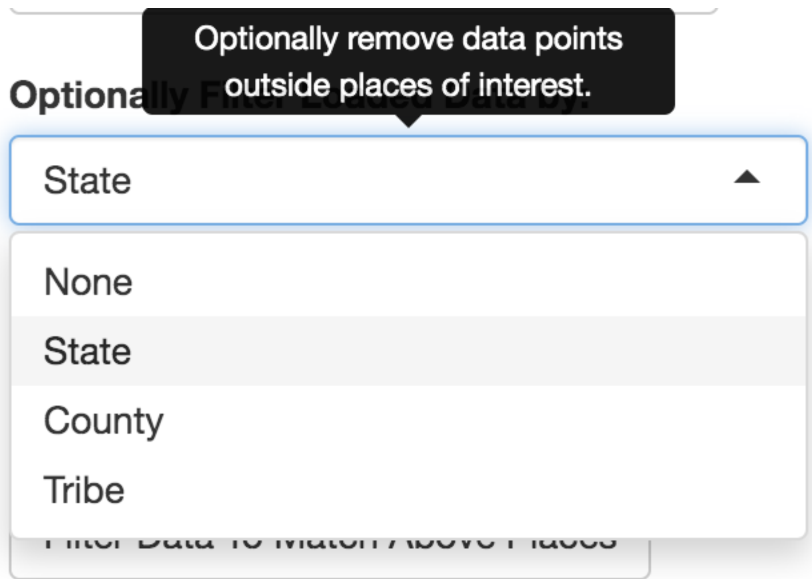
Quit/restart ASNAT and load one dataset:
Reset map
Click the '-' in the map to **zoom out**
Use the mouse to drag/**pan the map** to show the entire state of New York
Start Date: **2022-06-01** Days: **1**
Timestep: **hours**
Delete Loaded Data
Load Web:
  AQS.pm25
Retrieve Data
Click the 'Show mean values over all timesteps' button.

ASNAT should appear as shown in the next image.

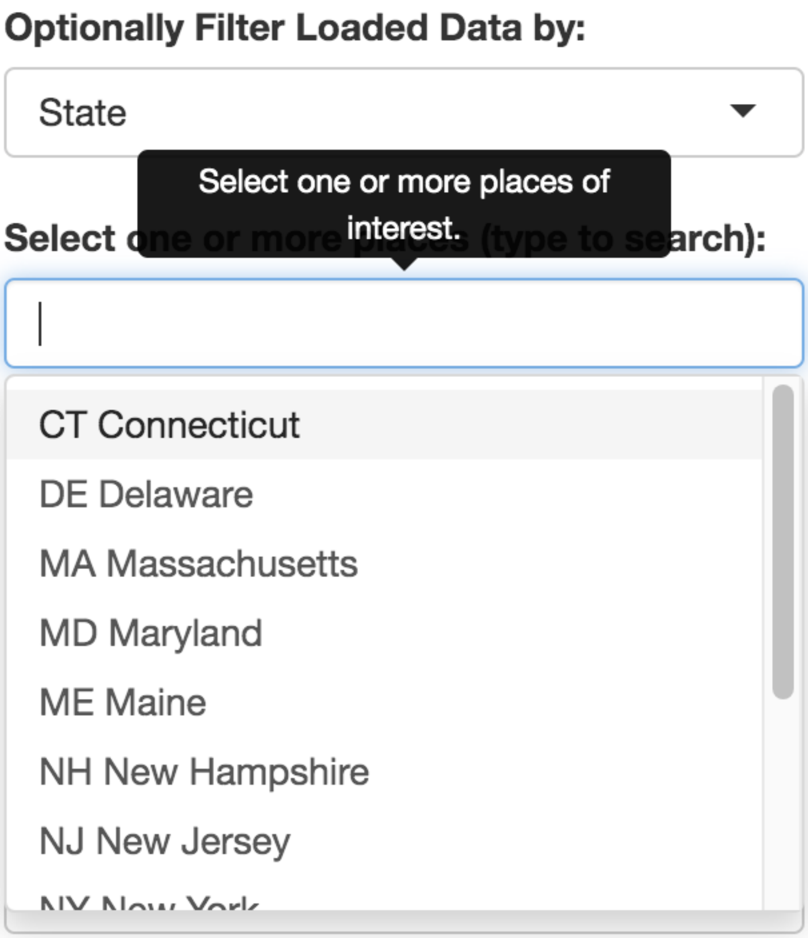


Note that the loaded data is for the entire viewing area - not just New York state.

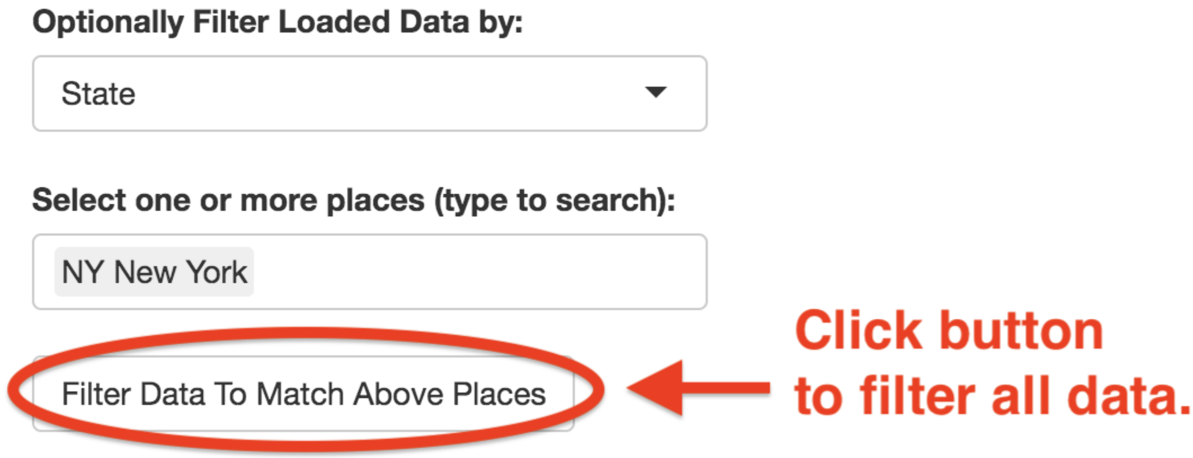
13.5 Scroll down to the '**Optionally Filter Loaded Data by**' menu and **select State**.



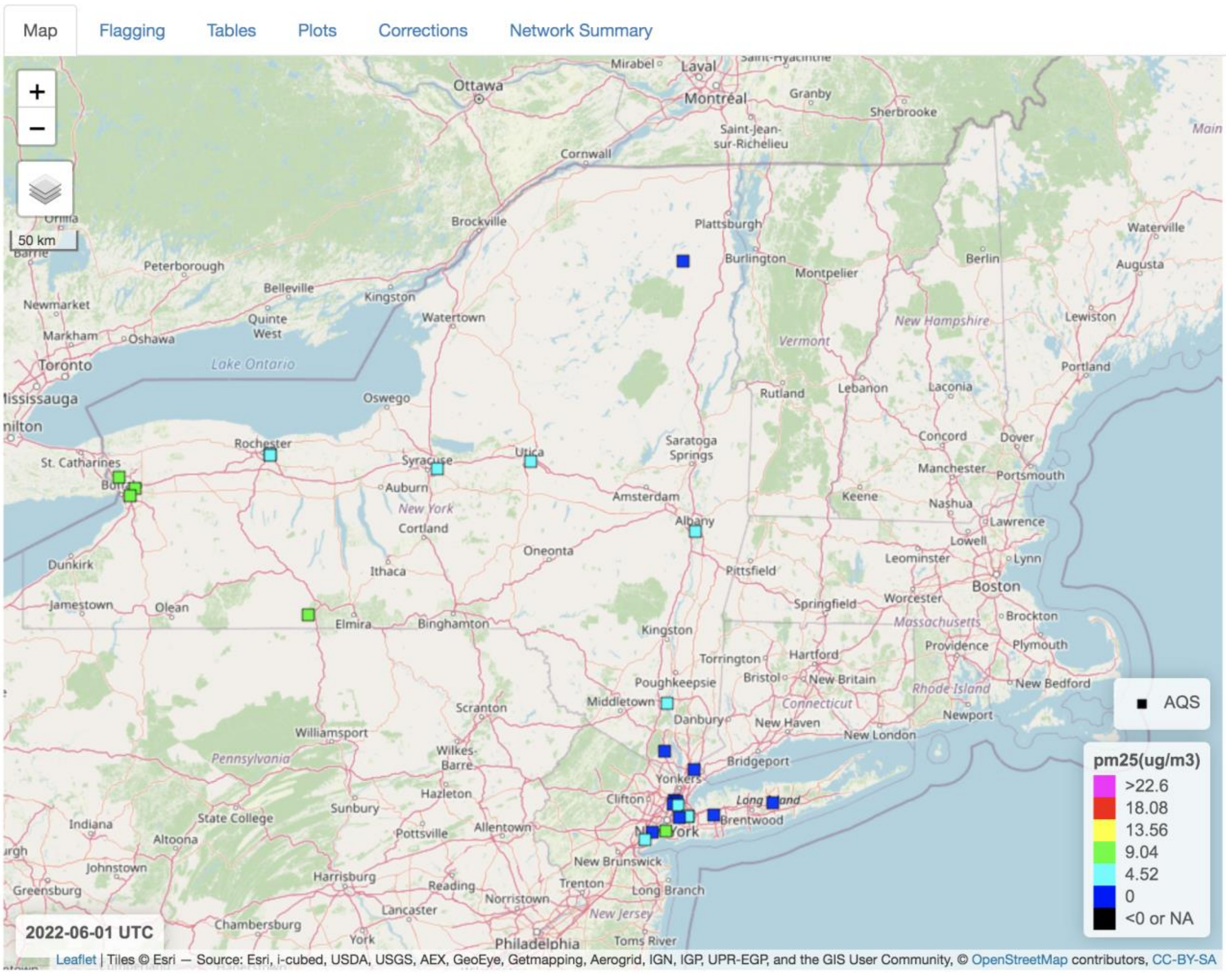
13.6 Click the menu '**Select one or more places (type to search)**'. Note that this menu lists places - *within the current view* - in a sorted scrollable/clickable list but also allows for *typing and searching*.



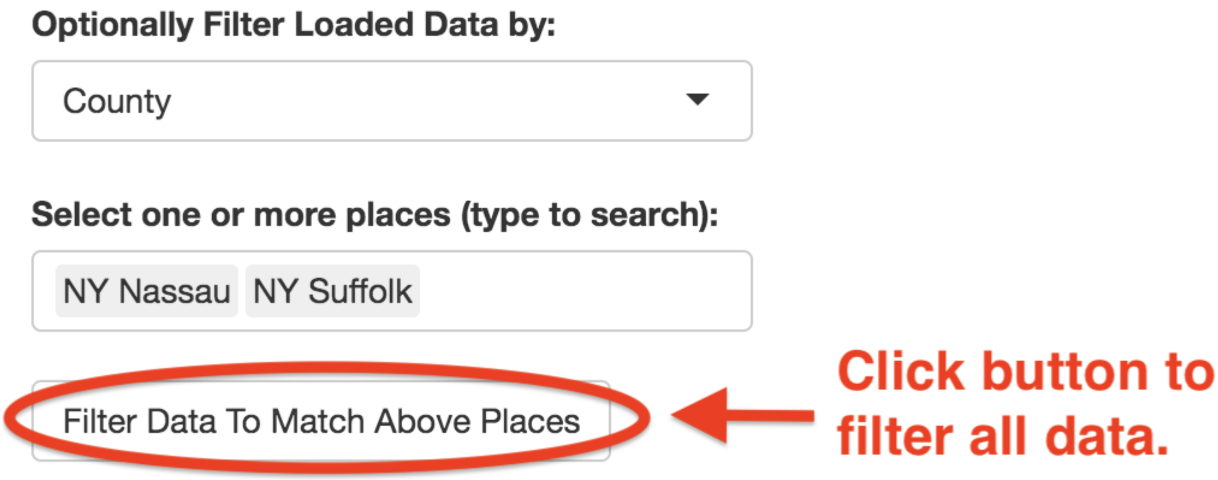
a. **Type ny** and notice that the menu now only lists **NY New York**.
c. Press the return/enter key to select NY New York.
This is faster and more convenient than scrolling through a potentially long menu.
d. **Click outside the menu in the blank area to the right** to finish the menu selection. This menu allows for *multiple* search and selections so clicking outside the menu when you are finished making selections is required.
e. **Click the 'Filter Data To Match Above Places' button**.
This will remove all data points that are outside the polygon(s) that define the border of the selected place(s) - in this example, New York. A given state or county can consist of multiple polygons such as islands.



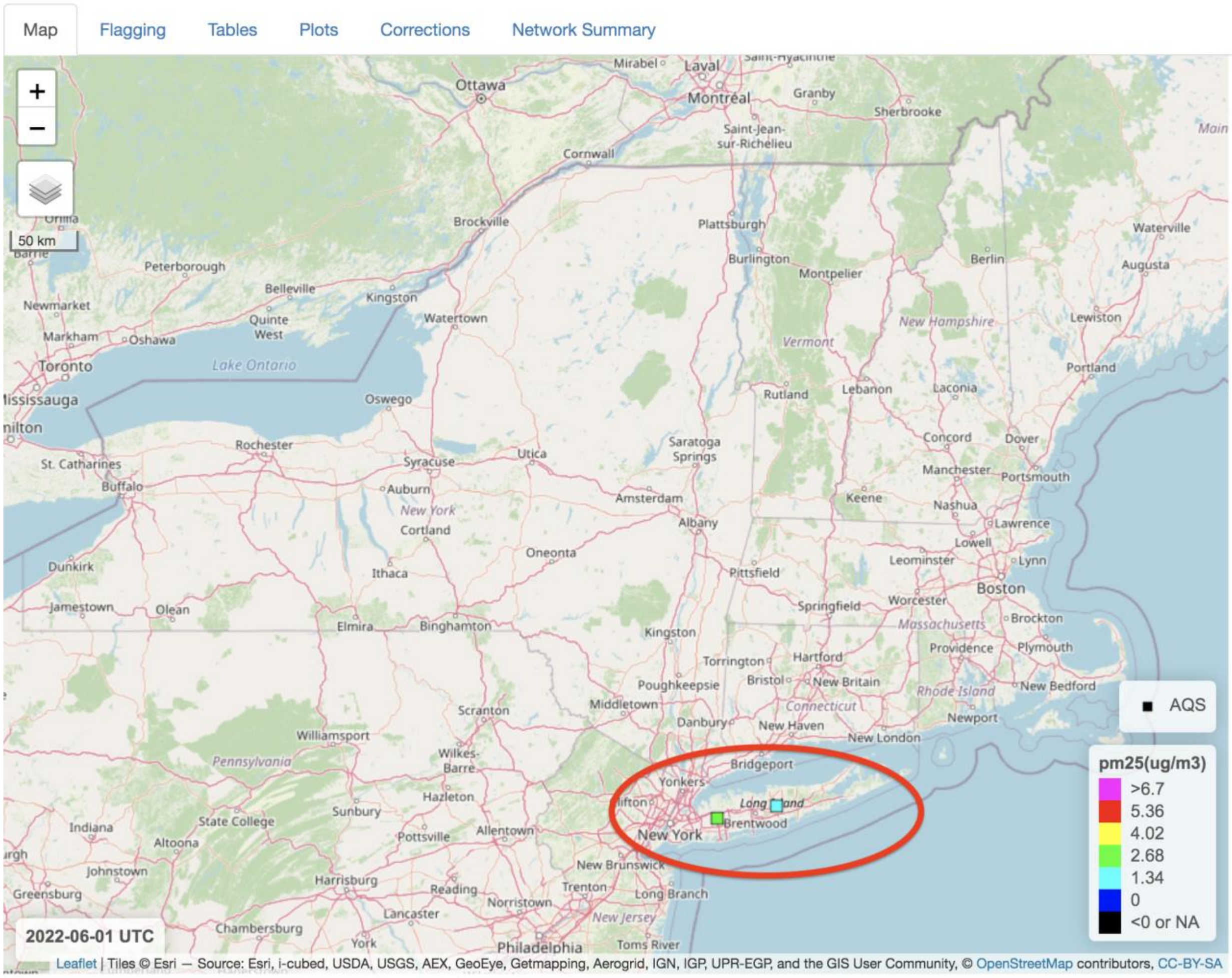
The map now shows the data points that remain - only those inside the state of New York.



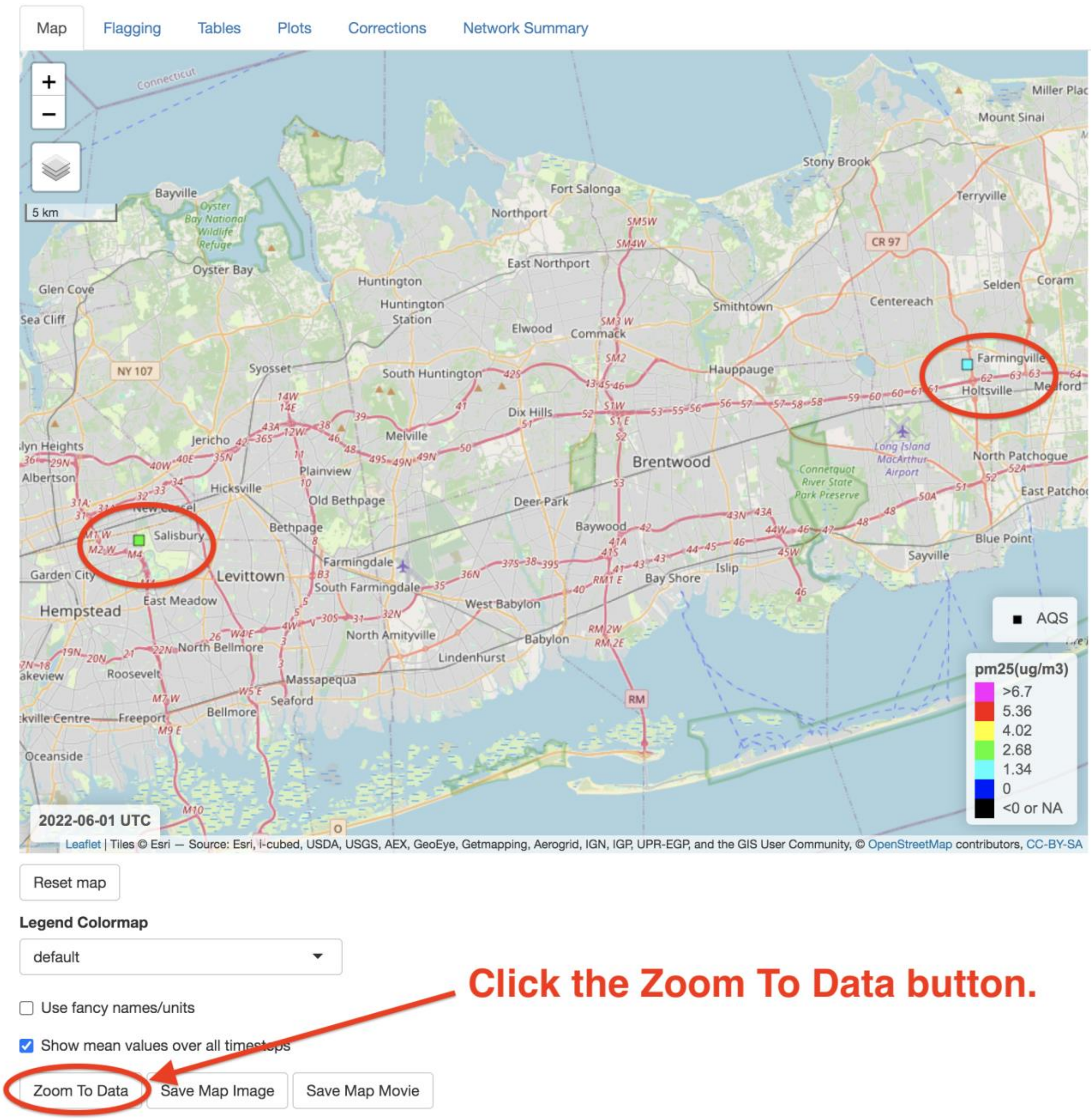
13.7 Now we will demonstrate multiple search/selections by further filtering the data by New York counties.
a. Click the menu '**Optionally Filter Loaded Data by**' and **select County**.
b. Click the menu '**Select one or more places**' menu and type
ny nas and press return/enter to select NY Nassau
then type **ny suf** to also select NY Suffolk.
c. **Click outside the menu** to finish the multiple menu selections.
d. **Click 'Filter Data To Match Above Places' button.**



This will leave only two sites on Long Island as shown.



13.8 **Click the 'Zoom To Data' button** to zoom the map to see these two sites. (Note that it, in this case, it would have been possible and easier to simply zoom the map to only Long Island and retrieve data for it rather than this filtering operation. However, not all desired places fit exclusively in a viewing area - e.g., filtering by entire state of New York.)



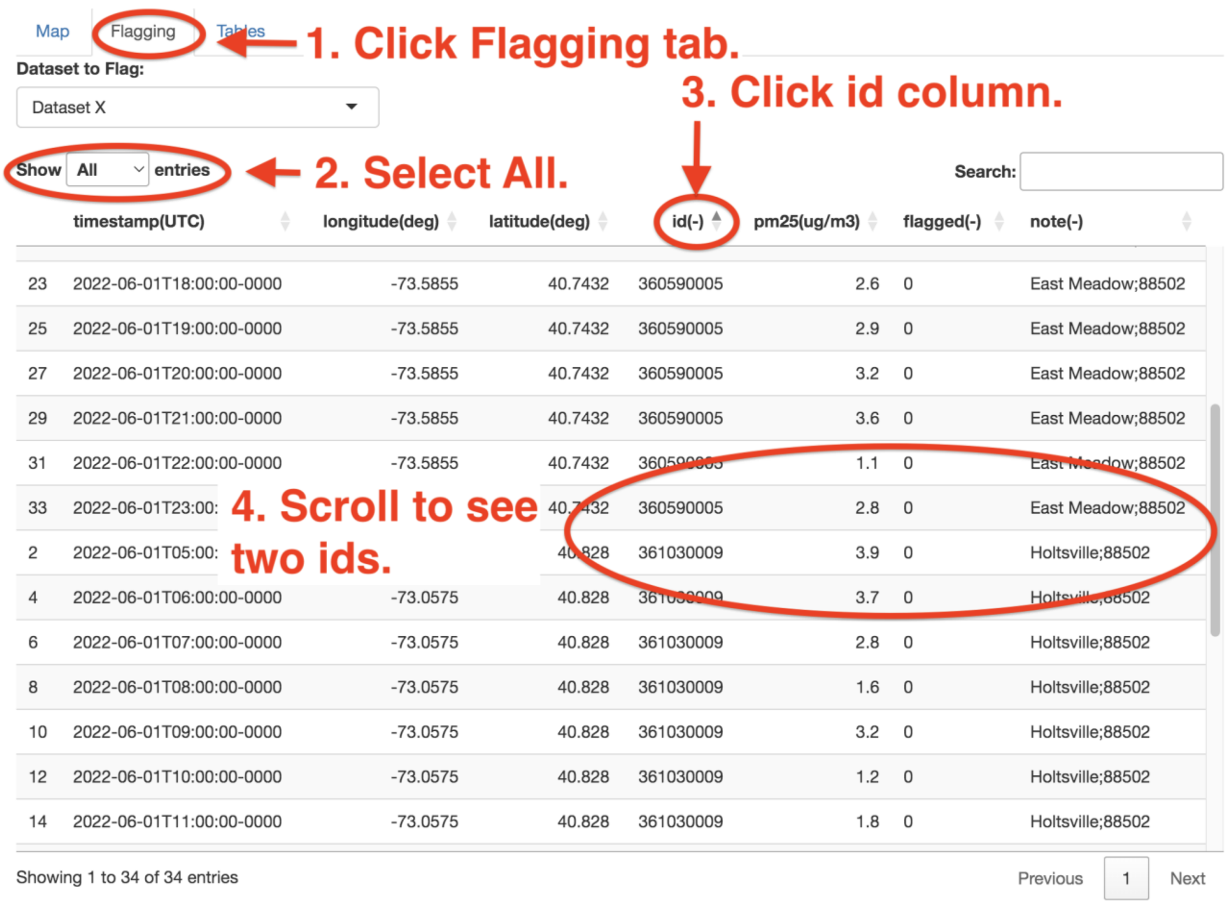
13.9 To see that the loaded data now only contains the remaining two sites,

a. **Click the Flagging tab.**

b. **Select All in the Show menu**.

c. **Click the id column** to sort the data by site id.

d. **Scroll down** to see that there are only these two sites remaining.



13.10 When you are done filtering data, **select None** in the menu.
This will avoid having ASNAT recompute the list of places when zooming and panning the map since this is an expensive computation. (For example, if you were to zoom out to view much of the US, there would be thousands of county polygon-rectangle intersection calculations done - each with hundreds of vertices!)

13.11 You can **click the Save Data button** to write filtered datasets to files.
Then **click the Quit button**.

Demo 14: Non-UTC timestamp plot labels

14.1 UTC Rules. All data in `ASNAT` are timestamped using the **ISO-8601** format **`yyyy-mm-ddThh:mm:ss-0000`**. The last part, **`-0000`** (-/+hhmm) indicates zero hours and minutes offset from **UTC** (Universal Time Coordinated). This is true of data read either from a webservice, or files. This is necessary to ensure interoperability - both within ANSAT and comparing these saved data (and plots) to other external datasets (and plots). However, ASNAT has a menu to allow selection of non-UTC timestamps *when displaying hourly data*. *This only affects the map display and timeseries plots and only if the data are hourly.* The data retain their UTC timestamp both within ASNAT and when saved. When reading daily aggregated data, the data was aggregated from UTC hours 0-23 so it would not be correct to label it otherwise. The same is true when viewing on-the-fly aggregated hourly data such as when checking the button 'Show mean values over all timesteps'.

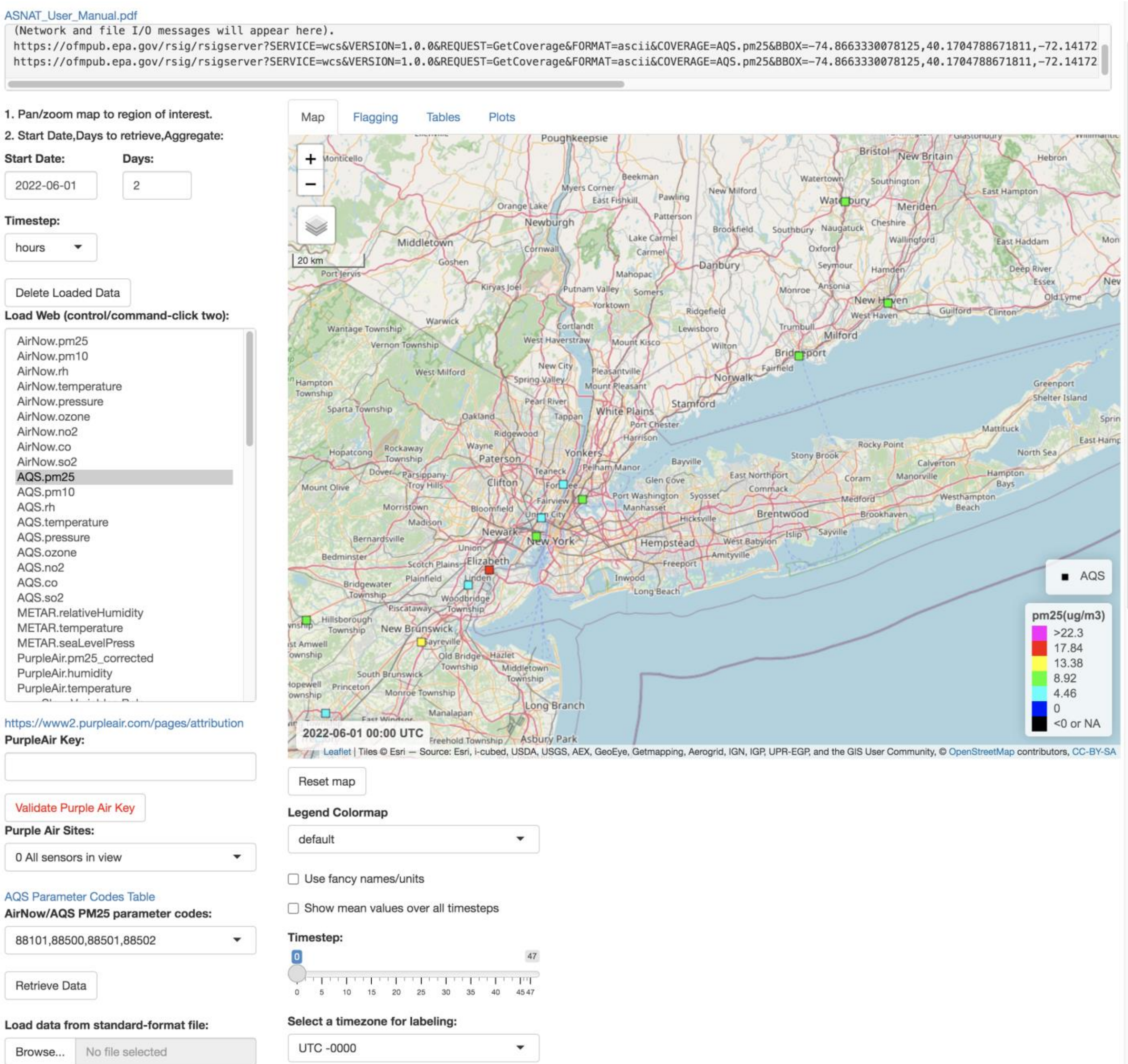
14.2 Caution. But it is recommended to leave the menu in the default selection: **UTC -0000** as non-UTC timezones are problematic: their boundaries change over the years with changing/conflicting policies of various nations so there is no static nor time-dependent definitive globally agreed upon set of timezone names/offsets/boundaries. Even within the US, as of the time of this writing, there are some areas that do not switch between 'local standard' time and *local daylight savings time.* For example, Arizona uses MST -0700 Mountain Standard Time and does not switch to/from MDT -0600 Mountain Daylight Time *except for areas of the Navajo Nation* (that cross several states). And there are proposals to change such switching policies nationally. In short, there is no correct set of timezone names/offsets/boundaries and labelling images/plots with a non-UTC timestamp over an area which can use multiple non-UTC timezones is misleading.

14.3 Compromise. ASNAT provides this feature to more conveniently relate the data to diurnal cycles. ASNAT uses the MazamaSpatialUtils::SimpleTimezones dataset
https://rdrr.io/cran/MazamaSpatialUtils/f/vignettes/MazamaSpatialUtils.Rmd
It contains (incomplete) names, offsets and coarse polygon boundaries for global timezones. Note *multiple timezones can have the same hours offset from UTC but different locally used names and different boundaries*. For example, AST -0400 Atlantic Standard Time (used in parts of Canada and the Caribbean) vs EST -0400 Eastern Standard Time (used in some US states). For 'more complete' names and offsets, ASNAT uses the table of timezones from
https://worldtimeconvert.com/timezone.

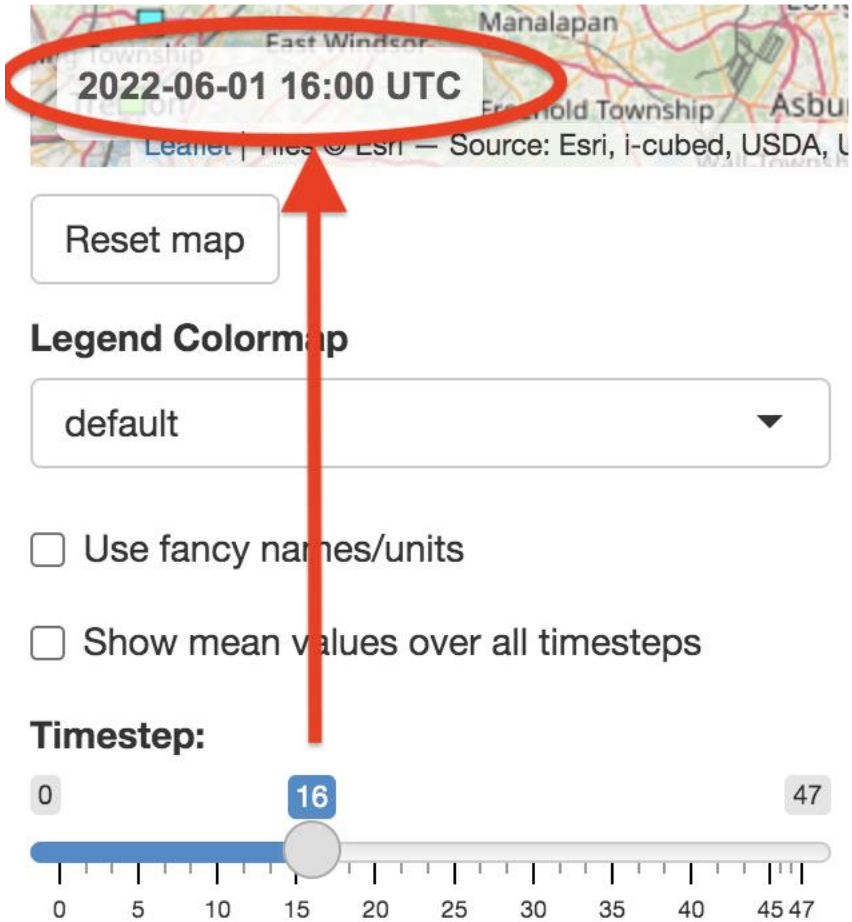
When panning and zooming the map the Timezone Menu is updated to show the sorted name and offset of every timezone intersecting the view area (plus UTC -0000 at the top always). It uses the polygon boundaries in SimpleTimezones to compute the intersection with the viewing area. But when the names in that dataset are incomplete (just hour offsets), it uses the hour offset to match with names in the more complete table referenced above. The result is the menu may contain some 'false positives' - timezone names that might not be locally used in the viewing area but happen to have the same offset (including daylight savings time offsets). Here the user must choose a timezone name and accept the responsibility for any misleading images/plots.

14.4 Demo. To demonstrate this feature, quit and restart ASNAT and make the following selections:
Reset map
Start Date: **2022-06-01** Days: **2**
Timestep: **hours**
Delete Loaded Data
Load Web:
  AQS.pm25
Retrieve Data
Uncheck the 'Show mean values over all timesteps' button.

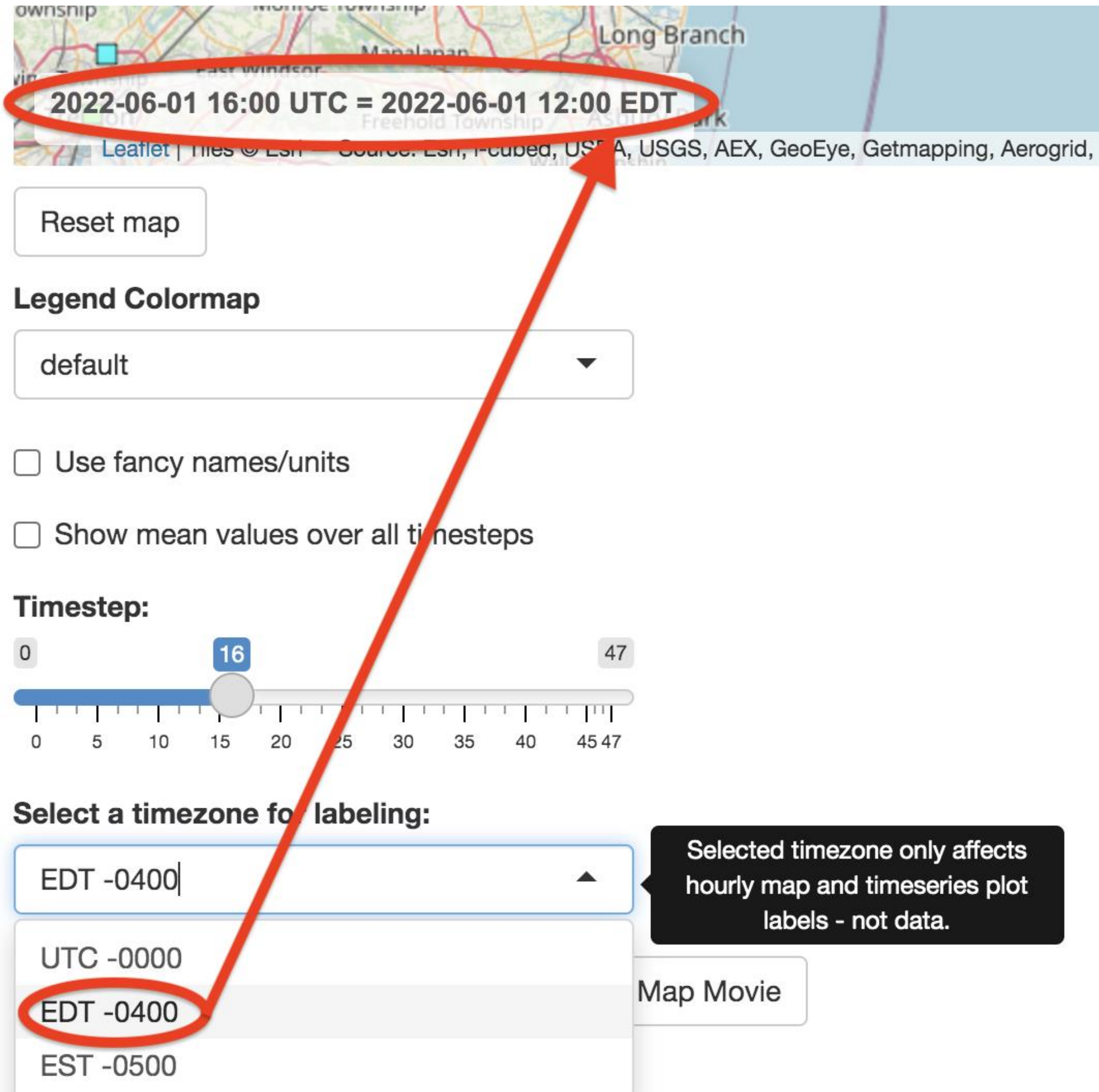
ASNAT should appear as shown in the next image.


ASNAT_User_Manual.pdf
(Network and file I/O messages will appear here).
https://ofmpub.epa.gov/rsig/rsigserver?SERVICE=wcs&VERSION=1.0.0&REQUEST=GetCoverage&FORMAT=ascii&COVERAGE=AQS.pm25&BBOX=-74.8663330078125,40.1704788671811,-72.14172
https://ofmpub.epa.gov/rsig/rsigserver?SERVICE=wcs&VERSION=1.0.0&REQUEST=GetCoverage&FORMAT=ascii&COVERAGE=AQS.pm25&BBOX=-74.8663330078125,40.1704788671811,-72.14172
1. Pan/zoom map to region of interest.
2. Start Date,Days to retrieve,Aggregate:
Start Date:
2022-06-01
Days:
2
Timestep:
hours
Delete Loaded Data
Load Web (control/command-click two):
AirNow.pm25
AirNow.pm10
AirNow.rh
AirNow.temperature
AirNow.pressure
AirNow.ozone
AirNow.no2
AirNow.co
AirNow.so2
AQS.pm25
AQS.pm10
AQS.rh
AQS.temperature
AQS.pressure
AQS.ozone
AQS.no2
AQS.co
AQS.so2
METAR.relativeHumidity
METAR.temperature
METAR.seaLevelPress
PurpleAir.pm25_corrected
PurpleAir.humidity
PurpleAir.temperature
https://www2.purpleair.com/pages/attribution
PurpleAir Key:
Validate Purple Air Key
Purple Air Sites:
0 All sensors in view
AQS Parameter Codes Table
AirNow/AQS PM25 parameter codes:
88101,88500,88501,88502
Retrieve Data
Load data from standard-format file:
Browse...
No file selected
Map
Flagging
Tables
Plots
20 km
AQS
pm25(ug/m3)
>22.3
17.84
13.38
8.92
4.46
0
<0 or NA
2022-06-01 00:00 UTC
Leaflet | Tiles © Esri — Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community, © OpenStreetMap contributors, CC-BY-SA
Reset map
Legend Colormap
default
Use fancy names/units
Show mean values over all timesteps
Timestep:
0
47
Select a timezone for labeling:
UTC -0000

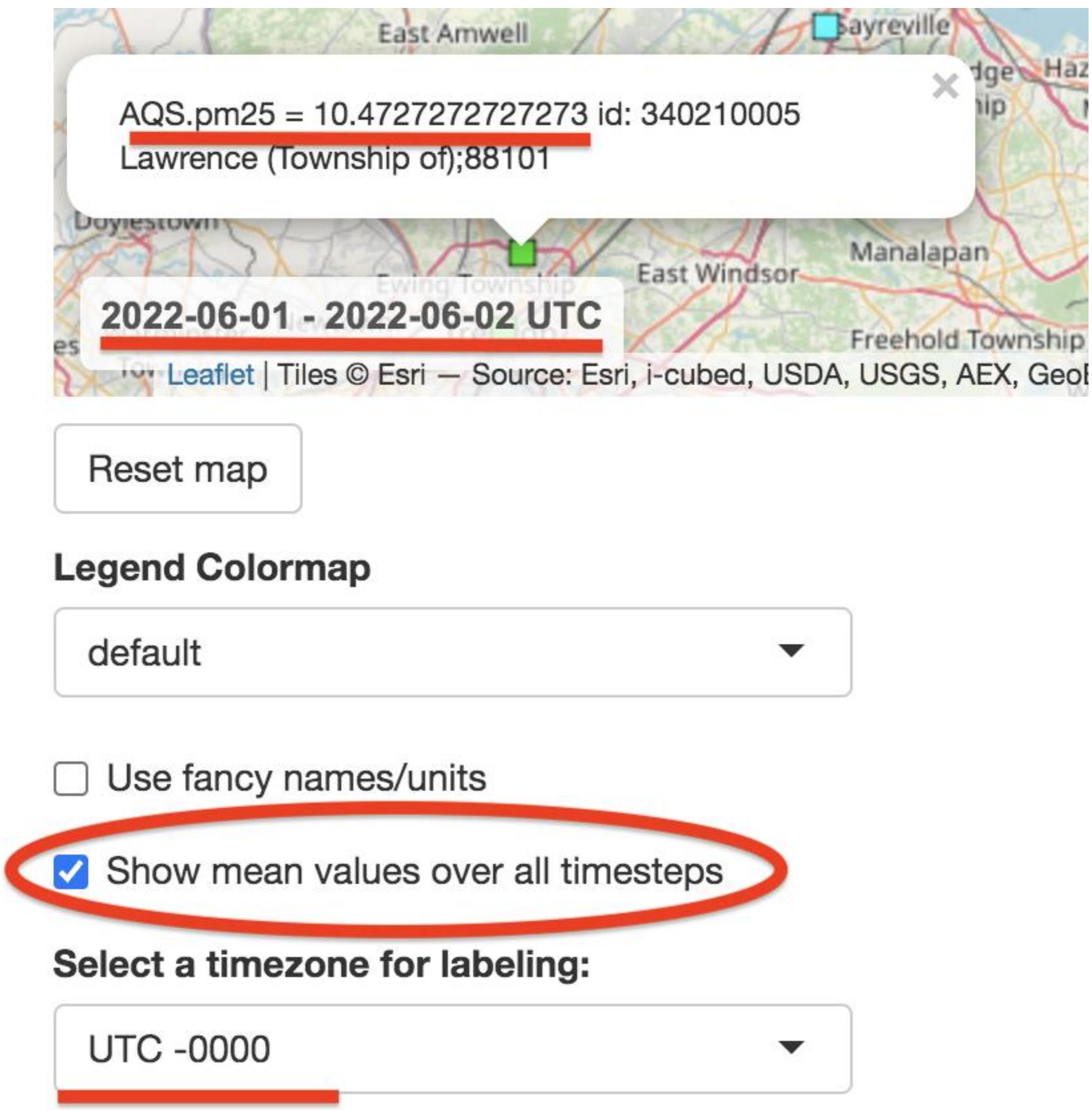
14.5 Note that the lower-left corner of the map has a timestamp of the date and time: **hh:mm UTC.**
Drag the Timestep slider to see the timestamp (and data) change to that hour UTC.



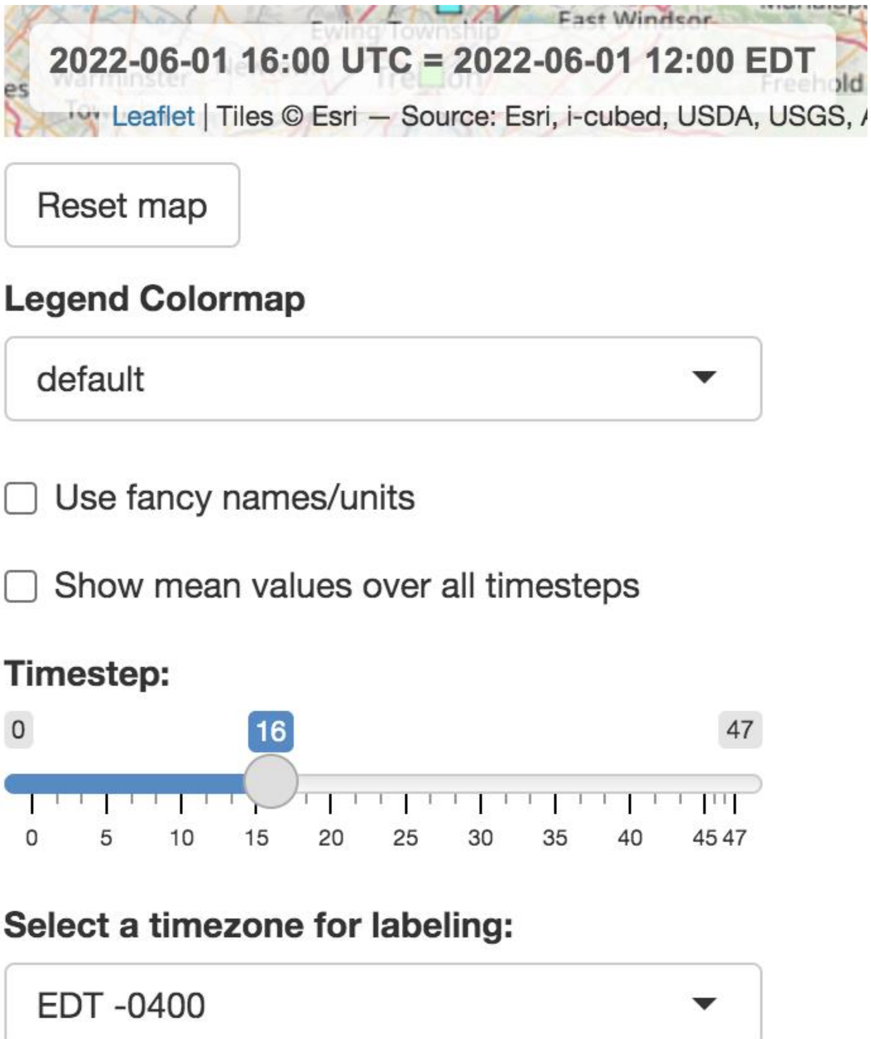
14.6 Note the menu for selecting a timezone for labeling.
It always lists UTC -0000 first followed by a sorted list of timezones within the viewing area.
Select EDT -0400.
Notice how the map timestamp now *appends* the equivalent timestamp of the selected non-UTC timezone. It retains the UTC timestamp first for interoperability and matching clicked site values to the table of the dataset shown in the Flagging tab. So *selecting a non-UTC timestamp does not change the data or tables*, just the map timestamp label and, as shown later, the Timeseries plot labels.



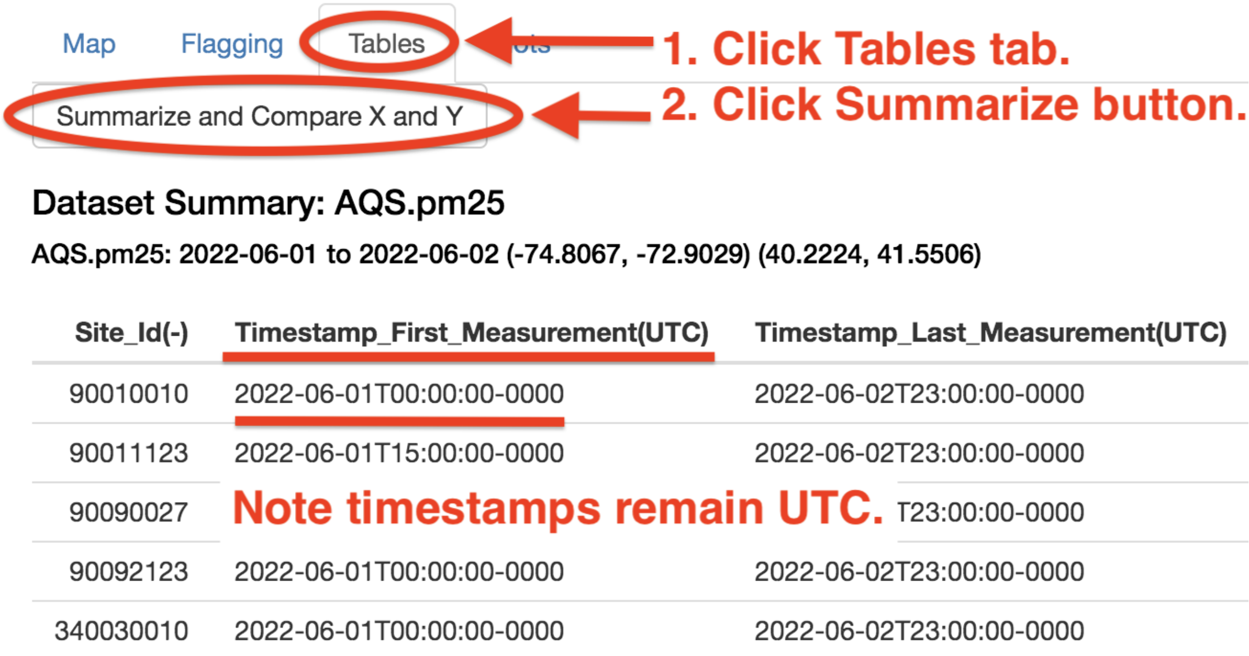
14.7 Click the checkbox Show mean values over all timesteps.
This computes and displays the mean value of each site measure over all timesteps (hours in this example).
Therefore the Timestep slider becomes hidden and the map timestamp label shows the date range as UTC days (because the data consists of *24 UTC hours per day*).
The site glyphs are also colored by the mean and, when clicked, show the mean.
The Timezone menu also reverts back to UTC -0000 because that matches the map timestamp label.



14.8 Now **uncheck Show mean values over all timesteps.**
The Timestep slider and Timezone menu reappear.
In the **Timezone menu, Select EDT -0400** again.
ASNAT appears as before:



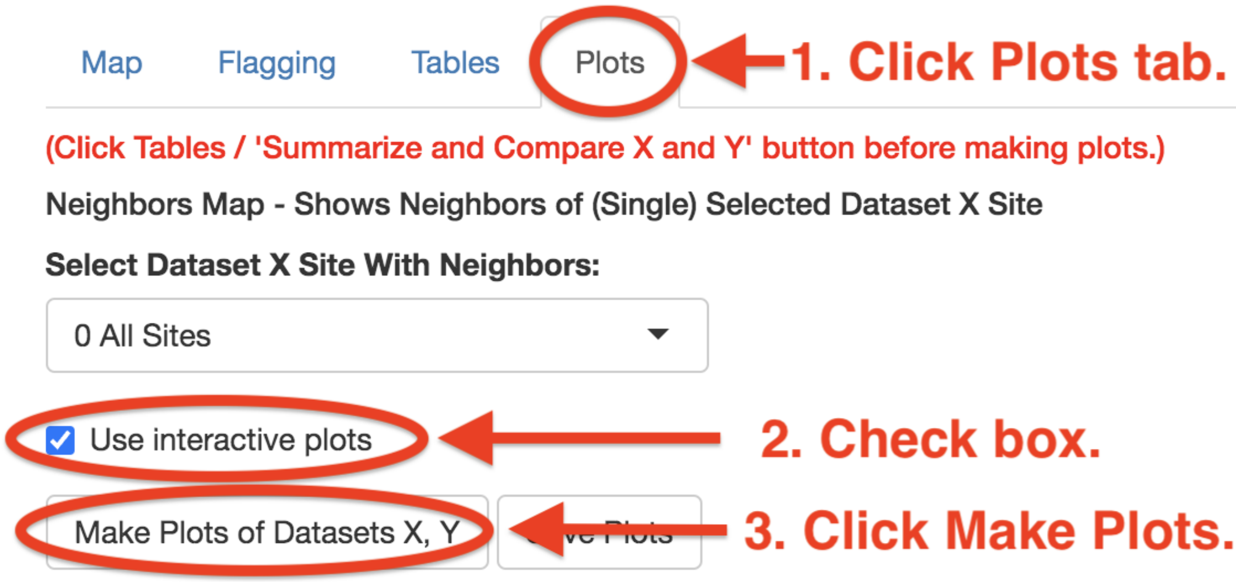
14.9 Now we'll see that the Timeseries plots can display the selected timezone hours.
a. **Click Tables tab.**
b. **Click Summarize button**



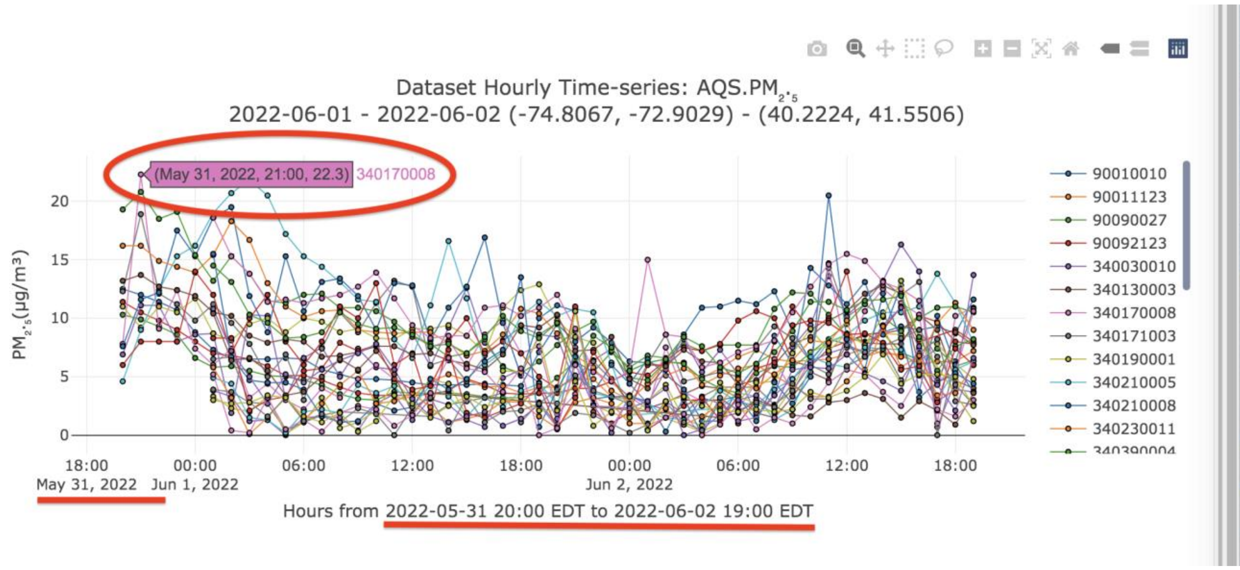
Site_Id(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)
90010010	2022-06-01T00:00:00-0000	2022-06-02T23:00:00-0000
90011123	2022-06-01T15:00:00-0000	2022-06-02T23:00:00-0000
90090027	[illegible]	T23:00:00-0000
90092123	2022-06-01T00:00:00-0000	2022-06-02T23:00:00-0000
340030010	2022-06-01T00:00:00-0000	2022-06-02T23:00:00-0000

The data retains their UTC timestamps.

14.10 Now we'll see that the Timeseries plots can display the selected timezone hours.
a. **Click Plots tab.**
b. **Click to check Use interactive plots.**



14.11 Scroll down to see **Time-series plot.**



Note the x-axis labels show EDT hours.

14.12 Local Daily Data.

a. So far all of the data shown in ASNAT has been either UTC hourly average or UTC daily average (i.e., mean of 24 UTC hours). This is what is most available from webservices and is most interoperable and comparable with other datasets. However, there are a few datasets available that are *local daily aggregates* -i.e., mean/max of 24 *local* hours. These are measured or pre-computed for regulatory purposes such as tracking non-attainment/exceedance of pollutants and human exposure.

ASNAT has a few of these as explained below:

AQS.ozone_8hour_average: Mean of 17 sliding/overlapping 'windows' of surface measured ozone (ppb) each over 8 consecutive local hours. The first window covers local hours 0-7, the second window covers local hours 1-8, the last (17th) window covers local hours 16-23. These 17 window averages are then averaged to this (local) daily measure.

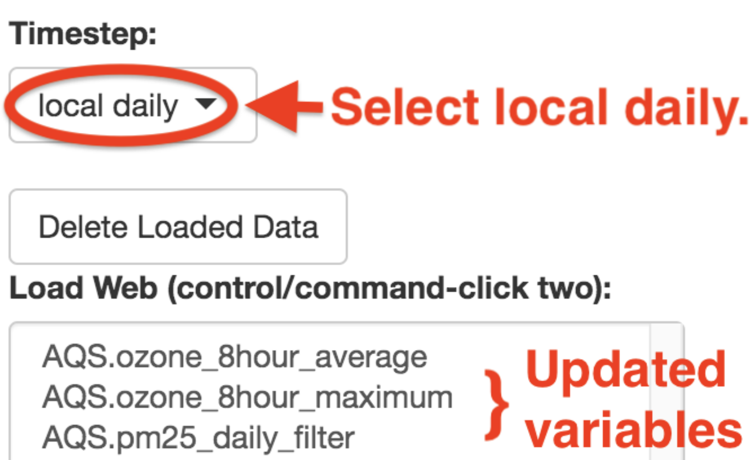
AQS.ozone_8hour_maximum: Like above except the result is the maximum of these 17 averaged windows.

AQS.pm25_daily_filter: This is local daily surface measured particulate matter (aerosols) 2.5 microns or smaller in diameter (units ug/m3) derived from a device filter examined daily.

b. **Caution**. Given that these measures are an average over 24 *local hours*, they are not comparable with sites in a different timezone because each timezone consists of a different set of (UTC) hours. So even displaying such data on a map area that includes multiple timezones is misleading. But it is useful, so ASNAT allows it.

c. **Demo**. To see these, **click the Timestep menu and select *local daily***.

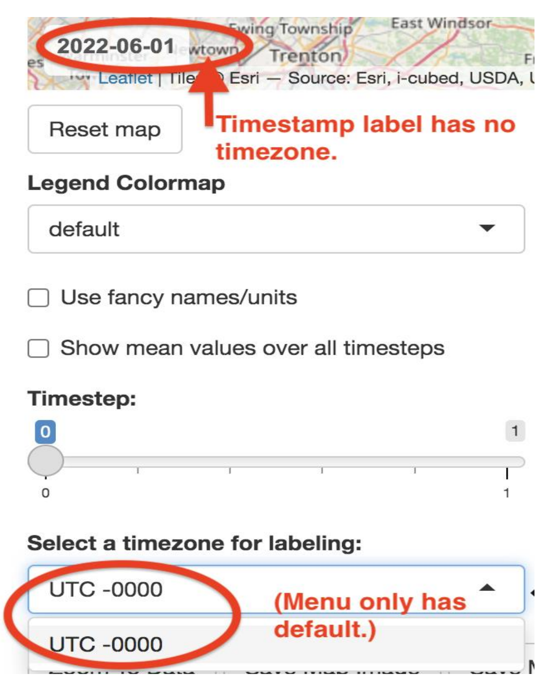
This will delete all loaded data (since it is not comparable) and update the Load Web list of variables as shown.



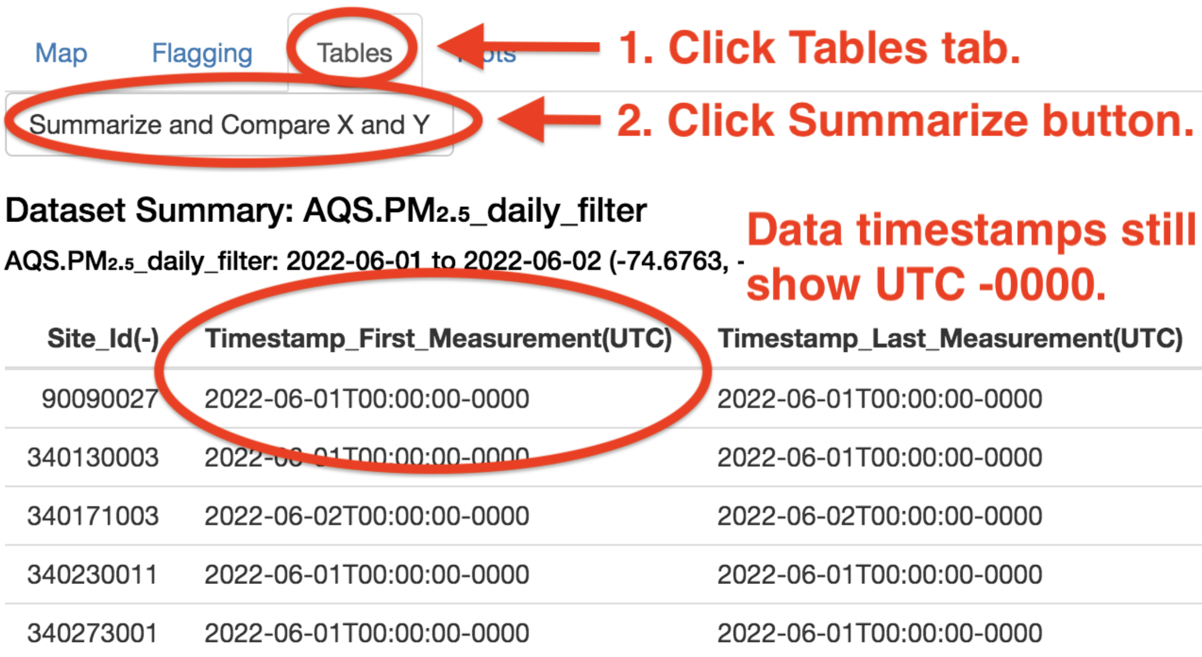
14.13 a. Click AQS.pm25_daily_filter.

b. Then **click the Retrieve Data button.**

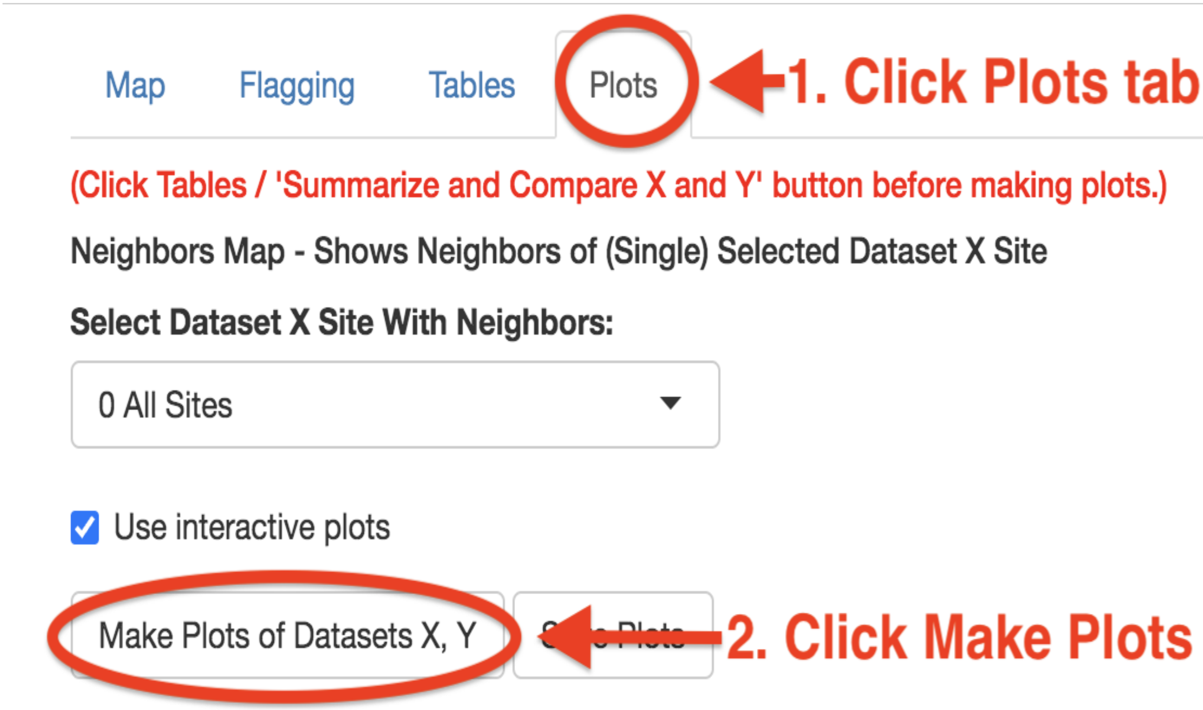
c. Note the lack of timezone in the timestamp.
This makes sense because each site could be in a different/unknown timezone. The Timezone menu only shows the default value (UTC -0000).



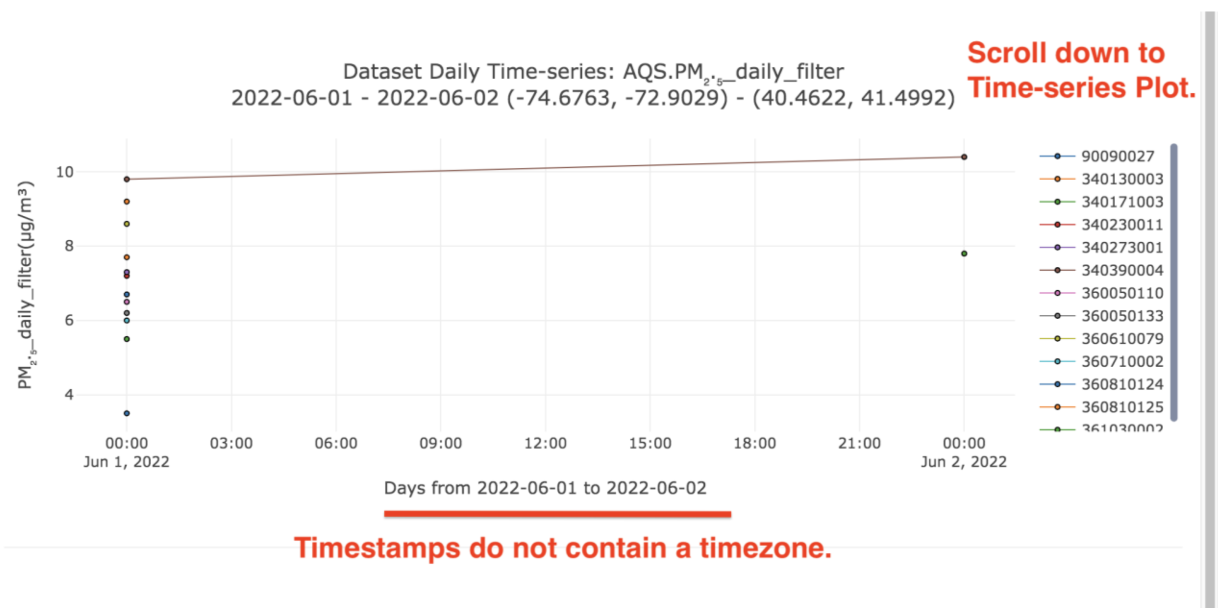
14.14 Click the **Tables** tab then **Click** the **Summarize** button**.**
Note that the (original and computed) data timestamps still have -0000 offset.



14.15a. Click the **Plots** tab then **Click** the **Make Plots** button**.**



b. **Scroll down** to the **Time-series Plot**.
As with the map timestamp, the x-axis *timestamp labels do not include a timezone*.



c. **Click Save Data** (to save both the original data and computed tables).
Then **click the Quit button.**

Specification of Standard Format Sensor Files:

`ASNAT` can load data from services, files in *standard format* and

Purple Air raw sensor files.
Standard format files can be loaded using the
Load data from a standard-format file Browse button.
This can be tested by saving retrieved data to files in tsv format
(tab-separated is preferred but comma-separated also works)
then later using the `ASNAT` browser to load them back into `ASNAT`.
The easiest way to learn the standard format is to simply use ASNAT to load data from a webservice (e.g., Airnow.pm25) and then click the Save Data button and examine the saved tsv (or csv) file.

The ***standard-format*** files must conform to the following description.
The first line is a tab-delimited or comma-delimited one-line header:

```
timestamp(UTC)   longitude(deg)   latitude(deg) id(-)
   pm25(ug/m3)   ...note(-)
```

The first 4 columns and the last column must exactly match the above.
There are no spaces in the header line.
If tab-delimited, there are no commas in the header line.
(Tab-delimited is preferred for human readability.)
The columns after `id(-)` are numeric variables with units,
for example, `pm25(ug/m3)` or `ozone(ppm).`
Variable names can only contain letters, digits, and internal underscores.
Every column contains parenthesized units.
Units cannot contain spaces. Shorter is better, e.g., `(ug/m3).`
Use '-' for non-dimensional values such `as id(-)` or `aod(-).`
The `id` is a unique integer id in [1, 2147483647] for each sensor in the file.
The remaining lines look like:

```
2022-06-01T00:00:00-0000   -74.05 40.72  202438586
   35.43  ...20220601.csv;mac_address=98:cd:ac:10:f7:ba
```

The data lines are *sorted by timestamp and then sensor id.*
Timestamps are in **UTC** using the **ISO-8601** format shown above.
Other time formats such as local time zones, t, Z, +00:00, etc., are invalid -
only **`yyyy-mm-ddThh:mm:ss-0000`** format is allowed.

A site can only have at most one row for each hour since all data to be loaded into ASNAT is aggregated to the hour or day.

The last column is called `note(-)` and can contain up to 80 characters of arbitrary useful information.
It cannot contain commas or tabs or control characters.
It is best if it does not contain whitespace or quotes or problematic text.
There is an optional `elevation(m)` column after column `latitude(deg)`.
There is an optional `count(-)` column after the `id(-)` column that contains the integer number of sample values that were aggregated in the hourly or daily value of the (user-designated) 'primary variable'.
The ***units must be SI*** so values such as temperature_F becomes `temperature(C)` and pressure in millibars becomes `pressure(hPa)`.
All of the variable columns appear after `id(-)` and `count(-)` columns and before the non-numeric column `note(-)`.
The last variable column (before flagged or note) will be the default selected variable in `ASNAT`.
The user can use the `ASNAT` menus to select any one of the variable columns.

For questions about `ASNAT` contact us using the email shown at the top of this document and displayed in the application.

Airnow/AQS Parameter codes:

https://aqs.epa.gov/aqsweb/documents/codetables/parameters.html
Here are the parameter codes used by the rsigserver web service (used by ASNAT):
pm25: 88101, 88500, 88501, 88502
pm10: 81102
rh: 62201, 62202, 68110
temperature: 62101
pressure: 64101
ozone: 44201

AirNow data are available near real-time.
AQS data are available on a 90-day delay because states have 90 days to report data.
Some data are often reported sooner than 90 days.
AQS data is quality-reviewed (and revised) more than AirNow data so prefer AQS data and, only if not available, try AirNow.
https://aqs.epa.gov/aqsweb/documents/about_aqs_data.html#timeliness-of-data

AirNow data are also QC-flagged and rsigserver webservice (used by ASNAT) filters-out data with a QC flag greater than 4.
https://www.airnowtech.org/resources/AIRNow-I_AQCSV-Final.pdf
Table 7, Page 28.
0 = Valid
1 = Adjusted
2 = Averaged
3 = interpolated
4 = Estimated
5 = Suspect
6 = Suspect (audit failure)
7 = Insufficient data
8 = Missing
9 = Invalid

Flagging Codes:

Neighbor-based Flags

- Flag 80 - Flags data points where the absolute difference between Dataset X and Y exceeds a specified threshold
- Flag 81 - Flags data points where the percent difference between Dataset X and Y exceeds a specified threshold
- Flag 82 - Flags data points where the R-squared value between Dataset X and Y is below a specified threshold

Additional Flags

- Flag 83 - Flags data points where values remain constant for more than a specified number of consecutive measurements
- Flag 84 - Flags data points where data is missing for more than a specified number of consecutive timesteps
- Flag 85 - Flags statistical outliers based on standard deviation from the mean
- Flag 86 - Flags outliers identified by the Hampel filter method (https://mazamascience.github.io/AirSensor/articles/outlier_detection.html)

Invalid Values Flags

- Flag 60 - Flags invalid negative values in pollutant measurements (O3, NO2, CO, etc.)

Cross-field Validation Flags

- Flag 65 - Flags temporal inconsistencies (e.g., end timestamps before start timestamps)

Suspicious Changes Flags

- Flag 70 - Flags sudden spikes in pollutant levels
- Flag 71 - Flags sudden drops in pollutant levels
- Flag 72 - Flags ozone measurements that deviate from typical daily patterns
- Flag 73 - Flags PM measurements that deviate from typical daily patterns

Redundancy check

- Flag 90 - Flags duplicate measurements at the same location and timestamp

Invalid formatting and units

- Flag 95 - Flags timestamps that don't conform to ISO 8601 format (YYYY-MM-DDTHH:MM:SS-ZZZZ)